

Zcim Project
Digital Research CP/M® 2.2
Operating System Manual

version 2025-07-31

Copyright

Copyright © 1976, 1977, 1978, 1979, 1982, and 1983 by Digital Research. All rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of

~~Digital Research, Post Office Box 579, Pacific Grove, California 93950.~~

~~<http://www.lineo.com>~~

DRDOS, Inc [Bryan Sparks]

Copyright © 2025 by James Burlingame.

License

The original Digital Research Inc assets license is available at
<http://cpm.z80.de/license.html>.

This *Zcim Project edition* is licensed under the
Creative Commons Attribution 4.0 International license. **CC BY 4.0**.

Trademarks

CP/M and CP/NET are registered trademarks of Digital Research. ASM, DESPOOL, DDT, LINK-80, MAC, MP/M, PL/I-80, and SID are trademarks of Digital Research. Intel is a registered trademark of Intel Corporation. TI Silent 700 is a trademark of Texas Instruments Incorporated. Zilog and Z80 are registered trademarks of Zilog, Inc.

Printing

The *CP/M Operating System Manual* **was** printed in the United States of America.

First Edition: 1976

Second Edition: July 1982

Third Edition: September 1983

Zcim Project edition: version 2025-07-31

Contents

1. CP/M Features and Facilities	1
1.1. Introduction	1
1.2. Functional Description	3
1.2.1. General Command Structure	3
1.2.2. File References	3
1.3. Switching Disks	5
1.4. Built-in Commands	6
1.4.1. ERA Command	6
1.4.2. DIR Command	7
1.4.3. REN Command	7
1.4.4. SAVE Command	8
1.4.5. TYPE Command	9
1.4.6. USER Command	9
1.5. Line Editing and Output Control	10
1.6. Transient Commands	11
1.6.1. STAT Command	12
1.6.2. ASM Command	19
1.6.3. LOAD Command	19
1.6.4. PIP Command	20
1.6.5. ED Command	29
1.6.6. SYSGEN Command	31
1.6.7. SUBMIT Command	32
1.6.8. DUMP Command	35
1.6.9. MOVCPM Command	35
1.7. BDOS Error Messages	37
1.8. Operation of CP/M on the MDS	38
2. The CP/M Editor	40
2.1. Introduction to ED	40
2.1.1. ED Operation	41
2.1.2. Text Transfer Commands	42
2.1.3. Memory Buffer Organization	43
2.1.4. Line Numbers and ED Start-up	44
2.1.5. Memory Buffer Operation	45
2.1.6. Command Strings	46
2.1.7. Text Search and Alteration	49

2.1.8.	Source Libraries	52
2.1.9.	Repetitive Command Execution	53
2.2.	ED Error Conditions	54
2.3.	Control Characters and Commands	55
3.	CP/M Assembler	58
3.1.	Introduction	58
3.2.	Program Format	59
3.3.	Forming the Operand	60
3.3.1.	Labels	61
3.3.2.	Numeric Constants	61
3.3.3.	Reserved Words	62
3.3.4.	String Constants	63
3.3.5.	Arithmetic and Logical Operators	63
3.3.6.	Precedence of Operators	65
3.4.	Assembler Directives	65
3.4.1.	The ORG Directive	66
3.4.2.	The END Directive	67
3.4.3.	The EQU Directive	67
3.4.4.	The SET Directive	68
3.4.5.	The IF and ENDIF Directive	68
3.4.6.	The DB Directive	69
3.4.7.	The DW Directive	69
3.4.8.	The DS Directive	70
3.5.	Operation Codes	70
3.5.1.	Jumps, Calls, and Returns	70
3.5.2.	Immediate Operand Instructions	72
3.5.3.	Increment and Decrement Instructions	73
3.5.4.	Data Movement Instructions	73
3.5.5.	Arithmetic Logic Unit	74
3.5.6.	Control Instructions	76
3.6.	Error Messages	76
3.7.	A Sample Session	78
4.	CP/M Dynamic Debugging Tool	87
4.1.	Introduction	87
4.2.	DDT Commands	89
4.2.1.	The A (Assembly) Command	89

4.2.2.	The D (Display) Command	90
4.2.3.	The F (Fill) Command	90
4.2.4.	The G (Go) Command	91
4.2.5.	The I (Input) Command	92
4.2.6.	The L (List) Command	92
4.2.7.	The M (Move) Command	92
4.2.8.	The R (Read) Command	93
4.2.9.	The S (Set) Command	93
4.2.10.	The T (Trace) Command	94
4.2.11.	The U (Untrace) Command	94
4.2.12.	The X (Examine) Command	95
4.3.	Implementation Notes	96
4.4.	A Sample Program	96
5.	CP/M 2 System Interface	110
5.1.	Introduction	110
5.2.	Operating System Call Conventions	112
5.2.1.	System Reset	118
5.2.2.	Console Input	118
5.2.3.	Console Output	118
5.2.4.	Reader Input	118
5.2.5.	Punch Output	119
5.2.6.	List Output	119
5.2.7.	Direct Console I/O	119
5.2.8.	Get IOBYTE	120
5.2.9.	Set IOBYTE	120
5.2.10.	Print String	120
5.2.11.	Read Console Buffer	120
5.2.12.	Get Console Status	121
5.2.13.	Return Version Number	122
5.2.14.	Reset Disk System	122
5.2.15.	Select Disk	122
5.2.16.	Open File	123
5.2.17.	Close File	123
5.2.18.	Search for First	124
5.2.19.	Search for Next	124
5.2.20.	Delete File	125

5.2.21. Read Sequential	125
5.2.22. Write Sequential	125
5.2.23. Make File	126
5.2.24. Rename File	126
5.2.25. Return Login Vector	127
5.2.26. Return Current Disk	127
5.2.27. Set DMA Address	127
5.2.28. Get Addr(ALLOC)	128
5.2.29. Write Protect Disk	128
5.2.30. Get R/O Vector	128
5.2.31. Set File Attributes	129
5.2.32. Get Addr(Disk Parameter Block)	129
5.2.33. Set/Get User Code	129
5.2.34. Read Random	130
5.2.35. Write Random	131
5.2.36. Compute File Size	132
5.2.37. Set Random Record	132
5.2.38. Reset Drive	133
5.2.39. Write Random with Zero Fill	133
5.3. A Sample File-to-File Copy Program	133
5.4. A Sample File Dump Utility	137
5.5. A Sample Random Access Program	142
5.6. System Function Summary	149
6. CP/M Alteration	152
6.1. Introduction	152
6.2. First-level System Regeneration	153
6.3. Second-level System Regeneration	156
6.4. Sample GETSYS and PUTSYS Program	159
6.5. Disk Organization	161
6.6. The BIOS Entry Points	163
6.7. A Sample BIOS	171
6.8. A Sample Cold Start Loader	171
6.9. Reserved Locations in Page Zero	172
6.10. Disk Parameter Tables	173
6.11. The DISKDEF Macro Library	178
6.12. Sector Blocking and Deblocking	182

A. The MDS Basic I/O System (BIOS)	184
B. A Skeletal CBIOS	197
C. A Skeletal GETSYS/PUTSYS Program	204
D. The MDS-800 Cold Start Loader for CP/M 2	207
E. A Skeletal Cold Start Loader	211
F. CP/M Disk Definition Library	214
G. Blocking and Deblocking Algorithms	219
H. Glossary	228
I. CP/M Error Messages	243
Notes on the Zcim Project Edition	255
License	255

List of Tables

Table 1-1	Line-editing Control Characters	10
Table 1-2	CP/M Transient Commands	11
Table 1-3	Physical Devices	15
Table 1-4	PIP Parameters	25
Table 2-5	ED Text Transfer Commands	42
Table 2-6	Editing Commands	46
Table 2-7	ED Line-editing Controls	47
Table 2-8	Error Message Symbols	54
Table 2-9	ED Control Characters	55
Table 2-10	ED Commands	55
Table 3-11	Reserved Characters	62
Table 3-12	Arithmetic and Logical Operators	63
Table 3-13	Assembler Directives	66
Table 3-14	Jumps, Calls, and Returns	70
Table 3-15	Immediate Operand Instructions	72
Table 3-16	Increment and Decrement Instructions	73
Table 3-17	Data Movement Instructions	73
Table 3-18	Arithmetic Logic Unit Operations	74
Table 3-19	ASM Error Codes	76
Table 3-20	ASM Error Messages	76
Table 4-21	DDT Line-editing Controls	87
Table 4-22	DDT Commands	88
Table 4-23	CPU Registers	95
Table 5-24	CP/M Memory Organization	110

Table 5-25 CP/M Filetypes	114
Table 5-26 File Control Block Format	116
Table 5-27 File Control Block Fields	116
Table 5-28 Command-line Layout	117
Function 0: System Reset	118
Function 1: Console Input	118
Function 2: Console Output	118
Function 3: Reader Input	118
Function 4: Punch Output	119
Function 5: List Output	119
Function 6: Direct Console I/O	119
Function 7: Get IOBYTE	120
Function 8: Set IOBYTE	120
Function 9: Print String	120
Function 10: Read Console Buffer	120
Table 5-29 Edit Control Characters	121
Function 11: Get Console Status	121
Function 12: Return Version Number	122
Function 13: Reset Disk System	122
Function 14: Select Disk	122
Function 15: Open File	123
Function 16: Close File	123
Function 17: Search for First	124
Function 18: Search for Next	124
Function 19: Delete File	125
Function 20: Read Sequential	125
Function 21: Write Sequential	125
Function 22: Make File	126
Function 23: Rename File	126
Function 24: Return Login Vector	127
Function 25: Return Current Disk	127
Function 26: Set DMA Address	127
Function 27: Get Addr(ALLOC)	128
Function 28: Write Protect Disk	128
Function 29: Get R/O Vector	128
Function 30: Set File Attributes	129

Function 31: Get Addr(Disk Parameter Block)	129
Function 32: Set/Get User Code	129
Function 33: Read Random	130
Function 34: Write Random	131
Function 35: Compute File Size	132
Function 36: Set Random Record	132
Function 37: Reset Drive	133
Function 40: Write Random with Zero Fill	133
Table 5-30 System Function Summary	149
Table 6-31 Standard Memory Size Values	153
Table 6-32 Common Value for CP/M Systems	157
Table 6-33 CP/M Disk Sector Allocation	161
Table 6-34 IOBYTE Fields	165
Table 6-35 IOBYTE Field Values	165
Table 6-36 BIOS Entry Points	166
Table 6-37 Reserved Locations in Page Zero	172
Table 6-38 Disk Parameter Header Format	173
Table 6-39 Disk Parameter Headers	173
Table 6-40 Disk Parameter Header Table	174
Table 6-41 Disk Parameter Block Format	175
Table 6-42 BSH and BLM Values	176
Table 6-43 EXM Values	176
Table 6-44 AL0 and AL1	176
Table 6-45 BLS Tabulation	177
Table 9-46 CP/M Error Messages	243

1. CP/M Features and Facilities

1.1. Introduction

CP/M is a monitor control program for microcomputer system development that uses floppy disks or Winchester hard disks for backup storage. Using a computer system based on the Intel 8080 microcomputer, CP/M provides an environment for program construction, storage, and editing, along with assembly and program checkout facilities. CP/M can be easily altered to execute with any computer configuration that uses a Zilog Z80 or an Intel 8080 Central Processing Unit (CPU) and has at least 20K bytes of main memory with up to 16 disk drives. A detailed discussion of the modifications required for any particular hardware environment is given in Section 6. Although the standard Digital Research version operates on a single-density Intel MDS 800, several different hardware manufacturers support their own input-output (I/O) drivers for CP/M.

The CP/M monitor provides rapid access to programs through a comprehensive file management package. The file subsystem supports a named file structure, allowing dynamic allocation of file space as well as sequential and random file access. Using this file system, a large number of programs can be stored in both source and machine executable form.

CP/M 2 is a high-performance, single console operating system that uses table-driven techniques to allow field reconfiguration to match a wide variety of disk capacities. All fundamental file restrictions are removed, maintaining upward compatibility from previous versions of release 1.

Features of CP/M 2 include field specification of one to sixteen logical drives, each containing up to eight megabytes. Any particular file can reach the full drive size with the capability of expanding to thirty-two megabytes in future releases. The directory size can be field-configured to contain any reasonable number of entries, and each file is optionally tagged with Read-Only and system attributes. Users of CP/M 2 are physically separated by user numbers, with facilities for file copy operations from one user area to another. Powerful relative-record random access functions are present in CP/M 2 that provide direct access to any of the 65536 records of an eight-megabyte file.

CP/M also supports ED, a powerful context editor, ASM, an Intel-compatible assembler, and DDT, debugger subsystems. Optional software includes a powerful Intel-compatible macro assembler, symbolic debugger, along with various high-level languages. When coupled with CP/M's Console Command Processor (CCP), the resulting facilities equal or exceed similar large computer facilities.

CP/M is logically divided into several distinct parts:

BIOS (Basic I/O System), hardware-dependent

BDOS (Basic Disk Operating System)

CCP (Console Command Processor)

TPA (Transient Program Area)

The BIOS provides the primitive operations necessary to access the disk drives and to interface standard peripherals: teletype, CRT, paper tape reader/punch, and user-defined peripherals. You can tailor peripherals for any particular hardware environment by patching this portion of CP/M. The BDOS provides disk management by controlling one or more disk drives containing independent file directories. The BDOS implements disk allocation strategies that provide fully dynamic file construction while minimizing head movement across the disk during access. The BDOS has entry points that include the following primitive operations, which the program accesses:

SEARCH looks for a particular disk file by name.

OPEN opens a file for further operations.

CLOSE closes a file after processing.

RENAME changes the name of a particular file.

READ reads a record from a particular file.

WRITE writes a record to a particular file.

SELECT selects a particular disk drive for further operations.

The CCP provides a symbolic interface between your console and the remainder of the CP/M system. The CCP reads the console device and processes commands, which include listing the file directory, printing the contents of files, and controlling the operation of transient programs, such as assemblers, editors, and debuggers. The standard commands that are available in the CCP are listed in Section 1.2.1.

The last segment of CP/M is the area called the Transient Program Area (TPA). The TPA holds programs that are loaded from the disk under command of the CCP. During program editing, for example, the TPA holds the CP/M text editor machine code and data areas. Similarly, programs created under CP/M can be checked out by loading and executing these programs in the TPA.

Any or all of the CP/M component subsystems can be overlaid by an executing program. That is, once a user's program is loaded into the TPA, the CCP, BDOS, and BIOS areas can be used as the program's data area. A bootstrap loader is programmatically accessible whenever the BIOS portion is not overlaid; thus, the user program need only branch to the bootstrap loader at the end of execution and the complete CP/M monitor is reloaded from disk.

The CP/M operating system is partitioned into distinct modules, including the BIOS portion that defines the hardware environment in which CP/M is executing. Thus, the standard system is

easily modified to any nonstandard environment by changing the peripheral drivers to handle the custom system.

1.2. Functional Description

You interact with CP/M primarily through the CCP, which reads and interprets commands entered through the console. In general, the CCP addresses one of several disks that are on-line. The standard system addresses up to sixteen different disk drives. These disk drives are labeled A through P. A disk is logged-in if the CCP is currently addressing the disk. To clearly indicate which disk is the currently logged disk, the CCP always prompts the operator with the disk name followed by the symbol >, indicating that the CCP is ready for another command. Upon initial start-up, the CP/M system is loaded from disk A, and the CCP displays the following message:

```
CP/M VER X.X
```

where x.x is the CP/M version number. All CP/M systems are initially set to operate in a 20K memory space, but can be easily reconfigured to fit any memory size on the host system (see Section 1.6.9). Following system sign-on, CP/M automatically logs in disk A, prompts you with the symbol A>, indicating that CP/M is currently addressing disk A, and waits for a command. The commands are implemented at two levels: built-in commands and transient commands.

1.2.1. General Command Structure

Built-in commands are a part of the CCP program, while transient commands are loaded into the TPA from disk and executed. The following are built-in commands:

ERA erases specified files.

DIR lists filenames in the directory.

REN renames the specified file.

SAVE saves memory contents in a file.

TYPE types the contents of a file on the logged disk.

Most of the commands reference a particular file or group of files. The form of a file reference is specified in Section 1.2.2.

1.2.2. File References

A file reference identifies a particular file or group of files on a particular disk attached to CP/M. These file references are either unambiguous (*ufn*) or ambiguous (*afn*). An unambiguous file reference uniquely identifies a single file, while an ambiguous file reference is satisfied by a number of different files.

File references consist of two parts: the primary filename and the filetype. Although the filetype is optional, it usually is generic. For example, the filetype *ASM* is used to denote that the file is an

assembly language source file, while the primary filename distinguishes each particular source file. The two names are separated by a period, as shown in the following example:

filename.typ

In this example, *filename* is the primary filename of eight characters or less, and *typ* is the filetype of no more than three characters. As mentioned above, the name

filename

is also allowed and is equivalent to a filetype consisting of three blanks. The characters used in specifying an unambiguous file reference cannot contain any of the following special characters:

< > . , ; : = ? * [] % | () / \

while all alphanumerics and remaining special characters are allowed.

An ambiguous file reference is used for directory search and pattern matching. The form of an ambiguous file reference is similar to an unambiguous reference, except the symbol ? can be interspersed throughout the primary and secondary names. In various commands throughout CP/M, the ? symbol matches any character of a filename in the ? position. Thus, the ambiguous reference

X?Z.C?M

matches the following unambiguous filenames:

XYZ.COM

and

X3Z.CAM

The wildcard character can also be used in an ambiguous file reference. The * character replaces all or part of a filename or filetype. Note that

.

equals the ambiguous file reference

?????????.???

while

filename.*

and

*.typ

are abbreviations for

filename.???

and

?????????.typ

respectively. As an example,

```
A>DIR *.*
```

is interpreted by the CCP as a command to list the names of all disk files in the directory. The following example searches only for a file by the name X.Y:

```
A>DIR X.Y
```

Similarly, the command

```
A>DIR X?Y.C?M
```

causes a search for all unambiguous filenames on the disk that satisfy this ambiguous reference.

The following file references are valid unambiguous file references:

```
X
X.Y
XYZ
XYZ.COM
GAMMA
GAMMA.1
```

As an added convenience, the programmer can generally specify the disk drive name along with the filename. In this case, the drive name is given as a letter A through P followed by a colon (:). The specified drive is then logged-in before the file operation occurs. Thus, the following are valid file references with disk name prefixes:

```
A:X.Y
P:XYZ.COM
B:XYZ
B:X.A?M
C:GAMMA
C:*.ASM
```

All alphabetic lower-case letters in file and drive names are translated to upper-case when they are processed by the CCP.

1.3. Switching Disks

The operator can switch the currently logged disk by typing the disk drive name, A through P, followed by a colon when the CCP is waiting for console input. The following sequence of prompts and commands can occur after the CP/M system is loaded from disk A:

CP/M VER 2.2

A>DIR↵ *List all the files on disk A*

A: SAMPLE ASM SAMPLE PRN

A>B:↵ *Switch to disk B*

B>DIR *.ASM↵ *List all ASM files on B*

B: DUMP ASM FILES ASM

B>A>↵ *Switch back to A*

A>

1.4. Built-in Commands

The file and device reference forms described can now be used to fully specify the structure of the built-in commands. Assume the following abbreviations in the description below:

ufn unambiguous file name reference

afn ambiguous file name reference

1.4.1. ERA Command

Syntax:

ERA *afn*

The ERA (erase) command removes files from the currently logged- in disk, for example, the disk name currently prompted by CP/M preceding the >. The files that are erased are those that satisfy the ambiguous file reference *afn*. The following examples illustrate the use of ERA:

ERA X.Y

The file named X.Y on the currently logged disk is removed from the disk directory and the space is returned.

ERA X.*

All files with primary name X are removed from the current disk.

ERA *.ASM

All files with secondary name ASM are removed from the current disk.

ERA X?Y.C?M

All files on the current disk that satisfy the ambiguous reference X?Y.C?M are deleted.

ERA *.*

Erase all files on the current disk. In this case, the CCP prompts the console with the message
ALL FILES (Y/N)?
which requires a Y response before files are actually removed.

ERA B:*.PRN

All files on drive B that satisfy the ambiguous reference ???????.PRN are deleted, independently of the currently logged disk.

1.4.2. DIR Command

Syntax:

DIR *afn*

The DIR (directory) command causes the names of all files that satisfy the ambiguous filename *afn* to be listed at the console device. As a special case, the command

DIR

lists the files on the currently logged disk (the command DIR is equivalent to the command DIR *.*). The following are valid DIR commands:

DIR X.Y

DIR X?Y.C?M

DIR ???.Y

Similar to other CCP commands, the *afn* can be preceded by a drive name. The following DIR commands cause the selected drive to be addressed before the directory search takes place:

DIR B:

DIR B:X.Y

DIR B:*.A?M

If no files on the selected disk satisfy the directory request, the message

NO FILE

appears at the console.

1.4.3. REN Command

Syntax

REN *ufn1=ufn2*

The REN (rename) command allows you to change the names of files on disk. The file satisfying *ufn2* is changed to *ufn1*. The currently logged disk is assumed to contain the file to rename (*ufn2*). You can also type a left-directed arrow (5FH) instead of the equal sign if the console supports this graphic character (normally rendered as an underscore “_” on modern systems). The following are examples of the REN command:

```
REN X.Y=Q.R
```

The file Q.R is changed to X.Y.

```
REN XYZ.COM=XYZ.XXX
```

The file XYZ.XXX is changed to XYZ.COM.

The operator precedes either *ufn1* or *ufn2* (or both) by an optional drive address. If *ufn1* is preceded by a drive name, then *ufn2* is assumed to exist on the same drive. Similarly, if *ufn2* is preceded by a drive name, then *ufn1* is assumed to exist on the drive as well. The same drive must be specified in both cases if both *ufn1* and *ufn2* are preceded by drive names. The following REN commands illustrate this format:

```
REN A:X.ASM=Y.ASM
```

The file Y.ASM is changed to X.ASM on drive A.

```
REN B:ZAP.BAS=ZOT.BAS
```

The file ZOT.BAS is changed to ZAP.BAS on drive B.

```
REN B:A.ASM=B:A.BAK
```

The file A.BAK is renamed to A.ASM on drive B.

If *ufn1* is already present, the REN command responds with the error

```
FILE EXISTS
```

and not perform the change. If *ufn2* does not exist on the specified disk, the message

```
NO FILE
```

is printed at the console.

1.4.4. SAVE Command

Syntax:

```
SAVE n ufn
```

The SAVE command places *n* pages (256-byte blocks) onto disk from the TPA and names this file *ufn*. In the CP/M distribution system, the TPA starts at 0100H (hexadecimal) which is the second page of memory. The SAVE command must specify 2 pages of memory if the user's program occupies the area from 0100H through 02FFH. The machine code file can be subsequently loaded and executed. The following are examples of the SAVE command:

```
SAVE 3 X.COM
```

Copies 0100H through 03FFH to X.COM.

SAVE 40 Q

Copies 0100H through 28FFH to Q. Note that 28 is the page count in 28FFH, and that $28H = 2 * 16 + 8 = 40$ decimal.

SAVE 4 X.Y

Copies 0100H through 04FFH to X.Y.

The SAVE command can also specify a disk drive in the *ufn* portion of the command, as shown in the following example:

SAVE 10 B:ZOT.COM

Copies 10 pages, 0100H through 0AFFH, to the file ZOT.COM on drive B.

1.4.5. TYPE Command

Syntax:

TYPE *ufn*

The TYPE command displays the content of the ASCII source file *ufn* on the currently logged disk at the console device. The following are valid TYPE commands:

TYPE X.Y

TYPE X.PLM

TYPE XXX

The TYPE command expands tabs, CTRL-I characters, assuming tab positions are set at every eighth column. The *ufn* can also reference a drive name.

TYPE B:X.PRN

The file X.PRN from drive B is displayed.

1.4.6. USER Command

Syntax:

USER *n*

The USER command allows maintenance of separate files in the same directory. In the syntax line, *n* is an Integer value in the range 0 to 15. On cold start, the operator is automatically logged into user area number 0, which is compatible with standard CP/M 1 directories. You can issue the USER command at any time to move to another logical area within the same directory. Drives that are logged-in while addressing one user number are automatically active when the operator moves to another. A user number is simply a prefix that accesses particular directory entries on the active disks.

The active user number is maintained until changed by a subsequent USER command, or until a cold start when user 0 is again assumed.

1.5. Line Editing and Output Control

The CCP allows certain line-editing functions while typing command lines. The CTRL-key sequences are obtained by pressing the control and letter keys simultaneously. Further, CCP command lines are generally up to 255 characters in length; they are not acted upon until the carriage return key is pressed.

Character	Meaning
CTRL-C	Reboots CP/M system when pressed at start of line.
CTRL-E	Physical end of line; carriage is returned, but line is not sent until the carriage return key is pressed.
CTRL-H	Backspaces one character position.
CTRL-J	Terminates current input (line-feed).
CTRL-M	Terminates current input (carriage return).
CTRL-P	Copies all subsequent console output to the currently assigned list device (see Section 1.6.1). Output is sent to the list device and the console device until the next CTRL-P is pressed.
CTRL-R	Retypes current command line; types a clean line following character deletion with rubouts.
CTRL-S	Stops the console output temporarily. Program execution and output continue when you press any character at the console, for example another CTRL-S. This feature stops output on high speed consoles, such as CRTs, in order to view a segment of output before continuing.
CTRL-U	Deletes the entire line typed at the console.
CTRL-X	Same as CTRL-U.
CTRL-Z	Ends input from the console (used in PIP and ED).
rubout/del	Deletes and echoes the last character typed at the console.

Table 1-1: Line-editing Control Characters

1.6. Transient Commands

Transient commands are loaded from the currently logged disk and executed in the TPA. The transient commands for execution under the CCP are below. Additional functions are easily defined by the user (see Section 1.6.3).

Command	Function
STAT	Lists the number of bytes of storage remaining on the currently logged disk, provides statistical information about particular files, and displays or alters device assignment.
ASM	Loads the CP/M assembler and assembles the specified program from disk.
LOAD	Loads the file in Intel HEX machine code format and produces a file in machine executable form which can be loaded into the TPA. This loaded program becomes a new command under the CCP.
DDT	Loads the CP/M debugger into TPA and starts execution.
PIP	Loads the Peripheral Interchange Program for subsequent disk file and peripheral transfer operations.
ED	Loads and executes the CP/M text editor program.
SYSGEN	Creates a new CP/M system disk.
SUBMIT	Submits a file of commands for batch processing.
DUMP	Dumps the contents of a file in hex.
MOVCPM	Regenerates the CP/M system for a particular memory size.

Table 1-2: CP/M Transient Commands

Transient commands are specified in the same manner as built-in commands, and additional commands are easily defined by the user. For convenience, the transient command can be preceded by a drive name which causes the transient to be loaded from the specified drive into the TPA for execution. Thus, the command

B:STAT

causes CP/M to temporarily log in drive B for the source of the STAT transient, and then return to the original logged disk for subsequent processing.

1.6.1. STAT Command

Syntax:

STAT

STAT *command line*

Command	Function
STAT	If you type an empty command line, the STAT transient calculates the storage remaining on all active drives, and prints one of the following messages: <i>d</i>: R/W, Space: <i>nnnk</i> <i>d</i>: R/O, Space: <i>nnnk</i> for each active drive <i>d</i>., where R/W indicates the drive can be read or written, and R/O indicates the drive is Read-Only (a drive becomes R/O by explicitly setting it to Read- Only, as shown below, or by inadvertently changing disks without performing a warm start). The space remaining on the disk in drive <i>d</i>: is given in kilobytes by <i>nnn</i>.
STAT <i>d</i> :	If a drive name is given, then the drive is selected before the storage is computed. Thus, the command STAT B: could be issued while logged into drive A, resulting in the message Bytes Remaining On B: <i>nnnk</i>
STAT <i>afn</i>	The command line can also specify a set of files to be scanned by STAT. The files that satisfy <i>afn</i> are listed in alphabetical order, with storage requirements for each file under the heading: Recs Bytes Ext Acc <i>rrrr bbbk ee acc d:filename.typ</i> where <i>rrrr</i> is the number of 128-byte records allocated to the file, <i>bbb</i> is the number of kilobytes allocated to the file $bbb = \frac{rrrr * 128}{1024}$ <i>ee</i> is the number of 16K extents $ee = \frac{bbb}{16}$ <i>acc</i> is either R/O, or R/W, <i>d</i> is the drive name containing the file (A ... P), <i>filename</i> is the eight-character primary filename, and <i>typ</i> is the three-character filetype. After listing the individual files, the storage usage is summarized.

Command	Function
STAT <i>d:afn</i>	The drive name can be given ahead of the <i>afn</i> . The specified drive is first selected, and the form STAT <i>afn</i> is executed.
STAT <i>d:=R/O</i>	This form sets the drive given by <i>d</i> to Read-Only, remaining in effect until the next warm or cold start takes place. When a disk is Read-Only, the message BDOS Err On <i>d:</i> R/O appears if there is an attempt to write to the Read-Only disk. CP/M waits until a key is pressed before performing an automatic warm start, at which time the disk becomes R/W.

The STAT command allows you to control the physical-to-logical device assignment. See the IOBYTE function described in Sections 5 and 6.

There are four logical peripheral devices that are, at any particular instant, each assigned one of several physical peripheral devices. The following is a list of the four logical devices:

CON: is the system console device, used by CCP for communication with the operator.

RDR: is the paper tape reader device.

PUN: is the paper tape punch device.

LST: is the output list device.

The actual devices attached to any particular computer system are driven by subroutines in the BIOS portion of CP/M. Thus, the logical RDR: device, for example, could actually be a high speed reader, teletype reader, or cassette tape. To allow some flexibility in device naming and assignment, several physical devices are defined in Table 1-3.

Device	Meaning
TTY:	Teletype device (slow speed console)
CRT:	Cathode ray tube device (high speed console)
BAT:	Batch processing (console is current RDR:, output goes to current LST: device)
UC1:	User-defined console
PTR:	Paper tape reader (high speed reader)
UR1:	User-defined reader #1
UR2:	User-defined reader #2
PTP:	Paper tape punch (high speed punch)
UP1:	User-defined punch #1
UP2:	User-defined punch #2
LPT:	Line printer
UL1:	User-defined list device #1

Table 1-3: Physical Devices

It is emphasized that the physical device names might not actually correspond to devices that the names imply. That is, you can implement the PTP: device as a cassette write operation. The exact correspondence and driving subroutine is defined in the BIOS portion of CP/M. In the standard distribution version of CP/M, these devices correspond to their names on the MDS 800 development system.

The command:

STAT VAL:

produces a summary of the available status commands, resulting in the output:

```
Temp R/O Disk d:=R/O
Set Indicator: d:filename.typ $R/O $R/W $SYS $DIR
Disk Status: DSK: d:DSK
User Status: USR:
```

which gives an instant summary of the possible STAT commands and shows the permissible logical-to-physical device assignments:

```

Iobyte Assign:
CON:=TTY:CRT:BAT:UCI:
RDR:=TTY:PTR:URI:UR2:
PUN:=TTY:PTP:UP1:UP2:
LST:=TTY:CRT:LPT:ULI:

```

The logical device to the left takes any of the four physical assignments shown to the right. The current logical-to-physical mapping is displayed by typing the command:

```
STAT DEV:
```

This command produces a list of each logical device to the left and the current corresponding physical device to the right. For example, the list might appear as follows:

```

CON:=CRT:
RDR:=URL:
PUN:=PTP:
LST:=TTY:

```

The current logical-to-physical device assignment is changed by typing a STAT command of the form:

```
STAT ld1=pd1,ld2=pd2,...ldn=pdn,
```

where *ld1* through *ldn* are logical device names and *pd1* through *pdn* are compatible physical device names. For example, *ld1* and *pd1* appear on the same line in the VAL: command shown above. The following example shows valid STAT commands that change the current logical-to-physical device assignments:

```

STAT CON:=CRT:
STAT PUN:=TTY:,LST:=LPT:,RDR:=TTY

```

The command form:

```
STAT d:filename.typ $S
```

where *d:* is an optional drive name and *filename.typ* is an unambiguous or ambiguous filename, produces the following output display format:

Size	Recs	Bytes	Ext	Acc
48	48	6K	1	R/O A:ED.COM
55	55	12K	1	R/O (A:PIP.COM)
65536	128	16K	2	R/W A:X.DAT

Bytes Remaining On A: 168k

where the \$S parameter causes the Size field to be displayed. Without the \$S, the Size field is skipped, but the remaining fields are displayed. The Size field lists the virtual file size in records, while the Recs field sums the number of virtual records in each extent. For files constructed sequentially, the Size and Recs fields are identical. The Bytes field lists the actual

number of bytes allocated to the corresponding file. The minimum allocation unit is determined at configuration time; thus, the number of bytes corresponds to the record count plus the remaining unused space in the last allocated block for sequential files. Random access files are given data areas only when written, so the `Bytes` field contains the only accurate allocation figure. In the case of random access, the `Size` field gives the logical end-of-file record position and the `Recs` field counts the logical records of each extent. Each of these extents, however, can contain unallocated holes even though they are added into the record count.

The `Ext` field counts the number of physical extents allocated to the file. The `Ext` count corresponds to the number of directory entries given to the file. Depending on allocation size, there can be up to 128K bytes (8 logical extents) directly addressed by a single directory entry. In a special case, there are actually 256K bytes that can be directly addressed by a physical extent.

The `Acc` field gives the R/O or R/W file indicator, which you can change using the commands shown. The four command forms,

```
STAT d:filename.typ $R/O
```

```
STAT d:filename.typ $R/W
```

```
STAT d:filename.typ $SYS
```

```
STAT d:filename.typ $DIR
```

set or reset various permanent file indicators. The R/O indicator places the file, or set of files, in a Read-Only status until changed by a subsequent `STAT` command. The R/O status is recorded in the directory with the file so that it remains R/O through intervening cold start operations. The R/W indicator places the file in a permanent Read-Write status. The `SYS` indicator attaches the system indicator to the file, while the `DIR` command removes the system indicator. The *filename.typ* may be ambiguous or unambiguous, but files whose attributes are changed are listed at the console when the change occurs. The drive name denoted by *d*: is optional.

When a file is marked R/O, subsequent attempts to erase or write into the file produce the following BDOS message at your screen:

```
BDOS Err On d: File R/O
```

lists the drive characteristics of the disk named by *d*: that is in the range A:, B:, ..., P:. The drive characteristics are listed in the following format:

```

d: Drive Characteristics
65536: 128 Byte Record Capacity
8192: Kilobyte Drive Capacity
128: 32 Byte Directory Entries
0: Checked Directory Entries
1024: Records/Extent
128: Records/Block
58: Sectors/Track
2: Reserved Tracks

```

where *d*: is the selected drive, followed by the total record capacity (65536 is an eight-megabyte drive), followed by the total capacity listed in kilobytes. The directory size is listed next, followed by the checked entries. The number of checked entries is usually identical to the directory size for removable media, because this mechanism is used to detect changed media during CP/M operation without an intervening warm start. For fixed media, the number is usually zero, because the media are not changed without at least a cold or warm start.

The number of records per extent determines the addressing capacity of each directory entry (1024 times 128 bytes, or 128K in the previous example). The number of records per block shows the basic allocation size (in the example, 128 records/block times 128 bytes per record, or 16K bytes per block). The listing is then followed by the number of physical sectors per track and the number of reserved tracks.

For logical drives that share the same physical disk, the number of reserved tracks can be quite large because this mechanism is used to skip lower-numbered disk areas allocated to other logical disks. The command form:

```
STAT DSK:
```

produces a drive characteristics table for all currently active drives. The final STAT command form is

```
STATUSR:
```

which produces a list of the user numbers that have files on the currently addressed disk. The display format is

```

Active User : 0
Active Files: 0 1 3

```

where the first line lists the currently addressed user number, as set by the last CCP USER command, followed by a list of user numbers scanned from the current directory. In this case, the active user number is 0 (default at cold start) with three user numbers that have active files on the current disk. The operator can subsequently examine the directories of the other user numbers by logging in with USER 1 or USER 3 commands, followed by a DIR command at the CCP level.

1.6.2. ASM Command

Syntax:

ASM *ufn*

The ASM command loads and executes the CP/M 8080 assembler. The *ufn* specifies a source file containing assembly language statements, where the filetype is assumed to be ASM and is not specified. The following ASM commands are valid:

ASM X

ASM GAMMA

The two-pass assembler is automatically executed. Assembly errors that occur during the second pass are printed at the console.

The assembler produces a file:

X.PRN

where X is the primary name specified in the ASM command. The PRN file contains a listing of the source program with embedded tab characters if present in the source program, along with the machine code generated for each statement and diagnostic error messages, if any. The PRN file is listed at the console using the TYPE command, or sent to a peripheral device using PIP (see Section 1.6.4). Note that the PRN file contains the original source program, augmented by miscellaneous assembly information in the leftmost 16 columns; for example, program addresses and hexadecimal machine code. The PRN file serves as a backup for the original source file. If the source file is accidentally removed or destroyed, the PRN file can be edited by removing the leftmost 16 characters of each line (see Section 2). This is done by issuing a single editor macro command. The resulting file is identical to the original source file and can be renamed from PRN to ASM for subsequent editing and assembly. The file

X.HEX

is also produced, which contains 8080 machine language in Intel HEX format suitable for subsequent loading and execution (see Section 1.6.3). For complete details of CP/M's assembly language program, see Section 3.

The source file for assembly is taken from an alternate disk by prefixing the assembly language filename by a disk drive name. The command

ASM B:ALPHA

loads the assembler from the currently logged drive and processes the source program ALPHA.ASM on drive B. The HEX and PRN files are also placed on drive B in this case.

1.6.3. LOAD Command

Syntax:

LOAD *ufn*

The `LOAD` command reads the file *ufn*, which is assumed to contain HEX format machine code, and produces a memory image file that can subsequently be executed. The filename *ufn* is assumed to be of the form:

`X.HEX`

and only the filename *X* need be specified in the command. The `LOAD` command creates a file named

`X.COM`

that marks it as containing machine executable code. The file is actually loaded into memory and executed when the user types the filename *x* immediately after the prompting character `>` printed by the CCP.

Generally, the CCP reads the filename *x* following the prompting character and looks for a built-in function name. If no function name is found, the CCP searches the system disk directory for a file by the name

`X.COM`

If found, the machine code is loaded into the TPA, and the program executes. Thus, the user need only `LOAD` a hex file once; it can be subsequently executed any number of times by typing the primary name. This way, you can invent new commands in the CCP. Initialized disks contain the transient commands as `COM` files, which are optionally deleted. The operation takes place on an alternate drive if the filename is prefixed by a drive name. Thus,

`LOAD B:BETA`

brings the `LOAD` program into the TPA from the currently logged disk and operates on drive `B` after execution begins.

Note: the `BETA.HEX` file must contain valid Intel format hexadecimal machine code records (as produced by the `ASM` program, for example) that begin at `0100H` of the TPA. The addresses in the hex records must be in ascending order; gaps in unfilled memory regions are filled with zeroes by the `LOAD` command as the hex records are read. Thus, `LOAD` must be used only for creating CP/M standard `COM` files that operate in the TPA. Programs that occupy regions of memory other than the TPA are loaded under `DDT`.

1.6.4. PIP Command

Syntax:

`PIP`

`PIP destination=source#1,source#2,...source#n`

PIP is the CP/M Peripheral Interchange Program that implements the basic media conversion operations necessary to load, print, punch, copy, and combine disk files. The PIP program is initiated by typing one of the following forms:

PIP

PIP command line

In both cases PIP is loaded into the TPA and executed. In the first form, PIP reads command lines directly from the console, prompted with the * character, until an empty command line is typed (for example, a single carriage return is issued by the operator). Each successive command line causes some media conversion to take place according to the rules shown below.

In the second form, the PIP command is equivalent to the first, except that the single command line given with the PIP command is automatically executed, and PIP terminates immediately with no further prompting of the console for input command lines. The form of each command line is

destination=source#1,source#2,...source#n

where *destination* is the file or peripheral device to receive the data, and *source#1, source#2, ... source#n* is a series of one or more files or devices that are copied from left to right to the destination.

When multiple files are given in the command line (for example, $n > 1$), the individual files are assumed to contain ASCII characters, with an assumed CP/M end-of-file character (CTRL-Z) at the end of each file (see the o parameter to override this assumption). Lower-case ASCII alphabetic characters are internally translated to upper-case to be consistent with CP/M file and device name conventions. Finally, the total command line length cannot exceed 255 characters. CTRL-E can be used to force a physical carriage return for lines that exceed the console width.

The destination and source elements are unambiguous references to CP/M source files with or without a preceding disk drive name. That is, any file can be referenced with a preceding drive name (A: through P:) that defines the particular drive where the file can be obtained or stored. When the drive name is not included, the currently logged disk is assumed. The destination file can also appear as one or more of the source files; in which case the source file is not altered until the entire concatenation is complete. If it already exists, the destination file is removed if the command line is properly formed. It is not removed if an error condition arises. The following command lines, with explanations to the right, are valid as input to PIP:

X=Y

Copies to file x from file y, where x and y are unambiguous filenames; y remains unchanged.

<code>X=Y,Z</code>	Concatenates files Y and Z and copies to file X, with Y and Z unchanged.
<code>X.ASM=Y.ASM,Z.ASM</code>	Creates the file X.ASM from the concatenation of the Y.ASM and Z.ASM files.
<code>NEW.ZOT=B:OLD.ZAP</code>	Moves a copy of OLD.ZAP from drive B to the currently logged disk; names the file NEW.ZOT.
<code>B:A.U=B:B.V,A:C.W,D.X</code>	Concatenates file B.Y from drive B with C.W from drive A and D.X from the logged disk; creates the file A.U on drive B.

For convenience, PIP allows abbreviated commands for transferring files between disk drives. The abbreviated PIP forms are

`PIP d:=afn`

`PIP d1:=d2:afn`

`PIP ufn:=d2`

`PIP d1:ufn=d2:`

The first form copies all files from the currently logged disk that satisfy the *afn* to the same files on drive *d*, where *d* = A...P. The second form is equivalent to the first, where the source for the copy is drive *d2* where *d2* = A ... P. The third form is equivalent to the command `PIP d1:ufn=d2:ufn` which copies the file given by *ufn* from drive *d2* to the file *ufn* on drive *d1*. The fourth form is equivalent to the third, where the source disk is explicitly given by *d2*.

The source and destination disks must be different in all of these cases. If an *afn* is specified, PIP lists each *ufn* that satisfies the *afn* as it is being copied. If a file exists by the same name as the destination file, it is removed after successful completion of the copy and replaced by the copied file.

The following PIP commands give examples of valid disk-to-disk copy operations:

<code>B=* .COM</code>	Copies all files that have the secondary name COM to drive B from the current drive.
<code>A:=B:ZAP.*</code>	Copies all files that have the primary name ZAP to drive A from drive B.
<code>ZAP.ASM=B:</code>	Same as <code>ZAP.ASM=B:ZAP.ASM</code>

B:ZOT.COM=A:	Same as B:ZOT.COM=A:ZOT.COM
B:=GAMMA.BAS	Same as B:GAMMA.BAS=GAMMA.BAS
B:=A:GAMMA.BAS	Same as B:GAMMA.BAS=A:GAMMA.BAS

PIP allows reference to physical and logical devices that are attached to the CP/M system. The device names are the same as given under the STAT command, along with a number of specially named devices. The following is a list of logical devices given in the STAT command

CON: console
RDR: reader
PUN: punch
LST: list

while the physical devices are

TTY: console, reader, punch or list
CRT: console, or list
UC1: console
PTR: reader
UR1: reader
UR2: reader
PTP: punch
UP1: punch
UP2: punch
LPT: list
UL1: list

The BAT: physical device is not included, because this assignment is used only to indicate that the RDR: and LST: devices are used for console input/output.

The RDR:, LST:, PUN:, and CON: devices are all defined within the BIOS portion of CP/M, and are easily altered for any particular I/O system. The current physical device mapping is defined by IOBYTE; see Section 6 for a discussion of this function. The destination device must be capable of receiving data, for example, data cannot be sent to the punch, and the source devices must be capable of generating data, for example, the LST: device cannot be read.

The following list describes additional device names that can be used in PIP commands.

NUL: sends 40 nulls (ASCII 0s) to the device. This can be issued at the end of punched output.
EOF: sends a CP/M end-of-file (ASCII CTRL-Z) to the destination device (sent automatically at the end of all ASCII data transfers through PIP).

PRN: is the same as **LST:**, except that tabs are expanded at every eighth character position, lines are numbered, and page ejects are inserted every 60 lines with an initial eject (same as using PIP options [**t8np**]).

INP: is a special PIP input source that can be patched into the PIP program. PIP gets the input data character-by-character, by **CALLing** location 0103H, with data returned in location 0109H (parity bit must be zero).

OUT: is a special PIP output destination that can be patched into the PIP program. PIP **CALLS** location 0106H with data in register C for each character to transmit.

Note: The Locations 010AH through 01FFH of the PIP memory image are not used and can be replaced by special purpose drivers using DDT (See Section 4).

File and device names can be interspersed in the PIP commands. In each case, the specific device is read until end-of-file (CTRL-Z for ASCII files, and end-of-data for non-ASCII disk files). Data from each device or file are concatenated from left to right until the last data source has been read.

The destination device or file is written using the data from the source files, and an end-of-file character, CTRL-Z, is appended to the result for ASCII files. If the destination is a disk file, a temporary file is created (\$\$\$ secondary name) that is changed to the actual filename only on successful completion of the copy. Files with the extension **COM** are always assumed to be non-ASCII.

The copy operation can be aborted at any time by pressing any key on the keyboard. PIP responds with the message

ABORTED

to indicate that the operation has not been completed. If any operation is aborted, or if an error occurs during processing, PIP removes any pending commands that were set up while using the **SUBMIT** command.

PIP performs a special function if the destination is a disk file with type **HEX** (an Intel hex-formatted machine code file), and the source is an external peripheral device, such as a paper tape reader. In this case, the PIP program checks to ensure that the source file contains a properly formed hex file, with legal hexadecimal values and checksum records.

When an invalid input record is found, PIP reports an error message at the console and waits for corrective action. Usually, you can open the reader and rerun a section of the tape (pull the tape back about 20 inches). When the tape is ready for the reread, a single carriage return is typed at the console, and PIP attempts another read. If the tape position cannot be properly read, continue the read by typing a return following the error message, and enter the record manually with the **ED** program after the disk file is constructed.

PIP allows the end-of-file to be entered from the console if the source file is an RDR: device. In this case, the PIP program reads the device and monitors the keyboard. If CTRL-Z is typed at the keyboard, the read operation is terminated normally.

The following are valid PIP commands:

PIP LST:=X.PRN	Copies X.PRN to the LST device and terminates the PIP program.
PIP	Starts PIP for a sequence of commands. PIP prompts with *. A single carriage return stops PIP.
PIP CON:=X.ASM,Y.ASM,Z.ASM	Concatenates three ASM files and copies to the CON device.
PIP X.HEX=CON: ,Y.HEX,PTR:	Creates a HEX file by reading the CON until a CTRL-Z is typed, followed by data from Y.HEX and PTR until a CTRL-Z is encountered.
PIP PUN:=NUL: ,X.ASM,EOF: ,NUL:	Sends 40 nulls to the punch device; copies the X.ASM file to the punch, followed by an end-of-file, CTRL-Z, and 40 more null characters.

You can also specify one or more PIP parameters, enclosed in left and right square brackets, separated by zero or more blanks. Each parameter affects the copy operation, and the enclosed list of parameters must immediately follow the affected file or device. Generally, each parameter can be followed by an optional decimal integer value (the s and q parameters are exceptions).

Parameter	Meaning
B	Blocks mode transfer. Data are buffered by PIP until an ASCII XOFF character, CTRL-S, is received from the source device. This allows transfer of data to a disk file from a continuous reading device, such as a cassette reader. Upon receipt of the XOFF, PIP clears the disk buffers and returns for more input data. The amount of data that can be buffered depends on the memory size of the host system. PIP issues an error message if the buffers overflow.
Dn	Deletes characters that extend past column <i>n</i> in the transfer of data to the destination from the character source. This parameter is generally used to truncate long lines that are sent to a narrow printer or console device.

Parameter	Meaning
E	Echoes all transfer operations to the console as they are being performed.
F	Filters form-feeds from the file. All embedded form-feeds are removed. The P parameter can be used simultaneously to insert new form-feeds.
Gn	Gets file from user number <i>n</i> (<i>n</i> in the range 0-15).
H	Transfers HEX data. All data are checked for proper Intel hex file format. Nonessential characters between hex records are removed during the copy operation. The console is prompted for corrective action in case errors occur.
I	Ignores :00 records in the transfer of Intel hex format file. The I parameter automatically sets the H parameter.
L	Translates upper-case alphabetic characters to lower-case.
N	Adds line numbers to each line transferred to the destination, starting at one and incrementing by 1. Leading zeroes are suppressed, and the number is followed by a colon. If N2 is specified, leading zeroes are included and a tab is inserted following the number. The tab is expanded if T is set.
O	Transfers non-ASCII object files. The normal CP/M end-of-file is ignored.
Pn	Includes page ejects at every <i>n</i> lines with an initial page eject. If <i>n</i> = 1 or is excluded altogether, page ejects occur every 60 lines. If the F parameter is used, form-feed suppression takes place before the new page ejects are inserted.
qs^Z	Quits copying from the source device or file when the string <i>s</i> , terminated by CTRL-Z, is encountered.
R	Reads system files.

Parameter	Meaning
ss^Z	Start copying from the source device when the string s, terminated by CTRL-Z, is encountered. The s and Q parameters can be used to abstract a particular section of a file, such as a subroutine. The start and quit strings are always included in the copy operation. If you specify a command line after the PIP command keyword, the CCP translates strings following the s and Q parameters to uppercase. If you do not specify a command line, PIP does not perform the automatic upper-case translation.
Tn	Expands tabs, CTRL-I characters, to every nth column during the transfer of characters to the destination from the source.
U	Translates lower-case alphabetic characters to upper-case during the copy operation.
V	Verifies that data have been copied correctly by rereading after the write operation (the destination must be a disk file).
W	Writes over R/O files without console interrogation.
Z	Zeros the parity bit on input for each ASCII character.

Table 1-4: PIP Parameters

The following examples show valid PIP commands that specify parameters in the file transfer.

PIP X.ASM=B:[V]

Copies X.ASM from drive B to the current drive and verifies that the data were properly copied.

PIP LPT:=X.ASM[NT8U]

Copies X.ASM to the LPT: device; numbers each line, expands tabs to every eighth column, and translates lower-case alphabetic characters to upper-case.

PIP PUN:=X.HEX[I],Y.ZOT[H]

First copies X.HEX to the PUN: device and ignores the trailing :00 record in X.HEX; continues the transfer of data by reading

Y.ZOT, which contains HEX records, including any :00 records it contains.

PIP X.LIB=Y.ASM[sSUBR1: ^zqJMP L3^z]

Copies from the file Y.ASM into the file X.LIB. The command starts the copy when the string SUBR1: has been found, and quits copying after the string JMP L3 is encountered.

PIP PRN:=X.ASM[p50]

Sends X.ASM to the LST: device with line numbers, expands tabs to every eighth column, and elects pages at every 50th line. The assumed parameter list for a PRN file is nt8p60; p50 overrides the default value.

Under normal operation, PIP does not overwrite a file that is set to a permanent R/O status. If an attempt is made to overwrite an R/O file, the following prompt appears:

DESTINATION FILE IS R/O, DELETE (Y/N)?

If you type Y, the file is overwritten. Otherwise, the following response appears:

** NOT DELETED **

The file transfer is skipped, and PIP continues with the next operation in sequence. To avoid the prompt and response in the case of R/O file overwrite, the command line can include the w parameter, as shown in this example:

PIP A:=B:*.COM[W]

The w parameter copies all non-system files to the A drive from the B drive and overwrites any R/O files in the process. If the operation involves several concatenated files, the w parameter need only be included with the last file in the list, as in this example:

PIP A.DAT=B.DAT,F:NEW.DAT,G:OLD.DAT[W]

Files with the system attribute can be included in PIP transfers if the R parameter is included; otherwise, system files are not recognized. For example, the command line:

PIP ED.COM=B:ED.COM[R]

reads the ED.COM file from the B drive, even if it has been marked as an R/O and system file. The system file attributes are copied, if present.

Downward compatibility with previous versions of CP/M is only maintained if the file does not exceed one megabyte, no file attributes are set, and the file is created by user 0. If compatibility

is required with nonstandard, for example, double-density versions of 1.4, it might be necessary to select 1.4 compatibility mode when constructing the internal disk parameter block. See Section 6 and refer to Section 6.10, which describes BIOS differences.

Note:

To copy files into another user area, PIP.COM must be located in that user area. Use the following procedure to make a copy of PIP.COM in another user area.

```
A>USER 0↵           Log in as user 0
A>DDT PIP.COM↵      Load PIP into memory. Note PIP size S
DDT VERS 2.2
NEXT PC
1E00 0100

-G0↵              Return to CCP
A>USER N↵          Log in as desired user
A>SAVE S PIP.COM↵   Save PIP.COM in new user area
A>
```

In this procedure, S is the integral number of memory pages, 256- byte segments, occupied by PIP.COM. The number S can be determined when PIP.COM is loaded under DDT, by referring to the value under the NEXT display. If, for example, the next available address is 1E00, then PIP.COM requires 1D hexadecimal pages, or 29 pages in decimal. So the value of S is 29 in the subsequent SAVE command. Once PIP is copied in this manner, it can be copied to another disk belonging to the same user number through normal PIP transfers.

1.6.5. ED Command

Syntax:

ED *ufn*

The ED program is the CP/M system context editor that allows creation and alteration of ASCII files in the CP/M environment. Complete details of operation are given in Section 2. ED allows the operator to create and operate upon source files that are organized as a sequence of ASCII characters, separated by end-of-line characters (a carriage return/line-feed sequence). There is no practical restriction on line length (no single line can exceed the size of the working memory) that is defined by the number of characters typed between carriage returns.

The ED program has a number of commands for character string searching, replacement, and insertion that are useful for creating and correcting programs or text files under CP/M. Although the CP/M has a limited memory work space area (approximately 5000 characters in a 20K CP/M system), the file size that can be edited is not limited, since data are easily paged through this work area.

If it does not exist, ED creates the specified source file and opens the file for access. If the source file does exist, the programmer appends data for editing (see the A command). The appended data can then be displayed, altered, and written from the work area back to the disk (see the W command). Particular points in the program can be automatically paged and located by context, allowing easy access to particular portions of a large file (see the N command).

If you type the following command line:

```
ED X.ASM
```

the ED program creates an intermediate work file with the name

```
X. $$$
```

to hold the edited data during the ED run. Upon completion of ED, the X.ASM file (original file) is renamed to X.BAK, and the edited work file is renamed to X.ASM. Thus, the X.BAK file contains the original unedited file, and the X.ASM file contains the newly edited file. The operator can always return to the previous version of a file by removing the most recent version and renaming the previous version. If the current X.ASM file has been improperly edited, the following sequence of commands reclaim the backup file.

DIR X.* Checks to see that .BAK file is available.

ERA X.ASM Erases the most recent version.

REN X.ASM=X.BAK Renames the .BAK file to .ASM

You can abort the edit at any point (reboot, power failure, CTRL-C, or CTRL-Q command) without destroying the original file. In this case, the .BAK file is not created and the original file is always intact.

The ED program allows the user to edit the source on one disk and create the back-up file on another disk. This form of the ED command is

```
ED ufn d:
```

where *ufn* is the name of the file to edit on the currently logged disk and *d* is the name of an alternate drive. The ED program reads and processes the source file and writes the new file to drive *d* using the name *ufn*. After processing, the original file becomes the back-up file. If the operator is addressing disk A, the following command is valid.

```
ED X.ASM B:
```

This edits the file X.ASM on drive A, creating the new file X. \$\$\$ on drive B. After a successful edit, A:X.ASM is renamed to A:X.BAK, and B:X. \$\$\$ is renamed to B:X.ASM. For convenience, the currently logged disk becomes drive B at the end of the edit. Note that if a file named B:X.ASM exists before the editing begins, the following message appears on the screen:

FILE EXISTS

This message is a precaution against accidentally destroying a source file. You should first erase the existing file and then restart the edit operation.

Similar to other transient commands, editing can take place on a drive different from the currently logged disk by preceding the source filename by a drive name. The following are examples of valid edit requests:

ED A:X.ASM

Edits the file X.ASM on drive A, with new file and back-up on drive A.

ED B:X.ASM A:

Edits the file X.ASM on drive B to the temporary file X. \$\$\$ on drive A. After editing, this command changes X.ASM on drive B to X.BAK and changes X. \$\$\$ on drive A to X.ASM

1.6.6. SYSGEN Command

Syntax:

SYSGEN

The SYSGEN transient command allows generation of an initialized disk containing the CP/M operating system. The SYSGEN program prompts the console for commands by interacting as shown.

SYSGEN

Initiates the SYSGEN program.

SYSGEN VERSION 2.0

SYSGEN sign-on message.

SOURCE DRIVE NAME (OR RETURN TO SKIP)

Respond with the drive name (one of the letters A, B, C, or D) of the disk containing a CP/M system, usually A. If a copy of CP/M already exists in memory due to a MOVCPM command, press only a carriage return. Typing a drive name *d* causes the response:

SOURCE ON <i>d</i> THEN TYPE RETURN	Place a disk containing the CP/M operating system on drive <i>d</i> (<i>d</i> is one of A, B, C, or D). Answer by pressing a carriage return when ready.
FUNCTION COMPLETE	System is copied to memory. SYSGEN then prompts with the following:
DESTINATION DRIVE NAME (OR RETURN TO REBOOT)	If a disk is being initialized, place the new disk into a drive and answer with the drive name. Otherwise, press a carriage return and the system reboots from drive A. Typing drive name <i>d</i> causes SYSGEN to prompt with the following message:
DESTINATION ON <i>d</i> , THEN TYPE RETURN	Place the new disk on drive <i>d</i> and press the RETURN key when ready.
FUNCTION COMPLETE	New disk is initialized in drive <i>d</i> .

The DESTINATION prompt is repeated until a single carriage return is pressed at the console, so that more than one disk can be initialized.

Upon completion of a successful system generation, the new disk contains the operating system, and only the built-in commands are available. An IBM-compatible disk appears to CP/M as a disk with an empty directory; therefore, the operator must copy the appropriate COM files from an existing CP/M disk to the newly constructed disk using the PIP transient.

You can copy all files from an existing disk by typing the following PIP command:

```
PIP B:=A:*.*[v]
```

This command copies all files from disk drive A to disk drive B and verifies that each file has been copied correctly. The name of each file is displayed at the console as the copy operation proceeds.

Note that a SYSGEN does not destroy the files that already exist on a disk; it only constructs a new operating system. If a disk is being used only on drives B through P and will never be the source of a bootstrap operation on drive A, the SYSGEN need not take place.

1.6.7. SUBMIT Command

Syntax:

`SUBMIT ufn parm#1 ... parm#n`

The `SUBMIT` command allows CP/M commands to be batched for automatic processing. The *ufn* given in the `SUBMIT` command must be the filename of a file that exists on the currently logged disk, with an assumed file type of `SUB`. The `.SUB` file contains CP/M prototype commands with possible parameter substitution. The actual parameters *parm#1 ... parm#n* are substituted into the prototype commands, and, if no errors occur, the file of substituted commands are processed sequentially by CP/M.

The prototype command file is created using the `ED` program, with interspersed `$` parameters of the form:

```
$1 $2 $3 ... $n
```

corresponding to the number of actual parameters that will be included when the file is submitted for execution. When the `SUBMIT` transient is executed, the actual parameters *parm#1 ... parm#n* are paired with the formal parameters `$1 ... $n` in the prototype commands. If the numbers of formal and actual parameters do not correspond, the `SUBMIT` function is aborted with an error message at the console. The `SUBMIT` function creates a file of substituted commands with the name

```
$$$ .SUB
```

on the logged disk. When the system reboots, at the termination of the `SUBMIT`, this command file is read by the CCP as a source of input rather than the console. If the `SUBMIT` function is performed on any disk other than drive A, the commands are not processed until the disk is inserted into drive A and the system reboots. You can abort command processing at any time by pressing the rubout key when the command is read and echoed. In this case, the `$$$ .SUB` file is removed and the subsequent commands come from the console. Command processing is also aborted if the CCP detects an error in any of the commands. Programs that execute under CP/M can abort processing of command files when error conditions occur by erasing any existing `$$$ .SUB` file.

To introduce dollar signs into a `SUBMIT` file, you can type a `$$` which reduces to a single `$` within the command file. A caret, `^`, precedes an alphabetic character `X`, which produces a single `CTRL-X` character within the file.

The last command in a `SUB` file can initiate another `SUB` file, allowing chained batch commands.

Suppose the file `ASMBL.SUB` exists on disk and contains the prototype commands:

```
ASM $1
DIR $1.*
ERA *.BAK
PIP $2:=$1.PRN
ERA $1.PRN
```

then, you issue the following command:

```
SUBMIT ASMBL X PRN
```

The SUBMIT program reads the ASMBL.SUB file, substituting X for all occurrences of \$1 and PRN for all occurrences of \$2. This results in a \$\$\$.SUB file containing the commands:

```
ASM X
DIR X.*
ERA *.BAK
PIP PRN:=X.PRN
ERA X.PRN
```

which are executed in sequence by the CCP.

The SUBMIT function can access a SUB file on an alternate drive by preceding the filename by a drive name. Submitted files are only acted upon when they appear on drive A. Thus, it is possible to create a submitted file on drive B that is executed at a later time when inserted in drive A.

An additional utility program called XSUB extends the power of the SUBMIT facility to include line input to programs as well as the CCP. The XSUB command is included as the first line of the SUBMIT file. When it is executed, XSUB self-relocates directly below the CCP. All subsequent SUBMIT command lines are processed by XSUB so that programs that read buffered console input, BDOS Function 10, receive their input directly from the SUBMIT file. For example, the file SAVER.SUB can contain the following SUBMIT lines:

```
XSUB
DDT
I $1.COM
R
G0
SAVE 1 $2.COM
```

a subsequent SUBMIT command, such as

```
A:SUBMIT SAVER PIP Y
```

substitutes PIP for \$1 and Y for \$2 in the command stream. The XSUB program loads, followed by DDT, which is sent to the command lines PIP.COM, R, and G0, thus returning to the CCP. The final command SAVE 1 Y.COM is processed by the CCP.

The XSUB program remains in memory and prints the message

```
(xsub active)
```

on each warm start operation to indicate its presence. Subsequent SUBMIT command streams do not require the XSUB, unless an intervening cold start occurs. Note that XSUB must be loaded after the optional CP/M DESPOOL utility, if both are to run simultaneously.

1.6.8. DUMP Command

Syntax:

DUMP *ufn*

The DUMP program types the contents of the disk file (*ufn*) at the console in hexadecimal form. The file contents are listed sixteen bytes at a time, with the absolute byte address listed to the left of each line in hexadecimal. Long type-outs can be aborted by pressing the rubout key during printout. The source listing of the DUMP program is given in Section 5 as an example of a program written for the CP/M environment.

1.6.9. MOVCPM Command

Syntax:

MOVCPM

MOVCPM *n*

MOVCPM *n* *

MOVCPM * *

The MOVCPM program allows you to reconfigure the CP/M system for any particular memory size. Two optional parameters can be used to indicate the desired size of the new system and the disposition of the new system at program termination. If the first parameter is omitted or an * is given, the MOVCPM program reconfigures the system to its maximum size, based upon the kilobytes of contiguous RAM in the host system (starting at 0000H). If the second parameter is omitted, the system is executed, but not permanently recorded; if * is given, the system is left in memory, ready for a SYSGEN operation. The MOVCPM program relocates a memory image of CP/M and places this image in memory in preparation for a system generation operation. The following is a list of MOVCPM command forms:

MOVCPM

Relocates and executes CP/M for management of the current memory configuration (memory is examined for contiguous RAM, starting at 100H). On completion of the relocation, the new system is executed but not permanently recorded on the disk. The system that is constructed contains a BIOS for the Intel MDS 800.

MOVCPM *n*

Creates a relocated CP/M system for management of an *n* kilobyte system (*n* must be in the range of 20 to 64), and executes the system as described.

MOVCPM * *

Constructs a relocated memory image for the current memory configuration, but leaves the memory image in memory in preparation for a SYSGEN operation.

MOVCPM *n* *

Constructs a relocated memory image for an *n* kilobyte memory system, and leaves the memory image in preparation for a SYSGEN operation.

For example, the command,

MOVCPM * *

constructs a new version of the CP/M system and leaves it in memory, ready for a SYSGEN operation. The message

READY FOR "SYSGEN" OR
"SAVE 34 CPMxx.COM"

appears at the console upon completion, where *xx* is the current memory size in kilobytes. You can then type the following sequence:

SYSGEN

This starts the system generation.

SOURCE DRIVE NAME (OR RETURN TO SKIP)

Respond with a carriage return to skip the CP/M read operation, because the system is already in memory as a result of the previous MOVCPM operation.

DESTINATION DRIVE NAME (OR RETURN TO REBOOT)

Respond with B to write new system to the disk in drive B. SYSGEN prompts with the following message:

DESTINATION ON B, THEN TYPE RETURN

Place the new disk on drive B and press the RETURN key when ready.

If you respond with A rather than B above, the system is written to drive A rather than B. SYSGEN continues to print this prompt:

DESTINATION DRIVE NAME (OR RETURN TO REBOOT)

until you respond with a single carriage return, which stops the SYSGEN program with a system reboot.

You can then go through the reboot process with the old or new disk. Instead of performing the SYSGEN operation, you can type a command of the form:

```
SAVE 34 CPMXX.COM
```

at the completion of the MOVCPM function, where *xx* is the value indicated in the SYSGEN message. The CP/M memory image on the currently logged disk is in a form that can be patched. This is necessary when operating in a nonstandard environment where the BIOS must be altered for a particular peripheral device configuration, as described in Section 6.

The following are valid MOVCPM commands:

MOVCPM 48	Constructs a 48K version of CP/M and starts execution.
MOVCPM 48 *	Constructs a 48K version of CP/M in preparation for permanent recording; the response is: READY FOR "SYSGEN" OR "SAVE 34 CPM48.COM"
MOVCPM	Constructs a maximum memory version of CP/M and starts execution.

The newly created system is serialized with the number attached to the original disk and is subject to the conditions of the Digital Research Software Licensing Agreement.

1.7. BDOS Error Messages

There are three error situations that the Basic Disk Operating System intercepts during file processing. When one of these conditions is detected, the BDOS prints the message:

```
BDOS Err On d: error
```

where *d* is the drive name and *error* is one of the three error messages:

- Bad Sector
- Select
- R/O

The Bad Sector message indicates that the disk controller electronics has detected an error condition in reading or writing the disk. This condition is generally caused by a malfunctioning disk controller or an extremely worn disk. If you find that CP/M reports this error more than once a month, the state of the controller electronics and the condition of the media should be checked.

You can also encounter this condition in reading files generated by a controller produced by a different manufacturer. Even though controllers claim to be IBM compatible, one often finds

small differences in recording formats. The MDS-800 controller, for example, requires two bytes of ones following the data CRC byte, which is not required in the IBM format. As a result, disks generated by the Intel MDS can be read by almost all other IBM- compatible systems, while disk files generated on other manufacturers' equipment produce the `Bad Sector` message when read by the MDS. To recover from this condition, press a CTRL-C to reboot (the safest course), or a return, which ignores the bad sector in the file operation.

Note:

pressing a return might destroy disk integrity if the operation is a directory write. Be sure you have adequate back-ups in this case.

The `Select` error occurs when there is an attempt to address a drive beyond the range supported by the BIOS. In this case, the value of `d` in the error message gives the selected drive. The system reboots following any input from the console.

The `R/O` (Read-Only) message occurs when there is an attempt to write to a disk or file that has been designated as Read-Only in a `STAT` command or has been set to Read-Only by the BDOS. Reboot CP/M by using the warm start procedure, CTRL-C, or by performing a cold start whenever the disks are changed. If a changed disk is to be read but not written, BDOS allows the disk to be changed without the warm or cold start, but internally marks the drive as Read-Only. The status of the drive is subsequently changed to Read- Write if a warm or cold start occurs. On issuing this message, CP/M waits for input from the console. An automatic warm start takes place following any input.

1.8. Operation of CP/M on the MDS

This section gives operating procedures for using CP/M on the Intel MDS microcomputer development system. Basic knowledge of the MDS hardware and software systems is assumed.

CP/M is initiated in essentially the same manner as the Intel ISIS operating system. The disk drives are labeled 0 through 3 on the MDS, corresponding to CP/M drives A through D, respectively. The CP/M system disk is inserted into drive 0, and the BOOT and RESET switches are pressed in sequence. The interrupt 2 light should go on at this point. The space bar is then pressed on the system console, and the light should go out. If it does not, the user should check connections and baud rates. The BOOT switch is turned off, and the CP/M sign-on message should appear at the selected console device, followed by the `A>` system prompt. You can then issue the various resident and transient commands.

The CP/M system can be restarted (warm start) at any time by pushing the `INT 0` switch on the front panel. The built-in Intel ROM monitor can be initiated by pushing the `INT 7` switch, which

generates an RST 7, except when operating under DDT, in which case the DDT program gets control instead.

Diskettes can be removed from the drives at any time, and the system can be shut down during operation without affecting data integrity. Do not remove a disk and replace it with another without rebooting the system (cold or warm start) unless the inserted disk is Read-Only.

As a result of hardware hang-ups or malfunctions, CP/M might print the following message:

```
BDOS Err On d: Bad Sector
```

where *d* is the drive that has a permanent error. This error can occur when drive doors are opened and closed randomly, followed by disk operations, or can be caused by a disk, drive, or controller failure. You can optionally elect to ignore the error by pressing a single return at the console. The error might produce a bad data record, requiring re-initialization of up to 128 bytes of data. You can reboot the CP/M system and try the operation again.

Termination of a CP/M session requires no special action, except that it is necessary to remove the disks before turning the power off to avoid random transients that often make their way to the drive electronics.

You should use IBM-compatible disks rather than disks that have previously been used with any ISIS version. In particular, the ISIS FORMAT operation produces nonstandard sector numbering throughout the disk. This nonstandard numbering seriously degrades the performance of CP/M, and causes CP/M to operate noticeably slower than the distribution version. If it becomes necessary to reformat a disk, which should not be the case for standard disks, a program can be written under CP/M that causes the MDS 800 controller to reformat with sequential sector numbering (1-26) on each track.

Generally, IBM-compatible 8-inch disks do not need to be formatted. However, 5 1/4-inch disks need to be formatted.

2. The CP/M Editor

2.1. Introduction to ED

ED is the context editor for CP/M, and is used to create and alter CP/M source files. To start ED, type a command of the following form:

ED *filename*

or

ED *filename.typ*

Generally, ED reads segments of the source file given by *filename* or *filename.typ* into the central memory, where you edit the file and it is subsequently written back to disk after alterations. If the source file does not exist before editing, it is created by ED and initialized to empty. The overall operation of ED is shown in Figure 2-1.

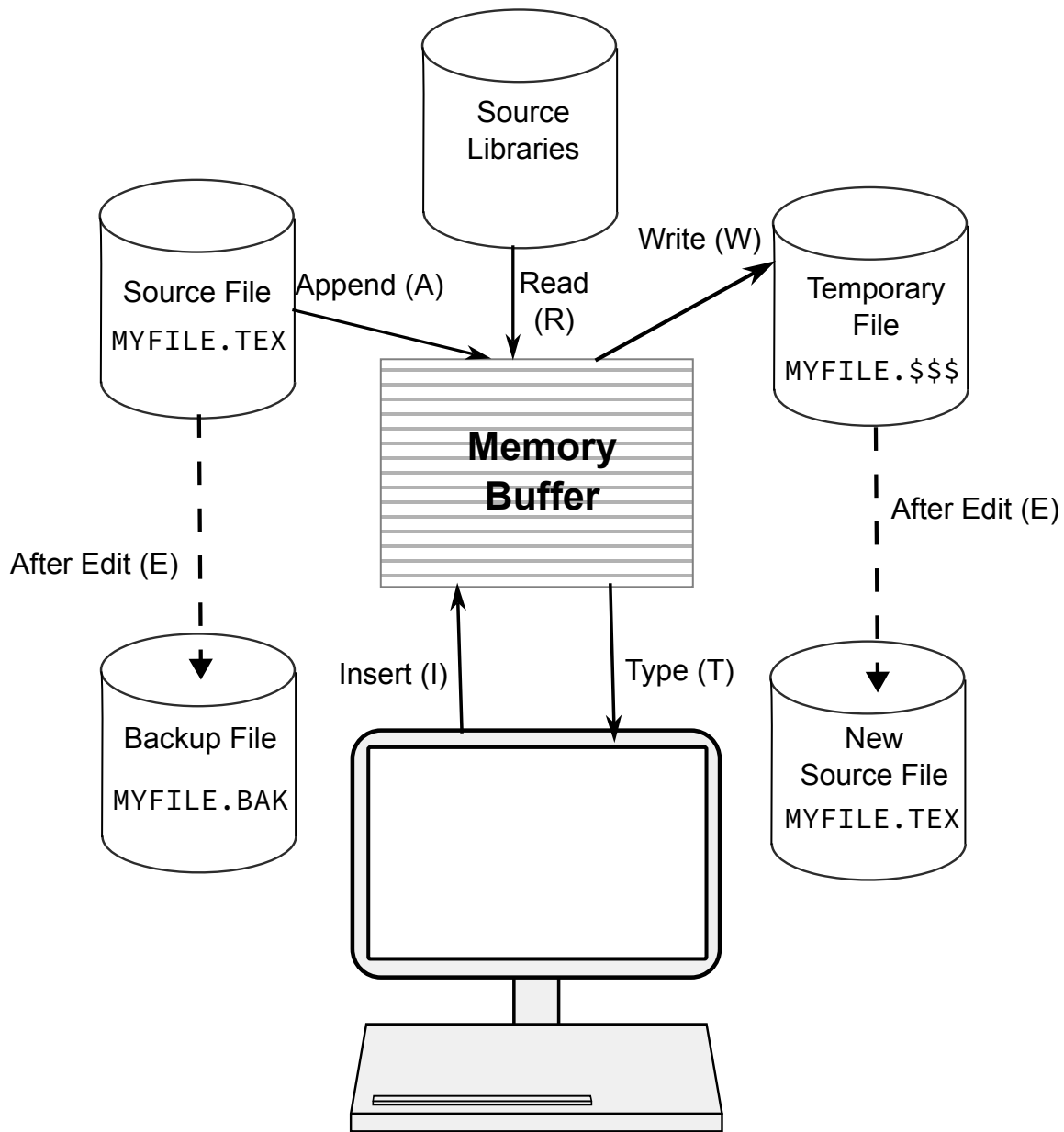


Figure 2-1: Overall ED Operation

2.1.1. ED Operation

ED operates upon the source file, shown in Figure 2-1 by MYFILE.TEX, and passes all text through a memory buffer where the text can be viewed or altered. The number of lines that can be maintained in the memory buffer varies with the line length, but has a total capacity of about 5000 characters in a 20K CP/M system.

Edited text material is written into a temporary work file under your command. Upon termination of the edit, the memory buffer is written to the temporary file, followed by any remaining (unread) text in the source file. The name of the original file is changed from MYFILE.TEX to MYFILE.BAK so that the most recent edited source file can be reclaimed if necessary. See the

CP/M commands ERASE and RENAME. The temporary file is then changed from MYFILE. \$\$\$ to MYFILE. TEX, which becomes the resulting edited file.

The memory buffer is logically between the source file and working file, as shown in Figure 2-2

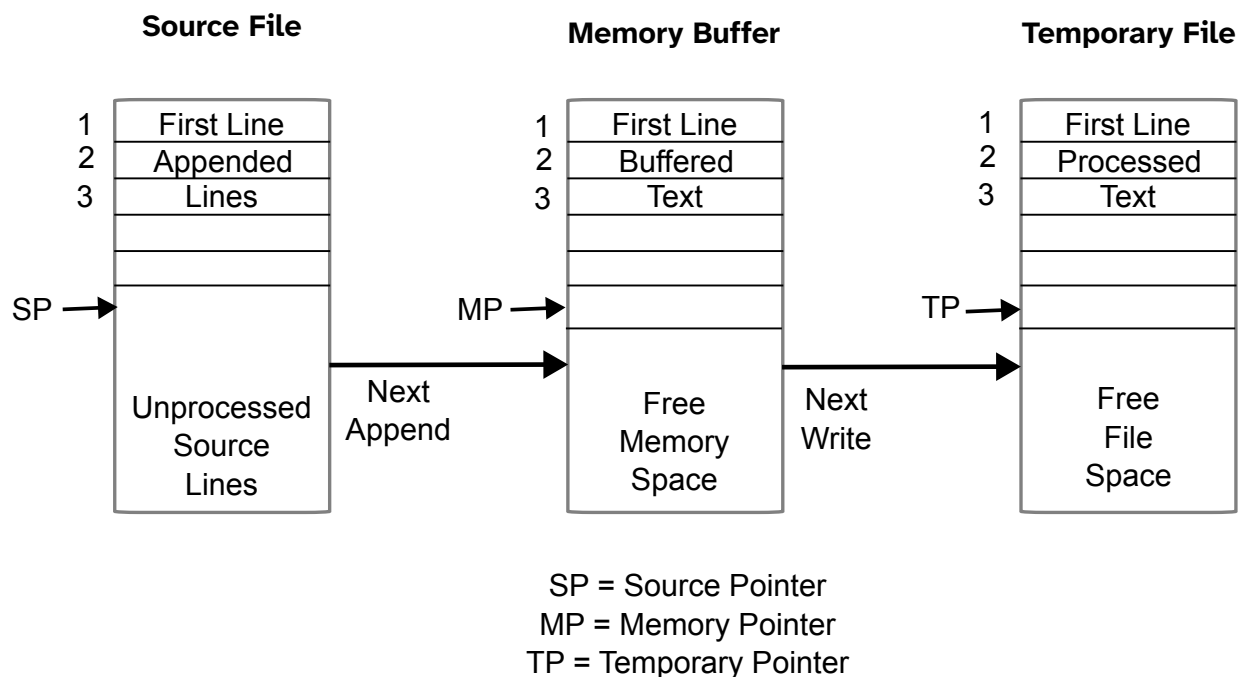


Figure 2-2: Memory Buffer Organization

2.1.2. Text Transfer Commands

Command	Result
<i>nA</i>	Appends the next <i>n</i> unprocessed source lines from the source file at SP to the end of the memory buffer at MP . Increment SP and MP by <i>n</i> . If upper-case translation is set (see the U command) and the A command is typed in upper-case, all input lines will automatically be translated to upper-case.
<i>nW</i>	Writes the first <i>n</i>) lines of the memory buffer to the temporary file free space. Shift the remaining lines <i>n</i> + 1 through MP to the top of the memory buffer. Increment TP by <i>n</i> .
E	Ends the edit. Copy all buffered text to temporary file and copy all unprocessed source lines to temporary file. Rename files.
H	Moves to head of new file by performing automatic E command. The temporary file becomes the new source file, the memory buffer is emptied, and a new temporary file is created. The effect is equivalent to issuing an E

Command	Result
	command, followed by a re-invocation of ED, using MYFILE.TEX as the file to edit.
O	Returns to original file. The memory buffer is emptied, the temporary file is deleted, and the SP is returned to position 1 of the source file. The effects of the previous editing commands are thus nullified.
Q	Quits edit with no file alterations, returns to CP/M.

Table 2-5: ED Text Transfer Commands

There are a number of special cases to consider. If the integer n is omitted in any ED command where an integer is allowed, then 1 is assumed. Thus, the commands A and W append one line and write one line, respectively. In addition, if a pound sign # is given in the place of n , then the integer 65535 is assumed (the largest value for n that is allowed). Because most source files can be contained entirely in the memory buffer, the command #A is often issued at the beginning of the edit to read the entire source file to memory. Similarly, the command #W writes the entire buffer to the temporary file.

Two special forms of the A and W commands are provided as a convenience. The command OA fills the current memory buffer at least half full, while OW writes lines until the buffer is at least half empty. An error is issued if the memory buffer size is exceeded. You can then enter any command, such as W, that does not increase memory requirements. The remainder of any partial line read during the overflow will be brought into memory on the next successful append.

2.1.3. Memory Buffer Organization

The memory buffer can be considered a sequence of source lines brought in with the A command from a source file. The memory buffer has an imaginary character pointer (**CP**) that moves throughout the memory buffer under command of the operator. The memory buffer appears logically as shown in Figure 2-3, where the dashes represent characters of the source line of indefinite length, terminated by carriage return (<cr>) and line-feed (<lf>) characters, and **CP** represents the imaginary character pointer. Note that the **CP** is always located ahead of the first character of the first line, behind the last character of the last line, or between two characters. The current line **CL** is the source line that contains the **CP**.

Memory Buffer

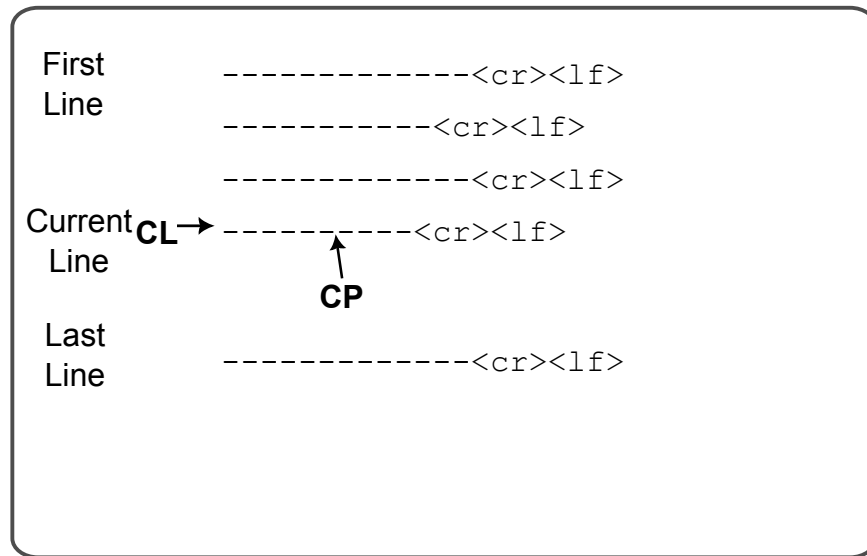


Figure 2-3: Logical Organization of Memory Buffer

2.1.4. Line Numbers and ED Start-up

ED produces absolute line number prefixes that are used to reference a line or range of lines. The absolute line number is displayed at the beginning of each line when ED is in insert mode (see the I command in Section 2.1.5). Each line number takes the form

`nnnnn:`

where *nnnnn* is an absolute line number in the range of 1 to 65535. If the memory buffer is empty or if the current line is at the end of the memory buffer, *nnnnn* appears as 5 blanks.

You can reference an absolute line number by preceding any command by a number followed by a colon, in the same format as the line number display. In this case, the ED program moves the current line reference to the absolute line number, if the line exists in the current memory buffer. The line denoted by the absolute line number must be in the memory buffer (see the A command). Thus, the command

`345:T`

is interpreted as move to absolute 345, and type the line. Absolute line numbers are produced only during the editing process and are not recorded with the file. In particular, the line numbers will change following a deleted or expanded section of text.

You can also reference an absolute line number as a backward or forward distance from the current line by preceding the absolute number by a colon. Thus, the command

`:400T`

is interpreted as type from the current line number through the line whose absolute number is 400. Combining the two line reference forms, the command

```
345 : 400T
```

is interpreted as move to absolute line 345, then type through absolute line 400. Absolute line references of this sort can precede any of the standard ED commands.

Line numbering is controlled by the v (Verify Line Numbers) command. Line numbering can be turned off by typing the -v command.

If the file to edit does not exist, ED displays the following message:

```
NEW FILE
```

To move text into the memory buffer, you must enter an i command before typing input lines and terminate each line with a carriage return. A single CTRL-Z character returns ED to command mode.

2.1.5. Memory Buffer Operation

When ED begins, the memory buffer is empty. You can either append lines from the source file with the A command, or enter the lines directly from the console with the insert command. The insert command takes the following form:

```
I
```

ED then accepts any number of input lines. You must terminate each line with a <cr> (also rendered as ↵ or C_R) (the <lf>, also rendered as L_F is supplied automatically). A single CTRL-Z, denoted by an up arrow ^Z, returns ED to command mode. The **CP** is positioned after the last character entered. The following sequence:

```
I↵
NOW IS THE↵
TIME FOR↵
ALL GOOD MEN↵
^Z
```

leaves the memory buffer as

```
NOW IS THE $C_R$  $L_F$ 
TIME FOR $C_R$  $L_F$ 
ALL GOOD MEN $C_R$  $L_F$ 
```

Generally, ED accepts command letters in upper- or lower-case. If the command is upper-case, all input values associated with the command are translated to upper-case. If the I command is typed, all input lines are automatically translated internally to upper-case. The lower-case form of the i command is most often used to allow both upper- and lower-case letters to be entered.

Various commands can be issued that control the **CP** or display source text in the vicinity of the **CP**. The commands shown below with a preceding n indicate that an optional unsigned value can be specified. When preceded by $+-$, the command can be unsigned, or have an optional preceding plus or minus sign. As before, the pound sign $\#$ is replaced by 65535. If an integer n is optional, but not supplied, then $n = 1$ is assumed. Finally, if a plus sign is optional, but none is specified, then $+$ is assumed.

Command	Result
$+B$ or $-B$	Move CP to beginning of memory buffer if $+B$ and to bottom if $-B$
$+nC$ or $-nC$	Move CP by n characters (moving ahead if $+$), counting the $C_R L_F$ as two characters.
$+nD$ or $-nD$	Delete n characters ahead of CP if plus and behind CP if minus.
$+nK$ or $-nK$	Kill (remove) n lines of source text using CP as the current reference. If CP is not at the beginning of the current line when K is issued, the characters before CP remain if $+$ is specified, while the characters after CP remain if $-$ is given in the command.
$+nL$ or $-nL$	If $n = 0$, move CP to the beginning of the current line, if it is not already there. If $n \neq 0$, first move the CP to the beginning of the current line and then move it to the beginning of the line that is n lines down (if $+$) or up (if $-$). The CP will stop at the top or bottom of the memory buffer if too large a value of n is specified.
$+nT$ or $-nT$	If $n = 0$, type the contents of the current line up to CP . If $n = 1$, type the contents of the current line from CP to the end of the line. If $n > 1$, type the current line along with $n + 1$ lines that follow, if $+$ is specified. Similarly, if $n > 1$ and $-$ is given, type the previous n lines up to the CP . Any key can be depressed to abort long type-outs.
$+n$ or $-n$	Equivalent to $+-nLT$, which moves up or down and types a single line.

Table 2-6: Editing Commands

2.1.6. Command Strings

Any number of commands can be typed contiguously (up to the capacity of the console buffer) and are executed only after you press the $\langle cr \rangle$. Table 2-7 summarizes the CP/M console line-editing commands used to control the input command line.

Character	Meaning
CTRL-C	Reboots CP/M system when pressed at start of line.
CTRL-E	Physical end of line; carriage is returned, but line is not sent until the carriage return key is pressed.
CTRL-H	Backspaces one character position.
CTRL-J	Terminates current input (line-feed).
CTRL-M	Terminates current input (carriage return).
CTRL-R	Retypes current command line; types a clean line following character deletion with rubouts.
CTRL-U	Deletes the entire line typed at the console.
CTRL-X	Same as CTRL-U.
CTRL-Z	Ends input from the console (used in PIP and ED).
Rubout/DEL	Deletes and echoes the last character typed at the console.

Table 2-7: ED Line-editing Controls

Suppose the memory buffer contains the characters shown in the previous section, with the **CP** “↑” following the last character of the buffer. In the following example, the command strings on the left produce the results shown to the right. Use lower-case command letters to avoid automatic translation of strings to upper-case.

Command String

B2T↵

Effect

Move to the beginning of the buffer and type two lines:

NOW IS THE↵
TIME FOR↵

The result in the memory buffer is

↑NOW IS THE^{C_RL_F}
TIME FOR^{C_RL_F}
ALL GOOD MEN^{C_RL_F}

Command String

5C0T↵

Effect

Move **CP** five characters and type the beginning of the line NOW I, The result in the memory buffer is

NOW I†S THE^{C_R}L_F
TIME FOR^{C_R}L_F
ALL GOOD MEN^{C_R}L_F

2L-T↵

Move two lines down and type the previous line TIME FOR. The result in the memory buffer is

NOW IS THE^{C_R}L_F
TIME FOR^{C_R}L_F
†ALL GOOD MEN^{C_R}L_F

-L#K↵

Move up one line, delete 65535 lines that follow. The result in the memory buffer is

NOW IS THE^{C_R}L_F†

I↵

Insert two lines of text with automatic translation to upper-case. The result in the memory buffer is

TIME TO↵

INSERT↵

^Z

NOW IS THE^{C_R}L_F
TIME TO^{C_R}L_F
INSERT^{C_R}L_F†

-2L#T↵

Move up two lines and type 65535 lines ahead of **CP** NOW IS THE. The result in the memory

NOW IS THE^{C_R}L_F
†TIME TO^{C_R}L_F
INSERT^{C_R}L_F

↵

Move down one line and type one line INSERT. The result in the memory buffer is

Command String

Effect

NOW IS THE^{C_RL_F}
TIME TO^{C_RL_F}
↑INSERT^{C_RL_F}

2.1.7. Text Search and Alteration

ED has a command that locates strings within the memory buffer. The command takes the form $nFs^{\wedge}Z$

where s represents the string to match, followed by either a <cr> ↵ or CTRL-Z, denoted by $^{\wedge}Z$. ED starts at the current position of **CP** and attempts to match the string. The match is attempted n times and, if successful, the **CP** is moved directly after the string. If the n matches are not successful, the **CP** is not moved from its initial position. Search strings can include CTRL-L, which is replaced by the pair of symbols ^{C_RL_F}.

The following commands illustrate the use of the F command:

Command String

Effect

B#T↵

Move to the beginning and type the entire buffer. The result in the memory buffer is

↑NOW IS THE^{C_RL_F}
TIME FOR^{C_RL_F}
ALL GOOD MEN^{C_RL_F}

FS T↵

Find the end of the string s T. The result in the memory buffer is

NOW IS T↑HE^{C_RL_F}
TIME FOR^{C_RL_F}
ALL GOOD MEN^{C_RL_F}

FI^s^Z0TT

Find the next I and type to the **CP**; then type the remainder of the current line ME FOR. The result in the memory buffer is

Command String**Effect**

NOW IS THE^{C_RL_F}
 TIME FOR^{C_RL_F}
 ALL GOOD MEN^{C_RL_F}

An abbreviated form of the insert command is also allowed, which is often used in conjunction with the **F** command to make simple textual changes. The form is

Is^Z

or

Is↵

where s is the string to insert. If the insertion string is terminated by a CTRL-Z, the string is inserted directly following the **CP**, and the **CP** is positioned directly after the string. The action is the same if the command is followed by a <cr> ↵ except that a ^{C_RL_F} is automatically inserted into the text following the string. The following command sequences are examples of the **F** and **I** commands:

Command String**Effect**

BITHIS IS ^Z↵

Insert THIS IS at the beginning of the text. The result in the memory buffer is

THIS IS ↵NOW IS THE^{C_RL_F}
 TIME FOR^{C_RL_F}
 ALL GOOD MEN^{C_RL_F}

FTIME^Z-DIPLACE^Z↵

Find TIME and delete it; then insert PLACE. The result in the memory buffer

THIS IS NOW IS THE^{C_RL_F}
 PLACE ↵FOR^{C_RL_F}
 ALL GOOD MEN^{C_RL_F}

3FO^Z-3D5DICHANGES^Z↵

Find third occurrence of O (that is, the second O in GOOD), delete previous 3 characters and the subsequent 5 characters; then insert CHANGES. The result in the memory buffer is

Command String

Effect

-8CISOURCE↵

THIS IS NOW IS THE^{C_RL_F}

PLACE FOR^{C_RL_F}

ALL CHANGES†^{C_RL_F}

Move back 8 characters and insert the line

SOURCE^{C_RL_F}. The result in the memory buffer is

THIS IS NOW IS THE^{C_RL_F}

PLACE FOR^{C_RL_F}

ALL SOURCE^{C_RL_F}

†CHANGES^{C_RL_F}

ED also provides a single command that combines the F and I commands to perform simple string substitutions. The command takes the following form:

*n*S1^ZS2↵

or

*n*S1^ZS2^Z

and has exactly the same effect as applying the following command string a total of *n* times:

F S1^Z - kDIS2↵

or

F S1^Z - kDIS2^Z

where *k* is the length of the string. ED searches the memory buffer starting at the current position of **CP** and successively substitutes the second string for the first string until the end of buffer, or until the substitution has been performed *n* times.

As a convenience, a command similar to F is provided by ED that automatically appends and writes lines as the search proceeds. The form is

*n*NS↵

or

*n*NS^Z

which searches the entire source file for the *n*th occurrence of the strings (you should recall that F fails if the string cannot be found in the current buffer). The operation of the N command is precisely the same as F except in the case that the string cannot be found within the current

memory buffer. In this case, the entire memory content is written (that is, an automatic #w is issued). Input lines are then read until the buffer is at least half full, or the entire source file is exhausted. The search continues in this manner until the string has been found n times, or until the source file has been completely transferred to the temporary file.

A final line editing function, called the Juxtaposition command, takes the form

$nJS1^ZS2^ZS3\downarrow$

or

$nJS1^ZS2^ZS3^Z$

with the following action applied n times to the memory buffer: search from the current **CP** for the next occurrence of the string $S1$. If found, insert the string $S2$, and move **CP** to follow $S2$. Then delete all characters following **CP** up to, but not including, the string $S3$, leaving **CP** directly after $S2$. If $S3$ cannot be found, then no deletion is made. If the current line is

NOW IS THE TIME $C_R^L F$

the command

$JW^ZWHAT^Z^L\downarrow$

results in

NOW WHAT $C_R^L F$

You should recall that a L (CTRL-L) represents the pair $C_R^L F$ in search and substitute strings.

The number of characters ED allows in the F, S, N, and J commands is limited to 100 symbols.

2.1.8. Source Libraries

ED also allows the inclusion of source libraries during the editing process with the R command. The form of this command is

$Rfilename\downarrow$

or

$Rfilename^Z$

where filename is the primary filename of a source file on the disk with an assumed filetype of LIB. ED reads the specified file, and places the characters into the memory buffer after **CP**, in a manner similar to the I command. Thus, if the command

$RMACRO\downarrow$

is issued by the operator, ED reads from the file MACRO.LIB until the end-of-file and automatically inserts the characters into the memory buffer.

ED also includes a block move facility implemented through the x (Transfer) command. The form

*n*X

transfers the next *n* lines from the current line to a temporary file called

X\$\$\$\$\$.LIB

which is active only during the editing process. You can reposition the current line reference to any portion of the source file and transfer lines to the temporary file. The transferred lines accumulate one after another in this file and can be retrieved by simply typing

R

which is the trivial case of the library read command. In this case, the entire transferred set of lines is read into the memory buffer. Note that the X command does not remove the transferred lines from the memory buffer, although a K command can be used directly after the X, and the R command does not empty the transferred LIB file. That is, given that a set of lines has been transferred with the X command, they can be reread any number of times back into the source file. The command

OX

is provided to empty the transferred line file.

Note that upon normal completion of the ED program through Q or E, the temporary LIB file is removed. If ED is aborted with a CTRL-C, the LIB file will exist if lines have been transferred, but will generally be empty (a subsequent ED invocation will erase the temporary file).

2.1.9. Repetitive Command Execution

The macro command M allows you to group ED commands together for repeated evaluation. The M command takes the following form:

*n*MCS↵

or

*n*MCS^Z

where CS represents a string of ED commands, not including another M command. ED executes the command string *n* times if *n* > 1. If *n* = 0 or *n* = 1, the command string is executed repetitively until an error condition is encountered (for example, the end of the memory buffer is reached with an F command).

As an example, the following macro changes all occurrences of GAMMA to DELTA within the current buffer, and types each line that is changed:

MFGAMMA^Z-5DIDELTA^ZOTT↵

or equivalently

MSGAMMA^ZDELTA^ZOTT↵

2.2. ED Error Conditions

On error conditions, ED prints the message `BREAK X AT C` where *X* is one of the error indicators shown in Table 2-8

Symbol	Meaning
?	Unrecognized command.
>	Memory buffer full (use one of the commands D, K, N, S, or W to remove characters); F, N, or S strings too long.
#	Cannot apply command the number of times specified (for example, in F command).
0	Cannot open LIB file in R command.

Table 2-8: Error Message Symbols

If there is a disk error, CP/M displays the following message:

```
BDOS Err On d: Bad Sector
```

You can choose to ignore the error by pressing RETURN at the console (in this case, the memory buffer data should be examined to see if they were incorrectly read), or you can reset the system with a CTRL-C and reclaim the backup file if it exists. The file can be reclaimed by first typing the contents of the BAK file to ensure that it contains the proper information. For example, type the following:

```
TYPE X.BAK
```

where *x* is the file being edited. Then remove the primary file

```
ERA X.Y
```

and rename the BAK file

```
REN X.Y=X.BAK
```

The file can then be reedited, starting with the previous version.

ED also takes file attributes into account. If you attempt to edit a Read-Only file, the message

```
** FILE IS READ/ONLY **
```

appears at the console. The file can be loaded and examined, but cannot be altered. You must end the edit session and use STAT to change the file attribute to R/W. If the edited file has the system attribute set, the following message:

```
"SYSTEM" FILE NOT ACCESSIBLE
```

is displayed and the edit session is aborted. Again, the STAT program can be used to change the system attribute, if desired.

2.3. Control Characters and Commands

Table 2-9 summarizes the control characters and commands available in ED.

Control Character	Function
CTRL-C	System reboot
CTRL-E	Physical <cr><lf> (not actually entered in command)
CTRL-H	Backspaces
CTRL-I	Logical tab (cols 1, 8, 16, ...)
CTRL-R	Repeat line
CTRL-U	Line delete
CTRL-X	Line delete
CTRL-Z	String terminator
rubout/del	Character delete

Table 2-9: ED Control Characters

Table 2-10 summarizes the commands used in ED.

Command	Result
<i>nA</i>	Append lines
<i>+B or -B</i>	Begin or bottom of buffer
<i>+nC or -nC</i>	Move character positions
<i>+nD or -nD</i>	Delete characters
E	End edit and close files (normal end)
<i>nF</i>	Find string
H	End edit, close and reopen files

Command	Result
I or i	Insert characters, use i if both upper- and lower-case characters are to be entered.
nJ	Place strings in juxtaposition
+nK or -nK	Kill lines
+nL or -nL	Move down/up lines
nM	Macro definition
nN	Find next occurrence with autoscan
O	Return to original file
+nP or -nP	Move and print pages
Q	Quit with no file changes
R	Read library file
nS	Substitute strings
+nT or -nT	Type lines
U or -U	Translate lower- to upper-case if U, no translation if -U
V	Verify line numbers, or show remaining free character space
ØV	A special case of the V command, ØV, prints the memory buffer statistics in the form <i>free/total</i> where <i>free</i> is the number of free bytes in the memory buffer (in decimal) and <i>total</i> is the size of the memory buffer
nW	Write lines
nZ	Wait (sleep) for approximately n seconds
+n or -n	Move and type

Table 2-10: ED Commands

Because of common typographical errors, ED requires several potentially disastrous commands to be typed as single letters, rather than in composite commands. The following commands:

E end

H head

O original

Q quit

must be typed as single letter commands.

The commands I, J, M, N, R, and S should be typed as i, j, m, n, r, and s if both upper- and lower-case characters are used in the operation, otherwise all characters are converted to upper-case. When a command is entered in upper-case, ED automatically converts the associated string to upper-case, and vice versa.

3. CP/M Assembler

3.1. Introduction

The CP/M assembler reads assembly-language source files from the disk and produces 8080 machine language in Intel hex format. To start the CP/M assembler, type a command in one of the following forms:

ASM filename

ASM filename.parms

In both cases, the assembler assumes there is a file on the disk with the name:

filename.asm

which contains an 8080 assembly-language source file. The first and second forms shown above differ only in that the second form allows parameters to be passed to the assembler to control source file access and hex and print file destinations.

In either case, the CP/M assembler loads and prints the message:

CP/M ASSEMBLER VER *n.n*

where *n.n* is the current version number. In the case of the first command, the assembler reads the source file with assumed filetype *ASM* and creates two output files

filename.hex

filename.prn

The *HEX* file contains the machine code corresponding to the original program in Intel hex format, and the *PRN* file contains an annotated listing showing generated machine code, error flags, and source lines. If errors occur during translation, they are listed in the *PRN* file and at the console.

The form *ASM filename.parms* is used to redirect input and output files from their defaults. In this case, the *parms* portion of the command is a three-letter group that defines the origin of the source file, the destination of the hex file, and the destination of the print file. The form is

filename.p1p2p3

where *p1*, *p2* and *p3* are single letters.

p1 can be A, B, ..., P which designates the disk name that contains the source file.

p2 can be A, B, ..., P, which designates the disk name that will receive the hex file; or, *p2* can be Z which skips the generation of the hex file.

p3 can be A, B, ..., P which designates the disk name that will receive the print file. *p3* can also be specified as X which places the listing at the console; or Z which skips generation of the print file.

Thus, the command

```
ASM X.AAA
```

indicates that the source, X.HEX and print, X.PRN files are also to be created on disk A. This form of the command is implied if the assembler is run from disk A. Given that you are currently addressing disk A, the above command is the same as

```
ASM X
```

The command

```
ASM X.ABX
```

indicates that the source file is to be taken from disk A, the hex file is to be placed on disk B, and the listing file is to be sent to the console.

The command

```
ASM X.BZZ
```

takes the source file from disk B and skips the generation of the hex and print files. This command is useful for fast execution of the assembler to check program syntax.

The source program format is compatible with the Intel 8080 assembler. Macros are not implemented in ASM; see the optional MAC macro assembler. There are certain extensions in the CP/M assembler that make it somewhat easier to use. These extensions are described below.

3.2. Program Format

An assembly-language program acceptable as input to the assembler consists of a sequence of statements of the form

line# label operation operand ;comment

where any or all of the fields may be present in a particular instance. Each assembly language statement is terminated with a carriage return and line-feed (the line-feed is inserted automatically by the ED program), or with the character !, which is treated as an end-of-line by the assembler. Thus, multiple assembly-language statements can be written on the same physical line if separated by exclamation point symbols.

The *line#* is an optional decimal integer value representing the source program line number, and ASM ignores this field if present.

The *label* field takes either of the following forms:

identifier

identifier:

The label field is optional, except where noted in particular statement types. The identifier is a sequence of alphanumeric characters where the first character is alphabetic. Identifiers can be

freely used by the programmer to label elements such as program steps and assembler directives, but cannot exceed 16 characters in length. All characters are significant in an identifier, except for the embedded dollar symbol \$, which can be used to improve readability of the name. Further, all lower-case alphabetic characters are treated as upper-case. The following are all valid instances of labels:

- x
- xy
- Long\$name
- x:
- yx1:
- Longer\$name'data:
- X1Y2
- X1x2
- x234\$5678\$9012#456:

The *operation* field contains either an **assembler directive** or **pseudo operation**, or an **8080 machine operation code**. The pseudo operations and machine operation codes are described in Section 3.3.

Generally, the *operand* field of the statement contains an expression formed out of constants and labels, along with arithmetic and logical operations on these elements. Again, the complete details of properly formed expressions are given in Section 3.3.

The *comment* field contains arbitrary characters following the semicolon symbol until the next real or logical end-of-line. These characters are read, listed, and otherwise ignored by the assembler. The CP/M assembler also treats statements that begin with an * in column one as comment statements that are listed and ignored in the assembly process.

The assembly-language program is formulated as a sequence of statements of the above form, terminated by an optional END statement. All statements following the END are ignored by the assembler.

3.3. Forming the Operand

To describe the operation codes and pseudo operations completely, it is necessary first to present the form of the operand field, since it is used in nearly all statements. Expressions in the operand field consist of simple operands, labels, constants, and reserved words, combined in properly formed sub-expressions by arithmetic and logical operators. The expression computation is carried out by the assembler as the assembly proceeds. Each expression must produce a 16-bit value during the assembly. Further, the number of significant digits in the result must not exceed the intended use. If an expression is to be used in a byte move immediate

instruction, the most significant 8 bits of the expression must be zero. The restriction on the expression significance is given with the individual instructions.

3.3.1. Labels

A label is an identifier that occurs on a particular statement. In general, the label is given a value determined by the type of statement that it precedes. If the label occurs on a statement that generates machine code or reserves memory space (for example, a `MOV` instruction or a `DS` pseudo operation), the label is given the value of the program address that it labels. If the label precedes an `EQU` or `SET`, the label is given the value that results from evaluating the operand field. Except for the `SET` statement, an identifier can label only one statement.

When a label appears in the operand field, its value is substituted by the assembler. This value can then be combined with other operands and operators to form the operand field for a particular instruction.

3.3.2. Numeric Constants

A numeric constant is a 16-bit value in one of several bases. The base, called the radix of the constant, is denoted by a trailing radix indicator. The following are radix indicators:

B is a binary constant (base 2).

O is a octal constant (base 8).

Q is a octal constant (base 8).

D is a decimal constant (base 10).

H is a hexadecimal constant (base 16).

Q is an alternate radix indicator for octal numbers because the letter **o** is easily confused with the digit **0**. Any numeric constant that does not terminate with a radix indicator is a decimal constant.

A constant is composed as a sequence of digits, followed by an optional radix indicator, where the digits are in the appropriate range for the radix. Binary constants must be composed of **0** and **1** digits, octal constants can contain digits in the range **0**-**7**, while decimal constants contain decimal digits. Hexadecimal constants contain decimal digits as well as hexadecimal digits **A**(10D), **B**(11D), **C**(12D), **D**(13D), **E**(14D), and **F**(15D). Note that the leading digit of a hexadecimal constant must be a decimal digit to avoid confusing a hexadecimal constant with an identifier. A leading **0** will always suffice. A constant composed in this manner must evaluate to a binary number that can be contained within a 16-bit counter, otherwise it is truncated on the right by the assembler.

Similar to identifiers, embedded **\$** signs are allowed within constants to improve their readability. Finally, the radix indicator is translated to upper-case if a lower-case letter is encountered. The following are all valid instances of numeric constants:

- 1234
- 1234D
- 1100B
- 1111\$0000\$1111\$0000B
- 1234H
- 0FFEh
- 33770
- 33\$77\$22Q
- 3377o
- 0fe3h
- 1234d
- 0ffffh

3.3.3. Reserved Words

There are several reserved character sequences that have predefined meanings in the operand field of a statement. The names of 8080 registers are given below. When they are encountered, they produce the values shown to the right.

Character	Value
A	7
B	0
C	1
D	2
E	3
H	4
L	5
M	6
SP	6
PSW	6

Table 3-11: Reserved Characters

Again, lower-case names have the same values as their upper-case equivalents. Machine instructions can also be used in the operand field; they evaluate to their internal codes. In the

case of instructions that require operands, where the specific operand becomes a part of the binary bit pattern of the instruction, for example, `MOV A, B`, the value of the instruction, in this case `MOV`, is the bit pattern of the instruction with zeros in the optional fields, for example, `MOV` produces 40H.

When the symbol `$` occurs in the operand field, not embedded within identifiers and numeric constants, its value becomes the address of the next instruction to generate, not including the instruction contained within the current logical line.

3.3.4. String Constants

String constants represent sequences of ASCII characters and are represented by enclosing the characters within apostrophe symbols. All strings must be fully contained within the current physical line (thus allowing exclamation point symbols within strings) and must not exceed 64 characters in length. The apostrophe character itself can be included within a string by representing it as a double apostrophe (the two keystrokes `' '`), which becomes a single apostrophe when read by the assembler. In most cases, the string length is restricted to either one or two characters (the `DB` pseudo operation is an exception), in which case the string becomes an 8- or 16-bit value, respectively. Two-character strings become a 16-bit constant, with the second character as the low-order byte, and the first character as the high-order byte. The value of a character is its corresponding ASCII code. There is no case translation within strings; both upper- and lower-case characters can be represented. You should note that only graphic printing ASCII characters are allowed within strings.

Valid String	How the Assembler Reads String
<code>'A' 'AB' 'ab' 'c'</code>	<code>A AB ab c</code>
<code>'''' 'a'''' '''''' ''''''</code>	<code>' a' '' ''</code>
<code>'Walla Walla Wash.'</code>	<code>Walla Walla Wash.</code>
<code>'She said "Hello" to me.'</code>	<code>She said "Hello" to me.</code>
<code>'I said "Hello" to her.'</code>	<code>I said "Hello" to her.</code>

3.3.5. Arithmetic and Logical Operators

The operands described in Section 3.3 can be combined in normal algebraic notation using any combination of properly formed operands, operators, and parenthesized expressions. The operators recognized in the operand field are described below in Table 3-12.

Operators	Meaning
$a + b$	unsigned arithmetic sum of a and b
$a - b$	unsigned arithmetic difference between a and b
$+ b$	unary plus (produces b)
$- b$	unary minus (identical to $0 - b$)
$a * b$	unsigned magnitude multiplication of a and b
a / b	unsigned magnitude division of a by b
$a \text{ MOD } b$	remainder after a / b
$\text{NOT } b$	logical inverse of b (all 0s become 1s, 1s become 0s), where b is considered a 16-bit value
$a \text{ AND } b$	bit-by-bit logical and of a and b
$a \text{ OR } b$	bit-by-bit logical or of a and b
$a \text{ XOR } b$	bit-by-bit logical exclusive or of a and b
$a \text{ SHL } b$	the value that results from shifting a to the left by an amount b , with zero fill
$a \text{ SHR } b$	the value that results from shifting a to the right by an amount b , with zero fill

Table 3-12: Arithmetic and Logical Operators

In each case, a and b represent simple operands (labels, numeric constants, reserved words, and one- or two-character strings) or fully enclosed parenthesized sub-expressions, like those shown in the following examples:

- $10+20$
- $10h+37Q$
- $L1/3$
- $(L2+4) \text{ SHR } 3$
- $('a' \text{ and } 5fh)+'0'$
- $('B'+B)\text{OR}(\text{PSW}+M)$
- $(1+(2+c))\text{shr}(A-(B+1))$

3.3.6. Precedence of Operators

As a convenience to the programmer, ASM assumes that operators have a relative precedence of application that allows the programmer to write expressions without nested levels of parentheses. The resulting expression has assumed parentheses that are defined by the relative precedence. The order of application of operators in unparenthesized expressions is listed below. Operators listed first have highest precedence (they are applied first in an unparenthesized expression), while operators listed last have lowest precedence. Operators listed on the same line have equal precedence, and are applied from left to right as they are encountered in an expression.

```
* / MOD SHL SHR
- +
NOT
AND
OR XOR
```

Thus, the expressions shown to the left below are interpreted by the assembler as the fully parenthesized expressions shown to the right.

Expression	How the Assembler Interprets It
$a*b+c$	$(a*b)+c$
$a+b*c$	$a+(b*c)$
$a \text{ MOD } b*c \text{ SHL } d$	$((a \text{ MOD } b) * c) \text{ SHL } d$
$a \text{ OR } b \text{ AND NOT } c+d \text{ SHL } e$	$a \text{ OR } (b \text{ AND } (\text{NOT } (c + (d \text{ SHL } e))))$

Balanced, parenthesized sub-expressions can always be used to override the assumed parentheses; thus, the last expression above could be rewritten to force application of operators in a different order, as shown:

```
( a OR b ) AND ( NOT c ) + d SHL e
```

This results in these assumed parentheses:

```
( a OR b ) AND ( (NOT c ) + ( d SHL e ) )
```

An unparenthesized expression is well-formed only if the expression that results from inserting the assumed parentheses is well-formed.

3.4. Assembler Directives

Assembler directives are used to set labels to specific values during the assembly, perform conditional assembly, define storage areas, and specify starting addresses in the program. Each

assembler directive is denoted by a pseudo operation that appears in the operation field of the line. The acceptable pseudo operations are shown below in Table 3-13.

Directive	Meaning
ORG	set the program or data origin
END	end program, optional start address
EQU	numeric equate
SET	numeric set
IF	begin conditional assembly
ENDIF	end of conditional assembly
DB	define data bytes
DW	define data words
DS	define data storage area

Table 3-13: Assembler Directives

3.4.1. The ORG Directive

The ORG statement takes the form:

label ORG *expression*

where *label* is an optional program identifier and *expression* is a 16-bit expression, consisting of operands that are defined before the ORG statement. The assembler begins machine code generation at the location specified in the expression. There can be any number of ORG statements within a particular program, and there are no checks to ensure that the programmer is not defining overlapping memory areas. Note that most programs written for the CP/M system begin with an ORG statement of the form:

```
ORG 100H
```

which causes machine code generation to begin at the base of the CP/M transient program area. If a label is specified in the ORG statement, the label is given the value of the expression. This label can then be used in the operand field of other statements to represent this expression.

3.4.2. The END Directive

The END statement is optional in an assembly-language program, but if it is present it must be the last statement. All subsequent statements are ignored in the assembly. The END statement takes the following two forms:

label END

label END *expression*

where the *label* is again optional. If the first form is used, the assembly process stops, and the default starting address of the program is taken as 0000. Otherwise, the *expression* is evaluated, and becomes the program starting address. This starting address is included in the last record of the Intel-formatted machine code hex file that results from the assembly. Thus, most CP/M assembly-language programs end with the statement:

```
END 100H
```

resulting in the default starting address of 100H (beginning of the transient program area).

3.4.3. The EQU Directive

The EQU (equate) statement is used to set up synonyms for particular numeric values. The EQU statement takes the form:

label EQU *expression*

where the *label* must be present and must not label any other statement. The assembler evaluates the *expression* and assigns this value to the identifier given in the *label* field. The identifier is usually a name that describes the value in a more human-oriented manner. Further, this name is used throughout the program to place parameters on certain functions. Suppose data received from a teletype appears on a particular input port, and data is sent to the teletype through the next output port in sequence. For example, you can use this series of equate statements to define these ports for a particular hardware environment:

```
TTYBASE    EQU 10H          ;BASE PORT NUMBER FOR TTY
TTYIN      EQU TTYBASE      ;TTY DATA IN
TTYOUT     EQU TTYBASE+1    ;TTY DATA OUT
```

At a later point in the program, the statements that access the teletype can appear as follows:

```
IN  TTYIN      ;READ TTY DATA TO REG-A
    ....

OUT TTYOUT     ;WRITE DATA TO TTY FROM REG-A
```

making the program more readable than if the absolute I/O ports are used. Further, if the hardware environment is redefined to start the teletype communications ports at 7FH instead of 10H, the first statement need only be changed to

```
TTYBASE    EQU 7FH          ;BASE PORT NUMBER FOR TTY
```

and the program can be reassembled without changing any other statements.

3.4.4. The SET Directive

The SET statement is similar to the EQU, taking the form:

label SET expression

except that the label can occur on other SET statements within the program. The expression is evaluated and becomes the current value associated with the label. Thus, the EQU statement defines a label with a single value, while the SET statement defines a value that is valid from the current SET statement to the point where the label occurs on the next SET statement. The use of the SET is similar to the EQU statement, but is used most often in controlling conditional assembly.

3.4.5. The IF and ENDIF Directive

The IF and ENDIF statements define a range of assembly-language statements that are to be included or excluded during the assembly process. These statements take on the form:

```
IF expression
statement#1
statement#2
...
statement#n
ENDIF
```

When encountering the IF statement, the assembler evaluates the expression following the IF. All operands in the expression must be defined ahead of the IF statement. If the expression evaluates to a nonzero value, then statement#1 through statement#n are assembled. If the expression evaluates to zero, the statements are listed but not assembled. Conditional assembly is often used write a single generic program that includes a number of possible run-time environments, with only a few specific portions of the program selected for any particular assembly. The following program segments, for example, might be part of a program that communicates with either a teletype or a CRT console (but not both) by selecting a particular value for TTY before the assembly begins.

```
TRUE    EQU    0FFFFH    ;DEFINE VALUE OF TRUE
FALSE   EQU    NOT TRUE  ;DEFINE VALUE OF FALSE
;
TTY     EQU    TRUE      ;TRUE IF TTY, FALSE IF CRT
;
TTYBASE EQU    10H       ;BASE OF TTY I/O PORTS
CRTBASE EQU    20H       ;BASE OF CRT I/O PORTS
        IF    TTY        ;ASSEMBLE RELATIVE TO
                        ;TTYBASE
CONIN   EQU    TTYBASE    ;CONSOLE INPUT
```

```

CONOUT EQU TTYBASE+1 ;CONSOLE OUTPUT
ENDIF
;
        IF NOT TTY ;ASSEMBLE RELATIVE TO
        ;CRTBASE
CONIN EQU CRTBASE ;CONSOLE INPUT
CONOUT EQU CRTBASE+1 ;CONSOLE OUTPUT
ENDIF
...
IN CONIN ;READ CONSOLE DATA
OUT CONOUT ;WRITE CONSOLE DATA

```

In this case, the program assembles for an environment where a teletype is connected, based at port 10H. The statement defining TTY can be changed to

```
TTY EQU FALSE
```

and, in this case, the program assembles for a CRT based at port 20H.

3.4.6. The DB Directive

The DB directive allows the programmer to define initialized storage areas in single precision byte format. The DB statement takes the form:

label DB *e#1*, *e#2*, ..., *e#n*

where *e#1* through *e#n* are either expressions that evaluate to 8-bit values (the high-order bit must be zero) or are ASCII strings of length no greater than 64 characters. There is no practical restriction on the number of expressions included on a single source line. The expressions are evaluated and placed sequentially into the machine code file following the last program address generated by the assembler. String characters are similarly placed into memory starting with the first character and ending with the last character. Strings of length greater than two characters cannot be used as operands in more complicated expressions.

Note: ASCII characters are always placed in memory with the parity bit reset (0)

Note Also, there is no translation from lower- to upper-case within strings.

The optional label can be used to reference the data area throughout the remainder of the program. The following are examples of valid DB statements:

```

data:      DB 0,1,2,3,4,5
           DB data and 0ffh,5,377Q,1+2+3+4
sign-on:   DB 'please type your name',CR,LF,0
           DB 'AB' SHR 8,'C','DE',AND 7FH

```

3.4.7. The DW Directive

The DW statement is similar to the DB statement except double- precision two-byte words of storage are initialized. The DW statement takes the form:

label DW *e#1*, *e#2*, ..., *e#n*

where *e#1* through *e#n* are expressions that evaluate to 16-bit results. Note that ASCII strings of one or two characters are allowed, but strings longer than two characters are disallowed. In all cases, the data storage is consistent with the 8080 processor; the least significant byte of the expression is stored first in memory, followed by the most significant byte. The following are examples of DW statements:

```
doub: DW 0ffefh,doub+4,signon-$,255+255
      DW 'a',5,'ab','CD',6 shl 8 or llb.
```

3.4.8. The DS Directive

The DS statement is used to reserve an area of uninitialized memory, and takes the form:

label DS *expression*

where the *label* is optional. The assembler begins subsequent code generation after the area reserved by the DS. Thus, the DS statement given above has exactly the same effect as the following statement:

```
label: EQU $           ;LABEL VALUE IS CURRENT CODE LOCATION
      ORG $+expression ;MOVE PAST RESERVED AREA
```

3.5. Operation Codes

Assembly-language operation codes form the principal part of assembly-language programs and form the operation field of the instruction. In general, ASM accepts all the standard mnemonics for the Intel 8080 microcomputer, which are given in detail in the Intel 8080 Assembly Language Programming Manual. Labels are optional on each input line. The individual operators are listed briefly in the following sections for completeness, although the Intel manuals should be referenced for exact operator details. In Table 3-14 through Table 3-18, bit values have the following meaning:

e3 represents a 3-bit value in the range 0-7 that can be one of the predefined registers A, B, C, D, E, H, L, M, SP, or PSW.

e8 represents an 8-bit value in the range 0-255.

e16 represents a 16-bit value in the range 0-65535.

These expressions can be formed from an arbitrary combination of operands and operators. In some cases, the operands are restricted to particular values within the allowable range, such as the PUSH instruction. These cases are noted as they are encountered.

3.5.1. Jumps, Calls, and Returns

The Jump, Call, and Return instructions allow several different forms that test the condition flags set in the 8080 microcomputer CPU. The forms are shown in Table 3-14.

Form	Bit Value	Example	Meaning
JMP	<i>e16</i>	JMP L1	Jump unconditionally to label
JNZ	<i>e16</i>	JNZ L2	Jump on nonzero condition to label
JZ	<i>e16</i>	JZ 100H	Jump on zero condition to label
JNC	<i>e16</i>	JNC L1+4	Jump no carry to label
JC	<i>e16</i>	JC L3	Jump on carry to label
JPO	<i>e16</i>	JPO \$+8	Jump on parity odd to label
JPE	<i>e16</i>	JPE L4	Jump on even parity to label
JP	<i>e16</i>	JP GAMMA	Jump on positive result to label
JM	<i>e16</i>	JM A1	Jump on minus to label
CALL	<i>e16</i>	CALL S1	Call subroutine unconditionally
CNZ	<i>e16</i>	CNZ S2	Call subroutine on nonzero condition
CZ	<i>e16</i>	CZ 100H	Call subroutine on zero condition
CNC	<i>e16</i>	CNC SI+4	Call subroutine if no carry set
CC	<i>e16</i>	CC S3	Call subroutine if carry set
CPO	<i>e16</i>	CPO \$+8\$	Call subroutine if parity odd
CPE	<i>e16</i>	CPE \$4	Call subroutine if parity even
CP	<i>e16</i>	CP GAMMA	Call subroutine if positive result
CM	<i>e16</i>	CM b1\$c2	Call subroutine if minus flag
RST	<i>e3</i>	RST 0	Programmed restart, equivalent to CALL 8 * <i>e3</i> , except one byte call
RET	-	RET	Return from subroutine
RNZ	-	RNZ	Return if nonzero flag set

Form	Bit Value	Example	Meaning
RZ	-	RZ	Return if zero flag set
RNC	-	RNC	Return if no carry
RC	-	RC	Return if carry flag set
RPO	-	RPO	Return if parity is odd
RPE	-	RPE	Return if parity is even
RP	-	RP	Return if positive result
RM	-	RM	Return if minus flag is set

Table 3-14: Jumps, Calls, and Returns

3.5.2. Immediate Operand Instructions

Several instructions are available that load single- or double-precision registers or single-precision memory cells with constant values, along with instructions that perform immediate arithmetic or logical operations on the accumulator (register A). Table 3-15 describes the immediate operand instructions.

Form and Bit Value	Example	Meaning
MVI e8,e8	MVI B,255	Move immediate data to register A, B, C, D, E, H, L, or M (memory)
ADI e8	ADI 1	Add immediate operand to A without carry
ACI e8	ACI 0FFH	Add immediate operand to A with carry
SUI e8	SUI L + 3	Subtract from A without borrow (carry)
SBI e8	SBI L AND 11B	Subtract from A with borrow (carry)
ANI e8	ANI \$ AND 7FH	Logical and A with immediate data
XRI e8	XRI 1111\$0000B	Exclusive-or A with immediate data
ORI e8	ORI L AND 1+1	Logical-or A with immediate data

Form and Bit Value	Example	Meaning
CPI e8	CPI 'a'	Compare A with immediate data, same as SUI except register A not changed.
LXI e3,e16	LXI B, 100H	Load extended immediate to register pair. e3 must be equivalent to B, D, H, or SP.

Table 3-15: Immediate Operand Instructions

3.5.3. Increment and Decrement Instructions

The 8080 provides instructions for incrementing or decrementing single- and double precision registers. The instructions are described in Table 3-16

Form and Bit Value	Example	Meaning
INR e3	INR E	Single-precision increment register. e3 produces one of A, B, C, D, E, H, L, or M (memory)
DCR e3	DCR A	Single-precision decrement register. e3 produces one of A, B, C, D, E, H, L, or M (memory)
INX e3	INX SP	Double-precision increment register pair. e3 must be equivalent to B, D, H, or SP.
DCX e3	DCX B	Double-precision decrement register pair. e3 must be equivalent to B, D, H, or SP.

Table 3-16: Increment and Decrement Instructions

3.5.4. Data Movement Instructions

Instructions that move data from memory to the CPU and from CPU to memory are given in Table 3-17.

Form and Bit Value	Example	Meaning
MOV e3,e3	MOV A, B	Move data to leftmost element from rightmost element. e3 produces one of A, B, C, D, E, H, L, or M (memory). Note: MOV M, M is disallowed
LDAX e3	LDAX B	Load register A from computed address. e3 must produce either B or D.

Form and Bit Value	Example	Meaning
STAX e3	STAX D	Store register A to computed address. e3 must produce either B or D.
LHLD e16	LHLD L1	Load HL direct from location e16. Double-precision load to H and L.
SHLD e16	SHLD L5+x	Store HL direct to location e16. Double-precision store from H and L to memory.
LDA e16	LDA GAMMA	Load register A from address e16.
STA e16	STA X3-5	Store register A into memory at e16.
POP e3	POP PSW	Load register pair from stack, set SP. e3 must produce one of B, D, H, or PSW.
PUSH e3	PUSH B	Store register pair into stack, set SP. e3 must produce one of B, D, H, or PSW.
IN e8	IN 0	Load register A with data from port e8.
OUT e8	OUT 255	Send data from register A to port e8.
XTHL	XTHL	Exchange data from top of stack with HL.
PCHL	PCHL	Fill program counter with data from HL.
SPHL	SPHL	Fill stack pointer with data from HL.
XCHG	XCHG	Exchange DE pair with HL pair.

Table 3-17: Data Movement Instructions

3.5.5. Arithmetic Logic Unit

Instructions that act upon the single-precision accumulator to perform arithmetic and logic operations are given in Table 3-18

Form and Bit Value	Example	Meaning
ADD e3	ADD B	Add register given by e3 to accumulator without carry. e3 must produce one of A, B, C, D, E, H, or L.

Form and Bit Value	Example	Meaning
ADC e3	ADC L	Subtract register e3 from A with carry, e3 defined as above.
SUB e3	SUB H	Subtract reg e3 from A without carry, e3 is defined as above.
SBB e3	SBB 2	Subtract register e3 from A with carry, e3 defined as above.
ANA e3	ANA 1+1	Logical-and reg with A, e3 as above.
XRA e3	XRA A	Exclusive-or with A, e3 as above.
ORA e3	ORA B	Logical-or with A, e3 defined as above.
CMP e3	CMP H	Compare register with A, e3 as above.
DAA	DAA	Decimal adjust register A based upon last arithmetic logic unit operation.
CMA	CMA	Complement the bits in register A.
STC	STC	Set the carry flag to 1.
CMC	CMC	Complement the carry flag.
RLC	RLC	Rotate bits left, (re)set carry as a side effect. High-order A bit becomes carry.
RRC	RRC	Rotate bits right, (re)set carry as side effect. Low-order A bit becomes carry.
RAL	RAL	Rotate carry/A register to left. Carry is involved in the rotate.
RAR	RAR	Rotate carry/A register to right. Carry is involved in the rotate.
DAD e3	DAD B	Double-precision add register pair e3 to HL. e3 must produce B, D, H, or SP.

Table 3-18: Arithmetic Logic Unit Operations

3.5.6. Control Instructions

The four remaining instructions, categorized as control instructions, are the following:

HLT halts the 8080 processor.

DI disables the interrupt system.

EI enables the interrupt system.

NOP means no operation.

3.6. Error Messages

When errors occur within the assembly-language program, they are listed as single character flags in the leftmost position of the source listing. The line in error is also echoed at the console so that the source listing need not be examined to determine if errors are present. The error codes are listed in Table 3-19.

Error Code	Meaning
D	Data error: element in data statement cannot be placed in the specified data area.
E	Expression error: expression is ill-formed and cannot be computed at assembly time.
L	Label error: label cannot appear in this context; might be duplicate label.
N	Not implemented: features that will appear in future ASM versions. For example, macros are recognized, but flagged in this version.
O	Overflow: expression is too complicated (too many pending operators) to be computed and should be simplified.
P	Phase error: label does not have the same value on two subsequent passes through the program.
R	Register error: the value specified as a register is not compatible with the operation code.
S	Syntax error: statement is not properly formed.
V	Value error: operand encountered in expression is improperly formed.

Table 3-19: ASM Error Codes

Table 3-20 lists the error messages that are due to terminal error conditions.

Message	Meaning
NO SOURCE FILE PRESENT	The file specified in the ASM command does not exist on disk.
NO DIRECTORY SPACE	The disk directory is full; erase files that are not needed and retry.
SOURCE FILE NAME ERROR	Improperly formed ASM filename, for example, It is specified with ? characters.
SOURCE FILE READ ERROR	Source file cannot be read properly by the assembler; execute a TYPE to determine the point of error.
OUTPUT FILE WRITE ERROR	Output files cannot be written properly; most likely cause is a full disk, erase and retry.
CANNOT CLOSE FILE	The file specified in the ASM command does not exist on disk.

Table 3-20: ASM Error Messages

3.7. A Sample Session

The following sample session shows interaction with the assembler and debugger in the development of a simple assembly-language program.

In the sample:

output text is in monospace

input text is in purple

carriage-return is displayed as ↵

rubout/DEL key is displayed as ☒.

comments displayed in 'hand-written' text.

A>ASM SORT↵

CP/M ASSEMBLER - VER 2.0

0015C

next free address

003H USE FACTOR

percent of table used 00 to ff (hexadecimal)

END OF ASSEMBLY

A>DIR SORT.*↵

SORT ASM

SORT BAK

Source file

SORT PRN

Back-up from last edit

SORT HEX

Print file (contains tab characters)

Machine code file

A>TYPE SORT.PRN

Source Program

```

;      SORT PROGRAM IN CP/M ASSEMBLY LANGUAGE
;      START AT THE BEGINNING OF THE TRANSIENT PROGRAM AREA

Machine code location
0100      ORG      100H

generated code
0100 214601 SORT:  LXI H,SW  ;ADDRESS SWITCH TOGGLE
0103 3601      MVI M,1    ;SET TO 1 FOR FIRST ITERATION
0105 214701      LXI H,I    ;ADDRESS INDEX
0108 3600      MVI M,0    ;I=0
;
;      COMPARE I WITH ARRAY SIZE
010A 7E      COMPL: MOV A,M    ;A REGISTER = I
010B FE09      CPI N-1    ;CY SET IF I<(N-1)
010D D21901      JNC CONT    ;CONTINUE IF I<=(N-2)
;
;      END OF ONE PASS THROUGH DATA
0110 214601      LXI H,SW    ;CHECK FOR ZERO SWITCHES
0113 7EB7C200001 MOV A, M! ORA A! JNZ SORT ;END OF SORT IF SW=0
;
0118 FF      RST 7    ;GO TO THE DEBUGGER INSTEAD OF REB
;
;      CONTINUE THIS PASS
;      ADDRESSING I, SO LOAD AV(I) INTO REGISTERS
0119 5F16002148CONT: MOV E, A! MVI D, 0! LXI H, AV! DAD D! DAD D
0121 4E792346      MOV C, M! MOV A, C! INX H! MOV B, M
;      LOW ORDER BYTE IN A AND C, HIGH ORDER BYTE IN B
;
;      MOV H AND L TO ADDRESS AV(I+1)
0125 23      INX H
;
;      COMPARE VALUE WITH REGS CONTAINING AV (I)
0126 965778239E      SUB M! MOV D, A! MOV A, B! INX H! SBB M ;SUBTRACT
;
;      BORROW SET IF AV(I+1)>AV(I)
012B DA3F01      JC INCI    ;SKIP IF IN PROPER ORDER

```

```

;                                (continued)
;                                CHECK FOR EQUAL VALUES
012E B2CA3F01    ORA D! JZ INCI ;SKIP IF AV(I) = AV(I+1)
0132 56702B5E    MOV D, M! MOV M, B! DCX H! MOV E, M
0136 712B722B73  MOV M, C! DCX H! MOV M, D! DCX H! MOV M, E
;
;                                INCREMENT SWITCH COUNT
013B 21460134    LXI H,SW! INR M
;
;                                INCREMENT I
013F 21470134C3INCI:LXI H,I! INR M! JMP COMP
;
;                                DATA DEFINITION SECTION
0146 00    SW:    DB 0          ;RESERVE SPACE FOR SWITCH COUNT
0147      I:    DS 1          ;SPACE FOR INDEX
0148 050064001EAV: DW 5, 100, 30, 50, 20, 7, 1000, 300, 100, -32767
000A =      N    EQU($-AV)/2    ;COMPUTE N INSTEAD OF PRE
015C      END
      ↗equate value
A>TYPE SORT.HEX↵

:10010000214601360121470136007EFE09D2190140
:100110002146017EB7C20001FF5F16002148011988
:10012000194E79234623965778239EDA3F01B2CAA7
:100130003F0156702B5E712B722B732146013421C7
:07014000470134C30A01006E
:10014800050064001E00320014000700E8032C01BB
:0401580064000180BE
:0000000000

```

*Machine Code
in Intel HEX format*

A>DDT SORT.HEX

Start debug run

DDT VER 2.2

NEXT PC

015C 0000

Default address (no address on END statement)

-XP

P=0000 100

Change PC to 100

-UFFFF

Untrace for 65535 Steps

⌘

Abort trace with Rubout/DEL

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 LXI H,0146*0100

-T10

Trace 10 hex/16 decimal steps

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0100 LXI H, 0146

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0103 MVI M, 01

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0105 LXI H, 0147

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0147 S=0100 P=0108 MVI M, 00

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0147 S=0100 P=010A MOV A, M

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=010B CPI 09

C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=010D JNC 0119

C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=0110 LXI H, 0146

C1Z0M1E0I0 A=00 B=0000 D=0000 H=0146 S=0100 P=0113 MOV A, M

C1Z0M1E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0114 ORA A

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0115 JNZ 0100

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0100 LXI H, 0146

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0103 MVI M, 01

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0146 S=0100 P=0105 LXI H, 0147

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0147 S=0100 P=0108 MVI M, 00

C0Z0M0E0I0 A=01 B=0000 D=0000 H=0147 S=0100 P=010A MOV A, M*010B

↙ Stopped at 10BH

-A10D

010D JC 119

0110

Change to Jump on Carry

-XP

P=010B 100

Reset the Program Counter

Back to beginning of program

-T10↓

Trace execution for 10h steps

```
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=0100 LXI H,0146
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0146 S=0100 P=0103 MVI M,01
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0146 S=0100 P=0105 LXI H,0147
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=0108 MVI M,00
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=010A MOV A,M
C0Z0M0E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=010B CPI 09
C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=010D JC 0119 Altered instruction
C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=0119 MOV E,A
C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=011A MVI D,00
C1Z0M1E0I0 A=00 B=0000 D=0000 H=0147 S=0100 P=011C LXI H,0148
C1Z0M1E0I0 A=00 B=0000 D=0000 H=0148 S=0100 P=011F DAD D
C0Z0M1E0I0 A=00 B=0000 D=0000 H=0148 S=0100 P=0120 DAD D
C0Z0M1E0I0 A=00 B=0000 D=0000 H=0148 S=0100 P=0121 MOV C,M
C0Z0M1E0I0 A=00 B=0005 D=0000 H=0148 S=0100 P=0122 MOV A,C
C0Z0M1E0I0 A=05 B=0005 D=0000 H=0148 S=0100 P=0123 INX H
C0Z0M1E0I0 A=05 B=0005 D=0000 H=0149 S=0100 P=0124 MOV B,M*0125 Automatic breakpoint
```

-L100↓

List some code from 100H

```
0100 LXI H,0146
0103 MVI M,01
0105 LXI H,0147
0108 MVI M,00
010A MOV A,M
010B CPI 09
010D JC 0119
0110 LXI H,0146
0113 MOV A,M
0114 ORA A
0115 JNZ 0100
```

-L↓

List some more

```
0118 RST 07
0119 MOV E,A
011A MVI D,00
011C LXI H,0148
```

<X

Abort list with Rubout/DEL

-G,11B↵

Start program from current PC (0125H)
and run in real time to IIBH

*0127

Front-panel interrupt (infinite loop)

-T4↵

Look at looping program in trace mode

```
C0Z0M0E0I0 A=38 B=0064 D=0006 H=0156 S=0100 P=0127 MOV D,A
C0Z0M0E0I0 A=38 B=0064 D=3806 H=0156 S=0100 P=0128 MOV A,B
C0Z0M0E0I0 A=00 B=0064 D=3806 H=0156 S=0100 P=0129 INX H
C0Z0M0E0I0 A=00 B=0064 D=3806 H=0157 S=0100 P=012A SBB M*012B
```

-D148↵

Examine memory

```
0148 05 00 07 00 14 00 1E 00 ..... Data are sorted, but program does not stop
0150 32 00 64 00 64 00 2C 01 E8 03 01 80 00 00 00 00 2.D.D.,.....
0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

-G0↵

Return to CP/M

A>DDT SORT.HEX↵

Reload the memory image

DDT VER 2.2

NEXT PC

015C 0000

-XP↵

P=0000 100↵

Set PC to beginning of program

-L10D↵

List Bad OPCODE

```
010D JNC 0119
0110 LXI H,0146
```

⌘

-A10D↵

Abort list with Rubout/DEL

Assemble new OPCODE

010D JC 119↵

0110 ↵

-L100↵

List starting section of program

```
0100 LXI H,0146
0103 MVI M,01
0105 LXI H,0147
0108 MVI M,00
```

↵

Abort list with Rubout/DEL

-A103↵

Change switch initialization to 00

```
0103 MVI M,0↵
0115 ↵
```

-^C

Return to CP/M with CTRL-C

A>SAVE 1 SORT.COM↵

Save 1 page (256 bytes, from 100H to 1FFH)
on disk in case there is a need to reload later

A>DDT SORT.COM↵

Restart with saved memory image

```
DDT VER 2.2
NEXT PC
0200 0100
```

COM file always starts with address 100H

-G↵

*0118

Run the program from PC=100H

-D148↵

Program stop (RST 7) encountered

```
0148 05 00 07 00 14 00 1E 00 .....
0150 32 00 64 00 64 00 2C 01 E8 03 01 80 00 00 00 00 2.D.D.....
0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

✓ Data properly sorted

-G0↵

Return to CP/M

```

A>ED SORT.ASM␣           Make changes to original program
*N,0^Z0TT␣              find next ",0" (^Z is CTRL-Z)
    MVI    M,0            ;I = 0
* -␣                    up one line in text
    LXI    H,I            ;ADDRESS INDEX
* -␣                    up another line
    MVI    M,1            ;SET TO 1 FOR FIRST ITERATION
*KT␣                   kill line and type next line
    LXI    H,I            ;ADDRESS INDEX
*I␣                    insert new line
    MVI    M,0            ;ZERO SW␣
*T␣
    LXI    H,I            ;ADDRESS INDEX
*NJNC^Z0T␣
    JNC* T␣
    CONT    ;CONTINUE IF I <= (N-2)
*-2DIC^Z0LT␣
    JC     CONT           ;CONTINUE IF I <= (N-2)
*E␣

```

A>ASM SORT.AAZ↵

Source from disk A, Hex to disk A, Skip PRN

CP/M ASSEMBLER - VER 2.0

015C

next address to assemble

003H USE FACTOR

END OF ASSEMBLY

A>DDT SORT.HEX↵

test program changes

DDT VER 2.2

NEXT PC

015C 0000

-G100↵

*0118

-D148↵

data sorted

0148 05 00 07 00 14 00 1E 00

0150 32 00 64 00 64 00 2C 01 E8 03 01 80 00 00 00 00 2.D.D.....

0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

⌘

-G0↵

abort with Rubout/DEL

Return to CP/M - program checks OK.

4. CP/M Dynamic Debugging Tool

4.1. Introduction

The DDT program allows dynamic interactive testing and debugging of programs generated in the CP/M environment. Invoke the debugger with a command of one of the following forms:

DDT

DDT *filename*.HEX

DDT *filename*.COM

where *filename* is the name of the program to be loaded and tested. In both cases, the DDT program is brought into main memory in place of the Console Command Processor (CCP) and resides directly below the Basic Disk Operating System (BDOS) portion of CP/M. Refer to Section 5 for standard memory organization. The BDOS starting address, located in the address field of the JMP instruction at location 0005H, is altered to reflect the reduced Transient Program Area (TPA) size.

The second and third forms of the DDT command perform the same actions as the first, except there is a subsequent automatic load of the specified HEX or COM file. The action is identical to the following sequence of commands:

DDT

I*filename*.HEX or I*filename*.COM

R

where the I and R commands set up and read the specified program to test. See the explanation of the I and R commands below for exact details.

Upon initiation, DDT prints a sign-on message in the form:

DDT VER *m.m*

where *m.m* is the revision number.

Following the sign-on message, DDT prompts you with the hyphen character, -, and waits for input commands from the console. You can type any of several single character commands, followed by a carriage return to execute the command. Each line of input can be line-edited using the following standard CP/M controls:

Character	Meaning
CTRL-C	Reboots CP/M system.
CTRL-H	Backspaces one character position.
CTRL-U	Deletes the entire line typed at the console.

Character	Meaning
CTRL-X	Same as CTRL-U.
rubout/del	Deletes and echoes the last character typed at the console.

Table 4-21: DDT Line-editing Controls

Any command can be up to 32 characters in length. An automatic carriage return is inserted as character 33, where the first character determines the command type. Table 4-22 describes DDT commands.

Character	Result
A	enters assembly-language mnemonics with operands.
D	displays memory in hexadecimal and ASCII.
F	fills memory with constant data.
G	begins execution with optional breakpoints.
I	sets up a standard input File Control Block.
L	lists memory using assembler mnemonics.
M	moves a memory segment from source to destination.
R	reads a program for subsequent testing.
S	substitutes memory values.
T	traces program execution.
U	untraced program monitoring.
X	examines and optionally alters the CPU state.

Table 4-22: DDT Commands

The command character, in some cases, is followed by zero, one, two, or three hexadecimal values, which are separated by commas or single blank characters. All DDT numeric output is in hexadecimal form. The commands are not executed until the carriage return is typed at the end of the command.

At any point in the debug run, you can stop execution of DDT by using either a CTRL-C or GO (jump to location 0000H) and save the current memory image by using a SAVE command of the form:

```
SAVE n filename.COM↵
```

where *n* is the number of pages (256 byte blocks) to be saved on disk. The number of blocks is determined by taking the high-order byte of the address in the TPA and converting this number to decimal. For example, if the highest address in the TPA is 134H, the number of pages is 12H or 18 in decimal. You could type a CTRL-C during the debug run, returning to the CCP level, followed by

```
SAVE 18 X.COM↵
```

The memory image is saved as X.COM on the disk and can be directly executed by typing the name X. If further testing is required, the memory image can be recalled by typing

```
DDT X.COM↵
```

which reloads the previously saved program from location 100H through page 18, 23FFH. The CPU state is not a part of the COM file; thus, the program must be restarted from the beginning to test it properly.

4.2. DDT Commands

The individual commands are detailed below. In each case, the operator must wait for the hyphen prompt character before entering the command. If control is passed to a program under test, and the program has not reached a breakpoint, control can be returned to DDT by executing a RST 7 from the front panel. In the explanation of each command, the command letter is shown in some cases with numbers separated by commas, the numbers are represented by lower-case letters. These numbers are always assumed to be in a hexadecimal radix and from one to four digits in length. Longer numbers are automatically truncated on the right.

Many of the commands operate upon a CPU state that corresponds to the program under test. The CPU state holds the registers of the program being debugged and initially contains zeros for all registers and flags except for the program counter, P, and stack pointer, S, which default to 100H. The program counter is subsequently set to the starting address given in the last record of a HEX file if a file of this form is loaded, see the I and R commands.

4.2.1. The A (Assembly) Command

DDT allows in-line assembly language to be inserted into the current memory image using the A command, which takes the form:

```
AS
```

where *s* is the hexadecimal starting address for the in-line assembly. DDT prompts the console with the address of the next instruction to fill and reads the console, looking for assembly-

language mnemonics followed by register references and operands in absolute hexadecimal form. See the Intel 8080 Assembly Language Reference Card for a list of mnemonics. Each successive load address is printed before reading the console. The A command terminates when the first empty line is input from the console.

Upon completion of assembly language input, you can review the memory segment using the DDT disassembler (see the L command).

Note that the assembler/disassembler portion of DDT can be overlaid by the transient program being tested, in which case the DDT program responds with an error condition when the A and L commands are used.

4.2.2. The D (Display) Command

The D command allows you to view the contents of memory in hexadecimal and ASCII formats. The D command takes the forms:

D

DS

DS, *f*

In the first form, memory is displayed from the current display address, initially 100H, and continues for 16 display lines. Each display line takes the following form:

aaaa bb bb bb bb bb bb bb bb bb bb bb bb bb bb ccccccccccccccc

where *aaaa* is the display address in hexadecimal and *bb* represents data present in memory starting at *aaaa*. The ASCII characters starting at *aaaa* are to the right (represented by the sequence of character *c*) where non-graphic characters are printed as a period. You should note that both upper- and lower-case alphabetic characters are displayed, and will appear as upper-case symbols on a console device that supports only upper-case. Each display line gives the values of 16 bytes of data, with the first line truncated so that the next line begins at an address that is a multiple of 16.

The second form of the D command is similar to the first, except that the display address is first set to address *s*.

The third form causes the display to continue from address *s* through address *f*. In all cases, the display address is set to the first address not displayed in this command, so that a continuing display can be accomplished by issuing successive D commands with no explicit addresses.

Excessively long displays can be aborted by pressing the return key.

4.2.3. The F (Fill) Command

The F command takes the form:

FS, *f*, *c*

where s is the starting address, f is the final address, and c is a hexadecimal byte constant. DDT stores the constant c at address s , increments the value of s and test against f . If s exceeds f , the operation terminates, otherwise the operation is repeated. Thus, the fill command can be used to set a memory block to a specific constant value.

4.2.4. The G (Go) Command

A program is executed using the G command, with up to two optional breakpoint addresses. The G command takes the forms:

G

GS

GS, b

GS, b , c

G, b

G, b , c

The first form executes the program at the current value of the program counter in the current machine state, with no breakpoints set. The only way to regain control in DDT is through a RST 7 execution. The current program counter can be viewed by typing an X or XP command.

The second form is similar to the first, except that the program counter in the current machine state is set to address s before execution begins.

The third form is the same as the second, except that program execution stops when address b is encountered (b must be in the area of the program under test). The instruction at location b is not executed when the breakpoint is encountered.

The fourth form is identical to the third, except that two breakpoints are specified, one at b and the other at c . Encountering either breakpoint causes execution to stop and both breakpoints are cleared. The last two forms take the program counter from the current machine state and set one and two breakpoints, respectively.

Execution continues from the starting address in real-time to the next breakpoint. There is no intervention between the starting address and the break address by DDT. If the program under test does not reach a breakpoint, control cannot return to DDT without executing a RST 7 instruction. Upon encountering a breakpoint, DDT stops execution and types

* d

where d is the stop address. The machine state can be examined at this point using the x (Examine) command. You must specify breakpoints that differ from the program counter address at the beginning of the G command. Thus, if the current program counter is 1234H, then the following commands:

G,1234

G400,400

both produce an immediate breakpoint without executing any instructions.

4.2.5. The I (Input) Command

The I command allows you to insert a filename into the default File Control Block (FCB) at 5CH. The FCB created by CP/M for transient programs is placed at this location (see Section 5). The default FCB can be used by the program under test as if it had been passed by the CP/M Console Processor. Note that this filename is also used by DDT for reading additional HEX and COM files. The I command takes the forms:

Ifilename

Ifilename.typ

If the second form is used and the filetype is either HEX or COM, subsequent R commands can be used to read the pure binary or hex format machine code. Section 4.2.8 gives further details.

4.2.6. The L (List) Command

The L command is used to list assembly-language mnemonics in a particular program region. The L command takes the forms:

L

LS

LS,*f*

The first form lists twelve lines of disassembled machine code from the current list address. The second form sets the list address to *s* and then lists twelve lines of code. The last form lists disassembled code from *s* through address *f*. In all three cases, the list address is set to the next unlisted location in preparation for a subsequent L command. Upon encountering an execution breakpoint, the list address is set to the current value of the program counter (G and T commands). Again, long type-outs can be aborted by pressing RETURN during the list process.

4.2.7. The M (Move) Command

The M command allows block movement of program or data areas from one location to another in memory. The M command takes the form:

MS,*f,d*

where *s* is the start address of the move, *f* is the final address, and *d* is the destination address. Data is first removed from *s* to *d*, and both addresses are incremented. If *s* exceeds *f*, the move operation stops; otherwise, the move operation is repeated.

4.2.8. The R (Read) Command

The R command is used in conjunction with the I command to read COM and HEX files from the disk into the transient program area in preparation for the debug run. The R command takes the forms:

R

R*b*

where *b* is an optional bias address that is added to each program or data address as it is loaded. The load operation must not overwrite any of the system parameters from 000H through 0FFH (that is, the first page of memory). If *b* is omitted, then *b* = 0000 is assumed. The R command requires a previous I command, specifying the name of a HEX or COM file. The load address for each record is obtained from each individual HEX record, while an assumed load address of 100H is used for COM files. Note that any number of R commands can be issued following the I command to reread the program under test, assuming the tested program does not destroy the default area at 5CH. Any file specified with the filetype COM is assumed to contain machine code in pure binary form (created with the LOAD or SAVE command), and all others are assumed to contain machine code in Intel hex format (produced, for example, with the ASM command).

Recall that the command,

DDT *filename.typ*↵

which initiates the DDT program, equals to the following commands:

DDT↵

-*filename.typ*↵ -R↵

Whenever the R command is issued, DDT responds with either the error indicator ? (file cannot be opened, or a checksum error occurred in a HEX file) or with a load message. The load message takes the form:

NEXT PC

nnnn pppp

where *nnnn* is the next address following the loaded program and *pppp* is the assumed program counter (100H for COM files, or taken from the last record if a HEX file is specified).

4.2.9. The S (Set) Command

The s command allows memory locations to be examined and optionally altered. The s command takes the form:

ss

where *s* is the hexadecimal starting address for examination and alteration of memory. DDT responds with a numeric prompt, giving the memory location, along with the data currently held in memory. If you type a carriage return, the data is not altered. If a byte value is typed, the value is stored at the prompted address. In either case, DDT continues to prompt with successive addresses and values until you type either a period or an invalid input value is detected.

4.2.10. The T (Trace) Command

The *T* command allows selective tracing of program execution for 1 to 65535 program steps. The *T* command takes the forms:

T

Tn

In the first form, the CPU state is displayed and the next program step is executed. The program terminates immediately, with the termination address displayed as

**hhhh*

where *hhhh* is the next address to execute. The display address (used in the *D* command) is set to the value of *H* and *L*, and the list address (used in the *L* command) is set to *hhhh*. The CPU state at program termination can then be examined using the *x* command.

The second form of the *T* command is similar to the first, except that execution is traced for *n* steps (*n* is a hexadecimal value) before a program breakpoint occurs. A breakpoint can be forced in the trace mode by typing a rubout character. The CPU state is displayed before each program step is taken in trace mode. The format of the display is the same as described in the *x* command.

You should note that program tracing is discontinued at the CP/M interface and resumes after return from CP/M to the program under test. Thus, CP/M functions that access I/O devices, such as the disk drive, run in real-time, avoiding I/O timing problems. Programs running in trace mode execute approximately 500 times slower than real-time because DDT gets control after each user instruction is executed. Interrupt processing routines can be traced, but commands that use the breakpoint facility (*G*, *T*, and *U*) accomplish the break using an *RST 7* instruction, which means that the tested program cannot use this interrupt location. Further, the trace mode always runs the tested program with interrupts enabled, which may cause problems if asynchronous interrupts are received during tracing.

To get control back to DDT during trace, press RETURN rather than executing an *RST 7*. This ensures that the trace for current instruction is completed before interruption.

4.2.11. The U (Untrace) Command

The *U* command is identical to the *T* command, except that intermediate program steps are not displayed. The untrace mode allows from 1 to 65535 (0FFFFH) steps to be executed in monitored

mode and is used principally to retain control of an executing program while it reaches steady state conditions. All conditions of the `T` command apply to the `U` command.

4.2.12. The X (Examine) Command

The `x` command allows selective display and alteration of the current CPU state for the program under test. The `x` command takes the forms:

`x`

`xr`

where *r* is one of the 8080 CPU registers listed in Table 4-23.

Register	Meaning	Value
C	Carry flag	0 or 1
Z	Zero flag	0 or 1
M	Minus flag	0 or 1
E	Even parity flag	0 or 1
I	Half-carry	0 or 1
A	Accumulator	0 ... FF
B	BC register pair	0 ... FFFF
D	DE register pair	0 ... FFFF
H	HL register pair	0 ... FFFF
S	Stack pointer	0 ... FFFF
P	Program counter	0 ... FFFF

Table 4-23: CPU Registers

In the first case, the CPU register state is displayed in the format:

`cf zf mf ef if A=bb B=dddd D=dddd H=dddd S=dddd P=dddd inst`

where *f* is a 0 or 1 flag value, *bb* is a byte value, and *dddd* is a double-byte quantity corresponding to the register pair. The *inst* field contains the disassembled instruction, that occurs at the location addressed by the CPU state's program counter.

The second form allows display and optional alteration of register values, where *r* is one of the registers given above (C, Z, M, E, I, A, B, D, H, S, or P). In each case, the flag or register value is first

displayed at the console. The DDT program then accepts input from the console. If a carriage return is typed, the flag or register value is not altered. If a value in the proper range is typed, the flag or register value is altered. You should note that BC, DE, and HL are displayed as register pairs. Thus, you must type the entire register pair when B, C, or the BC pair is altered.

4.3. Implementation Notes

The organization of DDT allows certain nonessential portions to be overlaid to gain a larger transient program area for debugging large programs. The DDT program consists of two parts: the DDT nucleus and the assembler/disassembler module. The DDT nucleus is loaded over the CCP and, although loaded with the DDT nucleus, the assembler/disassembler can be overlaid unless used to assemble or disassemble.

In particular, the BDOS address at location 0006H (address field of the JMP instruction at location 0005H) is modified by DDT to address the base location of the DDT nucleus, which, in turn, contains a JMP instruction to the BDOS. Thus, programs that use this address field to size memory see the logical end of memory at the base of the DDT nucleus rather than the base of the BDOS.

The assembler/disassembler module resides directly below the DDT nucleus in the transient program area. If the A, L, T, or X commands are used during the debugging process, the DDT program again alters the address field at 0006H to include this module, further reducing the logical end of memory. If a program loads beyond the beginning of the assembler/disassembler module, the A and L commands are lost (their use produces a ? in response) and the trace and display (T and X) commands list the inst field of the display in hexadecimal, rather than as a decoded instruction.

4.4. A Sample Program

The following example shows an edit, assemble, and debug for a simple program that reads a set of data values and determines the largest value in the set. The largest value is taken from the vector and stored into LARGE at the termination of the program.

In the sample:

output text is in a monospace font

input text is in purple

carriage-return is displayed as ↵

rubout/DEL key is displayed as ⌫.

CTRL-Z ^Z

comments are displayed in 'hand-written' text.

Create Source Program

```

: * I ↵

                                ORG                100H                ;START OF TRANSIENT↵
                                ;AREA↵
                                MVI                B, LEN                ;LENGTH OF VECTOR TO SCAN↵
                                MVI                C, 0                ;LARGER_RST VALUE SO FAR↵
                                LXI                H, VECT                ;BASE OF VECTOR↵
LOOP:                            MOV                A, M                ;GET VALUE↵
                                SUB                C                ;LARGER VALUE IN C?↵
                                JNC                NFOUND                ;JUMP IF LARGER VALUE NOT↵
                                ;FOUND↵
;                                NEW LARGEST VALUE, STORE IT TO C↵
                                MOV                C, A↵
NFOUND:                          INX                H                ;TO NEXT ELEMENT↵
                                DCR                B                ;MORE TO SCAN?↵
                                JNZ                LOOP                ;FOR ANOTHER↵
;↵
;                                END OF SCAN, STORE C↵
                                MOV                A, C                ;GET LARGEST VALUE↵
                                STA                LARGE↵
                                JMP                0                ;REBOOT↵
;↵
;                                TEST DATA↵
VECT:                            DB                2,0,4,3,5,6,1,5↵
LEN                               EQU                $-VECT                ;LENGTH↵
LARGE:                          DS                1                ;LARGEST VALUE ON EXIT↵
                                END↵

^Z
: *

```

```

: *BOP↵

1:          ORG          100H          ;START OF TRANSIENT
2:                                     ;AREA
3:          MVI          B, LEN        ;LENGTH OF VECTOR TO SCAN
4:          MVI          C, 0          ;LARGER_RST VALUE SO FAR
5:          LXI          H, VECT       ;BASE OF VECTOR
6:  LOOP:    MOV          A, M          ;GET VALUE
7:          SUB          C             ;LARGER VALUE IN C?
8:          JNC          NFOUND        ;JUMP IF LARGER VALUE NOT
9:                                     ;FOUND
10: ;          NEW LARGEST VALUE, STORE IT TO C
11:          MOV          C, A
12:  NFOUND  INX          H             ;TO NEXT ELEMENT
13:          DCR          B             ;MORE TO SCAN?
14:          JNZ          LOOP         ;FOR ANOTHER
15: ;
16: ;          END OF SCAN, STORE C
17:          MOV          A, C          ;GET LARGEST VALUE
18:          STA          LARGE
19:          JMP          0             ;REBOOT
20: ;
21: ;          TEST DATA
22:  VECT:    DB          2,0,4,3,5,6,1,5
23:  LEN      EQU          $-VECT       ;LENGTH

1: *E↵

```

End of edit

A>ASM SCAN↵

CP/M ASSEMBLER - VER 2.0 *Start Assembler*
0122
002H USE FACTOR
END OF ASSEMBLY

Assembly complete; look at program listing

A>TYPE SCAN.PRN

<i>//Code Address</i>	<i>//Source Program</i>	
0100	ORG 100H	;START OF TRANSIENT
		;AREA
0100 0608	MVI B, LEN	;LENGTH OF VECTOR TO SCAN
0102 0E00 <i>//Machine Code</i>	MVI C, 0	;LARGER_RST VALUE SO FAR
0104 211901	LXI H, VECT	;BASE OF VECTOR
0107 7E LOOP:	MOV A, M	;GET VALUE
0108 91	SUB C	;LARGER VALUE IN C?
0109 D20D01	JNC NFOUND	;JUMP IF LARGER VALUE NOT
		;FOUND
		; NEW LARGEST VALUE, STORE IT TO C
010C 4F	MOV C, A	
010D 23 NFOUND	INX H	;TO NEXT ELEMENT
010E 05	DCR B	;MORE TO SCAN?
010F C20701	JNZ LOOP	;FOR ANOTHER
		END OF SCAN, STORE C
0112 79	MOV A, C	;GET LARGEST VALUE
0113 322101	STA LARGE	
0116 C30000	JMP 0	;REBOOT
		TEST DATA
0119 0200040305VECT:	DB 2,0,4,3,5,6,1,5	
0008 = <i>//Listing truncated</i>	EQU \$-VECT	;LENGTH
0121 LARGE:	DS 1	;LARGEST VALUE ON EXIT
0122 <i>//Value of Equate</i>	END	

A>DDT SCAN.HEX↵ *Start debugger using hex format machine code*

DDT VERS 2.2

NEXT PC

0121 0000

-X↵ *↖ last load address + 1*

COZOM0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0000 JMP FA03

-XP↵ *Next instruction to execute at PC=0 ↗*

P=0000 100↵

-X↵ *Change PC to 100*

Look at registers again

COZOM0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08

-L100↵ *Next instruction to execute at PC=100 ↗*

0100 MVI B,08

0102 MVI C,00

0104 LXI H,0119

0107 MOV A,M

0108 SUB C

0109 JNC 010D

010C MOV C,A

010D INX H

010E DCR B

010F JNZ 0107

0112 MOV A,C

*Disassembled machine
code at 100H
(See Source listing
for comparison)*

-L↵

0113 STA 0121

0116 JMP 0000

0119 STAX B

011A NOP

011B INR B

011C INX B

011D DCR B

011E MVI B,01

0120 DCR B

0121 ??= 20

0122 MOV D,D

*A little more machine
code. (Note that the program
ends at location 116
with a JMP to 0000.)*

-A116↵

Enter inline assembly mode to change the
JMP to 0000 into a RST 7, which will
return to DDT if ever executed

0116 RST 7↵

0117 ↵

single <cr> stops assemble mode

-L113↵

List code at 113H to check change

0113 STA 0121

0116 RST 07

← in place of JMP

0117 NOP

0118 NOP

0119 STAX B

011A NOP

011B INR B

011C INX B

011D DCR B

011E MVI B,01

0120 DCR B

-X↵

COZOM0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08

Look at registers

-T↵

COZOM0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08*0102

Execute program for one step

initial CPU state, before execution

-T↵

COZOM0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0102 MVI C,00*0104

automatic break-point ↗

Trace one step again (note 08H in B)

-T↵

COZOM0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0104 LXI H,0119*0107

Trace again (Register C is cleared)

-T3↵

COZOM0E0I0 A=00 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M

Trace three steps

COZOM0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C

COZOM0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JNC 010D*010D

automatic break-point ↗

-D119↵ Display memory starting at 119H

```

0119 02 00 04 03 05 06 01 . . . . .
0120 05 11 00 22 21 00 02 7E EB 77 13 23 EB 0B 78 B1 . . . . . lower-case X . w . # . . X .
0130 C2 27 01 C3 03 29 00 00 00 00 00 00 00 00 00 . ' . . . . . ) . . . . .
0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . .
0150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . .
0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . Data is displayed
0170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . in ASCII with a '
0180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . in the position of
0190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . non-graphic
01A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . . characters
01B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . .
01C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 . . . .

```

-X↵

COZOM0E0I0 *Current CPU State* A=02 B=0800 D=0000 H=0119 S=0100 P=010D INX H

-T5↵

COZOM0E0I0 *Trace 5 steps from current CPU state* A=02 B=0800 D=0000 H=0119 S=0100 P=010D INX H
COZOM0E0I0 A=02 B=0800 D=0000 H=011A S=0100 P=010E DCR B
COZOM0E0I0 A=02 B=0700 D=0000 H=011A S=0100 P=010F JNZ 0107
COZOM0E0I0 A=02 B=0700 D=0000 H=011A S=0100 P=0107 MOV A,M
COZOM0E0I0 A=00 B=0700 D=0000 H=011A S=0100 P=0108 SUB C*0109

-U5↵

COZ1M0E0I0 *automatic break-point* A=00 B=0700 D=0000 H=011A S=0100 P=0109 JNC 010D*0108
Trace without listing intermediate states

-X↵

COZOM0E0I0 A=04 B=0600 D=0000 H=011B S=0100 P=0108 SUB C
CPU state at end of U5

-G↵

*0116

Run program from current PC until completion (in real-time)
breakpoint at 116H, caused by executing RST 7 in machine code

-X↵ CPU State at end of program

COZ1M0E0I0 A=00 B=0000 D=0000 H=0121 S=0100 P=0116 RST 07

-XP↵ Examine and change program counter

P=0116 100↵

-X↵

COZ1M0E0I0 A=00 B=0000 D=0000 H=0121 S=0100 P=0100 MVI B,08

-T10↵ Trace 10 (hexadecimal) steps

COZ1M0E0I0 A=00 B=0000 D=0000 H=0121 S=0100 P=0100 MVI B,08

COZ1M0E0I0 A=00 B=0800 D=0000 H=0121 S=0100 P=0102 MVI C,00

COZ1M0E0I0 A=00 B=0800 D=0000 H=0121 S=0100 P=0104 LXI H,0119

COZ1M0E0I0 A=00 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M

COZ1M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C

COZ0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JNC 010D

COZ0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=010D INX H ← A=02 C=00

COZ0M0E0I0 A=02 B=0800 D=0000 H=011A S=0100 P=010E DCR B

COZ0M0E0I0 A=02 B=0700 D=0000 H=011A S=0100 P=010F JNZ 0107

COZ0M0E0I0 A=02 B=0700 D=0000 H=011A S=0100 P=0107 MOV A,M

COZ0M0E0I0 A=00 B=0700 D=0000 H=011A S=0100 P=0108 SUB C

COZ1M0E0I0 A=00 B=0700 D=0000 H=011A S=0100 P=0109 JNC 010D

COZ1M0E0I0 A=00 B=0700 D=0000 H=011A S=0100 P=010D INX H

COZ1M0E0I0 A=00 B=0700 D=0000 H=011B S=0100 P=010E DCR B

COZ0M0E0I0 A=00 B=0600 D=0000 H=011B S=0100 P=010F JNZ 0107

COZ0M0E0I0 A=00 B=0600 D=0000 H=011B S=0100 P=0107 MOV A,M*0108

-A109↵

-JC 10D↵

insert a "hot patch"

010C↵

to change JNC to JC

-G0↵

A>SAVE 1 SCAN.COM↵

Stop DDT so program can be saved

Program resides on first page, so save 1 page

A>DDT SCAN.COM↵

restart DDT with the saved image to continue testing

DDT VERS 2.2

NEXT PC

0200 0100

-L100↵

List some code

0100 MVI B,08

0102 MVI C,00

0104 LXI H,0119

0107 MOV A,M

0108 SUB C

0109 JC 010D

010C MOV C,A

010D INX H

010E DCR B

010F JNZ 0107

0112 MOV A,C

← Previous patch is present in SCAN.COM

-XP↵

P=0100 ↵

-T10↵

Trace to see how patched version operates

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0102 MVI C,00

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0104 LXI H,0119

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C

C0Z0M0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JC 010D Data is moved

C0Z0M0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=010C MOV C,A from A to C

C0Z0M0E0I1 A=02 B=0802 D=0000 H=0119 S=0100 P=010D INX H

C0Z0M0E0I1 A=02 B=0802 D=0000 H=011A S=0100 P=010E DCR B

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=010F JNZ 0107

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=0107 MOV A,M

C0Z0M0E0I1 A=00 B=0702 D=0000 H=011A S=0100 P=0108 SUB C

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=0109 JC 010D

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=010D INX H

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011B S=0100 P=010E DCR B

C1Z0M0E1I1 A=FE B=0602 D=0000 H=011B S=0100 P=010F JNZ 0107*0107

Breakpoint after 16 steps ↗

-X↵

C1Z0M0E1I1 A=FE B=0602 D=0000 H=011B S=0100 P=0107 MOV A,M

-G,108↵

Run from current PC and breakpoint at 108H

*0108

-X↵

C1Z0M0E1I1 A=04 B=0602 ^{↗next data item} D=0000 H=011B S=0100 P=0108 SUB C

-T↵

Single step for a few cycles

C1Z0M0E1I1 A=04 B=0602 D=0000 H=011B S=0100 P=0108 SUB C*0109

-T↵

C0Z0M0E0I1 A=02 B=0602 D=0000 H=011B S=0100 P=0109 JC 010D*010C

-X↵

C0Z0M0E0I1 A=02 B=0602 D=0000 H=011B S=0100 P=010C MOV C,A

-G↵

run to completion

*0116

-X↵

C0Z1M0E1I1 A=03 B=0003 D=0000 H=0121 S=0100 P=0116 RST 07

-S121↵

0121 03 ↵

look at the value of "LARGE"

0122 52 .↵

wrong value!

-L100↵

↖End of the S command

0100 MVI B,08

0102 MVI C,00

0104 LXI H,0119

0107 MOV A,M

0108 SUB C

0109 JC 010D

010C MOV C,A

010D INX H

010E DCR B

010F JNZ 0107

0112 MOV A,C

Review the code

-L↵

```
0113 STA 0121
0116 RST 07
0117 NOP
0118 NOP
0119 STAX B
011A NOP
011B INR B
011C INX B
011D DCR B
011E MVI B,01
0120 DCR B
```

Review the code (continued)

-XP↵

Reset the PC

P=0116 100↵

-T↵

Single step, and watch data values

C0Z1M0E1I1 A=03 B=0003 D=0000 H=0121 S=0100 P=0100 MVI B,08*0102

-T↵

C0Z1M0E1I1 A=03 B=0803 D=0000 H=0121 S=0100 P=0102 MVI C,00*0104

-T↵

C0Z1M0E1I1 A=03 B=0800 D=0000 H=0121 S=0100 P=0104 LXI H,0119*0107
↗count set

-T↵

"largest" set ↗

C0Z1M0E1I1 A=03 B=0800 D=0000 H=0119 S=0100 P=0107 MOY A,M*0108
↗base address of data set

-T↵

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C*0109
↗first data item brought into A

-T↵

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JC 010D*010C

-T↵

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=010C MOV C,A*010D

-T↵

C0Z0M0E0I1 A=02 B=0802 D=0000 H=0119 S=0100 P=010D INX H*010E
↗first data item moved to C correctly

-T↵

C0Z0M0E0I1 A=02 B=0802 D=0000 H=011A S=0100 P=010E DCR B*010F

-T↵

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=010F JNZ 0107*0107

-T↵

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=0107 MOV A,M*0108

-T↵

C0Z0M0E0I1 A=00 *Second data item brought to A*
B=0702 D=0000 H=011A S=0100 P=0108 SUB C*0109

-T↵

C1Z0M1E0I0 A=FE *Subtract destroys data value that was loaded*
B=0702 D=0000 H=011A S=0100 P=0109 JC 010D*010D

-T↵

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=010D INX H*010E

-L100↵

0100	MVI	B,08	
0102	MVI	C,00	
0104	LXI	H,0119	
0107	MOV	A,M	
0108	SUB	C	
0109	JC	010D	← This should have been a CMP so that
010C	MOV	C,A	register A would not be destroyed
010D	INX	H	
010E	DCR	B	
010F	JNZ	0107	
0112	MOV	A,C	

-A108↵

-CMP C↵

Hot patch at 108H changes SUB to CMP

0109↵

-G0↵

A>SAVE 1 SCAN.COM↵

Stop DDT for save

A>DDT SCAN.COM↵

Save memory image

DDT VERS 2.2

Restart DDT

NEXT PC

0200 0100

-XP↵

P=0100 ↵

-L116↵

Look at code to see if it was properly loaded

```
0116 RST      07
0117 NOP
0118 NOP
0119 STAX     B
011A NOP
```

⌘

(long type-out aborted with rubout/DEL)

-G,116↵

Run from 100H to completion

*0116

-XC↵

Look at carry (typo)

C1 ↵

-X↵

C1Z1M0E1I1 A=06 B=0006 D=0000 H=0121 S=0100 P=0116 RST 07

-S121↵

Look at "LARGE"

0121 06 ↵

0122 00 ↵

it appears to be correct

0123 22 .↵

-G0↵

A>ED SCAN.ASM↵

Stop DDT

: *NSUB↵

Re-edit the source program

7: *0LT↵

7: SUB C ;LARGER VALUE IN C?

7: *SSUB^ZCMP^Z0LT↵

7: CMP C Change SUB;LARGER VALUE IN C?

7: *↵

8: JNC NFOUND ;JUMP IF LARGER VALUE NOT FOUND

8: *SNC^ZC^Z0LT↵

8: JC NFOUND Change JNC to JC

8: *E↵

A>ASM SCAN.AAZ↵ *Re-assemble, selecting source from disk A*
 CP/M ASSEMBLER - VER 2.0 *hex to disk A*
 0122 *Print to Z (selects no print file)*

002H USE FACTOR
 END OF ASSEMBLY

A>DDT SCAN.HEX↵ *Re-run debugger to check changes*

DDT VERS 2.2
 NEXT PC
 0121 0000

-L116↵ *check to ensure end is still at 116H*

0116 JMP 0000
 0119 STAX B
 011A NOP
 011B INR B

⌘
 -G100,116↵ *(long type-out aborted with rubout/DEL)*

*0116 *Run from beginning with breakpoint at end*

-D121↵ *Breakpoint reached*

0121 06 00 22 21 00 02 7E EB 77 13 23 EB 0B 78 B1 .. '!... W .#..X.
 0130 C2 27 01 C3 03 29 00 00 00 00 00 00 00 00 00 00 .').....
 0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Look at "LARGE"
Correct value computed

⌘
 -G0↵ *(rubout) aborts long type-out*
Stop DDT, debug session complete.

5. CP/M 2 System Interface

5.1. Introduction

This chapter describes CP/M (release 2) system organization including the structure of memory and system entry points. This section provides the information you need to write programs that operate under CP/M and that use the peripheral and disk I/O facilities of the system.

CP/M is logically divided into four parts, called the Basic Input/Output System (BIOS), the Basic Disk Operating System (BDOS), the Console Command Processor (CCP), and the Transient Program Area (TPA). The BIOS is a hardware-dependent module that defines the exact low level interface with a particular computer system that is necessary for peripheral device I/O. Although a standard BIOS is supplied by Digital Research, explicit instructions are provided for field reconfiguration of the BIOS to match nearly any hardware environment, see Section 6.

The BIOS and BDOS are logically combined into a single module with a common entry point and referred to as the FDOS. The CCP is a distinct program that uses the FDOS to provide a human-oriented interface with the information that is cataloged on the back-up storage device. The TPA is an area of memory, not used by the FDOS and CCP, where various nonresident operating system commands and user programs are executed. The lower portion of memory is reserved for system information and is detailed in later sections. Memory organization of the CP/M system is shown in Table 5-24

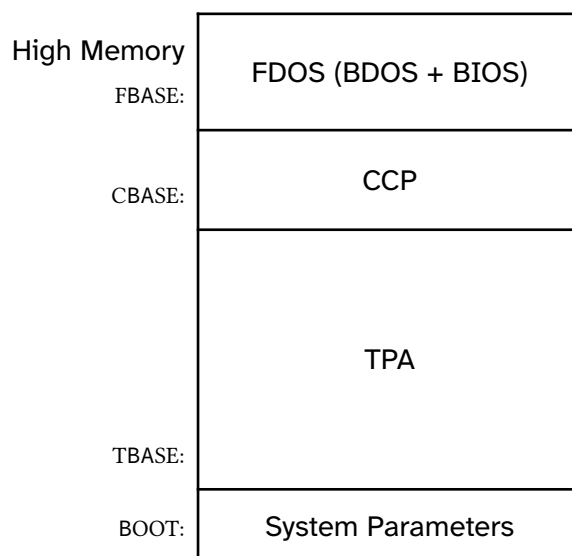


Table 5-24: CP/M Memory Organization

The exact memory addresses corresponding to `BOOT`, `TBASE`, `CBASE`, and `FBASE` vary from version to version and are described fully in Section 6.

All standard CP/M versions assume `BOOT=0000H`, which is the base of random access memory. The machine code found at location `BOOT` performs a system warm start, which loads and initializes

the programs and variables necessary to return control to the CCP. Thus, transient programs need only jump to location `BOOT` to return control to CP/M at the command level. further, the standard versions assume `TBASE=BOOT+0100H`, which is normally location `0100H`. The principal entry point to the FDOS is at location `BOOT+0005H` (normally `0005H`) where a jump to `FBASE` is found. The address field at `BOOT+0006H` (normally `0006H`) contains the value of `FBASE` and can be used to determine the size of available memory, assuming that the CCP is being overlaid by a transient program.

Transient programs are loaded into the TPA and executed as follows. The operator communicates with the CCP by typing command lines following each prompt. Each command line takes one of the following forms:

command

command file1

command file file2

where *command* is either a built-in function, such as `DIR` or `TYPE`, or the name of a transient command or program. If the *command* is a built-in function of CP/M, it is executed immediately. Otherwise, the CCP searches the currently addressed disk for a file by the name

command.COM

If the file is found, it is assumed to be a memory image of a program that executes in the TPA and thus implicitly originates at `TBASE` in memory. The CCP loads the `COM` file from the disk into memory starting at `TBASE` and can extend up to `CBASE`.

If the command is followed by one or two file specifications, the CCP prepares one or two File Control Block (FCB) names in the system parameter area. These optional FCBs are in the form necessary to access files through the FDOS and are described in Section 5.2.

The transient program receives control from the CCP and begins execution, using the I/O facilities of the FDOS. The transient program is called from the CCP. Thus, it can simply return to the CCP upon completion of its processing, or can Jump to `BOOT` to pass control back to CP/M. In the first case, the transient program must not use memory above `CBASE`, while in the latter case, memory up through `FBASE-1` can be used.

The transient program can use the CP/M I/O facilities to communicate with the operator's console and peripheral devices, including the disk subsystem. The I/O system is accessed by passing a function number and an information address to CP/M through the FDOS entry point at `BOOT+0005H`. In the case of a disk read, for example, the transient program sends the number corresponding to a disk read, along with the address of an FCB to the CP/M FDOS. The FDOS, in turn, performs the operation and returns with either a disk read completion indication or an error number indicating that the disk read was unsuccessful.

5.2. Operating System Call Conventions

This section provides detailed information for performing direct operating system calls from user programs. Many of the functions listed below, however, are accessed more simply through the I/O macro library provided with the `MAC` macro assembler and listed in the Digital Research manual entitled *Programmer's Utilities Guide for the CP/M Family of Operating Systems*.

CP/M facilities that are available for access by transient programs fall into two general categories: simple device I/O and disk file I/O. The simple device operations are

- read a console character
- write a console character
- read a sequential character
- write a sequential character
- get or set I/O status
- print console buffer
- interrogate console ready

The following FDOS operations perform disk I/O:

- disk system reset
- drive selection
- file creation
- file close
- directory search
- file delete
- file rename
- random or sequential read
- random or sequential write
- interrogate available disks
- interrogate selected disk
- set DMA address
- set/reset file indicators.

As mentioned above, access to the FDOS functions is accomplished by passing a function number and information address through the primary point at location `BOOT+0005H`. In general, the function number is passed in register `C` with the information address in the double byte pair `DE`. Single byte values are returned in register `A`, with double byte values returned in `HL`, a zero value is returned when the function number is out of range. For reasons of compatibility, register `A = L` and register `B = H` upon return in all cases. Note that the register passing conventions of CP/M agree with those of the Intel PL/M systems programming language. CP/M functions and their numbers are listed below.

Code	Function	Code	Function
0	System Reset	19	Delete File
1	Console Input	20	Read Sequential
2	Console Output	21	Write Sequential
3	Reader Input	22	Make File
4	Punch Output	23	Rename File
5	List Output	24	Return Login Vector
6	Direct Console I/O	25	Return Current Disk
7	Get IOBYTE	26	Set DMA Address
8	Set IOBYTE	27	Get Addr(Alloc)
9	Print String	28†	Write Protect Disk
10	Read Console Buffer	29	Get R/O Vector
11	Get Console Status	30	Set File Attributes
12	Return Version Number	31	Get Addr(Disk Params)
13	Reset Disk System	32†	Set/Get User Code
14	Select Disk	33	Read Random
15	Open File	34	Write Random
16	Close File	35	Compute File Size
17	Search for First	36	Set Random Record
18	Search for Next	37	Reset Drive
		40	Write Random with Zero Fill

†**Note** Function 28 and Function 32 should be avoided in application programs to maintain upward compatibility with CP/M.

Upon entry to a transient program, the CCP leaves the stack pointer set to an eight-level stack area with the CCP return address pushed onto the stack, leaving seven levels before overflow occurs. Although this stack is usually not used by a transient program (most transients return to the CCP through a jump to location 0000H) it is large enough to make CP/M system calls because the FDOS switches to a local stack at system entry. For example, the assembly-language program segment below reads characters continuously until an asterisk is encountered, at which time control returns to the CCP, assuming a standard CP/M system with BOOT=0000H.

```

BDOS    EQU 0005H    ;STANDARD CP/M ENTRY
CONIN   EQU 1        ;CONSOLE INPUT FUNCTION
;
        ORG 0100H    ;BASE OF TPA
NEXTC:  MVI C,CONIN  ;READ NEXT CHARACTER
        CALL BDOS    ;RETURN CHARACTER IN <A>
        CPI  '*'      ;END OF PROCESSING?
        JNZ  NEXTC    ;LOOP IF NOT
        RET          ;RETURN TO CCP
        END

```

CP/M implements a named file structure on each disk, providing a logical organization that allows any particular file to contain any number of records from completely empty to the full capacity of the drive. Each drive is logically distinct with a disk directory and file data area. The disk filenames are in three parts: the drive select code, the filename (consisting of one to eight non-blank characters), and the filetype (consisting of zero to three non-blank characters). The filetype names the generic category of a particular file, while the filename distinguishes individual files in each category. The filetypes listed in Table 5-25 name a few generic categories that have been established, although they are somewhat arbitrary.

Filetype	Meaning
ASM	Assembler Source
PRN	Printer Listing
HEX	Hex Machine Code
BAS	Basic Source File
INT	Intermediate Code
COM	Command File
PLI	PL/I Source File
REL	Relocatable Module
TEX	TEX Formatter Source
BAK	ED Source Backup
SYM	SID Symbol File

Filetype	Meaning
\$\$\$	Temporary File

Table 5-25: CP/M Filetypes

Source files are treated as a sequence of ASCII characters, where each line of the source file is followed by a carriage return, and line-feed sequence (0DH followed by 0AH). Thus, one 128-byte CP/M record can contain several lines of source text. The end of an ASCII file is denoted by a CTRL-Z character (1AH) or a real end-of-file returned by the CP/M read operation. CTRL-Z characters embedded within machine code files (for example, COM files) are ignored and the end-of-file condition returned by CP/M is used to terminate read operations.

Files in CP/M can be thought of as a sequence of up to 65536 records of 128 bytes each, numbered from 0 through 65535, thus allowing a maximum of 8 megabytes per file. Note, however, that although the records may be considered logically contiguous, they may not be physically contiguous in the disk data area. Internally, all files are divided into 16K byte segments called logical extents, so that counters are easily maintained as 8-bit values. The division into extents is discussed in the paragraphs that follow: however, they are not particularly significant for the programmer, because each extent is automatically accessed in both sequential and random access modes.

In the file operations starting with Function 15, DE usually addresses a FCB. Transient programs often use the default FCB area reserved by CP/M at location BOOT+005CH (normally 005CH) for simple file operations. The basic unit of file information is a 128-byte record used for all file operations. Thus, a default location for disk I/O is provided by CP/M at location BOOT+0080H (normally 0080H) which is the initial default DMA address. See Function 26.

All directory operations take place in a reserved area that does not affect write buffers as was the case in release 1, with the exception of **Search for First** and **Search for Next**, where compatibility is required.

The FCB data area consists of a sequence of 33 bytes for sequential access and a series of 36 bytes in the case when the file is accessed randomly. The default FCB, normally located at 005CH, can be used for random access files, because the three bytes starting at BOOT+007DH are available for this purpose.

Field	DR	F1 ... F8	T1	T2	T3	EX	S1	S2	RC	D0 ... D15	CR	R0	R1	R2
Decimal	00	01 ... 08	09	10	11	12	13	14	15	16 ... 31	32	33	34	35
Hex	00	01 ... 08	09	0A	0B	0C	0D	0E	0F	10 ... 1F	20	21	22	23

Table 5-26: File Control Block Format

Field	Definition
DR	drive code (0-16). 0=default, 1=Drive A:, 2=B:, ... 16=P:
F1 ... F8	contain the filename in ASCII upper-case, with high bits = f1' ... f8' normally zero
T1	contains the first character of the filetype in ASCII upper-case, with high bit = t1'. Set for Read/Only file.
T2	contains the second character of the filetype in ASCII upper-case with high bit = t2'. Set for SYS file, no DIR list.
T3	contains the third character of the filetype in ASCII upper-case with high bit = t3' normally zero.
EX	contains the current extent number, normally set to 00 by the user, but in range 0-31 during file I/O
S1	reserved for internal system use
S2	reserved for internal system use. Set to zero on calls to Open , Make and Search for First
RC	record count for extent EX ; takes on values from 0-127
D0 ... D15	filled in by CP/M; reserved for system use
CR	current record to read or write in a sequential file operation; normally set to zero by user
R0	Optional Random Record Number low byte
R1	Optional Random Record Number high byte
R2	Optional Random Record Number overflow

Table 5-27: File Control Block Fields

Each file being accessed through CP/M must have a corresponding FCB, which provides the name and allocation information for all subsequent file operations. When accessing files, it is the programmer's responsibility to fill the lower 16 bytes of the FCB and initialize the **CR** field. Normally, bytes 1 through 11 are set to the ASCII character values for the filename and filetype, while all other fields are zero.

FCBs are stored in a directory area of the disk, and are brought into central memory before the programmer proceeds with file operations (see the **Open** and **Make** functions). The memory copy of the FCB is updated as file operations take place and later recorded permanently on disk at the termination of the file operation, (see the **Close** command).

The CCP constructs the first 16 bytes of two optional FCBs for a transient by scanning the remainder of the line following the transient name, denoted by *file1* and *file2* in the prototype command line described above, with unspecified fields set to ASCII blanks. The first FCB is constructed at location `BOOT+005CH` and can be used as is for subsequent file operations. The second FCB occupies the **D0 ... Dn** portion of the first FCB and must be moved to another area of memory before use. If, for example, the following command line is typed:

```
PROGNAME B:X.ZOT Y.ZAP
```

the file `PROGNAME.COM` is loaded into the TPA, and the default FCB at `BOOT+005CH` is initialized to drive code 2, filename `X`, and filetype `ZOT`. The second drive code takes the default value `0`, which is placed at `BOOT+006CH`, with the filename `Y` placed into location `BOOT+006DH` and filetype `ZAP` located 8 bytes later at `BOOT+0075H`. All remaining fields through **CR** are set to zero. Note again that it is the programmer's responsibility to move this second filename and filetype to another area, usually a separate file control block, before opening the file that begins at `BOOT+005CH`, because the open operation overwrites the second name and type.

If no filenames are specified in the original command, the fields beginning at `BOOT+005DH` and `BOOT+006DH` contain blanks. In all cases, the CCP translates lower-case alphabetic characters to upper-case to be consistent with the CP/M file naming conventions

As an added convenience, the default buffer area at location `BOOT+0080H` is initialized to the command line tail typed by the operator following the program name. The first position contains the number of characters, with the characters themselves following the character count. Given the above command line, the area beginning at `BOOT+0080H` is initialized as follows:

Offset	80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	90 ... FF
Content	0EH	sp	'B'	':'	'X'	':'	'Z'	'O'	'T'	sp	'Y'	':'	'Z'	'A'	'P'	00H	???

Table 5-28: Command-line Layout

where the characters are translated to upper-case ASCII with uninitialized memory following the last valid character. Again, it is the responsibility of the programmer to extract the information

from this buffer before any file operations are performed, unless the default DMA address is explicitly changed.

Individual functions are described in detail in the pages that follow.

5.2.1. System Reset

Function 0: System Reset

Entry Parameters:

Register C: 0 = 00H

Returned Value: Function does **not return**

The **System Reset** function returns control to the CP/M operating system at the CCP level. The CCP re-initializes the disk subsystem by selecting and logging-in disk drive A. This function has exactly the same effect as a jump to location BOOT.

5.2.2. Console Input

Function 1: Console Input

Entry Parameters:

Register C: 1 = 01H

Returned Value:

Register A: ASCII Character

The **Console Input** function reads the next console character to register A. Graphic characters, along with carriage return, line-feed, and back space (CTRL-H) are echoed to the console. Tab characters, CTRL-I, move the cursor to the next tab stop. A check is made for start/stop scroll, CTRL-S, and start/stop printer echo, CTRL-P. The FDOS does not return to the calling program until a character has been typed, thus suspending execution if a character is not ready.

5.2.3. Console Output

Function 2: Console Output

Entry Parameters:

Register C: 2 = 02H

Register E: ASCII Character

Returned Value: none

The **Console Output** function sends the ASCII character from register E to the console device. As in Function 1, tabs are expanded and checks are made for start/stop scroll and printer echo.

5.2.4. Reader Input

Function 3: Reader Input

Entry Parameters:

Register C: 3 = 03H

Returned Value:

Register A: ASCII Character

The **Reader Input** function reads the next character from the logical reader into register A. See the IOBYTE definition in Section 6. Control does not return until the character has been read.

5.2.5. Punch Output**Function 4:** Punch Output**Entry Parameters:**

Register C: 4 = 04H

Register E: ASCII Character

Returned Value: none

The **Punch Output** function sends the character from register E to the logical punch device.

5.2.6. List Output**Function 5:** List Output**Entry Parameters:**

Register C: 5 = 05H

Register E: ASCII Character

Returned Value: none

The **List Output** function sends the ASCII character in register E to the logical listing device.

5.2.7. Direct Console I/O**Function 6:** Direct Console I/O**Entry Parameters:**

Register C: 6 = 06H

Register E: 0FFH (input) or char (output)

Returned Value:

Register A: char or status

Direct Console I/O is supported under CP/M for those specialized applications where basic console input and output are required. Use of this function should, in general, be avoided since it bypasses all of the CP/M normal control character functions (for example, CTRL-S and CTRL-P). Programs that perform direct I/O through the BIOS under previous releases of CP/M, however, should be changed to use direct I/O under BDOS so that they can be fully supported under future releases of MP/M and CP/M.

Upon entry to Function 6, register E either contains hexadecimal 0FFH, denoting a console input request, or an ASCII character. If the input value is 0FFH, Function 6 returns A = 00 if no character is ready, otherwise A contains the next console input character.

If the input value in E is not 0FFH, Function 6 assumes that E contains a valid ASCII character that is sent to the console.

5.2.8. Get IOBYTE

Function 7: Get IOBYTE

Entry Parameters:

Register C: 7 = 07H

Returned Value:

Register A: IOBYTE Value

The **Get IOBYTE** function returns the current value of IOBYTE in register A. See Section 6 for IOBYTE definition.

5.2.9. Set IOBYTE

Function 8: Set IOBYTE

Entry Parameters:

Register C: 8 = 08H

Register E: IOBYTE Value

Returned Value: none

The **Set IOBYTE** function changes the IOBYTE value to that given in register E.

5.2.10. Print String

Function 9: Print String

Entry Parameters:

Register C: 9 = 09H

Registers DE: String Address

Returned Value: none

The **Print String** function sends the character string stored in memory at the location given by DE to the console device, until a '\$' (24H) is encountered in the string. Tabs are expanded as in Function 2, and checks are made for start/stop scroll and printer echo.

5.2.11. Read Console Buffer

Function 10: Read Console Buffer

Entry Parameters:

Register C: 10 = 0AH

Registers DE: Buffer Address

Returned Value:

Console Characters in Buffer

The **Read Buffer** functions reads a line of edited console input into a buffer addressed by registers `DE`. Console input is terminated when either input buffer overflows or a carriage return or line-feed is typed. The Read Buffer takes the form:

DE:	+0	+1	+2	+3	+4	+5	+6	+7	+8	...	+n
	mx	nc	<i>c1</i>	<i>c2</i>	<i>c3</i>	<i>c4</i>	<i>c5</i>	<i>c6</i>	<i>c7</i>	...	??

where **mx** is the maximum number of characters that the buffer will hold, 1 to 255, and **nc** is the number of characters read (set by FDOS upon return) followed by the characters read from the console. If **nc** < **mx**, then uninitialized positions follow the last character, denoted by ?? in the above figure. A number of control functions, summarized in Table 5-29, are recognized during line editing.

Character	Meaning
CTRL-C	reboots when at beginning of line
CTRL-E	causes physical end of line
CTRL-H	backspaces one character position
CTRL-J	(line-feed) terminates input line
CTRL-M	(carriage return) terminates input line
CTRL-R	retypes the current line after new line
CTRL-U	removes current line
CTRL-X	Same as CTRL-U.
rubout/del	Deletes and echoes the last character typed at the console.

Table 5-29: Edit Control Characters

The user should also note that certain functions that return the carriage to the leftmost position (for example, CTRL-X) do so only to the column position where the prompt ended. In earlier releases, the carriage returned to the extreme left margin. This convention makes operator data input and line correction more legible.

5.2.12. Get Console Status

Function 11: Get Console Status

Entry Parameters:

Register C: 11 = 0BH

Returned Value:

Register A: Console Status

The **Console Status** function checks to see if a character has been typed at the console. If a character is ready, the value 0FFH is returned in register A. Otherwise a 00H value is returned.

5.2.13. Return Version Number**Function 12: Return Version Number****Entry Parameters:**

Register C: 12 = 0CH

Returned Value:

Registers HL: Version Number

Flags:

Function 12 provides information that allows version independent programming. A two-byte value is returned, with H = 00H designating the CP/M release (H = 01 for MP/M) and L = 00 for all releases previous to 2.0. CP/M 2.0 returns a hexadecimal 20 in register L, with subsequent version 2 releases in the hexadecimal range 21, 22, through 2F. Using Function 12, for example, the user can write application programs that provide both sequential and random access functions.

5.2.14. Reset Disk System**Function 13: Reset Disk System****Entry Parameters:**

Register C: 13 = 0DH

Returned Value: none

The **Reset Disk** function is used to programmatically restore the file system to a reset state where all disks are set to Read-Write. See functions 28 and 29, only disk drive A is selected, and the default DMA address is reset to BOOT+0080H. This function can be used, for example, by an application program that requires a disk change without a system reboot.

5.2.15. Select Disk**Function 14: Select Disk****Entry Parameters:**

Register C: 14 = 0EH

Register E: Selected Disk

Returned Value: none

The **Select Disk** function designates the disk drive named in register E as the default disk for subsequent file operations, with E = 0 for drive A, 1 for drive B, and so on through 15,

corresponding to drive P in a full 16 drive system. The drive is placed in an on-line status, which activates its directory until the next cold start, warm start, or disk system reset operation. If the disk medium is changed while it is on-line, the drive automatically goes to a Read-Only status in a standard CP/M environment, see Function 28. FCBs that specify drive code zero (**DR** = 00H) automatically reference the currently selected default drive. Drive code values between 1 and 16 ignore the selected default drive and directly reference drives A through P.

5.2.16. Open File

Function 15: Open File

Entry Parameters:

Register C: 15 = 0FH

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The **Open** File operation is used to activate a file that currently exists in the disk directory for the currently active user number. The FDOS scans the referenced disk directory for a match in positions 1 through 14 of the FCB referenced by DE (byte **S1** is automatically zeroed) where an ASCII question mark '?' (3FH) matches any directory character in any of these positions.

Normally, no question marks are included, and bytes **EX** and **S2** of the FCB are zero.

If a directory element is matched, the relevant directory information is copied into bytes **D0** through **Dn** of FCB, thus allowing access to the files through subsequent read and write operations. The user should note that an existing file must not be accessed until a successful open operation is completed. Upon return, the open function returns a directory code with the value 0 through 3 if the open was successful or 0FFH (255 decimal) if the file cannot be found. If question marks occur in the FCB, the first matching FCB is activated. Note that the current record, (**CR**) must be zeroed by the program if the file is to be accessed sequentially from the first record.

5.2.17. Close File

Function 16: Close File

Entry Parameters:

Register C: 16 = 10H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The **Close** File function performs the inverse of the **Open** File function. Given that the FCB addressed by DE has been previously activated through an **Open** or **Make** function, the **Close**

function permanently records the new FCB in the reference disk directory see functions 15 and 22. The FCB matching process for the close is identical to the open function. The directory code returned for a successful close operation is 0, 1, 2, or 3, while a 0FFH (255 decimal) is returned if the filename cannot be found in the directory. A file need not be closed if only read operations have taken place. If write operations have occurred, the close operation is necessary to record the new directory information permanently.

5.2.18. Search for First

Function 17: Search for First

Entry Parameters:

Register C: 17 = 11H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

Search for First scans the directory for a match with the file given by the FCB addressed by DE. The value 255 (hexadecimal 0FFH) is returned if the file is not found; otherwise, 0, 1, 2, or 3 is returned indicating the file is present. When the file is found, the current DMA address is filled with the record containing the directory entry, and the relative starting position is $A * 32$ (that is, rotate the A register left 5 bits, or ADD A five times). Although not normally required for application programs, the directory information can be extracted from the buffer at this position. An ASCII question mark '?' (63 decimal, 3F hexadecimal) in any position from **F1** through **EX** matches the corresponding field of any directory entry on the default or auto-selected disk drive. If the **DR** field contains an ASCII question mark, the auto disk select function is disabled and the default disk is searched, with the search function returning any matched entry, allocated or free, belonging to any user number. This latter function is not normally used by application programs, but it allows complete flexibility to scan all current directory values. If the **DR** field is not a question mark, the **S2** byte is automatically zeroed.

5.2.19. Search for Next

Function 18: Search for Next

Entry Parameters:

Register C: 18 = 12H

Returned Value:

Register A: Directory Code

The **Search for Next** function is similar to the **Search for First** function, except that the directory scan continues from the last matched entry. Similar to Function 17, Function 18 returns the decimal value 255 in A when no more directory items match.

5.2.20. Delete File

Function 19: Delete File

Entry Parameters:

Register C: 19 = 13H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The **Delete** File function removes files that match the FCB addressed by DE. The filename and type may contain ambiguous references (that is, question marks in various positions), but the drive select code (**DR**) cannot be ambiguous, as in the **Search for First** and **Search for Next** functions.

Function 19 returns a decimal 255 if the referenced file or files cannot be found; otherwise, a value in the range 0 to 3 returned.

5.2.21. Read Sequential

Function 20: Read Sequential

Entry Parameters:

Register C: 20 = 14H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

Given that the FCB addressed by DE has been activated through an **Open** or **Make** function, the **Read Sequential** function reads the next 128-byte record from the file into memory at the current DMA address. The record is read from position **CR** of the extent, and the **CR** field is automatically incremented to the next record position. If the **CR** field overflows, the next logical extent is automatically opened and the **CR** field is reset to zero in preparation for the next read operation. The value 00H is returned in the A register if the read operation was successful, while a nonzero value is returned if no data exist at the next record position (for example, end-of-file occurs).

5.2.22. Write Sequential

Function 21: Write Sequential

Entry Parameters:

Register C: 21 = 15H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

Given that the FCB addressed by DE has been activated through an **Open** or **Make** function, the **Write Sequential** function writes the 128-byte data record at the current DMA address to the file named by the FCB. The record is placed at position **CR** of the file, and the **CR** field is automatically incremented to the next record position. If the **CR** field overflows, the next logical extent is automatically opened and the **CR** field is reset to zero in preparation for the next write operation. Write operations can take place into an existing file, in which case, newly written records overlay those that already exist in the file. Register A = 00H upon return from a successful write operation, while a nonzero value indicates an unsuccessful write caused by a full disk.

5.2.23. Make File

Function 22: Make File

Entry Parameters:

Register C: 22 = 16H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The **Make** File operation is similar to the **Open** File operation except that the FCB must name a file that does not exist in the currently referenced disk directory (that is, the one named explicitly by a nonzero **DR** code or the default disk if **DR** is zero). The FDOS creates the file and initializes both the directory and main memory value to an empty file. The programmer must ensure that no duplicate filenames occur, and a preceding delete operation is sufficient if there is any possibility of duplication. Upon return, register A = 0, 1, 2, or 3 if the operation was successful and 0FFH (255 decimal) if no more directory space is available. The Make function has the side effect of activating the FCB and thus a subsequent open is not necessary.

5.2.24. Rename File

Function 23: Rename File

Entry Parameters:

Register C: 23 = 17H

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The **Rename** function uses the FCB addressed by DE to change all occurrences of the file named in the first 16 bytes to the file named in the second 16 bytes. The drive code **DR** at position 0 is used to select the drive, while the drive code for the new filename at position 16 of the FCB is assumed to be zero. Upon return, register A is set to a value between 0 and 3 if the

rename was successful and 0FFH (255 decimal) if the first filename could not be found in the directory scan.

5.2.25. Return Login Vector

Function 24: Return Login Vector

Entry Parameters:

Register C: 24 = 18H

Returned Value:

Registers HL: Log-in Vector

The log-in vector value returned by CP/M is a 16-bit value in HL, where the least significant bit of L corresponds to the first drive A and the high-order bit of H corresponds to the sixteenth drive, labeled P. A 0 bit indicates that the drive is not on-line, while a 1 bit marks a drive that is actively on-line as a result of an explicit disk drive selection or an implicit drive select caused by a file operation that specified a nonzero **DR** field. The user should note that compatibility is maintained with earlier releases, because registers A and L contain the same values upon return.

5.2.26. Return Current Disk

Function 25: Return Current Disk

Entry Parameters:

Register C: 25 = 19H

Returned Value:

Register A: Current Disk

Function 25 returns the currently selected default disk number in register A. The disk numbers range from 0 through 15 corresponding to drives A through P.

5.2.27. Set DMA Address

Function 26: Set DMA Address

Entry Parameters:

Register C: 26 = 1AH

Registers DE: DMA Address

Returned Value: none

DMA is an acronym for Direct Memory Address, which is often used in connection with disk controllers that directly access the memory of the mainframe computer to transfer data to and from the disk subsystem. Although many computer systems use non-DMA access (that is, the data is transferred through programmed I/O operations), the DMA address has, in CP/M, come to mean the address at which the 128-byte data record resides before a disk write and after a disk read. Upon cold start, warm start, or disk system reset, the DMA address is automatically

set to `BOOT+0080H`. The **Set DMA** function can be used to change this default value to address another area of memory where the data records reside. Thus, the DMA address becomes the value specified by `DE` until it is changed by a subsequent **Set DMA** function, cold start, warm start, or disk system reset.

5.2.28. Get Addr(ALLOC)

Function 27: Get Addr(ALLOC)

Entry Parameters:

Register C: 27 = 1BH

Returned Value:

Registers HL: ALLOC Address

An allocation vector (ALLOC) is maintained in main memory for each on-line disk drive. Various system programs use the information provided by the allocation vector to determine the amount of remaining storage (see the `STAT` program). Function 27 returns the base address of the allocation vector for the currently selected disk drive. However, the allocation information might be invalid if the selected disk has been marked Read-Only. Although this function is not normally used by application programs, additional details of the allocation vector are found in Section 6.

5.2.29. Write Protect Disk

Function 28: Write Protect Disk

Entry Parameters:

Register C: 28 = 1CH

Returned Value: none

The Write Protect Disk function provides temporary write protection for the currently selected disk. Any attempt to write to the disk before the next cold or warm start operation produces the message:

BDOS Err On d:R/O

5.2.30. Get R/O Vector

Function 29: Get R/O Vector

Entry Parameters:

Register C: 29 = 1DH

Returned Value:

Registers HL: R/O Vector Value

Function 29 returns a bit vector in register pair HL, which indicates drives that have the temporary Read-Only bit set. As in Function 24, the least significant bit corresponds to drive A,

while the most significant bit corresponds to drive P. The R/O bit is set either by an explicit call to Function 28 or by the automatic software mechanisms within CP/M that detect changed disks.

5.2.31. Set File Attributes

Function 30: Set File Attributes

Entry Parameters:

Register C: 30 = 1EH

Registers DE: FCB Address

Returned Value:

Register A: Directory Code

The Set File Attributes function allows programmatic manipulation of permanent indicators attached to files. In particular, the R/O and System attributes (t_1' and t_2') can be set or reset. The DE pair addresses an unambiguous filename with the appropriate attributes set or reset. Function 30 searches for a match and changes the matched directory entry to contain the selected indicators. Indicators f_1' through f_4' are not currently used, but may be useful for applications programs, since they are not involved in the matching process during file open and close operations. Indicators f_5' through f_8' and t_3' are reserved for future system expansion.

5.2.32. Get Addr(Disk Parameter Block)

Function 31: Get Addr(Disk Parameter Block)

Entry Parameters:

Register C: 31 = 1FH

Returned Value:

Registers HL: **DPB** Address

The address of the BIOS resident disk parameter block (**DPB**) is returned in HL as a result of this function call. This address can be used for either of two purposes. First, the disk parameter values can be extracted for display and space computation purposes, or transient programs can dynamically change the values of current disk parameters when the disk environment changes, if required. Normally, application programs will not require this facility.

5.2.33. Set/Get User Code

Function 32: Set/Get User Code

Entry Parameters:

Register C: 32 = 20H

Register E: 0FFH (get) or *User Code* (set)

Returned Value:

Register A: Current code (get) or none (set)

An application program can change or interrogate the currently active user number by calling Function 32. If register **E** = 0FFH, the value of the current user number is returned in register **A**, where the value is in the range of 0 to 15. If register **E** is not 0FFH, the current user number is changed to the value of **E**, modulo 16.

5.2.34. Read Random

Function 33: Read Random

Entry Parameters:

Register **C**: 33 = 21H

Registers **DE**: FCB Address

Returned Value:

Register **A**: Return code

The **Read Random** function is similar to the sequential file read operation of previous releases, except that the read operation takes place at a particular record number, selected by the 24-bit value constructed from the 3-byte field following the FCB (byte positions **R0** at 33, **R1** at 34, and **R3** at 35). The user should note that the sequence of 24 bits is stored with least significant byte first (**R0**), middle byte next (**R1**), and high byte last (**R2**). CP/M does not reference byte **R2**, except in computing the size of a file (Function 35). Byte **R2** must be zero, however, since a nonzero value indicates overflow past the end of file.

Thus, the **R0, R1** byte pair is treated as a double-byte, or word value, that contains the record to read. This value ranges from 0 to 65535, providing access to any particular record of the 8-megabyte file. To process a file using random access, the base extent (extent 0) must first be opened. Although the base extent might or might not contain any allocated data, this ensures that the file is properly recorded in the directory and is visible in **DIR** requests. The selected record number is then stored in the random record field (**R0, R1**), and the BDOS is called to read the record.

Upon return from the call, register **A** either contains an error code, as listed below, or the value 00, indicating the operation was successful. In the latter case, the current DMA address contains the randomly accessed record. Note that contrary to the sequential read operation, the record number is not advanced. Thus, subsequent random read operations continue to read the same record.

Upon each random read operation, the logical extent and current record values are automatically set. Thus, the file can be sequentially read or written, starting from the current randomly accessed position. However, note that, in this case, the last randomly read record will be reread as one switches from random mode to sequential read and the last record will be rewritten as one switches to a sequential write operation. The user can simply advance the

random record position following each random read or write to obtain the effect of sequential I/O operation.

Error codes returned in register A following a random read are listed below.

- 01 reading unwritten data
- 02 (not returned in random mode)
- 03 cannot close current extent
- 04 seek to unwritten extent
- 05 (not returned in read mode)
- 06 seek past physical end of disk

Error codes 01 and 04 occur when a random read operation accesses a data block that has not been previously written or an extent that has not been created, which are equivalent conditions. Error code 03 does not normally occur under proper system operation. If it does, it can be cleared by simply rereading or reopening extent zero as long as the disk is not physically write protected. Error code 06 occurs whenever byte **R2** is nonzero under the current 2.0 release. Normally, nonzero return codes can be treated as missing data, with zero return codes indicating operation complete.

5.2.35. Write Random

Function 34: Write Random

Entry Parameters:

Register C: 34 = 22H

Registers DE: FCB Address

Returned Value:

Register A: Return Code

The **Write Random** operation is initiated similarly to the Read Random call, except that data is written to the disk from the current DMA address. Further, if the disk extent or data block that is the target of the write has not yet been allocated, the allocation is performed before the write operation continues. As in the Read Random operation, the random record number is not changed as a result of the write. The logical extent number and current record positions of the FCB are set to correspond to the random record that is being written. Again, sequential read or write operations can begin following a random write, with the notation that the currently addressed record is either read or rewritten again as the sequential operation begins. You can also simply advance the random record position following each write to get the effect of a sequential write operation. Note that reading or writing the last record of an extent in random mode does not cause an automatic extent switch as it does in sequential mode.

The error codes returned by a random write are identical to the random read operation with the addition of error code 05, which indicates that a new extent cannot be created as a result of directory overflow.

5.2.36. Compute File Size

Function 35: Compute File Size

Entry Parameters:

Register C: 35 = 23H

Registers DE: FCB Address

Returned Value: Random Record Field Set

When computing the size of a file, the DE register pair addresses an FCB in random mode format (bytes **R0**, **R1**, and **R2** are present). The FCB contains an unambiguous filename that is used in the directory scan. Upon return, the random record bytes contain the virtual file size, which is, in effect, the record address of the record following the end of the file. Following a call to Function 35, if the high record byte **R2** is 01, the file contains the maximum record count 65536. Otherwise, bytes **R0** and **R1** constitute a 16-bit value as before (**R0** is the least significant byte), which is the file size.

Data can be appended to the end of an existing file by simply calling Function 35 to set the random record position to the end of file and then performing a sequence of random writes starting at the preset record address.

The virtual size of a file corresponds to the physical size when the file is written sequentially. If the file was created in random mode and holes exist in the allocation, the file might contain fewer records than the size indicates. For example, if only the last record of an 8-megabyte file is written in random mode (that is, record number 65535), the virtual size is 65536 records, although only one block of data is actually allocated.

5.2.37. Set Random Record

Function 36: Set Random Record

Entry Parameters:

Register C: 36 = 24H

Registers DE: FCB Address

Returned Value: Random Record Field Set

The Set Random Record function causes the BDOS automatically to produce the random record position from a file that has been read or written sequentially to a particular point. The function can be useful in two ways.

First, it is often necessary initially to read and scan a sequential file to extract the positions of various key fields. As each key is encountered, Function 36 is called to compute the random

record position for the data corresponding to this key. If the data unit size is 128 bytes, the resulting record position is placed into a table with the key for later retrieval. After scanning the entire file and tabulating the keys and their record numbers, the user can move instantly to a particular keyed record by performing a random read, using the corresponding random record number that was saved earlier. The scheme is easily generalized for variable record lengths, because the program need only store the buffer-relative byte position along with the key and record number to find the exact starting position of the keyed data at a later time.

A second use of Function 36 occurs when switching from a sequential read or write over to random read or write. A file is sequentially accessed to a particular point in the file, Function 36 is called, which sets the record number, and subsequent random read and write operations continue from the selected point in the file.

5.2.38. Reset Drive

Function 37: Reset Drive

Entry Parameters:

Register C: 37 = 25H

Registers DE: Drive Vector

Returned Value:

Register A: 00H

The Reset Drive function allows resetting of specified drives. The passed parameter is a 16-bit vector of drives to be reset; the least significant bit is drive A.

To maintain compatibility with MP/M, CP/M returns a zero value.

5.2.39. Write Random with Zero Fill

Function 40: Write Random with Zero Fill

Entry Parameters:

Register C: 40 = 26H

Registers DE: FCB Address

Returned Value:

Register A: Return Code

The Write With Zero Fill operation is similar to Function 34, with the exception that a previously unallocated block is filled with zeros before the data is written.

5.3. A Sample File-to-File Copy Program

The following program provides a relatively simple example of file operations. The program source file is created as COPY.ASM using the CP/M ED program and then assembled using ASM or MAC, resulting in a HEX file. The LOAD program is used to produce a COPY.COM file that executes directly under the CCP. The program begins by setting the stack pointer to a local area and

proceeds to move the second name from the default area at 006CH to a 33-byte File Control Block called DFCB. The DFCB is then prepared for file operations by clearing the current record field. At this point, the source and destination FCBs are ready for processing, because the SFCB at 005CH is properly set up by the CCP upon entry to the COPY program. That is, the first name is placed into the default FCB, with the proper fields zeroed, including the current record field at 007CH. The program continues by opening the source file, deleting any existing destination file, and creating the destination file. If all this is successful, the program loops at the label COPY until each record is read from the source file and placed into the destination file. Upon completion of the data transfer, the destination file is closed and the program returns to the CCP command level by jumping to BOOT.

```

;          sample file-to-file copy program
;
;          at the ccp level, the command
;
;          copy a:x.y b:u.v
;
;          copies the file named x.y from drive
;          a to a file named u.v. on drive b.
;
0000 =      boot      equ 0000h      ;system reboot
0005 =      bdos      equ 0005h      ;bdos entry point
005C =      fcbl      equ 005ch      ;first file name
005C =      sfcbl     equ fcbl       ;source fcb
006C =      fcb2      equ 006ch      ;second file name
0080 =      dbuff     equ 0080h      ;default buffer
0100 =      tpa       equ 0100h      ;beginning of tpa
;
0009 =      printf    equ 9          ;print buffer func#
000F =      openf     equ 15         ;open file func#
0010 =      closef    equ 16         ;close file func#
0013 =      deletf    equ 19         ;delete file func#
0014 =      readf     equ 20         ;sequential read
0015 =      writef    equ 21         ;sequential write
0016 =      makef     equ 22         ;make file func#
;
0100              org tpa           ;beginning of tpa
0100 311A02      lxi sp,stack      ;local stack
;
;          move second file name to dfcb
0103 0E10              mvi c,16      ;half an fcb
0105 116C00          lxi d,fcb2      ;source of move
0108 21D901          lxi h,dfcb      ;destination fcb
010B 1A            mfcbl:  ldax d      ;source fcb

```

```

010C 13          inx d          ;ready next
010D 77          mov m,a        ;dest fcb
010E 23          inx h          ;ready next
010F 0D          dcr c          ;count 16...0
0110 C20B01      jnz mfcb       ;loop 16 times
;
;      name has been removed, zero cr
0113 AF          xra a          ;a = 00h
0114 32F901      sta dfcbr      ;current rec = 0
;
;      source and destination fcbs ready
;
0117 115C00      lxi d,sfcb     ;source file
011A CD6901      call open      ;error if 255
011D 118701      lxi d,nofile   ;ready message
0120 3C          inr a          ;255 becomes 0
0121 CC6101      cz finis       ;done if no file
;
;      source file open, prep destination
0124 11D901      lxi d,dfcb     ;destination
0127 CD7301      call delete    ;remove if present
;
012A 11D901      lxi d,dfcb     ;destination
012D CD8201      call make      ;create the file
0130 119601      lxi d,nodir    ;ready message
0133 3C          inr a          ;255 becomes 0
0134 CC6101      cz finis       ;done if no dir space
;
;      source file open, dest file open
;      copy until end of file on source
;
0137 115C00      copy: lxi d,sfcb ;source
013A CD7801      call read      ;read next record
013D B7          ora a          ;end of file?
013E C25101      jnz eofile     ;skip write if so
;
;      not end of file, write the record
0141 11D901      lxi d,dfcb     ;destination
0144 CD7D01      call write     ;write record
0147 11A901      lxi d,space    ;ready message
014A B7          ora a          ;00 if write ok
014B C46101      cnz finis      ;end if so
014E C33701      jmp copy       ;loop until eof
;
eofile: ;end of file, close destination
0151 11D901      lxi d,dfcb     ;destination

```

```

0154 CD6E01      call close      ;255 if error
0157 21BA01      lxi  h,wrprot  ;ready message
015A 3C          inr  a          ;255 becomes 00
015B CC6101      cz   finis      ;should not happen
                ;
                ;      copy operation complete, end
015E 11CB01      lxi  d,normal  ;ready message
                ;
                finis  ;write message given by de, reboot
0161 0E09        mvi  c,printf
0163 CD0500      call bdos      ;write message
0166 C30000      jmp  boot      ;reboot system
                ;
                ;      system interface subroutines
                ;      (all return directly from bdos)
                ;
0169 0E0F      open:  mvi  c,openf
016B C30500      jmp  bdos
                ;
016E 0E10      close: mvi  c,closef
0170 C30500      jmp  bdos
                ;
0173 0E13      delete mvi  c,deletef
0175 C30500      jmp  bdos
                ;
0178 0E14      read:  mvi  c,readf
017A C30500      jmp  bdos
                ;
017D 0E15      write: mvi  c,writef
017F C30500      jmp  bdos
                ;
0182 0E16      make:  mvi  c,makef
0184 C30500      jmp  bdos
                ;
                ;      console messages
0187 6E6F20736Fnofile: db  'no source file$'
0196 6E6F206469nodir:  db  'no directory space$'
01A9 6F7574206Fspace:  db  'out of dat space$'
01BA 7772697465wrprot: db  'write protected?$$'
01CB 636F707920normal: db  'copy complete$'
                ;
                ;      data areas
01D9            dfcb:  ds   33      ;destination fcb
01F9 =          dfcbr equ  dfcb+32  ;current record
                ;
01FA            ds   32      ;16 level stack

```

```

                                stack:
021A                                end

```

Note that there are several simplifications in this particular program. First, there are no checks for invalid filenames that could contain ambiguous references. This situation could be detected by scanning the 32-byte default area starting at location 005CH for ASCII question marks. A check should also be made to ensure that the filenames have been included (check locations 005DH and 006DH for non-blank ASCII characters). Finally, a check should be made to ensure that the source and destination filenames are different. An improvement in speed could be obtained by buffering more data on each read operation. One could, for example, determine the size of memory by fetching FBASE from location 0006H and using the entire remaining portion of memory for a data buffer. In this case, the programmer simply resets the DMA address to the next successive 128-byte area before each read. Upon writing to the destination file, the DMA address is reset to the beginning of the buffer and incremented by 128 bytes to the end as each record is transferred to the destination file.

5.4. A Sample File Dump Utility

The following file dump program is slightly more complex than the simple copy program given in the previous section. The dump program reads an input file, specified in the CCP command line, and displays the content of each record in hexadecimal format at the console. Note that the dump program saves the CCP's stack upon entry, resets the stack to a local area, and restores the CCP's stack before returning directly to the CCP. Thus, the dump program does not perform a warm start at the end of processing.

```

                                ;      FILE DUMP PROGRAM, READS AN INPUT FILE AND PRINTS IN HEX
                                ;
                                ;      COPYRIGHT (C) 1975, 1976, 1977, 1978
                                ;      DIGITAL RESEARCH
                                ;      BOX 579, PACIFIC GROVE
                                ;      CALIFORNIA, 93950
                                ;
0100                                ORG      100H
0005 =      BDOS      EQU      0005H      ;DOS ENTRY POINT
0001 =      CONS      EQU      1          ;READ CONSOLE
0002 =      TYPEF     EQU      2          ;TYPE FUNCTION
0009 =      PRINTF    EQU      9          ;BUFFER PRINT ENTRY
000B =      BRKF      EQU      11         ;BREAK KEY FUNCTION (TRUE IF CHAR READY)
000F =      OPENF     EQU      15         ;FILE OPEN
0014 =      READF     EQU      20         ;READ FUNCTION
                                ;
005C =      FCB       EQU      5CH        ;FILE CONTROL BLOCK ADDRESS
0080 =      BUFF      EQU      80H        ;INPUT DISK BUFFER ADDRESS

```

```

;
;      NON GRAPHIC CHARACTERS
000D =   CR      EQU      0DH      ;CARRIAGE RETURN
000A =   LF      EQU      0AH      ;LINE FEED
;
;      FILE CONTROL BLOCK DEFINITIONS
005C =   FCBDN   EQU      FCB+0    ;DISK NAME
005D =   FCBFN   EQU      FCB+1    ;FILE NAME
0065 =   FCBFT   EQU      FCB+9    ;DISK FILE TYPE (3 CHARACTERS)
0068 =   FCBRL   EQU      FCB+12   ;FILE'S CURRENT REEL NUMBER
006B =   FCBRC   EQU      FCB+15   ;FILE'S RECORD COUNT (0 TO 128)
007C =   FCBRC   EQU      FCB+32   ;CURRENT (NEXT) RECORD NUMBER (0 TO 127)
007D =   FCBLN   EQU      FCB+33   ;FCB LENGTH
;
;      SET UP STACK
0100 210000      LXI      H,0
0103 39          DAD      SP
;      ENTRY STACK POINTER IN HL FROM THE CCP
0104 221502      SHLD     OLDSP
;      SET SP TO LOCAL STACK AREA (RESTORED AT FINIS)
0107 315702      LXI      SP,STKTOP
;      READ AND PRINT SUCCESSIVE BUFFERS
010A CDC101      CALL     SETUP      ;SET UP INPUT FILE
010D FEFF        CPI      255      ;255 IF FILE NOT PRESENT
010F C21B01      JNZ      OPENOK    ;SKIP IF OPEN IS OK
;
;      FILE NOT THERE, GIVE ERROR MESSAGE AND RETURN
0112 11F301      LXI      D,OPNMSG
0115 CD9C01      CALL     ERR
0118 C35101      JMP      FINIS     ;TO RETURN
;
OPENOK: ;OPEN OPERATION OK, SET BUFFER INDEX TO END
011B 3E80        MVI      A,80H
011D 321302      STA      IBP      ;SET BUFFER POINTER TO 80H
;      HL CONTAINS NEXT ADDRESS TO PRINT
0120 210000      LXI      H,0      ;START WITH 0000
;
GLOOP:
0123 E5          PUSH     H          ;SAVE LINE POSITION
0124 CDA201      CALL     GNB
0127 E1          POP      H          ;RECALL LINE POSITION
0128 DA5101      JC       FINIS     ;CARRY SET BY GNB IF END FILE
012B 47          MOV      B,A
;      PRINT HEX VALUES
;      CHECK FOR LINE FOLD
012C 7D          MOV      A,L

```

```

012D E60F          ANI      0FH      ;CHECK LOW 4 BITS
012F C24401        JNZ      NONUM
;                PRINT LINE NUMBER
0132 CD7201        CALL     CRLF
;
;                CHECK FOR BREAK KEY
0135 CD5901        CALL     BREAK
;                ACCUM LSB = 1 IF CHARACTER READY
0138 0F            RRC              ;INTO CARRY
0139 DA5101        JC       FINIS   ;DON'T PRINT ANY MORE
;
013C 7C            MOV      A,H
013D CD8F01        CALL     PHEX
0140 7D            MOV      A,L
0141 CD8F01        CALL     PHEX
NONUM:
0144 23            INX      H        ;TO NEXT LINE NUMBER
0145 3E20          MVI      A,' '
0147 CD6501        CALL     PCHAR
014A 78            MOV      A,B
014B CD8F01        CALL     PHEX
014E C32301        JMP      GLOOP
;
FINIS:
;                END OF DUMP, RETURN TO CCP
;                (NOTE THAT A JMP TO 0000H REBOOTS)
0151 CD7201        CALL     CRLF
0154 2A1502        LHLD     OLDSP
0157 F9            SPHL
;                STACK POINTER CONTAINS CCP'S STACK LOCATION
0158 C9            RET              ;TO THE CCP
;
;
;                SUBROUTINES
;
BREAK: ;CHECK BREAK KEY (ACTUALLY ANY KEY WILL DO)
0159 E5D5C5        PUSH H! PUSH D! PUSH B; ENVIRONMENT SAVED
015C 0E0B          MVI      C,BRKF
015E CD0500        CALL     BDOS
0161 C1D1E1        POP B! POP D! POP H; ENVIRONMENT RESTORED
0164 C9            RET
;
PCHAR: ;PRINT A CHARACTER
0165 E5D5C5        PUSH H! PUSH D! PUSH B; SAVED
0168 0E02          MVI      C,TYPEF
016A 5F            MOV      E,A

```

```

016B CD0500      CALL    BDOS
016E C1D1E1      POP B! POP D! POP H; RESTORED
0171 C9          RET

;
CRLF:
0172 3E0D        MVI     A,CR
0174 CD6501      CALL    PCHAR
0177 3E0A        MVI     A,LF
0179 CD6501      CALL    PCHAR
017C C9          RET

;
;
PNIB:  ;PRINT NIBBLE IN REG A
017D E60F        ANI     0FH      ;LOW 4 BITS
017F FE0A        CPI     10
0181 D28901      JNC     P10
;               LESS THAN OR EQUAL TO 9
0184 C630        ADI     '0'
0186 C38B01      JMP     PRN

;
;               GREATER OR EQUAL TO 10
0189 C637        P10:    ADI     'A' - 10
018B CD6501      PRN:    CALL    PCHAR
018E C9          RET

;
PHEX:  ;PRINT HEX CHAR IN REG A
018F F5          PUSH    PSW
0190 0F          RRC
0191 0F          RRC
0192 0F          RRC
0193 0F          RRC
0194 CD7D01      CALL    PNIB      ;PRINT NIBBLE
0197 F1          POP     PSW
0198 CD7D01      CALL    PNIB
019B C9          RET

;
ERR:    ;PRINT ERROR MESSAGE
;        D,E ADDRESSES MESSAGE ENDING WITH "$"
019C 0E09        MVI     C,PRINTF      ;PRINT BUFFER FUNCTION
019E CD0500      CALL    BDOS
01A1 C9          RET

;
;
GNB:    ;GET NEXT BYTE
01A2 3A1302      LDA     IBP
01A5 FE80        CPI     80H

```

```

01A7 C2B301      JNZ      G0
                  ;      READ ANOTHER BUFFER
                  ;
                  ;
01AA CDCE01      CALL     DISKR
01AD B7          ORA      A      ;ZERO VALUE IF READ OK
01AE CAB301      JZ       G0      ;FOR ANOTHER BYTE
                  ;      END OF DATA, RETURN WITH CARRY SET FOR EOF
01B1 37          STC
01B2 C9          RET
                  ;
G0:              ;READ THE BYTE AT BUFF+REG A
01B3 5F          MOV      E,A      ;LS BYTE OF BUFFER INDEX
01B4 1600        MVI      D,0      ;DOUBLE PRECISION INDEX TO DE
01B6 3C          INR      A      ;INDEX=INDEX+1
01B7 321302      STA      IBP      ;BACK TO MEMORY
                  ;      POINTER IS INCREMENTED
                  ;      SAVE THE CURRENT FILE ADDRESS
01BA 218000      LXI      H,BUFF
01BD 19          DAD      D
                  ;      ABSOLUTE CHARACTER ADDRESS IS IN HL
01BE 7E          MOV      A,M
                  ;      BYTE IS IN THE ACCUMULATOR
01BF B7          ORA      A      ;RESET CARRY BIT
01C0 C9          RET
                  ;
SETUP:           ;SET UP FILE
                  ;      OPEN THE FILE FOR INPUT
01C1 AF          XRA      A      ;ZERO TO ACCUM
01C2 327C00      STA      FCBCR    ;CLEAR CURRENT RECORD
                  ;
01C5 115C00      LXI      D,FCB
01C8 0E0F        MVI      C,OPENF
01CA CD0500      CALL     BDOS
                  ;      255 IN ACCUM IF OPEN ERROR
01CD C9          RET
                  ;
DISKR:           ;READ DISK FILE RECORD
01CE E5D5C5      PUSH H! PUSH D! PUSH B
01D1 115C00      LXI      D,FCB
01D4 0E14        MVI      C,READF
01D6 CD0500      CALL     BDOS
01D9 C1D1E1      POP B! POP D! POP H
01DC C9          RET
                  ;
                  ;      FIXED MESSAGE AREA

```

```

01DD 46494C4520SIGNON: DB      'FILE DUMP VERSION 1.4$'
01F3 0D0A4E4F20OPNMSG: DB      CR,LF,'NO INPUT FILE PRESENT ON DISK$'

;          VARIABLE AREA
0213      IBP:   DS      2      ;INPUT BUFFER POINTER
0215      OLDSP: DS      2      ;ENTRY SP VALUE FROM CCP
;
;          STACK AREA
0217      DS      64      ;RESERVE 32 LEVEL STACK
      STKTOP:
;
0257      END

```

5.5. A Sample Random Access Program

This chapter concludes with an extensive example of random access operation. The program listed below performs the simple function of reading or writing random records upon command from the terminal. When a program has been created, assembled, and placed into a file labeled `RANDOM.COM`, the CCP level command

```
RANDOM X.DAT
```

starts the test program. The program looks for a file by the name `X.DAT` and, if found, proceeds to prompt the console for input. If not found, the file is created before the prompt is given. Each prompt takes the form

```
next command?
```

and is followed by operator input, followed by a carriage return. The input commands take the form

```
nW nR Q
```

where n is an integer value in the range 0 to 65535, and W, R, and Q are simple command characters corresponding to random write, random read, and quit processing, respectively. If the W command is issued, the `RANDOM` program issues the prompt

```
type data:
```

The operator then responds by typing up to 127 characters, followed by a carriage return.

`RANDOM` then writes the character string into the `X.DAT` file at record n . If the R command is issued, `RANDOM` reads record number n and displays the string value at the console. If the Q command is issued, the `X.DAT` file is closed, and the program returns to the CCP. In the interest of brevity, the only error message is

```
error, try again.
```

The program begins with an initialization section where the input file is opened or created, followed by a continuous loop at the label ready where the individual commands are interpreted. The DFBC at 005CH and the default buffer at 0080H are used in all disk operations. The utility subroutines then follow, which contain the principal input line processor, called readc. This particular program shows the elements of random access processing, and can be used as the basis for further program development.

Sample Random Access Program for CP/M 2.0

```

0100          org      100h      ;base of tpa
          ;
0000 =      reboot    equ      0000h    ;system reboot
0005 =      bdos      equ      0005h    ;bdos entry point
          ;
0001 =      coninp    equ      1        ;console input function
0002 =      conout    equ      2        ;console output function
0009 =      pstring   equ      9        ;print string until '$'
000A =      rstring   equ      10       ;read console buffer
000C =      version   equ      12       ;return version number
000F =      openf     equ      15       ;file open function
0010 =      closef    equ      16       ;close function
0016 =      makef     equ      22       ;make file function
0021 =      readr     equ      33       ;read random
0022 =      writer    equ      34       ;write random
          ;
005C =      fcb       equ      005ch    ;default file control
          ;block
007D =      ranrec    equ      fcb+33    ;random record position
007F =      ranovf    equ      fcb+35    ;high order (overflow)
          ;byte
0080 =      buff      equ      0080h    ;buffer address
          ;
000D =      cr        equ      0dh      ;carriage return
000A =      lf        equ      0ah      ;line feed
          ;
          ;      Load SP, Set-Up File for Random Access
          ;
0100 31BC02          lxi      sp,stack
          ;
          ;      version 2.0
0103 0E0C          mvi      c,version
0105 CD0500          call     bdos
0108 FE20          cpi      20h        ;version 2.0 or better?
010A D21601          jnc      versok
          ;      bad version, message and go back
010D 111B02          lxi      d,badver

```

```

0110 CDDA01      call    print
0113 C30000      jmp     reboot
;
versok:
;      correct versionm for random access
0116 0E0F      mvi     c,openf    ;open default fcb
0118 115C00     lxi     d,fcbl
011B CD0500     call    bdos
011E 3C         inr     a          ;err 255 becomes zero
011F C23701     jnz     ready
;
;      cannot open file, so create it
0122 0E16      mvi     c,makef
0124 115C00     lxi     d,fcbl
0127 CD0500     call    bdos
012A 3C         inr     a          ;err 255 becomes zero
012B C23701     jnz     ready
;
;      cannot create file, directory full
012E 113A02     lxi     d,nospace
0131 CDDA01     call    print
0134 C30000     jmp     reboot    ;back to ccp
;
;      Loop Back to Ready After Each Command
;
ready:
;      file is ready for processing
;
0137 CDE501     call    readcom    ;read next command
013A 227D00     shld    ranrec    ;store input record#
013D 217F00     lxi     h,ranovf
0140 3600      mvi     m,0          ;clear high byte if set
0142 FE51      cpi     'Q'        ;quit?
0144 C25601     jnz     notq
;
;      quit processing, close file
0147 0E10      mvi     c,closef
0149 115C00     lxi     d,fcbl
014C CD0500     call    bdos
014F 3C         inr     a          ;err 255 becomes 0
0150 CAB901     jz      error      ;error message, retry
0153 C30000     jmp     reboot    ;back to ccp
;
;
;      End of Quit Command, Process Write
;

```

```

notq:
;       not the quit command, random write?
0156 FE57      cpi      'W'
0158 C28901    jnz      notw
;
;       this is a random write, fill buffer untill cr
015B 114D02    lxi      d,datmsg
015E CDDA01    call     print      ;data prompt
0161 0E7F      mvi      c,127      ;up to 127 characters
0163 218000    lxi      h,buff      ;destination
rloop: ;read next character to buff
0166 C5        push     b          ;save counter
0167 E5        push     h          ;next destination
0168 CDC201    call     getch      ;character to a
016B E1        pop      h          ;restore counter
016C C1        pop      b          ;restore next to fill
016D FE0D      cpi      cr         ;end of line?
016F CA7801    jz       erloop
;       not end, store character
0172 77        mov      m,a
0173 23        inx      h          ;next to fill
0174 0D        dcr      c          ;counter goes down
0175 C26601    jnz      rloop      ;end of buffer?
erloop:
;       end of read loop, store 00
0178 3600      mvi      m,0
;
;       write the record to selected record number
017A 0E22      mvi      c,writer
017C 115C00    lxi      d,fcbb
017F CD0500    call     bdos
0182 B7        ora      a          ;erro code zero?
0183 C2B901    jnz      error      ;message if not
0186 C33701    jmp      ready      ;for another record
;
;       End of Write Command, Process Read
;
notw:
;       not a write command, read record?
0189 FE52      cpi      'R'
018B C2B901    jnz      error      ;skip if not
;
;       read random record
018E 0E21      mvi      c,readr
0190 115C00    lxi      d,fcbb
0193 CD0500    call     bdos

```

```

0196 B7          ora    a          ;return code 00?
0197 C2B901      jnz     error

;
;      read was successful, write to console
019A CDCF01      call    crlf        ;new line
019D 0E80        mvi     c,128      ;max 128 characters
019F 218000      lxi     h,buff     ;next to get

wloop:
01A2 7E          mov     a,m        ;next character
01A3 23          inx     h          ;next to get
01A4 E67F        ani     7fh        ;mask parity
01A6 CA3701      jz      ready      ;for another command
                                ;if 00

01A9 C5          push    b          ;save counter
01AA E5          push    h          ;save next to get
01AB FE20        cpi     ' '        ;graphic?
01AD D4C801      cnc     putchr     ;skip output if not
01B0 E1          pop     h
01B1 C1          pop     b
01B2 0D          dcr     c          ;count=count-1
01B3 C2A201      jnz     wloop
01B6 C33701      jmp     ready

;
;      End of Read Command, All Errors End Up Here
;
error:
01B9 115902      lxi     d,errmsg
01BC CDDA01      call    print
01BF C33701      jmp     ready

;

getchr:
                                ;read next console character to a
01C2 0E01        mvi     c,coninp
01C4 CD0500      call    bdos
01C7 C9          ret

;

putchr:
                                ;write character from a to console
01C8 0E02        mvi     c,conout
01CA 5F          mov     e,a        ;character to send
01CB CD0500      call    bdos      ;send character
01CE C9          ret

;

crlf:
                                ;send carriage return line feed

```

```

01CF 3E0D          mvi    a,cr      ;carriage return
01D1 CDC801        call   putchar
01D4 3E0A          mvi    a,lf      ;line feed
01D6 CDC801        call   putchar
01D9 C9            ret

;
print:
;print the buffer addressed by de untill $
01DA D5            push   d
01DB CDCF01        call   crlf
01DE D1            pop    d          ;new line
01DF 0E09          mvi    c,pstring
01E1 CD0500        call   bdos       ;print the string
01E4 C9            ret

;
readcom:
;read the next command line to the conbuf
01E5 116B02        lxi    d,prompt
01E8 CDDA01        call   print     ;command?
01EB 0E0A          mvi    c,rstring
01ED 117A02        lxi    d,conbuf
01F0 CD0500        call   bdos       ;read command line
;
; command line is present, scan it
01F3 210000        lxi    h,0       ;start with 0000
01F6 117C02        lxi    d,conlin  ;command line
01F9 1A            readc: ldax   d          ;next command
;character
01FA 13            inx    d          ;to next command
;position
01FB B7            ora    a          ;cannot be end of
;command
01FC C8            rz
;
; not zero, numeric?
01FD D630          sui    '0'
01FF FE0A          cpi    10        ;carry if numeric
0201 D21302        jnc    endrd
;
; add-in next digit
0204 29            dad    h          ;*2
0205 4D            mov    c,l
0206 44            mov    b,h       ;bc = value * 2
0207 29            dad    h          ;*4
0208 29            dad    h          ;*8
0209 09            dad    b          ;*2 + *8 = *10
020A 85            add    l          ;*digit
020B 6F            mov    l,a
020C D2F901        jnc    readc     ;for another char

```

```

020F 24          inr    h          ;overflow
0210 C3F901      jmp    readc      ;for another char
                        endrd:
                        ;          end of read, restore value in a
0213 C630        adi    '0'        ;command
0215 FE61        cpi    'a'        ;translate case?
0217 D8          rc
                        ;          lower case, mask lower case bits
0218 E65F        ani    101$1111b
021A C9          ret
                        ;
                        ;          String Data Area for Console Messages
                        ;
badver:
021B 736F727279  db      'sorry, you need cp/m version 2$'
nospace:
023A 6E6F206469  db      'no directory space$'
datmsg:
024D 7479706520  db      'type data: $'
errmsg:
0259 6572726F72  db      'error, try again.$'
prompt:
026B 6E65787420  db      'next command? $'
                        ;
                        ;          Fixed and Variable Data Area
                        ;
027A 21          conbuf: db      conlen      ;length of console buffer
027B            consiz: ds      1          ;resulting size after read
027C            conlin: ds      32         ;length 32 buffer
0021 =          conlen  equ      $-consiz
                        ;
029C            ds      32          ;16 level stack
                        stack:
02BC            end

```

Major improvements could be made to this particular program to enhance its operation. In fact, with some work, this program could evolve into a simple data base management system. One could, for example, assume a standard record size of 128 bytes, consisting to arbitrary fields within the record. A program, called GETKEY, could be developed that first reads a sequential file and extracts a specific field defined by the operator. For example, the command

```
GETKEY NAMES.DAT LASTNAME 10 20
```

would cause GETKEY to read the data base file NAMES.DAT and extract the LASTNAME field from each record, starting in position 10 and ending at character 20. GETKEY builds a table in memory consisting of each particular LASTNAME field, along with its 16-bit record number location within

the file. The `GETKEY` program then sorts this list and writes a new file, called `LASTNAME.KEY`, which is an alphabetical list of `LASTNAME` fields with their corresponding record numbers. This list is called an inverted index in information retrieval parlance.

If the programmer were to rename the program shown above as `QUERY` and modify it so that it reads a sorted key file into memory, the command line might appear as

```
QUERY NAMES.DAT LASTNAME.KEY
```

Instead of reading a number, the `QUERY` program reads an alphanumeric string that is a particular key to find in the `NAMES.DAT` data base. Because the `LASTNAME.KEY` list is sorted, one can find a particular entry rapidly by performing a binary search, similar to looking up a name in the telephone book. Starting at both ends of the list, one examines the entry halfway in between and, if not matched, splits either the upper half or the lower half for the next search. You will quickly reach the item you are looking for and find the corresponding record number. You should fetch and display this record at the console, just as was done in the program shown above.

With some more work, you can allow a fixed grouping size that differs from the 128-byte record shown above. This is accomplished by keeping track of the record number and the byte offset within the record. Knowing the group size, you randomly access the record containing the proper group, offset to the beginning of the group within the record read sequentially until the group size has been exhausted.

Finally, you can improve `QUERY` considerably by allowing boolean expressions, which compute the set of records that satisfy several relationships, such as a `LASTNAME` between `HARDY` and `LAUREL` and an `AGE` lower than 45. Display all the records that fit this description. Finally, if your lists are getting too big to fit into memory, randomly access key files from the disk as well.

5.6. System Function Summary

Decimal	Hex	Function Name	Input	Output
0	00	System Reset		<i>no return</i>
1	01	Console Input		A = ASCII character
2	02	Console Output	E = character to output	none
3	03	Reader Input		A = ASCII character
4	04	Punch Output	E = character to output	none
5	05	List Output	E = character to output	none
6	06	Direct Console I/O	E = 0FFH for input, otherwise character to output	A = ASCII character on input or requested status, none on output
7	07	Get IOBYTE		A = IOBYTE value
8	08	Set IOBYTE	E = IOBYTE	

Decimal	Hex	Function Name	Input	Output
9	09	Print String	DE = String Address	none
10	0A	Read Console Buffer	DE = Buffer Address	characters in buffer
11	0B	Get Console Status		A = Status
12	0C	Return Version Number		HL = Version number
13	0D	Reset Disk System		none
14	0E	Select Disk	E = disk number	none
15	0F	Open File	DE = FCB Address	A = 0FFH if not found
16	10	Close File	DE = FCB Address	0FFH if not found
17	11	Search for First	DE = FCB Address	A = Directory Code
18	12	Search for Next		A = Directory Code
19	13	Delete File	DE = FCB Address	none
20	14	Read Sequential	DE = FCB Address	A = Error Code
21	15	Write Sequential	DE = FCB Address	A = Error Code
22	16	Make File	DE = FCB Address	A = 0FFH if no space
23	17	Rename File	DE = FCB Address	A = 0FFH if not found
24	18	Return Login Vector		HL = Login vector
25	19	Return Current Disk		A = Current Disk
26	1A	Set DMA Address	DE = DMA Address	none
27	1B	Get ADDR (ALLOC)		HL = ALLOC Address
28	1C	Write Protect Disk		none
29	1D	Get Read/Only Vector		HL = R/O Vector
30	1E	Set File Attributes	DE = FCB Address	A = Error Code
31	1F	Get ADDR (Disk Params)		HL = DPB Address
32	20	Set/Get User Code	E = 0FFH to get, otherwise value to set mod 16	A = User Number
33	21	Read Random	DE = FCB Address	A = Error Code
34	22	Write Random	DE = FCB Address	A = Error Code
35	23	Compute File Size	DE = FCB Address	random record number in FCB
36	24	Set Random Record	DE = FCB Address	random record number in FCB
37	25	Reset Drive	DE = Drive Vector	A = 0
38	26	Access Drive	<i>not supported</i>	<i>not supported</i>
39	27	Free Drive	<i>not supported</i>	<i>not supported</i>

Decimal	Hex	Function Name	Input	Output
40	28	Write Random with Zero Fill	DE = FCB Address	A = Error Code

Table 5-30: System Function Summary

6. CP/M Alteration

6.1. Introduction

The standard CP/M system assumes operation on an Intel MDS-800 microcomputer development system, but is designed so you can alter a specific set of subroutines that define the hardware operating environment.

Although standard CP/M 2 is configured for single-density floppy disks, field alteration features allow adaptation to a wide variety of disk subsystems from single drive minidisks to high-capacity, hard disk systems. To simplify the following adaptation process, it is assumed that CP/M 2 is first configured for single-density floppy disks where minimal editing and debugging tools are available. If an earlier version of CP/M is available, the customizing process is eased considerably. In this latter case, you might want to review the system generation process and skip to later sections that discuss system alteration for nonstandard disk systems.

To achieve device independence, CP/M is separated into three distinct modules:

BIOS is the Basic I/O System, which is environment dependent.

BDOS is the Basic Disk Operating System, which is not dependent upon the hardware configuration.

CCP is the Console Command Processor, which uses the BDOS.

Of these modules, only the BIOS is dependent upon the particular hardware. You can patch the distribution version of CP/M to provide a new BIOS that provides a customized interface between the remaining CP/M modules and the hardware system. This document provides a step-by-step procedure for patching a new BIOS into CP/M.

All disk-dependent portions of CP/M 2 are placed into a BIOS, a resident disk parameter block, which is either hand coded or produced automatically using the disk definition macro library provided with CP/M 2. The end user need only specify the maximum number of active disks, the starting and ending sector numbers, the data allocation size, the maximum extent of the logical disk, directory size information, and reserved track values. The macros use this information to generate the appropriate tables and table references for use during CP/M 2 operation.

Deblocking information is provided, which aids in assembly or disassembly of sector sizes that are multiples of the fundamental 128-byte data unit, and the system alteration manual includes general purpose subroutines that use the deblocking information to take advantage of larger sector sizes. Use of these subroutines, together with the table-drive data access algorithms, makes CP/M 2 a universal data management system.

File expansion is achieved by providing up to 512 logical file extents, where each logical extent contains 16K bytes of data. CP/M 2 is structured, however, so that as much as 128K bytes of data are addressed by a single physical extent, corresponding to a single directory entry, maintaining

compatibility with previous versions while taking advantage of directory space. If CP/M is being tailored to a computer system for the first time, the new BIOS requires some simple software development and testing. The standard BIOS is listed in Appendix A and can be used as a model for the customized package. A skeletal version of the BIOS given in Appendix B can serve as the basis for a modified BIOS.

In addition to the BIOS, you must write a simple memory loader, called GETSYS, which brings the operating system into memory. To patch the new BIOS into CP/M, you must write the reverse of GETSYS, called PUTSYS, which places an altered version of CP/M back onto the disk. PUTSYS can be derived from GETSYS by changing the disk read commands into disk write commands. Sample skeletal GETSYS and PUTSYS programs are described in Section 6.4 and listed in Appendix C.

To make the CP/M system load automatically, you must also supply a cold start loader, similar to the one provided with CP/M, listed in Appendices A and D. A skeletal form of a cold start loader is given in Appendix E, which serves as a model for the loader.

6.2. First-level System Regeneration

The procedure to patch the CP/M system is given below. Address references in each step are shown with H denoting the hexadecimal radix, and are given for a 20K CP/M system. For larger CP/M systems, a **bias** is added to each address that is shown with a '+*b*' following it, where *b* is equal to the memory size – 20K. Values for *b* in various standard memory sizes are listed in Table 6-31

Memory Size	Value
24K	$b = 24K - 20K = 4K = 1000H$
32K	$b = 32K - 20K = 12K = 3000H$
40K	$b = 40K - 20K = 20K = 5000H$
48K	$b = 48K - 20K = 28K = 7000H$
56K	$b = 56K - 20K = 36K = 9000H$
62K	$b = 62K - 20K = 42K = A800H$
64K	$b = 64K - 20K = 44K = B000H$

Table 6-31: Standard Memory Size Values

Note that the standard distribution version of CP/M is set for operation within a 20K CP/M system. Therefore, you must first bring up the 20K CP/M system, then configure it for actual memory size (see Section 6.3).

Follow these steps to patch your CP/M system:

1. Read Section 6.4 and write a GETSYS program that reads the first two tracks of a disk into memory. The program from the disk must be loaded starting at location 3380H. GETSYS is coded to start at location 100H (base of the TPA) as shown in Appendix C.

2. Test the GETSYS program by reading a blank disk into memory, and check to see that the data has been read properly and that the disk has not been altered in any way by the GETSYS program.
3. Run the GETSYS program using an initialized CP/M disk to see if GETSYS loads CP/M starting at 3380H (the operating system actually starts 128 bytes later at 3400H).
4. Read Section 6.4 and write the PUTSYS program. This writes memory starting at 3380H back onto the first two tracks of the disk. The PUTSYS program should be located at 200H, as shown in Appendix C.
5. Test the PUTSYS program using a blank, uninitialized disk by writing a portion of memory to the first two tracks; clear memory and read it back using GETSYS. Test PUTSYS completely, because this program will be used to alter CP/M on disk.
6. Study Section 6.5, Section 6.6, and Section 6.7 along with the distribution version of the BIOS given in Appendix A and write a simple version that performs a similar function for the customized environment. Use the program given in Appendix B as a model. Call this new BIOS by name CBIOS (customized BIOS). Implement only the primitive disk operations on a single drive and simple console input/output functions in this phase.
7. Test CBIOS completely to ensure that it properly performs console character I/O and disk reads and writes. Be careful to ensure that no disk write operations occur during read operations and check that the proper track and sectors are addressed on all reads and writes. Failure to make these checks might cause destruction of the initialized CP/M system after it is patched.
8. Referring to Table 6-33 in Section 6.5, note that the BIOS is placed between locations 4A00H and 4FFFH. Read the CP/M system using GETSYS and replace the BIOS segment by the CBIOS developed in step 6 and tested in step 7. This replacement is done in memory.
9. Use PUTSYS to place the patched memory image of CP/M onto the first two tracks of a blank disk for testing.
10. Use GETSYS to bring the copied memory image from the test disk back into memory at 3380H and check to ensure that it has loaded back properly (clear memory, 1 if possible, before the load). Upon successful load, branch to the cold start code at location 4A00H. The cold start routine initializes page zero, then jumps to the CCP at location 3400H, which calls the BDOS, which calls the CBIOS. The CCP asks the CBIOS to read sixteen sectors on track 2, and CP/M types A>, the system prompt.
If difficulties are encountered, use whatever debug facilities are available to trace and breakpoint the CBIOS.
11. Upon completion of step 10, CP/M has prompted the console for a command input. To test the disk write operation, type

SAVE 1 X.COM

All commands must be followed by a carriage return. CP/M responds with another prompt after several disk accesses:

A>

If it does not, debug the disk write functions and retry.

12. Test the directory command by typing

DIR

CP/M responds with

A:X COM

13. Test the erase command by typing

ERA X.COM

CP/M responds with the A prompt. This is now an operational system that only requires a bootstrap loader to function completely.

14. Write a bootstrap loader that is similar to GETSYS and place it on track 00, sector 01, using PUTSYS (again using the test disk, not the distribution disk). See Section 6.5, and Section 6.8 for more information on the bootstrap operation.

15. Retest the new test disk with the bootstrap loader installed by executing steps 11, 12, and 13. Upon completion of these tests, type a CTRL-C. The system executes a warm start, which reboots the system, and types the A prompt.

16. At this point, there is probably a good version of the customized CP/M system on the test disk. Use GETSYS to load CP/M from the test disk. Remove the test disk, place the distribution disk, or a legal copy, into the drive, and use PUTSYS to replace the distribution version with the customized version. Do not make this replacement if you are unsure of the patch because this step destroys the system that was obtained from Digital Research.

17. Load the modified CP/M system and test it by typing

DIR

CP/M responds with a list of files that are provided on the initialized disk. The file DDT.COM is the memory image for the debugger. Note that from now on, you must always reboot the CP/M system (CTRL-C is sufficient) when the disk is removed and replaced by another disk, unless the new disk is to be Read-Only.

18. Load and test the debugger by typing

DDT

See Section 4 for operating procedures.

19. Before making further CBIOS modifications, practice using the editor (Section 2), and assembler (see Section 3). Recode and test the GETSYS, PUTSYS, and CBIOS programs using ED, ASM, and DDT. Code and test a COPY program that does a sector-to-sector copy from one disk to another to obtain back-up copies of the original disk. Read the CP/M Licensing

Agreement specifying legal responsibilities when copying the CP/M system. Place the following copyright notice:

Copyright (c), 1983

Digital Research

on each copy that is made with the COPY program.

20. Modify the CBIOS to include the extra functions for punches, readers, and sign-on messages, and add the facilities for additional disk drives, if desired. These changes can be made with the GETSYS and PUTSYS programs or by referring to the regeneration process in Section 6.3.

You should now have a good copy of the customized CP/M system. Although the CBIOS portion of CP/M belongs to the user, the modified version cannot be legally copied.

It should be noted that the system remains file-compatible with all other CP/M systems (assuming media compatibility) which allows transfer of nonproprietary software between CP/M users.

6.3. Second-level System Regeneration

Once the system is running, the next step is to configure CP/M for the desired memory size. Usually, a memory image is first produced with the MOVCPM program (system relocater) and then placed into a named disk file. The disk file can then be loaded, examined, patched, and replaced using the debugger and the system generation program (refer to Section 1).

The CBIOS and BOOT are modified using ED and assembled using ASM, producing files called CBIOS.HEX and BOOT.HEX, which contain the code for CBIOS and BOOT in Intel hex format.

To get the memory image of CP/M into the TPA configured for the desired memory size, type the command:

```
MOVCPM xx *
```

where xx is the memory size in decimal K bytes, for example, 32 for 32K. The response is as follows:

```
CONSTRUCTING xxK CP/M VERS 2.0
```

```
READY FOR "SYSGEN" OR
```

```
"SAVE 34 CPMxx.COM"
```

An image of CP/M in the TPA is configured for the requested memory size. The memory image is at location 0900H through 227FH, that is, the BOOT is at 0900H, the CCP is at 0980H, the BDOS starts at 1180H, and the BIOS is at 1F80H. Note that the memory image has the standard Model 800 BIOS and BOOT on it. It is now necessary to save the memory image in a file so that you can patch the CBIOS and CBOOT into it:

SAVE 34 CPMXX.COM

The memory image created by the MOVCPM program is offset by a negative bias so that it loads into the free area of the TPA, and thus does not interfere with the operation of CP/M in higher memory. This memory image can be subsequently loaded under DDT and examined or changed in preparation for a new generation of the system. DDT is loaded with the memory image by typing:

DDT CPMXX.COM

Which loads DDT, then reads the CP/M image.

DDT should respond with the following:

DDT VERS 2.2

NEXT PC

2300 0100

-

Where - is the DDT prompt.

You can then give the display and disassembly commands to examine portions of the memory image between 900H and 227FH. Note, however, that to find any particular address within the memory image, you must apply the negative bias to the CP/M address to find the actual address. Track 00, sector 01, is loaded to location 900H (the user should find the cold start loader at 900H to 97FH); track 00, sector 02, is loaded into 980H (this is the base of the CCP); and so on through the entire CP/M system load. In a 20K system, for example, the CCP resides at the CP/M address 3400H, but is placed into memory at 980H by the SYSGEN program. Thus, the negative bias, denoted by n , satisfies

$$3400H + n = 980H, \text{ or } n = 980H - 3400H$$

Assuming two's complement arithmetic, $n = D580H$, which can be checked by

$$3400H + D580H = 10980H = 0980H$$

when ignoring high-order overflows.

Note that for larger systems, n satisfies:

$$(3400H + b) + n = 0980H, \text{ or}$$

$$n = 0980H - (3400H + b), \text{ or}$$

$$n = D580H - b$$

The value for n for common CP/M systems is given in Table 6-32.

Memory Size	Bias b	Negative Offset n
20K	1000H	$n = D580H - 0000H = D580H$
24K	1000H	$n = D580H - 1000H = C580H$

Memory Size	Bias b	Negative Offset n
32K	3000H	$n = D580H - 3000H = A580H$
40K	5000H	$n = D580H - 5000H = 8580H$
48K	7000H	$n = D580H - 7000H = 6580H$
56K	9000H	$n = D580H - 9000H = 4580H$
62K	A800H	$n = D580H - A800H = 2D80H$
64K	B000H	$n = D580H - B000H = 2580H$

Table 6-32: Common Value for CP/M Systems

If you want to locate the address x within the memory image loaded under DDT in a 20K system, first type

HX, n

for the Hexadecimal sum and difference command.

and DDT responds with the value of $x + n$ (sum) and $x - n$ (difference). The first number printed by DDT is the actual memory address in the image where the data or code is located. For example, the following DDT command:

H3400,D580

produces 980H as the sum, which is where the CCP is located in the memory image under DDT.

Type the L command to disassemble portions of the BIOS located at $(4A00H+b)-n$, which, when one uses the H command, produces an actual address of 1F80H. The disassembly command would thus be as follows:

L1F80

It is now necessary to patch in the CBOOT and CBIOS routines. The BOOT resides at location 0900H in the memory image. If the actual load address is n , then to calculate the bias (m), type the command:

H900, n

The second number typed by DDT in response to the command is the desired bias (m). For example, if the BOOT executes at 0080H, the command

H900,80

produces the sum and difference in hex

0980 0880

Therefore, the bias m would be 0880H. To read-in the BOOT, give the command:

ICBOOT.HEX

R m

to read CBOOT.HEX with a bias of $m (=900H-n)$

Examine the CBOOT with

L900

You are now ready to replace the CBIOS by examining the area at 1F80H, where the original version of the CBIOS resides, and then typing

ICBIOS.HEX Ready the hex file for loading

Assume that the CBIOS is being integrated into a 20K CP/M system and thus originates at location 4A00H. To locate the CBIOS properly in the memory image under DDT, you must apply the negative bias n for a 20K system when loading the hex file. This is accomplished by typing

RD580 Read the file with bias of D580H

Upon completion of the read, reexamine the area where the CBIOS has been loaded (use an L1F80 command) to ensure that it is properly loaded. When you are satisfied that the change has been made, return from DDT using a CTRL-C or, G0 command.

6.4. Sample GETSYS and PUTSYS Program

The following program provides a framework for the GETSYS and PUTSYS programs referenced in Sections 6.1 and 6.2. To read and write the specific sectors, you must insert the READSEC and WRITESEC subroutines.

```
; GETSYS PROGRAM -- READ TRACKS 0 AND 1 TO MEMORY AT 3380H
; REGISTER                USE

;      A                  (SCRATCH REGISTER)

;      B                  TRACK COUNT (0, 1)

;      C                  SECTOR COUNT (1,2,...,26)

;      DE                 (SCRATCH REGISTER PAIR)

;      HL                 LOAD ADDRESS

;      SP                 SET TO STACK ADDRESS

;
START:  LXI  SP,3380H      ;SET STACK POINTER TO SCRATCH
                        ;AREA
        LXI  H,3380H      ;SET BASE LOAD ADDRESS
        MVI  B,0          ;START WITH TRACK 0
RDTRK:  ;READ NEXT TRACK (INITIALLY 0)
        MVI  C,1          ;READ STARTING WITH SECTOR 1

RDSEC:  ;READ NEXT SECTOR
```

```

CALL READSEC      ;USER-SUPPLIED SUBROUTINE
LXI  D,128        ;MOVE LOAD ADDRESS TO NEXT 1/2
                    ;PAGE
DAD  D            ;HL = HL + 128
INR  C            ;SECTOR = SECTOR + 1
MOV  A,C          ;CHECK FOR END OF TRACK
CPI  27
JC   RDSEC        ;CARRY GENERATED IF SECTOR <27

;
;  ARRIVE HERE AT END OF TRACK, MOVE TO NEXT TRACK
INR  B
MOV  A,B          ;TEST FOR LAST TRACK
CPI  2
JC   RDTRK        ;CARRY GENERATED IF TRACK <2

;
;  USER-SUPPLIED SUBROUTINE TO READ THE DISK
READSEC:
;  ENTER WITH TRACK NUMBER IN REGISTER B,
;  SECTOR NUMBER IN REGISTER C, AND
;  ADDRESS TO FILL IN HL

;

PUSH B            ;SAVE B AND C REGISTERS
PUSH H            ;SAVE HL REGISTERS

;
;  *****
;  perform disk read at this point, branch to
;  label START if an error occurs
;  *****
POP H             ;RECOVER HL
POP B             ;RECOVER B AND C REGISTERS
RET              ;BACK TO MAIN PROGRAM

END START

```

This program is assembled and listed in Appendix B for reference purposes, with an assumed origin of 100H. The hexadecimal operation codes that are listed on the left might be useful if the program has to be entered through the panel switches.

The PUTSYS program can be constructed from GETSYS by changing only a few operations in the GETSYS program given above, as shown in Appendix C. The register pair HL becomes the dump address, next address to write, and operations on these registers do not change within the program. The READSEC subroutine is replaced by a WRITESEC subroutine, which performs the

opposite function; data from address HL is written to the track given by register B and sector given by register C. It is often useful to combine GETSYS and PUTSYS into a single program during the test and development phase, as shown in Appendix C.

6.5. Disk Organization

The sector allocation for the standard distribution version of CP/M is given here for reference purposes. The first sector contains an optional software boot section (see the table on the following page. Disk controllers are often set up to bring track 0, sector 1, into memory at a specific location, often location 0000H. The program in this sector, called BOOT, has the responsibility of bringing the remaining sectors into memory starting at location 3400H+b. If the controller does not have a built-in sector load, the program in track 00, sector 01 can be ignored. In this case, load the program from track 00, sector 02, to location 3400H+b.

As an example, the Intel Model 800 hardware cold start loader brings track 00, sector 01, into absolute address 3000H. Upon loading this sector, control transfers to location 3000H, where the bootstrap operation commences by loading the remainder of track 00 and all of track 01 into memory, starting at 3400H+b. Note that this bootstrap loader is of little use in a non-microcomputer development system environment, although it is useful to examine it because some of the boot actions will have to be duplicated in the user's cold start loader.

Track #	Sector #	Page #	Memory Address	CP/M Module name
00	01		(boot address)	Cold Start Loader
00	02	00	3400H + <i>b</i>	CCP
00	03	00	3480H + <i>b</i>	CCP
00	04	01	3500H + <i>b</i>	CCP
00	05	01	3580H + <i>b</i>	CCP
00	06	02	3600H + <i>b</i>	CCP
00	07	02	3680H + <i>b</i>	CCP
00	08	03	3700H + <i>b</i>	CCP
00	09	03	3780H + <i>b</i>	CCP
00	10	04	3800H + <i>b</i>	CCP
00	11	04	3880H + <i>b</i>	CCP
00	12	05	3900H + <i>b</i>	CCP
00	13	05	3980H + <i>b</i>	CCP
00	14	06	3A00H + <i>b</i>	CCP
00	15	06	3A80H + <i>b</i>	CCP
00	16	07	3B00H + <i>b</i>	CCP
00	17	07	3B80H + <i>b</i>	CCP
00	18	08	3C00H + <i>b</i>	BDOS

Track #	Sector #	Page #	Memory Address	CP/M Module name
00	19	08	3C80H + <i>b</i>	BDOS
00	20	09	3D00H + <i>b</i>	BDOS
00	21	09	3D80H + <i>b</i>	BDOS
00	22	10	3E00H + <i>b</i>	BDOS
00	23	10	3E80H + <i>b</i>	BDOS
00	24	11	3F00H + <i>b</i>	BDOS
00	25	11	3F80H + <i>b</i>	BDOS
00	26	12	4000H + <i>b</i>	BDOS
01	01	12	4080H + <i>b</i>	BDOS
01	02	13	4100H + <i>b</i>	BDOS
01	03	13	4180H + <i>b</i>	BDOS
01	04	14	4200H + <i>b</i>	BDOS
01	05	14	4280H + <i>b</i>	BDOS
01	06	15	4300H + <i>b</i>	BDOS
01	07	15	4380H + <i>b</i>	BDOS
01	08	16	4400H + <i>b</i>	BDOS
01	09	16	4480H + <i>b</i>	BDOS
01	10	17	4500H + <i>b</i>	BDOS
01	11	17	4580H + <i>b</i>	BDOS
01	12	18	4600H + <i>b</i>	BDOS
01	13	18	4680H + <i>b</i>	BDOS
01	14	19	4700H + <i>b</i>	BDOS
01	15	19	4780H + <i>b</i>	BDOS
01	16	20	4800H + <i>b</i>	BDOS
01	17	20	4880H + <i>b</i>	BDOS
01	18	21	4900H + <i>b</i>	BDOS
01	19	21	4980H + <i>b</i>	BDOS
01	20	22	4A00H + <i>b</i>	BIOS
01	21	22	4A80H + <i>b</i>	BIOS
01	22	23	4B00H + <i>b</i>	BIOS
01	23	23	4B80H + <i>b</i>	BIOS
01	24	24	4C00H + <i>b</i>	BIOS
01	25	24	4C80H + <i>b</i>	BIOS
01	26	25	4D00H + <i>b</i>	BIOS

Table 6-33: CP/M Disk Sector Allocation

6.6. The BIOS Entry Points

The entry points into the BIOS from the cold start loader and BDOS are detailed below. Entry to the BIOS is through a jump vector located at $4A00H + b$, as shown below. See Appendixes A and B. The jump vector is a sequence of 17 jump instructions that send program control to the individual BIOS subroutines. The BIOS subroutines might be empty for certain functions (they might contain a single `RET` operation) during reconfiguration of CP/M, but the entries must be present in the jump vector.

The jump vector at $4A00H + b$ takes the form shown below, where the individual jump addresses are given to the left:

4A00H+b	JMP BOOT	;ARRIVE HERE FROM COLD START LOAD
4A03H+b	JMP WBOOT	;ARRIVE HERE FOR WARM START
4A06H+b	JMP CONST	;CHECK FOR CONSOLE CHAR READY
4A09H+b	JMP CONIN	;READ CONSOLE CHARACTER IN
4A0CH+b	JMP CONOUT	;WRITE CONSOLE CHARACTER OUT
4A0FH+b	JMP LIST	;WRITE LISTING CHARACTER OUT
4A12H+b	JMP PUNCH	;WRITE CHARACTER TO PUNCH DEVICE
4A15H+b	JMP READER	;READ READER DEVICE
4A18H+b	JMP HOME	;MOVE TO TRACK 00 ON SELECTED DISK
4A1BH+b	JMP SELDSK	;SELECT DISK DRIVE
4A1EH+b	JMP SETTRK	;SET TRACK NUMBER
4A21H+b	JMP SETSEC	;SET SECTOR NUMBER
4A24H+b	JMP SETDMA	;SET DMA ADDRESS
4A27H+b	JMP READ	;READ SELECTED SECTOR
4A2AH+b	JMP WRITE	;WRITE SELECTED SECTOR
4A2DH+b	JMP LISTST	;RETURN LIST STATUS
4A30H+b	JMP SECTTRAN	;SECTOR TRANSLATE SUBROUTINE

Each jump address corresponds to a particular subroutine that performs the specific function, as outlined below. There are three major divisions in the jump table: the system re-initialization, which results from calls on `BOOT` and `WBOOT`; simple character I/O, performed by calls on `CONST`, `CONIN`, `CONOUT`, `LIST`, `PUNCH`, `READER`, and `LISTST`; and disk I/O, performed by calls on `HOME`, `SELDSK`, `SETTRK`, `SETSEC`, `SETDMA`, `READ`, `WRITE`, and `SECTTRAN`.

All simple character I/O operations are assumed to be performed in ASCII, upper- and lower-case, with high-order (parity bit) set to zero. An end-of-file condition for an input device is given by an ASCII CTRL-Z (1AH). Peripheral devices are seen by CP/M as logical devices and are assigned to physical devices within the BIOS.

To operate, the BDOS needs only the `CONST`, `CONIN`, and `CONOUT` subroutines. `LIST`, `PUNCH`, and `READER` can be used by PIP, but not the BDOS. Further, the `LISTST` entry is currently used only by `DESPOOL`, the print spooling utility. Thus, the initial version of CBIOS can have empty subroutines for the remaining ASCII devices.

The following list describes the characteristics of each device.

CONSOLE is the principal interactive console that communicates with the operator and it is accessed through `CONST`, `CONIN`, and `CONOUT`. Typically, the `CONSOLE` is a device such as a CRT or teletype.

LIST is the principal listing device. If it exists on the user's system, it is usually a hard-copy device, such as a printer or teletype.

PUNCH is the principal tape punching device. If it exists, it is normally a high-speed paper tape punch or teletype.

READER is the principal tape reading device, such as a simple optical reader or teletype.

A single peripheral can be assigned as the `LIST`, `PUNCH`, and `READER` device simultaneously. If no peripheral device is assigned as the `LIST`, `PUNCH`, or `READER` device, the CBIOS gives an appropriate error message so that the system does not hang if the device is accessed by PIP or some other user program. Alternately, the `PUNCH` and `LIST` routines can just simply return, and the `READER` routine can return with a 1AH (CTRL-Z) in register A to indicate immediate end-of-file.

For added flexibility, you can optionally implement the `IOBYTE` function, which allows reassignment of physical devices. The `IOBYTE` function creates a mapping of logical-to-physical devices that can be altered during CP/M processing, see the `STAT` command in Section 1.6.1.

The definition of the `IOBYTE` function corresponds to the Intel standard as follows: a single location in memory, currently location 0003H, is maintained, called `IOBYTE`, which defines the logical-to-physical device mapping that is in effect at a particular time. The mapping is performed by splitting the `IOBYTE` into four distinct fields of two bits each, called the `CONSOLE`, `READER`, `PUNCH`, and `LIST` fields, as shown in Table 6-34

Most Significant		Bits				Least Significant	
List		Punch		Reader		Console	
7	6	5	4	3	2	1	0

Table 6-34: IOBYTE Fields

Value	Bits	Meaning
	1 0	Console field
0	0 0	CONSOLE is assigned to the console printer device (TTY:)
1	0 1	CONSOLE is assigned to the CRT device (CRT:)
2	1 0	Batch mode: use the READER as the CONSOLE input, and the LIST device as the CONSOLE output (BAT:)
3	1 1	CONSOLE is assigned to the user-defined console device (UC1:)
	3 2	Reader field
0	0 0	READER is the teletype device (TTY:)
1	0 1	READER is the high speed reader device (PTR:)
2	1 0	READER is the user-defined reader #1 (UR1:)
3	1 1	READER is the user-defined reader #2 (UR2:)
	5 4	Punch field
0	0 0	PUNCH is the teletype device (TTY:)
1	0 1	PUNCH is the high speed punch device (PTP:)
2	1 0	PUNCH is the user-defined punch #1 (UP1:)
3	1 1	PUNCH is the user-defined punch #2 (UP2:)
	7 6	List field
0	0 0	LIST is the teletype device (TTY:)
1	0 1	LIST is the CRT device (CRT:)
2	1 0	LIST is the line printer device (LPT:)
3	1 1	LIST is the user-defined list device (UL1:)

Table 6-35: IOBYTE Field Values

The implementation of the IOBYTE is optional and effects only the organization of the CBIOS. No CP/M systems use the IOBYTE (although they tolerate the existence of the IOBYTE at location 0003H) except for PIP, which allows access to the physical devices, and STAT, which allows logical-physical assignments to be make or displayed. For more information see Section 1. In any case the IOBYTE implementation should be omitted until the basic CBIOS is fully implemented and tested; then you should add the IOBYTE to increase the facilities.

Disk I/O is always performed through a sequence of calls on the various disk access subroutines that set up the disk number to access, the track and sector on a particular disk, and the Direct

Memory Access (DMA) address involved in the I/O operation. After all these parameters have been set up, a call is made to the `READ` or `WRITE` function to perform the actual I/O operation.

There is often a single call to `SELDSK` to select a disk drive, followed by a number of read or write operations to the selected disk before selecting another drive for subsequent operations.

Similarly, there might be a single call to set the DMA address, followed by several calls that read or write from the selected DMA address before the DMA address is changed. The track and sector subroutines are always called before the `READ` or `WRITE` operations are performed.

The `READ` and `WRITE` routines should perform several retries (10 is standard) before reporting the error condition to the BDOS. If the error condition is returned to the BDOS, it reports the error to the user. The `HOME` subroutine might or might not actually perform the track 00 seek, depending upon controller characteristics; the important point is that track 00 has been selected for the next operation and is often treated in exactly the same manner as `SETTRK` with a parameter of 00.

The following table describes the exact responsibilities of each BIOS entry point subroutine.

Entry Point	Function
BOOT	The <code>BOOT</code> entry point gets control from the cold start loader and is responsible for basic system initialization, including sending a sign-on message, which can be omitted in the first version. If the <code>IOBYTE</code> function is implemented, it must be set at this point. The various system parameters that are set by the <code>WBOOT</code> entry point must be initialized, and control is transferred to the CCP at $3400H + b$ for further processing. Note that register C must be set to zero to select drive A.

Entry Point	Function
WBOOT	The WBOOT entry point gets control when a warm start occurs. A warm start is performed whenever a user program branches to location 0000H, or when the CPU is reset from the front panel. The CP/M system must be loaded from the first two tracks of drive A up to, but not including, the BIOS, or CBIOS, if the user has completed the patch. System parameters must be initialized as follows: locations 0, 1, and 2 Set to JMP WBOOT for warm starts (000H: JMP 4A03H+b) location 3 Set initial value of IOBYTE, if implemented in the CBIOS. location 4 High nibble = current user number; low nibble = current drive locations 5, 6, and 7 Set to JMP BDOS, which is the primary entry point to CP/M for transient programs. (0005H: JMP 3C06H+b) Refer to Section 6.9 for complete details of page zero use. Upon completion of the initialization, the WBOOT program must branch to the CCP at 3400H+b to restart the system. Upon entry to the CCP, register C is set to the drive to select after system initialization. The WBOOT routine should read location 4 in memory, verify that is a legal drive, and pass it to the CCP in register C.
CONST	You should sample the status of the currently assigned console device and return 0FFH in register A if a character is ready to read and 00H in register A if no console characters are ready.
CONIN	The next console character is read into register A, and the parity bit is set, high-order bit, to zero. If no console character is ready, wait until a character is typed before returning.
CONOUT	The character is sent from register C to the console output device. The character is in ASCII, with high-order parity bit set to zero. You might want to include a time-out on a line-feed or carriage return, if the console device requires some time interval at the end of the line (such as a TI Silent 700 terminal). You can filter out control characters that cause the console device to react in a strange way (CTRL-Z causes the Lear-Siegler terminal to clear the screen, for example).
LIST	The character is sent from register C to the currently assigned listing device. The character is in ASCII with zero parity bit.

Entry Point	Function
PUNCH	The character is sent from register C to the currently assigned punch device. The character is in ASCII with zero parity.
READER	The next character is read from the currently assigned reader device into register A with zero parity (high-order bit must be zero); an end-of-file condition is reported by returning an ASCII CTRL-Z(1AH).
HOME	The disk head of the currently selected disk (initially disk A) is moved to the track 00 position. If the controller allows access to the track 0 flag from the drive, the head is stepped until the track 0 flag is detected. If the controller does not support this feature, the HOME call is translated into a call to SETTRK with a parameter of 0.
SELDSK	The disk drive given by register C is selected for further operations, where register C contains 0 for drive A, 1 for drive B, and so on up to 15 for drive P (the standard CP/M distribution version supports four drives). On each disk select, SELDSK must return in HL the base address of a 16-byte area, called the Disk Parameter Header, described in Section 6.10. For standard floppy disk drives, the contents of the header and associated tables do not change; thus, the program segment included in the sample CBIOS performs this operation automatically. If there is an attempt to select a nonexistent drive, SELDSK returns HL=0000H as an error indicator. Although SELDSK must return the header address on each call, it is advisable to postpone the physical disk select operation until an I/O function (seek, read, or write) is actually performed, because disk selects often occur without ultimately performing any disk I/O, and many controllers unload the head of the current disk before selecting the new drive. This causes an excessive amount of noise and disk wear. The least significant bit of register E is zero if this is the first occurrence of the drive select since the last cold or warm start.

Entry Point	Function
SETTRK	Register BC contains the track number for subsequent disk accesses on the currently selected drive. The sector number in BC is the same as the number returned from the SECTRN entry point. You can choose to seek the selected track at this time or delay the seek until the next read or write actually occurs. Register BC can take on values in the range 0-76 corresponding to valid track numbers for standard floppy disk drives and 0-65535 for nonstandard disk subsystems.
SETSEC	Register BC contains the sector number, 01 through 26, for subsequent disk accesses on the currently selected drive. The sector number in BC is the same as the number returned from the SECTRN entry point. You can choose to send this information to the controller at this point or delay sector selection until a read or write operation occurs.
SETDMA	Register BC contains the DMA (Disk Memory Access) address for subsequent read or write operations. For example, if B = 00H and C = 80H when SETDMA is called, all subsequent read operations read their data into 80H through 0FFH and all subsequent write operations get their data from 80H through 0FFH, until the next call to SETDMA occurs. The initial DMA address is assumed to be 80H. The controller need not actually support Direct Memory Access. If, for example, all data transfers are through I/O ports, the CBIOS that is constructed uses the 128-byte area starting at the selected DMA address for the memory buffer during the subsequent read or write operations.

Entry Point	Function
READ	Assuming the drive has been selected, the track has been set, and the DMA address has been specified, the READ subroutine attempts to read one sector based upon these parameters and returns the following error codes in register A: 0 no errors occurred 1 nonrecoverable error condition occurred Currently, CP/M responds only to a zero or nonzero value as the return code. That is, if the value in register A is 0, CP/M assumes that the disk operation was completed properly. If an error occurs the CBIOS should attempt at least 10 retries to see if the error is recoverable. When an error is reported the BDOS prints the message <pre>BDOS Err On X: Bad Sector</pre> The operator then has the option of pressing a carriage return to ignore the error, or CTRL-C to abort.
WRITE	Data is written from the currently selected DMA address to the currently selected drive, track, and sector. For floppy disks, the data should be marked as non-deleted data to maintain compatibility with other CP/M systems. The error codes given in the READ command are returned in register A, with error recovery attempts as described above.
LISTST	You return the ready status of the list device used by the DESPOOL program to improve console response during its operation. The value 00 is returned in A if the list device is not ready to accept a character and 0FFH if a character can be sent to the printer. A 00 value should be returned if LIST status is not implemented.

Entry Point	Function
SECTRAN	Logical-to-physical sector translation is performed to improve the overall response of CP/M. Standard CP/M systems are shipped with a skew factor of 6, where six physical sectors are skipped between each logical read operation. This skew factor allows enough time between sectors for most programs to load their buffers without missing the next sector. In particular computer systems that use fast processors, memory, and disk subsystems, the skew factor might be changed to improve overall response. However, the user should maintain a single-density IBM-compatible version of CP/M for information transfer into and out of the computer system, using a skew factor of 6. In general, SECTRAN receives a logical sector number relative to zero in BC and a translate table address in DE. The sector number is used as an index into the translate table, with the resulting physical sector number in HL. For standard systems, the table and indexing code is provided in the CBIOS and need not be changed.

Table 6-36: BIOS Entry Points

6.7. A Sample BIOS

The program shown in Appendix B can serve as a basis for your first BIOS. The simplest functions are assumed in this BIOS, so that you can enter it through a front panel, if absolutely necessary. You must alter and insert code into the subroutines for CONST, CONIN, CONOUT, READ, WRITE, and WAITIO subroutines. Storage is reserved for user-supplied code in these regions. The scratch area reserved in page zero (see Section 6.9) for the BIOS is used in this program, so that it could be implemented in ROM, if desired.

Once operational, this skeletal version can be enhanced to print the initial sign-on message and perform better error recovery. The subroutines for LIST, PUNCH, and READER can be filled out and the IOBYTE function can be implemented.

6.8. A Sample Cold Start Loader

The program shown in Appendix E can serve as a basis for a cold start loader. The disk read function must be supplied by the user, and the program must be loaded somehow starting at location 0000. Space is reserved for the patch code so that the total amount of storage required for the cold start loader is 128 bytes.

Eventually, you might want to get this loader onto the first disk sector (track 00, sector 01) and cause the controller to load it into memory automatically upon system start up. Alternatively, the cold start loader can be placed into ROM, and above the CP/M system. In this case, it is

necessary to originate the program at a higher address and key in a jump instruction at system start up that branches to the loader. Subsequent warm starts do not require this key-in operation, because the entry point `WBOOT` gets control, thus bringing the system in from disk automatically. The skeletal cold start loader has minimal error recovery, which might be enhanced in later versions.

6.9. Reserved Locations in Page Zero

Locations	Contents
0000H ... 0002H	Contains a jump instruction to the warm start entry location <code>4A03H + b</code> . This allows a simple programmed restart (<code>JMP 0000H</code>) or manual restart from the front panel.
0003H	Contains the Intel standard <code>IOBYTE</code> is optionally included in the user's CBIOS (refer to Section 6.6)
0004H	High nibble is user number (0 ... 15). Low nibble is current default drive number (0=A, 1=B, ... 15=P)
0005H ... 0007H	Contains a jump instruction to the BDOS and serves two purposes: <code>JMP 0005H</code> provides the primary entry point to the BDOS, as described in Chapter 5, and <code>LHLD 0006H</code> brings the address field of the instruction to the HL register pair. This value is the lowest address in memory used by CP/M, assuming the CCP is being overlaid. The DDT program changes the address field to reflect the reduced memory size in debug mode.
0008H ... 0027H	Interrupt locations 1 through 5 not used.
0030H ... 0037H	Interrupt location 6 (not currently used) is reserved.
0038H ... 003AH	Restart 7; contains a jump instruction into the DDT or SID program when running in debug mode for programmed breakpoints, but is not otherwise used by CP/M.
003BH ... 003FH	Not currently used; reserved.
0040H ... 004FH	A 16-byte area reserved for scratch by CBIOS, but is not used for any purpose in the distribution version of CP/M.
0050H ... 005BH	Not currently used; reserved.
005CH ... 007CH	Default File Control Block produced for a transient program by the CCP.
007DH ... 007FH	Optional default random record position.
0080H ... 00FFH	Default 128-byte disk buffer, also filled with the command line when a transient is loaded under the CCP.

Table 6-37: Reserved Locations in Page Zero

This information is set up for normal operation under the CP/M system, but can be overwritten by a transient program if the BDOS facilities are not required by the transient.

If, for example, a particular program performs only simple I/O and must begin execution at location 0, it can first be loaded into the TPA, using normal CP/M facilities, with a small memory move program that gets control when loaded. The memory move program must get control from location 0100H, which is the assumed beginning of all transient programs. The move program can then proceed to the entire memory image down to location 0 and pass control to the starting address of the memory load.

If the BIOS is overwritten or if location 0, containing the warm start entry point, is overwritten, the operator must bring the CP/M system back into memory with a cold start sequence.

6.10. Disk Parameter Tables

Tables are included in the BIOS that describe the particular characteristics of the disk subsystem used with CP/M. These tables can be either hand-coded, as shown in the sample CBIOS in Appendix B, or automatically generated using the DISKDEF macro library, as shown in Appendix F. The purpose here is to describe the elements of these tables.

In general, each disk drive has an associated (16-byte) disk parameter header that contains information about the disk drive and provides a scratch pad area for certain BDOS operations. The format of the disk parameter header for each drive is shown in Table 6-38, where each element is a word (16-bit) value.

XLT	0000	0000	0000	DIRBUF	DPB	CSV	ALV
16b	16b	16b	16b	16b	16b	16b	16b

Table 6-38: Disk Parameter Header Format

The meaning of each Disk Parameter Header (**DPH**) element is detailed in Table 6-39.

Parameter	Meaning
XLT	Address of the logical-to-physical translation vector, if used for this particular drive, or the value 0000H if no sector translation takes place (that is, the physical and logical sector numbers are the same). Disk drives with identical sector skew factors share the same translate tables.
0000	Scratch pad values for use within the BDOS, initial value is unimportant.
DIRBUF	Address of a 128-byte scratch pad area for directory operations within BDOS. All DPHs address the same scratch pad area.

Parameter	Meaning
DPB	Address of a disk parameter block for this drive. Drives with identical disk characteristics address the same disk parameter block.
CSV	Address of a scratch pad area used for software check for changed disks. This address is different for each DPH.
ALV	Address of a scratch pad area used by the BDOS to keep disk storage allocation information. This address is different for each DPH.

Table 6-39: Disk Parameter Headers

Given n disk drives, the **DPH**'s are arranged in a table whose first row of 16 bytes corresponds to drive 0, with the last row corresponding to drive $n - 1$. In the following figure the label **DPBASE** defines the base address of the **DPH** table.

DPBASE:

+00	XLT00	0000	0000	0000	DIRBUF	DPB00	CSV00	ALV00
+01	XLT00	0000	0000	0000	DIRBUF	DPB00	CSV00	ALV00
<i>(and so on through)</i>								
$n - 1$	XLT $n-1$	0000	0000	0000	DIRBUF	DPB $n-1$	CSV $n-1$	ALV $n-1$

Table 6-40: Disk Parameter Header Table

A responsibility of the **SELDSK** subroutine is to return the base address of the **DPH** for the selected drive. The following sequence of operations returns the table address, with a 0000H returned if the selected drive does not exist.

```
NDISKS      EQU      4           ;NUMBER OF DISK DRIVES
```

.....

```
SELDSK:      ;SELECT DISK GIVEN BY BC
              LSI      H,0000H    ;ERROR CODE
              MOV      A,C        ;DRIVE OK?
              CPI      NDISKS     ;CY IF SO
              RNC                      ;RET IF ERROR
              ;NO ERROR, CONTINUE
              MOV      L,C        ;LOW(DISK)
              MOV      H,B        ;HIGH(DISK)
              DAD      H          ;*2
              DAD      H          ;*4
              DAD      H          ;*8
              DAD      H          ;*16
              LXI      D,DPBASE   ;FIRST DPH
```

```

DAD      D      ;DPH(DISK)
RET

```

The translation vectors, XLT_{00} through $XLT_n - 1$, are located elsewhere in the BIOS, and simply correspond one-for-one with the logical sector numbers zero through the sector count $- 1$. The Disk Parameter Block (**DPB**) for each drive is more complex. As shown in Table 6-41, a particular **DPB**, that is addressed by one or more **DPHs**, takes the general form:

SPT	BSH	BLM	EXM	DSM	DRM	AL0	AL1	CHK	OFF
16b	8b	8b	8b	16b	16b	8b	8b	16b	16b

Table 6-41: Disk Parameter Block Format

where each is a byte or word value, as shown by the 8b or 16b indicator below the field.

The following field abbreviations are used in Table 6-41:

SPT is the total number of sectors per track.

BSH is the data allocation block shift factor, determined by the data block allocation size.

BLM is the data allocation block mask ($2^{BSH} - 1$)

EXM is the extent mask, determined by the data block allocation size and the number of disk blocks.

DSM determines the total storage capacity of the disk drive.

DRM determines the total number of directory entries that can be stored on this drive.

AL0 and AL1 determine reserved directory blocks.

CKS is the size of the directory check vector.

OFF is the number of reserved tracks at the beginning of the (logical) disk.

The values of **BSH** and **BLM** determine the data allocation size **BLS**, which is not an entry in the **DPB**. Given that the designer has selected a value for **BLS**, the values of **BSH** and **BLM** are shown in Table 6-42.

BLS	BSH	BLM
1024	3	7
2048	4	15
4096	5	31
8192	6	63
16384	7	127

Table 6-42: **BSH** and **BLM** Values

where all values are in decimal. The value of **EXM** depends upon both the **BLS** and whether the **DSM** value is less than 256 or greater than 255, as shown in Table 6-43.

BLS	DSM < 256	DSM > 255
1024	0	n/a
2048	1	0
4096	3	1
8192	7	3
16384	15	7

Table 6-43: **EXM** Values

The value of **DSM** is the maximum data block number supported by this particular drive, measured in **BLS** units. The product (**DSM + 1**) is the total number of bytes held by the drive and must be within the capacity of the physical disk, not counting the reserved operating system tracks.

The **DRM** entry is the one less than the total number of directory entries that can take on a 16-bit value. The values of **AL0** and **AL1**, however, are determined by **DRM**. The values **AL0** and **AL1** can together be considered a string of 16-bits, as shown in Table 6-44.

AL0								AL1							
07	06	05	04	03	02	01	00	07	06	05	04	03	02	01	00
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15

Table 6-44: **AL0** and **AL1**

Position 00 corresponds to the high-order bit of the byte labeled **AL0** and 15 corresponds to the low-order bit of the byte labeled **AL1**. Each bit position reserves a data block for number of directory entries, thus allowing a total of 16 data blocks to be assigned for directory entries (bits are assigned starting at 00 and filled to the right until position 15). Each directory entry occupies 32 bytes, resulting in Table 6-45:

BLS	Directory Entries
1024	32 * number of bits
2048	64 * number of bits
4096	128 * number of bits
8192	256 * number of bits
16384	512 * number of bits

Table 6-45: **BLS** Tabulation

Thus, if **DRM** = 127 (128 directory entries) and **BLS** = 1024, there are 32 directory entries per block, requiring 4 reserved blocks. In this case, the 4 high-order bits of **AL0** are set, resulting in the values **AL0** = 0F0H and **AL1** = 00H.

The **CKS** value is determined as follows: if the disk drive media is removable, then **CKS** = $\frac{\text{DRM}+1}{4}$, where **DRM** is the last directory entry number. If the media are fixed, then set **CKS** = 0 (no directory records are checked in this case).

Finally, the **OFF** field determines the number of tracks that are skipped at the beginning of the physical disk. This value is automatically added whenever **SETTRK** is called and can be used as a mechanism for skipping reserved operating system tracks or for partitioning a large disk into smaller segmented sections.

To complete the discussion of the **DPB**, several **DPHs** can address the same **DPB** if their drive characteristics are identical. Further, the **DPB** can be dynamically changed when a new drive is addressed by simply changing the pointer in the **DPH**; because the BDOS copies the **DPB** values to a local area whenever the **SELDSK** function is invoked.

Returning back to **DPH** for a particular drive, the two address values **CSV** and **ALV** remain. Both addresses reference an area of uninitialized memory following the BIOS. The areas must be unique for each drive, and the size of each area is determined by the values in the **DPB**.

The size of the area addressed by **CSV** is **CKS** bytes, which is sufficient to hold the directory check information for this particular drive. If **CKS** = $\frac{\text{DRM}+1}{4}$, you must reserve $\frac{\text{DRM}+1}{4}$ bytes for directory check use. If **CKS** = 0, no storage is reserved.

The size of the area addressed by **ALV** is determined by the maximum number of data blocks allowed for this particular disk and is computed as $\left(\frac{DSM}{8}\right) + 1$.

The CBIOS shown in Appendix B demonstrates an instance of these tables for standard 8-inch, single-density drives. It might be useful to examine this program and compare the tabular values with the definitions given above.

6.11. The DISKDEF Macro Library

A macro library called `DISKDEF` (shown in Appendix F), greatly simplifies the table construction process. You must have access to the `MAC` macro assembler, of course, to use the `DISKDEF` facility, while the macro library is included with all CP/M 2 distribution disks.

A BIOS disk definition consists of the following sequence of macro statements:

```
MACLIB      DISKDEF
.....
DISKS        n
DISKDEF      0,...
DISKDEF      1,...
.....
DISKDEF      n-1
.....
ENDEF
```

where the `MACLIB` statement loads the `DISKDEF.LIB` file, on the same disk as the BIOS, into `MAC`'s internal tables. The `DISKS` macro call follows, which specifies the number of drives to be configured with the user's system, where n is an integer in the range 1 to 16. A series of `DISKDEF` macro calls then follow that define the characteristics of each logical disk, 0 through $n - 1$, corresponding to logical drives A through P. The `DISKS` and `DISKDEF` macros generate the in-line fixed data tables described in the previous section and thus must be placed in a non-executable portion of the BIOS, typically directly following the BIOS jump vector.

The remaining portion of the BIOS is defined following the `DISKDEF` macros, with the `ENDEF` macro call immediately preceding the `END` statement. The `ENDEF` (End of `DISKDEF`) macro generates the necessary uninitialized RAM areas that are located in memory above the BIOS.

The `DISKDEF` macro call takes the form:

```
DISKDEF dn,fsc,lsc,[skf],bls dks,dir,cks,ofs,[0]
```

where

dn is the logical disk number, 0 to $n - 1$.

fsc is the first physical sector number (0 or 1).

lsc is the last sector number.

skf optional sector skew factor

bls is the data allocation block size.

dks is the number of blocks on the disk.

dir is the number of directory entries.

cks is the number of checked directory entries.

ofs is the track offset to logical track 00.

[0] is an optional 1.4 compatibility flag.

The value *dn* is the drive number being defined with this DISKDEF macro invocation. The *fsc* parameter accounts for differing sector numbering systems and is usually 0 to 1. The *lsc* is the last numbered sector on a track. When present, the *skf* parameter defines the sector skew factor, which is used to create a sector translation table according to the skew.

If the number of sectors is less than 256, a single-byte table is created, otherwise each translation table element occupies two bytes. No translation table is created if the *skf* parameter is omitted, or equal to 0.

The *bls* parameter specifies the number of bytes allocated to each data block, and takes on the values 1024, 2048, 4096, 8192, or 16384. Generally, performance increases with larger data block sizes because there are fewer directory references, and logically connected data records are physically close on the disk. Further, each directory entry addresses more data and the BIOS-resident RAM space is reduced.

The *dks* parameter specifies the total disk size in *bls* units. That is, if the *bls* = 2048 and *dks* = 1000, the total disk capacity is 2,048,000 bytes. If *dks* is greater than 255, the block size parameter *bls* must be greater than 1024. The value of *dir* is the total number of directory entries that might exceed 255, if desired.

The *cks* parameter determines the number of directory items to check on each directory scan and is used internally to detect changed disks during system operation, where an intervening cold or warm start has not occurred. When this situation is detected, CP/M automatically marks the disk Read-Only so that data is not subsequently destroyed.

As stated in the previous section, the value of *cks* = *dir* when the medium is easily changed, as is the case with a floppy disk subsystem. If the disk is permanently mounted, the value of *cks* is typically 0, because the probability of changing disks without a restart is low.

The *ofs* value determines the number of tracks to skip when this particular drive is addressed, which can be used to reserve additional operating system space or to simulate several logical drives on a single large capacity physical drive.

Finally, the [0] parameter is included when file compatibility is required with versions of 1.4 that have been modified for higher density disks. This parameter ensures that only 16K is allocated

for each directory record, as was the case for previous versions. Normally, this parameter is not included.

For convenience and economy of table space, the special form:

DISKDEF *i,j*

gives disk *i* the same characteristics as a previously defined drive *j*. A standard four-drive, single-density system, which is compatible with version 1.4, is defined using the following macro invocations:

```
DISKS          4
DISKDEF        0,1,26,6,1024,243,64,2
DISKDEF        1,0
DISKDEF        2,0
DISKDEF        3,0
....
ENDEF
```

with all disks having the same parameter values of 26 sectors per track, numbered 01 through 26, with 6 sectors skipped between each access, 1024 bytes per data block, 243 data blocks for a total of 243K-byte disk capacity, 64 checked directory entries, and two operating system tracks.

The DISKS macro generates *n* **DPHs**, starting at the **DPH** table address DPBASE generated by the macro. Each disk header block contains sixteen bytes, as described above, and correspond one-for-one to each of the defined drives. In the four-drive standard system, for example, the DISKS macro generates a table of the form:

```
DPBASE      EQU  $
DPE0:       DW  XLT0,0000H,0000H,0000H,DIRBUF,DPB0,CSV0,ALV0
DPE1:       DW  XLT0,0000H,0000H,0000H,DIRBUF,DPB0,CSV1,ALV1
DPE2:       DW  XLT0,0000H,0000H,0000H,DIRBUF,DPB0,CSV2,ALV2
DPE3:       DW  XLT0,0000H,0000H,0000H,DIRBUF,DPB0,CSV3,ALV3
```

where the **DPH** labels are included for reference purposes to show the beginning table addresses for each drive 0 through 3. The values contained within the **DPH** are described in detail in the previous section. The check and allocation vector addresses are generated by the ENDEF macro in the ram area following the BIOS code and tables.

Note that if the *skf* (skew factor) parameter is omitted, or equal to 0, the translation table is omitted and a 0000H value is inserted in the **XLT** position of the **DPH** for the disk. In a subsequent call to perform the logical-to-physical translation, SECTRAN receives a translation table address of DE = 0000H and simply returns the original logical sector from BC in the HL register pair.

A translate table is constructed when the *skf* parameter is present, and the (nonzero) table address is placed into the corresponding **DPHs**. The following for example, is constructed when the standard skew factor *skf* = 6 is specified in the DISKDEF macro call:

```
XLT0:    DB      1,7,13,19,25,5,11,17,23,3,9,15,21
          DB      2,8,14,20,26,6,12,18,24,4,10,16,22
```

Following the ENDEF macro call, a number of uninitialized data areas are defined. These data areas need not be a part of the BIOS that is loaded upon cold start, but must be available between the BIOS and the end of memory. The size of the uninitialized RAM area is determined by EQU statements generated by the ENDEF macro. For a standard four-drive system, the ENDEF macro might produce the following EQU statement:

```
4C72 =      BEGDAT EQU $
          (data areas)

4DB0 =      ENDDAT EQU $

013C =      DATSIZ EQU $-BEGDAT
```

which indicates that uninitialized RAM begins at location 4C72H, ends at 4DB0H-1, and occupies 013CH bytes. You must ensure that these addresses are free for use after the system is loaded.

After modification, you can use the STAT program to check drive characteristics, because STAT uses the disk parameter block to decode the drive information. A STAT command of the form:

STAT *d*:DSK:

decodes the disk parameter block for drive *d* (*d*=A,...,P) and displays the following values:

```
r: 128-byte record capacity
k: kilobyte drive capacity
d: 32-byte directory entries
c: checked directory entries
e: records/extent
b: records/block
s: sectors/track
t: reserved tracks
```

Three examples of DISKDEF macro invocations are shown below with corresponding STAT parameter values. The last example produces a full 8-megabyte system.

```
DISKDEF 0,1,58,,2048,256,128,128,2
```

- r=4096, k=512, d=128, c=128, e=256, b=16, s=58, t=2

```
DISKDEF 0,1,58,,2048,1024,300,0,2
```

- r=16348, k=2048, d=300, c=0, e=128, b=16, s=58, t=2

```
DISKDEF 0,1,58,,16348,512,128,128,2
```

- $r=65536, k=8192, d=128, c=128, e=1024, b=128, s=58, t=2$

6.12. Sector Blocking and Deblocking

Upon each call to BIOS `WRITE` entry point, the CP/M BDOS includes information that allows effective sector blocking and deblocking where the host disk subsystem has a sector size that is a multiple of the basic 128-byte unit. The purpose here is to present a general-purpose algorithm that can be included within the BIOS and that uses the BDOS information to perform the operations automatically.

On each call to `WRITE`, the BDOS provides the following information in register `C`:

- 0 normal sector write
- 1 write to directory sector
- 2 write to the first sector of a new block

Condition 0 occurs whenever the next write operation is into a previously written area, such as a random mode record update; when the write is to other than the first sector of an unallocated block; or when the write is not into the directory area. Condition 1 occurs when a write into the directory area is performed. Condition 2 occurs when the first record (only) of a newly allocated data block is written. In most cases, application programs read or write multiple 128-byte sectors in sequence; thus, there is little overhead involved in either operation when blocking and deblocking records, because pre-read operations can be avoided when writing records.

Appendix G lists the blocking and deblocking algorithms in skeletal form; this file is included on your CP/M disk. Generally, the algorithms map all CP/M sector read operations onto the host disk through an intermediate buffer that is the size of the host disk sector. Throughout the program, values and variables that relate to the CP/M sector involved in a seek operation are prefixed by `sek`, while those related to the host disk system are prefixed by `hst`. The equate statements beginning on line 29 of Appendix G define the mapping between CP/M and the host system, and must be changed if other than the sample host system is involved.

The entry points `BOOT` and `WBOOT` must contain the initialization code starting on line 57, while the `SELDSK` entry point must be augmented by the code starting on line 65. Note that although the `SELDSK` entry point computes and returns the Disk Parameter Header address, it does not physically select the host disk at this point (it is selected later at `READHST` or `WRITEHST`). Further, `SETTRK`, `SETSEC`, and `SETDMA` simply store the values, but do not take any other action at this point. `SECTTRAN` performs a trivial function of returning the physical sector number.

The principal entry points are `READ` and `WRITE`, starting on lines 110 and 125, respectively. These subroutines take the place of your previous `READ` and `WRITE` operations.

The actual physical read or write takes place at either `WRITEHST` or `READHST`, where all values have been prepared: `hstdsk` is the host disk number, `hstrk` is the host track number, and `hstsec` is the host

sector number, which may require translation to physical sector number. You must insert code at this point that performs the full sector read or write into or out of the buffer at `hstbuf` of length `hstsiz`. All other mapping functions are performed by the algorithms.

This particular algorithm was tested using an 80-megabyte hard disk unit that was originally configured for 128-byte sectors, producing approximately 35 megabytes of formatted storage. When configured for 512-byte host sectors, usable storage increased to 57 megabytes, with a corresponding 400% improvement in overall response. In this situation, there is no apparent overhead involved in deblocking sectors, with the advantage that user programs still maintain 128-byte sectors. This is primarily because of the information provided by the BDOS, which eliminates the necessity for pre-read operations.

A. The MDS Basic I/O System (BIOS)

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1          ; MDS-800 I/O DRIVERS FOR CP/M 2.2
2          ; (FOUR DRIVE SINGLE DENSITY VERSION)
3          ;
4          ; VERSION 2.2 FEBRUARY, 1980
5          ;
6 0016 =    VERS EQU 22 ;VERSION 2.2
7          ;
8          ; COPYRIGHT (C) 1980
9          ; DIGITAL RESEARCH
10         ; BOX 579, PACIFIC GROVE
11         ; CALIFORNIA, 93950
12         ;
13         ;
14 FFFF =    TRUE EQU 0FFFFH ;VALUE OF "TRUE"
15 0000 =    FALSE EQU NOT TRUE ;"FALSE"
16 0000 =    TEST EQU FALSE ;TRUE IF TEST BIOS
17         ;
18         IF TEST
19         BIAS EQU 03400H ;BASE OF CCP IN TEST SYSTEM
20         ENDIF
21         IF NOT TEST
22 0000 =    BIAS EQU 0000H ;GENERATE RELOCATABLE CP/M SYSTEM
23         ENDIF
24         ;
25 1600 =    PATCH EQU 1600H
26         ;
27 1600      ORG PATCH
28 0000 =    CPMB EQU $-PATCH ;BASE OF CPM CONSOLE PROCESSOR
29 0806 =    BDOS EQU 806H+CPMB ;BASIC DOS (RESIDENT PORTION)
30 1600 =    CPML EQU $-CPMB ;LENGTH (IN BYTES) OF CPM SYSTEM
31 002C =    NSECTS EQU CPML/128 ;NUMBER OF SECTORS TO LOAD
32 0002 =    OFFSET EQU 2 ;NUMBER OF DISK TRACKS USED BY CP/M
33 0004 =    CDISK EQU 0004H ;ADDRESS OF LAST LOGGED DISK ON WARM START
34 0080 =    BUFF EQU 0080H ;DEFAULT BUFFER ADDRESS
35 000A =    RETRY EQU 10 ;MAX RETRIES ON DISK I/O BEFORE ERROR
36         ;
37         ; PERFORM FOLLOWING FUNCTIONS
38         ; BOOT COLD START
39         ; WBOOT WARM START (SAVE I/O BYTE)
40         ; (BOOT AND WBOOT ARE THE SAME FOR MDS)
41         ; CONST CONSOLE STATUS
42         ; REG-A = 00 IF NO CHARACTER READY
43         ; REG-A = FF IF CHARACTER READY
44         ; CONIN CONSOLE CHARACTER IN (RESULT IN REG-A)
```

```

45      ; CONOUT  CONSOLE CHARACTER OUT (CHAR IN REG-C)
46      ; LIST   LIST OUT (CHAR IN REG-C)
47      ; PUNCH  PUNCH OUT (CHAR IN REG-C)
48      ; READER  PAPER TAPE READER IN (RESULT TO REG-A)
49      ; HOME   MOVE TO TRACK 00
50      ;
51      ; (THE FOLLOWING CALLS SET-UP THE IO PARAMETER BLOCK FOR THE
52      ; MDS, WHICH IS USED TO PERFORM SUBSEQUENT READS AND WRITES)
53      ; SELDSK  SELECT DISK GIVEN BY REG-C (0,1,2,...)
54      ; SETTRK  SET TRACK ADDRESS (0,...76) FOR SUBSEQUENT READ/WRITE
55      ; SETSEC  SET SECTOR ADDRESS (1,...,26) FOR SUBSEQUENT READ/WRITE
56      ; SETDMA  SET SUBSEQUENT DMA ADDRESS (INITIALLY 80H)
57      ;
58      ; (READ AND WRITE ASSUME PREVIOUS CALLS TO SET UP THE IO PARAMETERS)
59      ; READ   READ TRACK/SECTOR TO PRESET DMA ADDRESS
60      ; WRITE  WRITE TRACK/SECTOR FROM PRESET DMA ADDRESS
61      ;
62      ; JUMP VECTOR FOR INDIVIDUAL ROUTINES
63 1600 C3B316      JMP BOOT
64 1603 C3C316      WBOOTE: JMP WBOOT
65 1606 C36117      JMP CONST
66 1609 C36417      JMP CONIN
67 160C C36A17      JMP CONOUT
68 160F C36D17      JMP LIST
69 1612 C37217      JMP PUNCH
70 1615 C37517      JMP READER
71 1618 C37817      JMP HOME
72 161B C37D17      JMP SELDSK
73 161E C3A717      JMP SETTRK
74 1621 C3AC17      JMP SETSEC
75 1624 C3BB17      JMP SETDMA
76 1627 C3C117      JMP READ
77 162A C3CA17      JMP WRITE
78 162D C37017      JMP LISTST ;LIST STATUS
79 1630 C3B117      JMP SECTAN
80      ;
81      MACLIB  DISKDEF ;LOAD THE DISK DEFINITION LIBRARY
82      DISKS 4 ;FOUR DISKS
83 1633+=          DPBASE EQU $ ;BASE OF DISK PARAMETER BLOCKS
84 1633+82160000   DPE0: DW  XLT0,0000H ;TRANSLATE TABLE
85 1637+00000000   DW  0000H,0000H ;SCRATCH AREA
86 163B+6E187316   DW  DIRBUF,DPB0 ;DIR BUFF,PARM BLOCK
87 163F+0D19EE18   DW  CSV0,ALV0 ;CHECK, ALLOC VECTORS
88 1643+82160000   DPE1: DW  XLT1,0000H ;TRANSLATE TABLE
89 1647+00000000   DW  0000H,0000H ;SCRATCH AREA
90 164B+6E187316   DW  DIRBUF,DPB1 ;DIR BUFF,PARM BLOCK
91 164F+3C191D19   DW  CSV1,ALV1 ;CHECK, ALLOC VECTORS
92 1653+82160000   DPE2: DW  XLT2,0000H ;TRANSLATE TABLE
93 1657+00000000   DW  0000H,0000H ;SCRATCH AREA
94 165B+6E187316   DW  DIRBUF,DPB2 ;DIR BUFF,PARM BLOCK
95 165F+6B194C19   DW  CSV2,ALV2 ;CHECK, ALLOC VECTORS
96 1663+82160000   DPE3: DW  XLT3,0000H ;TRANSLATE TABLE
97 1667+00000000   DW  0000H,0000H ;SCRATCH AREA
98 166B+6E187316   DW  DIRBUF,DPB3 ;DIR BUFF,PARM BLOCK
99 166F+9A197B19   DW  CSV3,ALV3 ;CHECK, ALLOC VECTORS
100      DISKDEF 0,1,26,6,1024,243,64,64,OFFSET
101 1673+=          DPB0 EQU $ ;DISK PARM BLOCK

```

```

102 1673+1A00      DW 26      ;SEC PER TRACK
103 1675+03        DB 3       ;BLOCK SHIFT
104 1676+07        DB 7       ;BLOCK MASK
105 1677+00        DB 0       ;EXTNT MASK
106 1678+F200      DW 242     ;DISK SIZE-1
107 167A+3F00      DW 63      ;DIRECTORY MAX
108 167C+C0        DB 192     ;ALLOC0
109 167D+00        DB 0       ;ALLOC1
110 167E+1000      DW 16      ;CHECK SIZE
111 1680+0200      DW 2       ;OFFSET
112 1682+=         XLT0 EQU $   ;TRANSLATE TABLE
113 1682+01        DB 1
114 1683+07        DB 7
115 1684+0D        DB 13
116 1685+13        DB 19
117 1686+19        DB 25
118 1687+05        DB 5
119 1688+0B        DB 11
120 1689+11        DB 17
121 168A+17        DB 23
122 168B+03        DB 3
123 168C+09        DB 9
124 168D+0F        DB 15
125 168E+15        DB 21
126 168F+02        DB 2
127 1690+08        DB 8
128 1691+0E        DB 14
129 1692+14        DB 20
130 1693+1A        DB 26
131 1694+06        DB 6
132 1695+0C        DB 12
133 1696+12        DB 18
134 1697+18        DB 24
135 1698+04        DB 4
136 1699+0A        DB 10
137 169A+10        DB 16
138 169B+16        DB 22
139               DISKDEF 1,0
140 1673+=         DPB1 EQU DPB0 ;EQUIVALENT PARAMETERS
141 001F+=         ALS1 EQU ALS0 ;SAME ALLOCATION VECTOR SIZE
142 0010+=         CSS1 EQU CSS0 ;SAME CHECKSUM VECTOR SIZE
143 1682+=         XLT1 EQU XLT0 ;SAME TRANSLATE TABLE
144               DISKDEF 2,0
145 1673+=         DPB2 EQU DPB0 ;EQUIVALENT PARAMETERS
146 001F+=         ALS2 EQU ALS0 ;SAME ALLOCATION VECTOR SIZE
147 0010+=         CSS2 EQU CSS0 ;SAME CHECKSUM VECTOR SIZE
148 1682+=         XLT2 EQU XLT0 ;SAME TRANSLATE TABLE
149               DISKDEF 3,0
150 1673+=         DPB3 EQU DPB0 ;EQUIVALENT PARAMETERS
151 001F+=         ALS3 EQU ALS0 ;SAME ALLOCATION VECTOR SIZE
152 0010+=         CSS3 EQU CSS0 ;SAME CHECKSUM VECTOR SIZE
153 1682+=         XLT3 EQU XLT0 ;SAME TRANSLATE TABLE
154               ; ENDEF OCCURS AT END OF ASSEMBLY
155               ;
156               ; END OF CONTROLLER - INDEPENDENT CODE, THE REMAINING SUBROUTINES
157               ; ARE TAILORED TO THE PARTICULAR OPERATING ENVIRONMENT, AND MUST
158               ; BE ALTERED FOR ANY SYSTEM WHICH DIFFERS FROM THE INTEL MDS.

```

```

159          ;
160          ; THE FOLLOWING CODE ASSUMES THE MDS MONITOR EXISTS AT 0F800H
161          ; AND USES THE I/O SUBROUTINES WITHIN THE MONITOR
162          ;
163          ; WE ALSO ASSUME THE MDS SYSTEM HAS FOUR DISK DRIVES
164 00FD =    REVRT EQU 0FDH ;INTERRUPT REVERT PORT
165 00FC =    INTC  EQU 0FCH ;INTERRUPT MASK PORT
166 00F3 =    ICON  EQU 0F3H ;INTERRUPT CONTROL PORT
167 007E =    INTE  EQU 0111$1110B ;ENABLE RST 0(WARM BOOT), RST 7 (MONITOR)
168          ;
169          ; MDS MONITOR EQUATES
170 F800 =    MON80 EQU 0F800H ;MDS MONITOR
171 FF0F =    RMON80 EQU 0FF0FH ;RESTART MON80 (BOOT ERROR)
172 F803 =    CI  EQU 0F803H ;CONSOLE CHARACTER TO REG-A
173 F806 =    RI  EQU 0F806H ;READER IN TO REG-A
174 F809 =    CO  EQU 0F809H ;CONSOLE CHAR FROM C TO CONSOLE OUT
175 F80C =    PO  EQU 0F80CH ;PUNCH CHAR FROM C TO PUNCH DEVICE
176 F80F =    LO  EQU 0F80FH ;LIST FROM C TO LIST DEVICE
177 F812 =    CSTS EQU 0F812H ;CONSOLE STATUS 00/FF TO REGISTER A
178          ;
179          ; DISK PORTS AND COMMANDS
180 0078 =    BASE EQU 78H ;BASE OF DISK COMMAND IO PORTS
181 0078 =    DSTAT EQU BASE ;DISK STATUS (INPUT)
182 0079 =    RTYPE EQU BASE+1 ;RESULT TYPE (INPUT)
183 007B =    RBYTE EQU BASE+3 ;RESULT BYTE (INPUT)
184          ;
185 0079 =    ILOW EQU BASE+1 ;IOPB LOW ADDRESS (OUTPUT)
186 007A =    IHIGH EQU BASE+2 ;IOPB HIGH ADDRESS (OUTPUT)
187          ;
188 0004 =    READF EQU 4H ;READ FUNCTION
189 0006 =    WRITF EQU 6H ;WRITE FUNCTION
190 0003 =    RECAL EQU 3H ;RECALIBRATE DRIVE
191 0004 =    IORDY EQU 4H ;I/O FINISHED MASK
192 000D =    CR  EQU 0DH ;CARRIAGE RETURN
193 000A =    LF  EQU 0AH ;LINE FEED
194          ;
195          SIGNON: ;SIGNON MESSAGE: XXK CP/M VERS YY
196 169C 0D0A0A    DB CR,LF,LF
197                IF TEST
198                DB '32' ;32K EXAMPLE BIOS
199                ENDIF
200                IF NOT TEST
201 169F 3030      DB '00' ;MEMORY SIZE FILLED BY RELOCATOR
202                ENDIF
203 16A1 6B2043502F DB 'k CP/M vers '
204 16AD 322E32    DB VERS/10+'0';,VERS MOD 10+'0'
205 16B0 0D0A00    DB CR,LF,0
206          ;
207          BOOT: ;PRINT SIGNON MESSAGE AND GO TO CCP
208          ; (NOTE: MDS BOOT INITIALIZED IOBYTE AT 0003H)
209 16B3 310001    LXI SP,BUFF+80H
210 16B6 219C16    LXI H,SIGNON
211 16B9 CDD317    CALL PRMSG ;PRINT MESSAGE
212 16BC AF        XRA A ;CLEAR ACCUMULATOR
213 16BD 320400    STA CDISK ;SET INITIALLY TO DISK A
214 16C0 C30F17    JMP GOCPM ;GO TO CP/M
215          ;

```

```

216      ;
217      WBOOT:; LOADER ON TRACK 0, SECTOR 1, WHICH WILL BE SKIPPED FOR WARM
218      ; READ CP/M FROM DISK - ASSUMING THERE IS A 128 BYTE COLD START
219      ; START.
220      ;
221      16C3 318000      LXI SP,BUFF ;USING DMA - THUS 80 THRU FF AVAILABLE FOR STACK
222      ;
223      16C6 0E0A      MVI C,RETRY ;MAX RETRIES
224      16C8 C5      PUSH B
225      WBOOT0: ;ENTER HERE ON ERROR RETRIES
226      16C9 010000      LXI B,CPMB ;SET DMA ADDRESS TO START OF DISK SYSTEM
227      16CC CDBB17      CALL SETDMA
228      16CF 0E00      MVI C,0 ;BOOT FROM DRIVE 0
229      16D1 CD7D17      CALL SELDSK
230      16D4 0E00      MVI C,0
231      16D6 CDA717      CALL SETTRK ;START WITH TRACK 0
232      16D9 0E02      MVI C,2 ;START READING SECTOR 2
233      16DB CDAC17      CALL SETSEC
234      ;
235      ; READ SECTORS, COUNT NSECTS TO ZERO
236      16DE C1      POP B ;10-ERROR COUNT
237      16DF 062C      MVI B,NSECTS
238      RDSEC: ;READ NEXT SECTOR
239      16E1 C5      PUSH B ;SAVE SECTOR COUNT
240      16E2 CDC117      CALL READ
241      16E5 C24917      JNZ BOOTERR ;RETRY IF ERRORS OCCUR
242      16E8 2A6C18      LHLD IOD ;INCREMENT DMA ADDRESS
243      16EB 118000      LXI D,128 ;SECTOR SIZE
244      16EE 19      DAD D ;INCREMENTED DMA ADDRESS IN HL
245      16EF 44      MOV B,H
246      16F0 4D      MOV C,L ;READY FOR CALL TO SET DMA
247      16F1 CDBB17      CALL SETDMA
248      16F4 3A6B18      LDA IOS ;SECTOR NUMBER JUST READ
249      16F7 FE1A      CPI 26 ;READ LAST SECTOR?
250      16F9 DA0517      JC RD1
251      ; MUST BE SECTOR 26, ZERO AND GO TO NEXT TRACK
252      16FC 3A6A18      LDA IOT ;GET TRACK TO REGISTER A
253      16FF 3C      INR A
254      1700 4F      MOV C,A ;READY FOR CALL
255      1701 CDA717      CALL SETTRK
256      1704 AF      XRA A ;CLEAR SECTOR NUMBER
257      1705 3C      RD1: INR A ;TO NEXT SECTOR
258      1706 4F      MOV C,A ;READY FOR CALL
259      1707 CDAC17      CALL SETSEC
260      170A C1      POP B ;RECALL SECTOR COUNT
261      170B 05      DCR B ;DONE?
262      170C C2E116      JNZ RDSEC
263      ;
264      ; DONE WITH THE LOAD, RESET DEFAULT BUFFER ADDRESS
265      GOCPM: ;(ENTER HERE FROM COLD START BOOT)
266      ; ENABLE RST0 AND RST7
267      170F F3      DI
268      1710 3E12      MVI A,12H ;INITIALIZE COMMAND
269      1712 D3FD      OUT REVRT
270      1714 AF      XRA A
271      1715 D3FC      OUT INTC ;CLEARED
272      1717 3E7E      MVI A,INTE ;RST0 AND RST7 BITS ON

```

```

273 1719 D3FC      OUT INTC
274 171B AF        XRA A
275 171C D3F3      OUT ICON ;INTERRUPT CONTROL
276                ;
277                ; SET DEFAULT BUFFER ADDRESS TO 80H
278 171E 018000     LXI B,BUFF
279 1721 CDBB17     CALL SETDMA
280                ;
281                ; RESET MONITOR ENTRY POINTS
282 1724 3EC3       MVI A,JMP
283 1726 320000     STA 0
284 1729 210316     LXI H,WBOOTE
285 172C 220100     SHLD 1 ;JMP WBOOT AT LOCATION 00
286 172F 320500     STA 5
287 1732 210608     LXI H,BDOS
288 1735 220600     SHLD 6 ;JMP BDOS AT LOCATION 5
289                IF NOT TEST
290 1738 323800     STA 7*8 ;JMP TO MON80 (MAY HAVE BEEN CHANGED BY DDT)
291 173B 2100F8     LXI H,MON80
292 173E 223900     SHLD 7*8+1
293                ENDIF
294                ; LEAVE IOBYTE SET
295                ; PREVIOUSLY SELECTED DISK WAS B, SEND PARAMETER TO CPM
296 1741 3A0400     LDA CDISK ;LAST LOGGED DISK NUMBER
297 1744 4F         MOV C,A ;SEND TO CCP TO LOG IT IN
298 1745 FB         EI
299 1746 C30000     JMP CPMB
300                ;
301                ; ERROR CONDITION OCCURRED, PRINT MESSAGE AND RETRY
302 BOOTERR:
303 1749 C1         POP B ;RECALL COUNTS
304 174A 0D         DCR C
305 174B CA5217     JZ  BOOTER0
306                ; TRY AGAIN
307 174E C5         PUSH B
308 174F C3C916     JMP WBOOT0
309                ;
310 BOOTER0:
311                ; OTHERWISE TOO MANY RETRIES
312 1752 215B17     LXI H,BOOTMSG
313 1755 CDD317     CALL PRMSG
314 1758 C30FFF     JMP RMON80 ;MDS HARDWARE MONITOR
315                ;
316 BOOTMSG:
317 175B 3F626F6F74 DB '?boot',0
318                ;
319                ;
320 CONST: ;CONSOLE STATUS TO REG-A
321                ; (EXACTLY THE SAME AS MDS CALL)
322 1761 C312F8     JMP CSTS
323                ;
324 CONIN: ;CONSOLE CHARACTER TO REG-A
325 1764 CD03F8     CALL CI
326 1767 E67F       ANI 7FH ;REMOVE PARITY BIT
327 1769 C9         RET
328                ;
329 CONOUT: ;CONSOLE CHARACTER FROM C TO CONSOLE OUT

```

```

330 176A C309F8    JMP CO
331                ;
332                LIST: ;LIST DEVICE OUT
333                ; (EXACTLY THE SAME AS MDS CALL)
334 176D C30FF8    JMP LO
335                ;
336                LISTST:
337                ;RETURN LIST STATUS
338 1770 AF         XRA A
339 1771 C9         RET    ;ALWAYS NOT READY
340                ;
341                PUNCH: ;PUNCH DEVICE OUT
342                ; (EXACTLY THE SAME AS MDS CALL)
343 1772 C30CF8    JMP PO
344                ;
345                READER: ;READER CHARACTER IN TO REG-A
346                ; (EXACTLY THE SAME AS MDS CALL)
347 1775 C306F8    JMP RI
348                ;
349                HOME: ;MOVE TO HOME POSITION
350                ; TREAT AS TRACK 00 SEEK
351 1778 0E00       MVI C,0
352 177A C3A717     JMP SETTRK
353                ;
354                SELDSK: ;SELECT DISK GIVEN BY REGISTER C
355 177D 210000     LXI H,0000H ;RETURN 0000 IF ERROR
356 1780 79         MOV A,C
357 1781 FE04       CPI NDISKS ;TOO LARGE?
358 1783 D0         RNC    ;LEAVE HL = 0000
359                ;
360 1784 E602       ANI 10B ;00 00 FOR DRIVE 0,1 AND 10 10 FOR DRIVE 2,3
361 1786 326618     STA DBANK ;TO SELECT DRIVE BANK
362 1789 79         MOV A,C ;00, 01, 10, 11
363 178A E601       ANI 1B ;MDS HAS 0,1 AT 78, 2,3 AT 88
364 178C B7         ORA A ;RESULT 00?
365 178D CA9217     JZ  SETDRIVE
366 1790 3E30       MVI A,00110000B ;SELECTS DRIVE 1 IN BANK
367                SETDRIVE:
368 1792 47         MOV B,A ;SAVE THE FUNCTION
369 1793 216818     LXI H,IOP ;IO FUNCTION
370 1796 7E         MOV A,M
371 1797 E6CF       ANI 11001111B ;MASK OUT DISK NUMBER
372 1799 B0         ORA B ;MASK IN NEW DISK NUMBER
373 179A 77         MOV M,A ;SAVE IT IN IOPB
374 179B 69         MOV L,C
375 179C 2600       MVI H,0 ;HL=DISK NUMBER
376 179E 29         DAD H ;*2
377 179F 29         DAD H ;*4
378 17A0 29         DAD H ;*8
379 17A1 29         DAD H ;*16
380 17A2 113316     LXI D,DPBASE
381 17A5 19         DAD D ;HL=DISK HEADER TABLE ADDRESS
382 17A6 C9         RET
383                ;
384                ;
385                SETTRK: ;SET TRACK ADDRESS GIVEN BY C
386 17A7 216A18     LXI H,IOT

```

```

387 17AA 71      MOV M,C
388 17AB C9      RET
389              ;
390              SETSEC: ;SET SECTOR NUMBER GIVEN BY C
391 17AC 216B18   LXI H,IOS
392 17AF 71      MOV M,C
393 17B0 C9      RET
394              SECTTRAN:
395              ;TRANSLATE SECTOR BC USING TABLE AT DE
396 17B1 0600     MVI B,0 ;DOUBLE PRECISION SECTOR NUMBER IN BC
397 17B3 EB      XCHG      ;TRANSLATE TABLE ADDRESS TO HL
398 17B4 09      DAD B ;TRANSLATE(SECTOR) ADDRESS
399 17B5 7E      MOV A,M ;TRANSLATED SECTOR NUMBER TO A
400 17B6 326B18   STA IOS
401 17B9 6F      MOV L,A ;RETURN SECTOR NUMBER IN L
402 17BA C9      RET
403              ;
404              SETDMA: ;SET DMA ADDRESS GIVEN BY REGS B,C
405 17BB 69      MOV L,C
406 17BC 60      MOV H,B
407 17BD 226C18   SHLD IOD
408 17C0 C9      RET
409              ;
410              READ: ;READ NEXT DISK RECORD (ASSUMING DISK/TRK/SEC/DMA SET)
411 17C1 0E04     MVI C,READF ;SET TO READ FUNCTION
412 17C3 CDE017   CALL SETFUNC
413 17C6 CDF017   CALL WAITIO ;PERFORM READ FUNCTION
414 17C9 C9      RET      ;MAY HAVE ERROR SET IN REG-A
415              ;
416              ;
417              WRITE: ;DISK WRITE FUNCTION
418 17CA 0E06     MVI C,WRITF
419 17CC CDE017   CALL SETFUNC ;SET TO WRITE FUNCTION
420 17CF CDF017   CALL WAITIO
421 17D2 C9      RET      ;MAY HAVE ERROR SET
422              ;
423              ;
424              ; UTILITY SUBROUTINES
425              PRMSG: ;PRINT MESSAGE AT H,L TO 0
426 17D3 7E      MOV A,M
427 17D4 B7      ORA A ;ZERO?
428 17D5 C8      RZ
429              ; MORE TO PRINT
430 17D6 E5      PUSH H
431 17D7 4F      MOV C,A
432 17D8 CD6A17   CALL CONOUT
433 17DB E1      POP H
434 17DC 23      INX H
435 17DD C3D317   JMP PRMSG
436              ;
437              SETFUNC:
438              ; SET FUNCTION FOR NEXT I/O (COMMAND IN REG-C)
439 17E0 216818   LXI H,IOF ;IO FUNCTION ADDRESS
440 17E3 7E      MOV A,M ;GET IT TO ACCUMULATOR FOR MASKING
441 17E4 E6F8     ANI 11111000B ;REMOVE PREVIOUS COMMAND
442 17E6 B1      ORA C ;SET TO NEW COMMAND
443 17E7 77      MOV M,A ;REPLACED IN IOPB

```

```

444                ; THE MDS-800 CONTROLLER REQUIRES DISK BANK BIT IN SECTOR BYTE
445                ; MASK THE BIT FROM THE CURRENT I/O FUNCTION
446 17E8 E620      ANI 00100000B ;MASK THE DISK SELECT BIT
447 17EA 216B18    LXI H,IOS      ;ADDRESS THE SECTOR SELECT BYTE
448 17ED B6        ORA M          ;SELECT PROPER DISK BANK
449 17EE 77        MOV M,A        ;SET DISK SELECT BIT ON/OFF
450 17EF C9        RET
451                ;
452                WAITIO:
453 17F0 0E0A      MVI C,RETRY ;MAX RETRIES BEFORE PERM ERROR
454                REWAIT:
455                ; START THE I/O FUNCTION AND WAIT FOR COMPLETION
456 17F2 CD3F18    CALL INTYPE ;IN RTYPE
457 17F5 CD4C18    CALL INBYTE ;CLEARS THE CONTROLLER
458                ;
459 17F8 3A6618    LDA DBANK      ;SET BANK FLAGS
460 17FB B7        ORA A          ;ZERO IF DRIVE 0,1 AND NZ IF 2,3
461 17FC 3E67      MVI A,IOPB AND 0FFH ;LOW ADDRESS FOR IOPB
462 17FE 0618      MVI B,IOPB SHR 8 ;HIGH ADDRESS FOR IOPB
463 1800 C20B18    JNZ IODR1 ;DRIVE BANK 1?
464 1803 D379      OUT ILOW       ;LOW ADDRESS TO CONTROLLER
465 1805 78        MOV A,B
466 1806 D37A      OUT IHIGH ;HIGH ADDRESS
467 1808 C31018    JMP WAIT0      ;TO WAIT FOR COMPLETE
468                ;
469                IODR1: ;DRIVE BANK 1
470 180B D389      OUT ILOW+10H ;88 FOR DRIVE BANK 10
471 180D 78        MOV A,B
472 180E D38A      OUT IHIGH+10H
473                ;
474 1810 CD5918    WAIT0: CALL INSTAT ;WAIT FOR COMPLETION
475 1813 E604      ANI IORDY ;READY?
476 1815 CA1018    JZ WAIT0
477                ;
478                ; CHECK IO COMPLETION OK
479 1818 CD3F18    CALL INTYPE ;MUST BE IO COMPLETE (00) UNLINKED
480                ; 00 UNLINKED I/O COMPLETE, 01 LINKED I/O COMPLETE (NOT USED)
481                ; 10 DISK STATUS CHANGED 11 (NOT USED)
482 181B FE02      CPI 10B ;READY STATUS CHANGE?
483 181D CA3218    JZ WREADY
484                ;
485                ; MUST BE 00 IN THE ACCUMULATOR
486 1820 B7        ORA A
487 1821 C23818    JNZ WERROR ;SOME OTHER CONDITION, RETRY
488                ;
489                ; CHECK I/O ERROR BITS
490 1824 CD4C18    CALL INBYTE
491 1827 17        RAL
492 1828 DA3218    JC WREADY ;UNIT NOT READY
493 182B 1F        RAR
494 182C E6FE      ANI 1111110B ;ANY OTHER ERRORS? (DELETED DATA OK)
495 182E C23818    JNZ WERROR
496                ;
497                ; READ OR WRITE IS OK, ACCUMULATOR CONTAINS ZERO
498 1831 C9        RET
499                ;
500                WREADY: ;NOT READY, TREAT AS ERROR FOR NOW

```

```

501 1832 CD4C18      CALL INBYTE      ;CLEAR RESULT BYTE
502 1835 C33818      JMP TRYCOUNT
503
504                ;
505                ;ERROR: ;RETURN HARDWARE MALFUNCTION (CRC, TRACK, SEEK, ETC.)
506                ; THE MDS CONTROLLER HAS RETURNED A BIT IN EACH POSITION
507                ; OF THE ACCUMULATOR, CORRESPONDING TO THE CONDITIONS:
508                ; 0 - DELETED DATA (ACCEPTED AS OK ABOVE)
509                ; 1 - CRC ERROR
510                ; 2 - SEEK ERROR
511                ; 3 - ADDRESS ERROR (HARDWARE MALFUNCTION)
512                ; 4 - DATA OVER/UNDER FLOW (HARDWARE MALFUNCTION)
513                ; 5 - WRITE PROTECT (TREATED AS NOT READY)
514                ; 6 - WRITE ERROR (HARDWARE MALFUNCTION)
515                ; 7 - NOT READY
516                ; (ACCUMULATOR BITS ARE NUMBERED 7 6 5 4 3 2 1 0)
517                ;
518                ; IT MAY BE USEFUL TO FILTER OUT THE VARIOUS CONDITIONS,
519                ; BUT WE WILL GET A PERMANENT ERROR MESSAGE IF IT IS NOT
520                ; RECOVERABLE. IN ANY CASE, THE NOT READY CONDITION IS
521                ; TREATED AS A SEPARATE CONDITION FOR LATER IMPROVEMENT
522                TRYCOUNT:
523                ; REGISTER C CONTAINS RETRY COUNT, DECREMENT 'TIL ZERO
524 1838 0D           DCR C
525 1839 C2F217       JNZ REWAIT      ;FOR ANOTHER TRY
526                ;
527                ; CANNOT RECOVER FROM ERROR
528 183C 3E01         MVI A,1 ;ERROR CODE
529 183E C9           RET
530                ;
531                ; INTYPE, INBYTE, INSTAT READ DRIVE BANK 00 OR 10
532 183F 3A6618       INTYPE: LDA DBANK
533 1842 B7           ORA A
534 1843 C24918       JNZ INTYP1     ;SKIP TO BANK 10
535 1846 DB79         IN RTYPE
536 1848 C9           RET
537 1849 DB89         INTYP1: IN RTYPE+10H ;78 FOR 0,1 88 FOR 2,3
538 184B C9           RET
539                ;
540 184C 3A6618       INBYTE: LDA DBANK
541 184F B7           ORA A
542 1850 C25618       JNZ INBYT1
543 1853 DB7B         IN RBYTE
544 1855 C9           RET
545 1856 DB8B         INBYT1: IN RBYTE+10H
546 1858 C9           RET
547                ;
548 1859 3A6618       INSTAT: LDA DBANK
549 185C B7           ORA A
550 185D C26318       JNZ INSTA1
551 1860 DB78         IN DSTAT
552 1862 C9           RET
553 1863 DB88         INSTA1: IN DSTAT+10H
554 1865 C9           RET
555                ;
556                ;
557                ; DATA AREAS (MUST BE IN RAM)

```

```

558 1866 00      DBANK: DB 0 ;DISK BANK 00 IF DRIVE 0,1
559              ;      10 IF DRIVE 2,3
560              IOPB: ;IO PARAMETER BLOCK
561 1867 80      DB 80H ;NORMAL I/O OPERATION
562 1868 04      IOF: DB READF ;IO FUNCTION, INITIAL READ
563 1869 01      ION: DB 1 ;NUMBER OF SECTORS TO READ
564 186A 02      IOT: DB OFFSET ;TRACK NUMBER
565 186B 01      IOS: DB 1 ;SECTOR NUMBER
566 186C 8000    IOD: DW BUFF ;IO ADDRESS
567              ;
568              ;
569              ; DEFINE RAM AREAS FOR BDOS OPERATION
570              ENDEF
571 186E+=       BEGDAT EQU $
572 186E+       DIRBUF: DS 128 ;DIRECTORY ACCESS BUFFER
573 18EE+       ALV0: DS 31
574 190D+       CSV0: DS 16
575 191D+       ALV1: DS 31
576 193C+       CSV1: DS 16
577 194C+       ALV2: DS 31
578 196B+       CSV2: DS 16
579 197B+       ALV3: DS 31
580 199A+       CSV3: DS 16
581 19AA+=      ENDDAT EQU $
582 013C+=      DATSIZ EQU $-BEGDAT
583 19AA        END

ALS1          001F 141#
ALS2          001F 146#
ALS3          001F 151#
ALV0          18EE 87 573#
ALV1          191D 91 575#
ALV2          194C 95 577#
ALV3          197B 99 579#
BASE          0078 180# 181 182 183 185 186
BDOS          0806 29# 287
BEGDAT        186E 571# 582
BIAS          0000 19# 22#
BOOT          16B3 63 207#
BOOTER0       1752 305 310#
BOOTERR       1749 241 302#
BOOTMSG       175B 312 316#
BUFF          0080 34# 209 221 278 566
CDISK         0004 33# 213 296
CI            F803 172# 325
CO            F809 174# 330
CONIN         1764 66 324#
CONOUT        176A 67 329# 432
CONST         1761 65 320#
CPMB          0000 28# 29 30 226 299
CPML          1600 30# 31
CR            000D 192# 196 205
CSS1          0010 142#
CSS2          0010 147#
CSS3          0010 152#
CSTS          F812 177# 322
CSV0          190D 87 574#
CSV1          193C 91 576#

```

CSV2	196B	95	578#				
CSV3	199A	99	580#				
DATSIK	013C	582#					
DBANK	1866	361	459	531	539	547	558#
DIRBUF	186E	86	90	94	98	572#	
DPB0	1673	86	101#	140	145	150	
DPB1	1673	90	140#				
DPB2	1673	94	145#				
DPB3	1673	98	150#				
DPBASE	1633	83#	380				
DPE0	1633	84#					
DPE1	1643	88#					
DPE2	1653	92#					
DPE3	1663	96#					
DSTAT	0078	181#	550	552			
ENDDAT	19AA	581#					
FALSE	0000	15#	16				
GOCPM	170F	214	265#				
HOME	1778	71	349#				
ICON	00F3	166#	275				
IHIGH	007A	186#	466	472			
ILOW	0079	185#	464	470			
INBYT1	1856	541	544#				
INBYTE	184C	457	490	501	539#		
INSTA1	1863	549	552#				
INSTAT	1859	474	547#				
INTC	00FC	165#	271	273			
INTE	007E	167#	272				
INTYP1	1849	533	536#				
INTYPE	183F	456	479	531#			
IOD	186C	242	407	566#			
IODR1	180B	463	469#				
IOF	1868	369	439	562#			
ION	1869	563#					
IOPB	1867	461	462	560#			
IORDY	0004	191#	475				
IOS	186B	248	391	400	447	565#	
IOT	186A	252	386	564#			
LF	000A	193#	196	196	205		
LIST	176D	68	332#				
LISTST	1770	78	336#				
LO	F80F	176#	334				
MON80	F800	170#	291				
NSECTS	002C	31#	237				
OFFSET	0002	32#	100	564			
PATCH	1600	25#	27	28			
PO	F80C	175#	343				
PRMSG	17D3	211	313	425#	435		
PUNCH	1772	69	341#				
RBYTE	007B	183#	542	544			
RD1	1705	250	257#				
RDSEC	16E1	238#	262				
READ	17C1	76	240	410#			
READER	1775	70	345#				
READF	0004	188#	411	562			
RECAL	0003	190#					
RETRY	000A	35#	223	453			

REVRT	00FD	164#	269				
REWAIT	17F2	454#	524				
RI	F806	173#	347				
RMON80	FF0F	171#	314				
RTYPE	0079	182#	534	536			
SECTRAN	17B1	79	394#				
SELDSK	177D	72	229	354#			
SETDMA	17BB	75	227	247	279	404#	
SETDRIVE	1792	365	367#				
SETFUNC	17E0	412	419	437#			
SETSEC	17AC	74	233	259	390#		
SETTRK	17A7	73	231	255	352	385#	
SIGNON	169C	195#	210				
TEST	0000	16#	18	21	197	200	289
TRUE	FFFF	14#	15				
TRYCOUNT	1838	502	521#				
VERS	0016	6#	204	204			
WAIT0	1810	467	474#	476			
WAITIO	17F0	413	420	452#			
WBOOT	16C3	64	217#				
WBOOT0	16C9	225#	308				
WBOOTE	1603	64#	284				
WERROR	1838	487	495	504#			
WREADY	1832	483	492	500#			
WRITE	17CA	77	417#				
WRITF	0006	189#	418				
XLT0	1682	84	112#	143	148	153	
XLT1	1682	88	143#				
XLT2	1682	92	148#				
XLT3	1682	96	153#				

B. A Skeletal CBIOS

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1          ; SKELETAL CBIOS FOR FIRST LEVEL OF CP/M 2.0 ALTERATION
2          ;
3 0014 =    MSIZE EQU 20      ;CP/M VERSION MEMORY SIZE IN KILOBYTES
4          ;
5          ; "BIAS" IS ADDRESS OFFSET FROM 3400H FOR MEMORY SYSTEMS
6          ; THAN 16K (REFERRED TO AS"B" THROUGHOUT THE TEXT)
7          ;
8 0000 =    BIAS EQU (MSIZE-20)*1024
9 3400 =    CCP EQU 3400H+BIAS ;BASE OF CCP
10 3C06 =    BDOS EQU CCP+806H ;BASE OF BDOS
11 4A00 =    BIOS EQU CCP+1600H ;BASE OF BIOS
12 0004 =    CDISK EQU 0004H   ;CURRENT DISK NUMBER 0=A,... L5=P
13 0003 =    IOBYTE EQU 0003H  ;INTEL I/O BYTE
14          ;
15 4A00      ORG BIOS          ;ORIGIN OF THIS PROGRAM
16 002C =    NSECTS EQU ($-CCP)/128 ;WARM START SECTOR COUNT
17          ;
18          ; JUMP VECTOR FOR INDIVIDUAL SUBROUTINES
19          ;
20 4A00 C39C4A    JMP BOOT ;COLD START
21 4A03 C3A64A    WBOOTE: JMP WBOOT ;WARM START
22 4A06 C3114B    JMP CONST ;CONSOLE STATUS
23 4A09 C3244B    JMP CONIN ;CONSOLE CHARACTER IN
24 4A0C C3374B    JMP CONOUT ;CONSOLE CHARACTER OUT
25 4A0F C3494B    JMP LIST ;LIST CHARACTER OUT
26 4A12 C34D4B    JMP PUNCH ;PUNCH CHARACTER OUT
27 4A15 C34F4B    JMP READER ;READER CHARACTER OUT
28 4A18 C3544B    JMP HOME ;MOVE HEAD TO HOME POSITION
29 4A1B C35A4B    JMP SELDSK ;SELECT DISK
30 4A1E C37D4B    JMP SETTRK ;SET TRACK NUMBER
31 4A21 C3924B    JMP SETSEC ;SET SECTOR NUMBER
32 4A24 C3AD4B    JMP SETDMA ;SET DMA ADDRESS
33 4A27 C3C34B    JMP READ ;READ DISK
34 4A2A C3D64B    JMP WRITE ;WRITE DISK
35 4A2D C34B4B    JMP LISTST ;RETURN LIST STATUS
36 4A30 C3A74B    JMP SECTTRAN ;SECTOR TRANSLATE
37          ;
38          ; FIXED DATA TABLES FOR FOUR-DRIVE STANDARD
39          ; IBM-COMPATIBLE 8" DISKS
40          ;
41          ; DISK PARAMETER HEADER FOR DISK 00
42 4A33 734A0000  DPBASE: DW TRANS, 0000H
43 4A37 00000000  DW 0000H, 0000H
44 4A3B F04C8D4A  DW DIRBF, DPBLK
45 4A3F EC4D704D  DW CHK00, ALL00
```

```

46          ; DISK PARAMETER HEADER FOR DISK 01
47 4A43 734A0000 DW TRANS, 0000H
48 4A47 00000000 DW 0000H, 0000H
49 4A4B F04C8D4A DW DIRBF, DPBLK
50 4A4F FC4D8F4D DW CHK01, ALL01
51          ; DISK PARAMETER HEADER FOR DISK 02
52 4A53 734A0000 DW TRANS, 0000H
53 4A57 00000000 DW 0000H, 0000H
54 4A5B F04C8D4A DW DIRBF, DPBLK
55 4A5F 0C4EAE4D DW CHK02, ALL02
56          ; DISK PARAMETER HEADER FOR DISK 03
57 4A63 734A0000 DW TRANS, 0000H
58 4A67 00000000 DW 0000H, 0000H
59 4A6B F04C8D4A DW DIRBF, DPBLK
60 4A6F 1C4ECD4D DW CHK03, ALL03
61          ;
62          ; SECTOR TRANSLATE VECTOR
63 4A73 01070D13 TRANS: DB 1, 7, 13, 19 ;SECTORS 1, 2, 3, 4
64 4A77 19050B11 DB 25, 5, 11, 17 ;SECTORS 5, 6, 7, 6
65 4A7B 1703090F DB 23, 3, 9, 15 ;SECTORS 9, 10, 11, 12
66 4A7F 1502080E DB 21, 2, 8, 14 ;SECTORS 13, 14, 15, 16
67 4A83 141A060C DB 20, 26, 6, 12 ;SECTORS 17, 18, 19, 20
68 4A87 1218040A DB 18, 24, 4, 10 ;SECTORS 21, 22, 23, 24
69 4A8B 1016 DB 16, 22 ;SECTORS 25, 26
70          ;
71          DPBLK: ;DISK PARAMETER BLOCK, COMMON TO ALL DISKS
72 4A8D 1A00 DW 26 ;SECTORS PER TRACK
73 4A8F 03 DB 3 ;BLOCK SHIFT FACTOR
74 4A90 07 DB 7 ;BLOCK MASK
75 4A91 00 DB 0 ;NULL MASK
76 4A92 F200 DW 242 ;DISK SIZE-1
77 4A94 3F00 DW 63 ;DIRECTORY MAX
78 4A96 C0 DB 192 ;ALLOC 0
79 4A97 00 DB 0 ;ALLOC 1
80 4A98 1000 DW 16 ;CHECK SIZE
81 4A9A 0200 DW 2 ;TRACK OFFSET
82          ;
83          ; END OF FIXED TABLES
84          ;
85          ; INDIVIDUAL SUBROUTINES TO PERFORM EACH FUNCTION
86          BOOT: ;SIMPLEST CASE IS TO JUST PERFORM PARAMETER INITIALIZATION
87 4A9C AF XRA A ;ZERO IN THE ACCUM
88 4A9D 320300 STA IOBYTE ;CLEAR THE IOBYTE
89 4AA0 320400 STA CDISK ;SELECT DISK ZERO
90 4AA3 C3EF4A JMP GOCPM ;INITIALIZE AND GO TO CP/M
91          ;
92          WBOOT: ;SIMPLEST CASE IS TO READ THE DISK UNTIL ALL SECTORS LOADED
93 4AA6 318000 LXI SP, 80H ;USE SPACE BELOW BUFFER FOR STACK
94 4AA9 0E00 MVI C, 0 ;SELECT DISK 0
95 4AAB CD5A4B CALL SELDSK
96 4AAE CD544B CALL HOME ;GO TO TRACK 00
97          ;
98 4AB1 062C MVI B, NSECTS ;B COUNTS * OF SECTORS TO LOAD
99 4AB3 0E00 MVI C, 0 ;C HAS THE CURRENT TRACK NUMBER
100 4AB5 1602 MVI D, 2 ;D HAS THE NEXT SECTOR TO READ
101          ; NOTE THAT WE BEGIN BY READING TRACK 0, SECTOR 2 SINCE SECTOR 1
102          ; CONTAINS THE COLD START LOADER, WHICH IS SKIPPED IN A WARM START

```

```

103 4AB7 210034 LXI H, CCP ;BASE OF CP/M (INITIAL LOAD POINT)
104 LOAD1: ;LOAD ONE MORE SECTOR
105 4ABA C5 PUSH B ;SAVE SECTOR COUNT, CURRENT TRACK
106 4ABB D5 PUSH D ;SAVE NEXT SECTOR TO READ
107 4ABC E5 PUSH H ;SAVE DMA ADDRESS
108 4ABD 4A MOV C, D ;GET SECTOR ADDRESS TO REGISTER C
109 4ABE CD924B CALL SETSEC ;SET SECTOR ADDRESS FROM REGISTER C
110 4AC1 C1 POP B ;RECALL DMA ADDRESS TO B, C
111 4AC2 C5 PUSH B ;REPLACE ON STACK FOR LATER RECALL
112 4AC3 CDAD4B CALL SETDMA ;SET DMA ADDRESS FROM B, C
113 ;
114 ; DRIVE SET TO 0, TRACK SET, SECTOR SET, DMA ADDRESS SET
115 4AC6 CDC34B CALL READ
116 4AC9 FE00 CPI 00H ;ANY ERRORS?
117 4ACB C2A64A JNZ WBOOT ;RETRY THE ENTIRE BOOT IF AN ERROR OCCURS
118 ;
119 ; NO ERROR, MOVE TO NEXT SECTOR
120 4ACE E1 POP H ;RECALL DMA ADDRESS
121 4ACF 118000 LXI D, 128 ;DMA=DMA+128
122 4AD2 19 DAD D ;NEW DMA ADDRESS IS IN H, L
123 4AD3 D1 POP D ;RECALL SECTOR ADDRESS
124 4AD4 C1 POP B ;RECALL NUMBER OF SECTORS REMAINING, AND CURRENT TRK
125 4AD5 05 DCR B ;SECTORS=SECTORS-1
126 4AD6 CAEF4A JZ GOCPM ;TRANSFER TO CP/M IF ALL HAVE BEEN LOADED
127 ;
128 ; MORE SECTORS REMAIN TO LOAD, CHECK FOR TRACK CHANGE
129 4AD9 14 INR D
130 4ADA 7A MOV A,D ;SECTOR=27?, IF SO, CHANGE TRACKS
131 4ADB FE1B CPI 27
132 4ADD DABA4A JC LOAD1 ;CARRY GENERATED IF SECTOR<27
133 ;
134 ; END OF CURRENT TRACK, GO TO NEXT TRACK
135 4AE0 1601 MVI D, 1 ;BEGIN WITH FIRST SECTOR OF NEXT TRACK
136 4AE2 0C INR C ;TRACK=TRACK+1
137 ;
138 ; SAVE REGISTER STATE, AND CHANGE TRACKS
139 4AE3 C5 PUSH B
140 4AE4 D5 PUSH D
141 4AE5 E5 PUSH H
142 4AE6 CD7D4B CALL SETTRK ;TRACK ADDRESS SET FROM REGISTER C
143 4AE9 E1 POP H
144 4AEA D1 POP D
145 4AEB C1 POP B
146 4AEC C3BA4A JMP LOAD1 ;FOR ANOTHER SECTOR
147 ;
148 ; END OF LOAD OPERATION, SET PARAMETERS AND GO TO CP/M
149 GOCPM:
150 4AEF 3EC3 MVI A, 0C3H ;C3 IS A JMP INSTRUCTION
151 4AF1 320000 STA 0 ;FOR JMP TO WBOOT
152 4AF4 21034A LXI H, WBOOTE ;WBOOT ENTRY POINT
153 4AF7 220100 SHLD 1 ;SET ADDRESS FIELD FOR JMP AT 0
154 ;
155 4AFA 320500 STA 5 ;FOR JMP TO BDOS
156 4AFD 21063C LXI H, BDOS ;BDOS ENTRY POINT
157 4B00 220600 SHLD 6 ;ADDRESS FIELD OF JUMP AT 5 TO BDOS
158 ;
159 4B03 018000 LXI B, 80H ;DEFAULT DMA ADDRESS IS 80H

```

```

160 4B06 CDAD4B      CALL  SETDMA
161                  ;
162 4B09 FB          EI           ;ENABLE THE INTERRUPT SYSTEM
163 4B0A 3A0400      LDA  CDISK   ;GET CURRENT DISK NUMBER
164 4B0D 4F          MOV  C, A    ;SEND TO THE CCP
165 4B0E C30034      JMP  CCP     ;GO TO CP/M FOR FURTHER PROCESSING
166                  ;
167                  ;
168                  ; SIMPLE I/O HANDLERS (MUST BE FILLED IN BY USER)
169                  ; IN EACH CASE, THE ENTRY POINT IS PROVIDED, WITH SPACE RESERVED
170                  ; TO INSERT YOUR OWN CODE
171                  ;
172 CONST:           ;CONSOLE STATUS, RETURN 0FFH IF CHARACTER READY, 00H IF NOT
173 4B11              DS      10H   ;SPACE FOR STATUS SUBROUTINE
174 4B21 3E00        MVI     A, 00H
175 4B23 C9          RET
176                  ;
177 CONIN:           ;CONSOLE CHARACTER INTO REGISTER A
178 4B24              DS      10H   ;SPACE FOR INPUT ROUTINE
179 4B34 E67F        ANI  7FH     ;STRIP PARITY BIT
180 4B36 C9          RET
181                  ;
182 CONOUT:          ;CONSOLE CHARACTER OUTPUT FROM REGISTER C
183 4B37 79          MOV  A, C     ;GET TO ACCUMULATOR
184 4B38              DS      10H   ;SPACE FOR OUTPUT ROUTINE
185 4B48 C9          RET
186                  ;
187 LIST:            ;LIST CHARACTER FROM REGISTER C
188 4B49 79          MOV  A, C     ;CHARACTER TO REGISTER A
189 4B4A C9          RET          ;NULL SUBROUTINE
190                  ;
191 LISTST:          ;RETURN LIST STATUS (0 IF NOT READY, 1 IF READY)
192 4B4B AF          XRA  A        ;0 IS ALWAYS OK TO RETURN
193 4B4C C9          RET
194                  ;
195 PUNCH:           ;PUNCH CHARACTER FROM REGISTER C
196 4B4D 79          MOV  A, C     ;CHARACTER TO REGISTER A
197 4B4E C9          RET          ;NULL SUBROUTINE
198                  ;
199                  ;
200 READER:          ;READER CHARACTER INTO REGISTER A FROM READER DEVICE
201 4B4F 3E1A        MVI     A, 1AH ;ENTER END OF FILE FOR NOW (REPLACE LATER)
202 4B51 E67F        ANI  7FH     ;REMEMBER TO STRIP PARITY BIT
203 4B53 C9          RET
204                  ;
205                  ;
206                  ; I/O DRIVERS FOR THE DISK FOLLOW
207                  ; FOR NOW, WE WILL SIMPLY STORE THE PARAMETERS AWAY FOR USE
208                  ; IN THE READ AND WRITE SUBROUTINES
209                  ;
210 HOME:            ;MOVE TO THE TRACK 00 POSITION OF CURRENT DRIVE
211                  ; TRANSLATE THIS CALL INTO A SETTRK CALL WITH PARAMETER 00
212 4B54 0E00        MVI     C, 0   ;SELECT TRACK 0
213 4B56 CD7D4B      CALL  SETTRK
214 4B59 C9          RET          ;WE WILL MOVE TO 00 ON FIRST READ/WRITE
215                  ;
216 SELDSK:          ;SELECT DISK GIVEN BY REGISTER C

```

```

217 4B5A 210000 LXI H, 0000H ;ERROR RETURN CODE
218 4B5D 79      MOV A, C
219 4B5E 32EF4C  STA DISKNO
220 4B61 FE04    CPI 4 ;MUST BE BETWEEN 0 AND 3
221 4B63 D0      RNC ;NO CARRY IF 4, 5,...
222            ; DISK NUMBER IS IN THE PROPER RANGE
223 4B64         DS 10 ;SPACE FOR DISK SELECT
224            ; COMPUTE PROPER DISK PARAMETER HEADER ADDRESS
225 4B6E 3AEF4C  LDA DISKNO
226 4B71 6F      MOV L, A ;L=DISK NUMBER 0, 1, 2, 3
227 4B72 2600    MVI H, 0 ;HIGH ORDER ZERO
228 4B74 29      DAD H ;*2
229 4B75 29      DAD H ;*4
230 4B76 29      DAD H ;*8
231 4B77 29      DAD H ;*16 (SIZE OF EACH HEADER)
232 4B78 11334A LXI D, DPBASE
233 4B7B 09      DAD 0 ;HL=,DPBASE (DISKNO*16)
234 4B7C C9      RET
235            ;
236            SETTRK: ;SET TRACK GIVEN BY REGISTER C
237 4B7D 79      MOV A, C
238 4B7E 32E94C  STA TRACK
239 4B81         DS 10H ;SPACE FOR TRACK SELECT
240 4B91 C9      RET
241            ;
242            SETSEC: ;SET SECTOR GIVEN BY REGISTER C
243 4B92 79      MOV A, C
244 4B93 32EB4C  STA SECTOR
245 4B96         DS 10H ;SPACE FOR SECTOR SELECT
246 4BA6 C9      RET
247            ;
248            ;
249            SECTRAN:
250            ;TRANSLATE THE SECTOR GIVEN BY BC USING THE
251            ;TRANSLATE TABLE GIVEN BY DE
252 4BA7 EB      XCHG ;HL=.TRANS
253 4BA8 09      DAD B ;HL=.TRANS (SECTOR)
254 4BA9 6E      MOV L, M ;L=.TRANS (SECTOR)
255 4BAA 2600    MVI H, 0 ;HL=.TRANS (SECTOR)
256 4BAC C9      RET ;WITH VALUE IN HL
257            ;
258            SETDMA: ;SET DMA ADDRESS GIVEN BY REGISTERS B AND C
259 4BAD 69      MOV L, C ;LOW ORDER ADDRESS
260 4BAE 60      MOV H, B ;HIGH ORDER ADDRESS
261 4BAF 22ED4C  SHLD DMAAD ;SAVE THE ADDRESS
262 4BB2         DS 10H ;SPACE FOR SETTING THE DMA ADDRESS
263 4BC2 C9      RET
264            ;
265            READ: ;PERFORM READ OPERATION (USUALLY THIS IS SIMILAR TO WRITE
266            ; SO WE WILL ALLOW SPACE TO SET UP READ COMMAND, THEN USE
267            ; COMMON CODE IN WRITE)
268 4BC3         DS 10H ;SET UP READ COMMAND
269 4BD3 C3E64B  JMP WAITIO ;TO PERFORM THE ACTUAL I/O
270            ;
271            WRITE: ;PERFORM A WRITE OPERATION
272 4BD6         DS 10H ;SET UP WRITE COMMAND
273            ;

```

```

274          WAITIO: ;ENTER HERE FROM READ AND WRITE TO PERFORM THE ACTUAL I/O
275          ; OPERATION. RETURN A 00H IN REGISTER A IF THE OPERATION COMPLETES
276          ; PROPERLY, AND 0LH IF AN ERROR OCCURS DURING THE READ OR WRITE
277          ;
278          ; IN THIS CASE, WE HAVE SAVED THE DISK NUMBER IN 'DISKNO' (0, 1)
279          ; THE TRACK NUMBER IN 'TRACK' (0-76)
280          ; THE SECTOR NUMBER IN 'SECTOR' (1-26)
281          ; THE DMA ADDRESS IN 'DMAAD' (0-65535)
282 4BE6          DS 256 ;SPACE RESERVED FOR I/O DRIVERS
283 4CE6 3E01     MVI A, 1 ;ERROR CONDITION
284 4CE8 C9       RET ;REPLACED WHEN FILLED-IN
285          ;
286          ; THE REMAINDER OF THE CBIOS IS RESERVED UNINITIALIZED
287          ; DATA AREA, AND DOES NOT NEED TO BE A PART OF THE
288          ; SYSTEM MEMORY IMAGE (THE SPACE MUST BE AVAILABLE,
289          ; HOWEVER, BETWEEN"BEGDAT" AND"ENDDAT").
290          ;
291 4CE9          TRACK: DS 2 ;TWO BYTES FOR EXPANSION
292 4CEB          SECTOR: DS 2 ;TWO BYTES FOR EXPANSION
293 4CED          DMAAD: DS 2 ;DIRECT MEMORY ADDRESS
294 4CEF          DISKNO: DS 1 ;DISK NUMBER 0-15
295          ;
296          ; SCRATCH RAM AREA FOR BDOS USE
297 4CF0 =        BEGDAT EQU $ ;BEGINNING OF DATA AREA
298 4CF0          DIRBF: DS 128 ;SCRATCH DIRECTORY AREA
299 4D70          ALL00: DS 31 ;ALLOCATION VECTOR 0
300 4D8F          ALL01: DS 31 ;ALLOCATION VECTOR 1
301 4DAE          ALL02: DS 31 ;ALLOCATION VECTOR 2
302 4DCD          ALL03: DS 31 ;ALLOCATION VECTOR 3
303 4DEC          CHK00: DS 16 ;CHECK VECTOR 0
304 4DFC          CHK01: DS 16 ;CHECK VECTOR 1
305 4E0C          CHK02: DS 16 ;CHECK VECTOR 2
306 4E1C          CHK03: DS 16 ;CHECK VECTOR 3
307          ;
308 4E2C =        ENDDAT EQU $ ;END OF DATA AREA
309 013C =        DATSIZ EQU $-BEGDAT; ;SIZE OF DATA AREA
310 4E2C          END

```

ALL00	4D70	45	299#				
ALL01	4D8F	50	300#				
ALL02	4DAE	55	301#				
ALL03	4DCD	60	302#				
BDOS	3C06	10#	156				
BEGDAT	4CF0	297#	309				
BIAS	0000	8#	9				
BIOS	4A00	11#	15				
BOOT	4A9C	20	86#				
CCP	3400	9#	10	11	16	103	165
CDISK	0004	12#	89	163			
CHK00	4DEC	45	303#				
CHK01	4DFC	50	304#				
CHK02	4E0C	55	305#				
CHK03	4E1C	60	306#				
CONIN	4B24	23	177#				
CONOUT	4B37	24	182#				
CONST	4B11	22	172#				
DATSIZ	013C	309#					
DIRBF	4CF0	44	49	54	59	298#	

DISKNO	4CEF	219	225	294#		
DMAAD	4CED	261	293#			
DPBASE	4A33	42#	232			
DPBLK	4A8D	44	49	54	59	71#
ENDDAT	4E2C	308#				
GOCPM	4AEF	90	126	149#		
HOME	4B54	28	96	210#		
IOBYTE	0003	13#	88			
LIST	4B49	25	187#			
LISTST	4B4B	35	191#			
LOAD1	4ABA	104#	132	146		
MSIZE	0014	3#	8			
NSECTS	002C	16#	98			
PUNCH	4B4D	26	195#			
READ	4BC3	33	115	265#		
READER	4B4F	27	200#			
SECTOR	4CEB	244	292#			
SECTRAN	4BA7	36	249#			
SELDSK	4B5A	29	95	216#		
SETDMA	4BAD	32	112	160	258#	
SETSEC	4B92	31	109	242#		
SETTRK	4B7D	30	142	213	236#	
TRACK	4CE9	238	291#			
TRANS	4A73	42	47	52	57	63#
WAITIO	4BE6	269	274#			
WBOOT	4AA6	21	92#	117		
WBOOTE	4A03	21#	152			
WRITE	4BD6	34	271#			

C. A Skeletal GETSYS/PUTSYS Program

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1          ; COMBINED GETSYS AND PUTSYS PROGRAMS FROM
2          ; SEC 6.4
3          ;
4          ; START THE PROGRAMS AT THE BASE OF THE TPA
5 0100      ORG 0100H
6
7 0014 =    MSIZE EQU 20      ;SIZE OF CP/M IN KBYTES
8
9          ;"BIAS" IS THE AMOUNT TO ADD TO ADDRESSES FOR > 20K
10         ; (REFERRED TO AS"B" THROUGHOUT THE TEXT)
11 0000 =    BIAS EQU (MSIZE-20)*1024
12 3400 =    CCP EQU 3400H+BIAS
13 3C00 =    BDOS EQU CCP+0800H
14 4A00 =    BIOS EQU CCP+1600H
15
16         ; GETSYS PROGRAMS TRACKS 0 AND 1 TO MEMORY AT 3880H + BIAS
17         ; REGISTER      USAGE
18         ; A  (SCRATCH REGISTER)
19         ; B  TRACK COUNT (0...76)
20         ; C  SECTOR COUNT (1...26)
21         ; D,E (SCRATCH REGISTER PAIR)
22         ; H,L  LOAD ADDRESS
23         ; SP   SET TO TRACK ADDRESS
24
25         GSTART: ;START OF GETSYS
26 0100 318033 LXI SP,CCP-0080H ;CONVENIENT PLACE
27 0103 218033 LXI H,CCP-0080H ;SET INITIAL LOAD
28 0106 0600   MVI B,0      ;START WITH TRACK
29          RD$TRK: ;READ NEXT TRACK
30 0108 0E01   MVI C,1      ;EACH TRACK START
31          RD$SEC:
32 010A CD0003 CALL READ$SEC ;GET THE NEXT SECTOR
33 010D 118000 LXI D,128    ;OFFSET BY ONE SECTOR
34 0110 19     DAD D      ; (HL=HL+128)
35 0111 0C     INR C      ;NEXT SECTOR
36 0112 79     MOV A,C     ;FETCH SECTOR NUMBER
37 0113 FE1B   CPI 27      ;AND SEE IF LAST
38 0115 DA0A01 JC RDSEC    ;<, DO ONE MORE
39
40         ;ARRIVE HERE AT END OF TRACK, MOVE TO NEXT TRACK
41
42 0118 04     INR B      ;TRACK = TRACK+1
43 0119 78     MOV A,B     ;CHECK FOR LAST
44 011A FE02   CPI 2      ;TRACK = 2 ?
45 011C DA0801 JC RD$TRK   ;<, DO ANOTHER
```

```

46
47 ;ARRIVE HERE AT END OF LOAD, HALT FOR LACK OF ANYTHING
48 ;BETTER
49
50 011F FB EI
51 0120 76 HLT
52
53 ; PUTSYS PROGRAM, PLACES MEMORY IMAGE
54 ; STARTING AT
55 ; 3880H + BIAS BACK TO TRACKS 0 AND 1
56 ; START THIS PROGRAM AT THE NEXT PAGE BOUNDARY
57 0200 ORG ($+0100H) AND 0FF00H
58 PUT$SYS:
59 0200 318033 LXI SP,CCP-0080H ;CONVENIENT PLACE
60 0203 218033 LXI H,CCP-0080H ;START OF DUMP
61 0206 0600 MVI B,0 ;START WITH TRACK
62 WR$TRK:
63 0208 0605 MVI B,L ;START WITH SECTOR
64 WR$SEC:
65 020A CD0004 CALL WRITE$SEC ;WRITE ONE SECTOR
66 020D 118000 LXI D,128 ;LENGTH OF EACH
67 0210 19 DAD D ;<HL>=<HL> + 128
68 0211 0C INR C ; <C>=<C> + 1
69 0212 79 MOV A,C ;SEE IF
70 0213 FE1B CPI 27 ;PAST END OF TRACK
71 0215 DA0A02 JC WR$SEC ;NO, DO ANOTHER
72
73 ;ARRIVE HERE AT END OF TRACK, MOVE TO NEXT TRACK
74
75 0218 04 INR B ;TRACK = TRACK+1
76 0219 78 MOV A,B ;SEE IF
77 021A FE02 CPI 2 ;LAST TRACK
78 021C DA0802 JC WR$TRK ;NO, DO ANOTHER
79
80
81 ; DONE WITH PUTSYS, HALT FOR LACK OF ANYTHING
82 ; BETTER
83 021F FB EI
84 0220 76 HLT
85
86
87 ;USER SUPPLIED SUBROUTINES FOR SECTOR READ AND WRITE
88
89 ; MOVE TO NEXT PAGE BOUNDARY
90 0300 ORG ($+0100H) AND 0FF00H
91
92 READ$SEC:
93 ;READ THE NEXT SECTOR
94 ;TRACK IN <B>;
95 ;SECTOR IN <C>;
96 ;DMAADDR IN<HL>;
97
98 0300 C5 PUSH B
99 0301 E5 PUSH H
100
101 ;USER DEFINED READ OPERATION GOES HERE
102 0302 DS 64

```

```

103 0342 E1      POP H
104 0343 C1      POP B
105 0344 C9      RET
106
107 0400          ORG ($+100H) AND 0FF00H ;ANOTHER PAGE
108              ; BOUNDARY
109 WRITE$SEC:
110
111              ;SAME PARAMETERS AS READ$SEC
112
113 0400 C5      PUSH B
114 0401 E5      PUSH H
115
116              ;USER DEFINED WRITE OPERATION GOES HERE
117 0402          DS 64
118 0442 E1      POP H
119 0443 C1      POP B
120 0444 C9      RET
121
122              ;END OF GETSYS/PUTSYS PROGRAM
123
124 0445          END

```

BDOS	3C00	13#							
BIAS	0000	11#	12						
BIOS	4A00	14#							
CCP	3400	12#	13	14	26	27	59	60	
GSTART	0100	25#							
MSIZE	0014	7#	11						
PUTSYS	0200	58#							
RDSEC	010A	31#	38						
RDTRK	0108	29#	45						
READSEC	0300	32	92#						
WRITESEC	0400	65	109#						
WRSEC	020A	64#	71						
WRTRK	0208	62#	78						

D. The MDS-800 Cold Start Loader for CP/M 2

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1          TITLE    'mds cold start loader at 3000h'
2          ;
3          ; MDS-800 COLD START LOADER FOR CP/M 2.0
4          ;
5          ; VERSION 2.0 AUGUST, 1979
6          ;
7 0000 =      FALSE EQU 0
8 FFFF =      TRUE  EQU NOT FALSE
9 0000 =      TESTING EQU FALSE      ;IF TRUE, THEN GO TO MON80 ON  ERRORS
10         ;
11         IF TESTING
12         BIAS EQU 03400H
13         ENDIF
14         IF NOT TESTING
15 0000 =      BIAS EQU 0000H
16         ENDIF
17 0000 =      CPMB EQU BIAS      ;BASE OF DOS LOAD
18 0806 =      BDOS EQU 806H+BIAS ;ENTRY TO DOS FOR CALLS
19 1880 =      BDOSE EQU 1880H+BIAS ;END OF DOS LOAD
20 1600 =      BOOT EQU 1600H+BIAS ;COLD START ENTRY POINT
21 1603 =      RBOOT EQU BOOT+3    ;WARM START ENTRY POINT
22         ;
23 3000      ORG 03000H      ;LOADED DOWN FROM HARDWARE BOOT AT 3000H
24         ;
25 1880 =      BDOSL EQU BDOSE-CPMB
26 0002 =      NTRKS EQU 2      ;NUMBER OF TRACKS TO READ
27 0031 =      BDOSS EQU BDOSL/128 ;NUMBER OF SECTORS IN DOS
28 0019 =      BDOSO EQU 25      ;NUMBER OF BDOS SECTORS ON TRACK 0
29 0018 =      BDOS1 EQU BDOSS-BDOS0 ;NUMBER OF SECTORS ON TRACK 1
30         ;
31 F800 =      MON80 EQU 0F800H    ;INTEL MONITOR BASE
32 FF0F =      RMON80 EQU 0FF0FH   ;RESTART LOCATION FOR MON80
33 0078 =      BASE EQU 078H      ;'BASE' USED BY CONTROLLER
34 0079 =      RTYPE EQU BASE+1    ;RESULT TYPE
35 007B =      RBYTE EQU BASE+3    ;RESULT BYTE
36 007F =      RESET EQU BASE+7    ;RESET CONTROLLER
37         ;
38         ;
39 0078 =      DSTAT EQU BASE      ;DISK STATUS PORT
40 0079 =      ILOW EQU BASE+1     ;LOW IOPB ADDRESS
41 007A =      IHIGH EQU BASE+2    ;HIGH IOPB ADDRESS
42 00FF =      BSW EQU 0FFH       ;BOOT SWITCH
43 0003 =      RECAL EQU 3H       ;RECALIBRATE SELECTED DRIVE
44 0004 =      READF EQU 4H       ;DISK READ FUNCTION
45 0100 =      STACK EQU 100H     ;USE END OF BOOT FOR STACK
```

```

46      ;
47      RSTART:
48      3000 310001    LXI SP,STACK; ;IN CASE OF CALL TO MON80
49      ; CLEAR DISK STATUS
50      3003 DB79      IN  RTYPE
51      3005 DB7B      IN  RBYTE
52      ; CHECK IF BOOT SWITCH IS OFF
53      COLDSTART:
54      3007 DBFF      IN  BSW
55      3009 E602      ANI 02H    ;SWITCH ON?
56      300B C20730    JNZ COLDSTART
57      ; CLEAR THE CONTROLLER
58      300E D37F      OUT RESET  ;LOGIC CLEARED
59      ;
60      ;
61      3010 0602      MVI B,NTRKS    ;NUMBER OF TRACKS TO READ
62      3012 214230    LXI H,IOPBO
63      ;
64      START:
65      ;
66      ; READ FIRST/NEXT TRACK INTO CPMB
67      3015 7D        MOV A,L
68      3016 D379      OUT ILOW
69      3018 7C        MOV A,H
70      3019 D37A      OUT IHIGH
71      301B DB78      WAITO: IN  DSTAT
72      301D E604      ANI 4
73      301F CA1B30    JZ  WAITO
74      ;
75      ; CHECK DISK STATUS
76      3022 DB79      IN  RTYPE
77      3024 E603      ANI 11B
78      3026 FE02      CPI 2
79      ;
80      IF TESTING
81      CNC RMON80      ;GO TO MONITOR IF 11 OR 10
82      ENDIF
83      IF NOT TESTING
84      3028 D20030    JNC RSTART    ;RETRY THE LOAD
85      ENDIF
86      ;
87      302B DB7B      IN  RBYTE    ;I/O COMPLETE, CHECK STATUS
88      ; IF NOT READY, THEN GO TO MON80
89      302D 17        RAL
90      302E DC0FFF    CC  RMON80    ;NOT READY BIT SET
91      3031 1F        RAR          ;RESTORE
92      3032 E61E      ANI 11110B    ;OVERRUN/ADDR ERR/SEEK/CRC/XXXX
93      ;
94      IF TESTING
95      CNZ RMON80      ;GO TO MONITOR
96      ENDIF
97      IF NOT TESTING
98      3034 C20030    JNZ RSTART    ;RETRY THE LOAD
99      ENDIF
100     ;
101     ;
102     3037 110700    LXI D,IOPBL    ;LENGTH OF IOPB

```

```

103 303A 19      DAD D      ;ADDRESSING NEXT IOPB
104 303B 05      DCR B      ;COUNT DOWN TRACKS
105 303C C21530  JNZ START
106              ;
107              ;
108              ; JMP TO BOOT TO PRINT INITIAL MESSAGE, AND SET UP JMPs
109 303F C30016  JMP BOOT
110              ;
111              ; PARAMETER BLOCKS
112 3042 80      IOPBO: DB 80H ;IOCW, NO UPDATE
113 3043 04      DB READF ;READ FUNCTION
114 3044 19      DB BDOSO ;*SECTORS TO READ ON TRACK 0
115 3045 00      DB 0 ;TRACK 0
116 3046 02      DB 2 ;START WITH SECTOR 2 ON TRACK 0
117 3047 0000    DW CPMB ;START AT BASE OF BDOS
118 0007 =      IOPBL EQU $-IOPBO
119              ;
120 3049 80      IOPB1: DB 80H
121 304A 04      DB READF
122 304B 18      DB BDOS1 ;SECTORS TO READ ON TRACK 1
123 304C 01      DB 1 ;TRACK 1
124 304D 01      DB 1 ;SECTOR 1
125 304E 800C    DW CPMB+BDOSO*128 ;BASE OF SECOND READ
126              ;
127 3050          END

```

BASE	0078	33#	34	35	36	39	40	41
BDOS	0806	18#						
BDOS1	0018	29#	122					
BDOSE	1880	19#	25					
BDOSL	1880	25#	27					
BDOSO	0019	28#	29	114	125			
BDOSs	0031	27#	29					
BIAS	0000	12#	15#	17	18	19	20	
BOOT	1600	20#	21	109				
BSW	00FF	42#	54					
COLDSTART	3007	53#	56					
CPMB	0000	17#	25	117	125			
DSTAT	0078	39#	71					
FALSE	0000	7#	8	9				
IHIGH	007A	41#	70					
ILOW	0079	40#	68					
IOPB1	3049	120#						
IOPBL	0007	102	118#					
IOPBO	3042	62	112#	118				
MON80	F800	31#						
NTRKS	0002	26#	61					
RBOOT	1603	21#						
RBYTE	007B	35#	51	87				
READF	0004	44#	113	121				
RECAL	0003	43#						
RESET	007F	36#	58					
RMON80	FF0F	32#	81	90	95			
RSTART	3000	47#	84	98				
RTYPE	0079	34#	50	76				
STACK	0100	45#	48					
START	3015	64#	105					
TESTING	0000	9#	11	14	80	83	94	97

TRUE	FFFF	8#	
WAITO	301B	71#	73

E. A Skeletal Cold Start Loader

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1      ;THIS IS A SAMPLE COLD START LOADER, WHICH, WHEN
2      ;MODIFIED
3      ;RESIDES ON TRACK 00, SECTOR 01 (THE FIRST SECTOR ON THE
4      ;DISKETTE), WE ASSUME THAT THE CONTROLLER HAS LOADED
5      ;THIS SECTOR INTO MEMORY UPON SYSTEM START-UP (THIS
6      ;PROGRAM CAN BE KEYED-IN, OR CAN EXIST IN READ-ONLY
7      ;MEMORY
8      ;BEYOND THE ADDRESS SPACE OF THE CP/M VERSION YOU ARE
9      ;RUNNING). THE COLD START LOADER BRINGS THE CP/M SYSTEM
10     ;INTO MEMORY AT"LOADP" (3400H +"BIAS"). IN A 20K
11     ;MEMORY SYSTEM, THE VALUE OF"BIAS" IS 000H, WITH
12     ;LARGE
13     ;VALUES FOR INCREASED MEMORY SIZES (SEE SECTION 2).
14     ;AFTER
15     ;LOADING THE CP/M SYSTEM, THE COLD START LOADER
16     ;BRANCHES
17     ;TO THE "BOOT" ENTRY POINT OF THE BIOS, WHICH BEGINS AT
18     ; "BIOS" +"BIAS". THE COLD START LOADER IS NOT USED UN-
19     ;TIL THE SYSTEM IS POWERED UP AGAIN, AS LONG AS THE BIOS
20     ;IS NOT OVERWRITTEN. THE ORIGIN IS ASSUMED AT 0000H, AND
21     ;MUST BE CHANGED IF THE CONTROLLER BRINGS THE COLD START
22     ;LOADER INTO ANOTHER AREA, OR IF A READ-ONLY MEMORY
23     ;AREA
24     ;IS USED.
25 0000      ORG 0      ;BASE OF RAM IN
26            ;CP/M
27 0014 =    MSIZE EQU 20      ;MIN MEM SIZE IN
28            ;KBYTES
29 0000 =    BIAS  EQU (MSIZE-20)*1024 ;OFFSET FROM 20K
30            ;SYSTEM
31 3400 =    CCP EQU 3400H+BIAS ;BASE OF THE CCP
32 4A00 =    BIOS EQU CCP+1600H ;BASE OF THE BIOS
33 0300 =    BIOSL EQU 0300H    ;LENGTH OF THE BIOS
34 4A00 =    BOOT EQU BIOS
35 1900 =    SIZE EQU BIOS+BIOSL-CCP ;SIZE OF CP/M
36            ;SYSTEM
37 0032 =    SECTS EQU SIZE/128  ;# OF SECTORS TO LOAD
38          ;
39          ; BEGIN THE LOAD OPERATION
40
41          COLD:
42 0000 010200 LXI B,2      ;B=0, C=SECTOR 2
43 0003 1632   MVI D,SECTS  ;D=# SECTORS TO
44            ;LOAD
45 0005 210034 LXI H,CCP    ;BASE TRANSFER
```

```

46                ;ADDRESS
47    LSECT:  ;LOAD THE NEXT SECTOR
48
49                ; INSERT INLINE CODE AT THIS POINT TO
50                ; READ ONE 128 BYTE SECTOR FROM THE
51                ; TRACK GIVEN IN REGISTER B, SECTOR
52                ; GIVEN IN REGISTER C,
53                ; INTO THE ADDRESS GIVEN BY <HL>;
54                ;BRANCH TO LOCATION "COLD" IF A READ ERROR OCCURS
55                ;
56                ;
57                ;
58                ;
59                ; USER SUPPLIED READ OPERATION GOES
60                ; HERE...
61                ;
62                ;
63                ;
64                ;
65    0008 C36B00    JMP PAST$PATCH  ;REMOVE THIS
66                ;WHEN PATCHED
67    000B          DS    60H
68
69                PAST$PATCH:
70                ;GO TO NEXT SECTOR IF LOAD IS INCOMPLETE
71    006B 15       DCR D    ;SECTS=SECTS-1
72    006C CA004A   JZ    BOOT    ;HEAD. FOR THE BIOS
73
74                ; MORE SECTORS TO LOAD
75                ;
76
77                ;WE AREN'T USING A STACK, SO USE <SP> AS SCRATCH
78                ;REGISTER
79                ; TO HOLD THE LOAD ADDRESS INCREMENT
80    006F 318000   LXI SP,128    ;128 BYTES PER
81                ;SECTOR
82    0072 39       DAD SP    ;<HL> = <HL> + 128
83    0073 0C       INR C    ;SECTOR=SECTOR + 1
84    0074 79       MOV A,C
85    0075 FE1B     CPI 27      ;LAST SECTOR OF
86                ;TRACK?
87    0077 DA0800   JC  LSECT    ;NO, GO READ
88                ;ANOTHER
89
90                ;END OF TRACK, INCREMENT TO NEXT TRACK
91
92    007A 0E01     MVI C,1    ;SECTOR = 1
93    007C 04       INR B    ;TRACK = TRACK + 1
94    007D C30800   JMP LSECT  ;FOR ANOTHER GROUP
95    0080          END        ;OF BOOT LOADER

```

BIAS	0000	29#	31		
BIOS	4A00	32#	34	35	
BIOSL	0300	33#	35		
BOOT	4A00	34#	72		
CCP	3400	31#	32	35	45
COLD	0000	41#			
LSECT	0008	47#	87	94	

MSIZE	0014	27#	29
PASTPATCH	006B	65	69#
SECTS	0032	37#	43
SIZE	1900	35#	37

F. CP/M Disk Definition Library

Note

This file is intended to be included with your CBIOS to provide disk definitions.

```
;      CP/M 2.0 disk re-definition library
;
;      Copyright (c) 1979
;      Digital Research
;      Box 579
;      Pacific Grove, CA
;      93950
;
;      CP/M logical disk drives are defined using the
;      macros given below, where the sequence of calls
;      is:
;
;      disks      n
;      diskdef parameter-list-0
;      diskdef parameter-list-1
;      ...
;      diskdef parameter-list-n
;      endef
;
;      where n is the number of logical disk drives attached
;      to the CP/M system, and parameter-list-i defines the
;      characteristics of the ith drive (i=0,1,...,n-1)
;
;      each parameter-list-i takes the form
;
;          dn,fsc,lsc,[skf],bls,dks,dir,cks,ofs,[0]
;      where
;      dn      is the disk number 0,1,...,n-1
;      fsc     is the first sector number (usually 0 or 1)
;      lsc     is the last sector number on a track
;      skf     is optional "skew factor" for sector translate
;      bls     is the data block size (1024,2048,...,16384)
;      dks     is the disk size in bls increments (word)
;      dir     is the number of directory elements (word)
;      cks     is the number of dir elements to checksum
;      ofs     is the number of tracks to skip (word)
;      [0]     is an optional 0 which forces 16K/directory entry
;
;      for convenience, the form
;
;          dn,dm
;      defines disk dn as having the same characteristics as
;      a previously defined disk dm.
;
;      a standard four drive CP/M system is defined by
;
;          disks      4
;          diskdef 0,1,26,6,1024,243,64,64,2
;      dsk      set      0
```

```

;           rept      3
;       dsk      set      dsk+1
;           diskdef %dsk,0
;           endm
;       endef
;
;       the value of "begdat" at the end of assembly defines the
;       beginning of the uninitialized ram area above the bios,
;       while the value of "enddat" defines the next location
;       following the end of the data area.  the size of this
;       area is given by the value of "datsiz" at the end of the
;       assembly.  note that the allocation vector will be quite
;       large if a large disk size is defined with a small block
;       size.
;
dskhdr macro    dn
;;      define a single disk header list
dpe&dn: dw      xlt&dn,0000h      ;translate table
        dw      0000h,0000h      ;scratch area
        dw      dirbuf,dpb&dn      ;dir buff,param block
        dw      csv&dn,alv&dn      ;check, alloc vectors
        endm
;
disks macro    nd
;;      define nd disks
ndisks set      nd                ;;for later reference
dpbase equ      $                ;base of disk parameter blocks
;;      generate the nd elements
dsknxt set      0
        rept      nd
        dskhdr    %dsknxt
dsknxt set      dsknxt+1
        endm
        endm
;
dpbhdr macro    dn
dpb&dn equ      $                ;disk parm block
        endm
;
ddb macro    data,comment
;;      define a db statement
        db      data                comment
        endm
;
ddw macro    data,comment
;;      define a dw statement
        dw      data                comment
        endm
;
gcd macro    m,n
;;      greatest common divisor of m,n
;;      produces value gcdn as result
;;      (used in sector translate table generation)
gcdm set      m                ;;variable for m
gcdn set      n                ;;variable for n
gcdr set      0                ;;variable for r
        rept      65535

```

```

gcdx    set      gcdm/gcdn
gcdr    set      gcdm - gcdx*gcdn
        if      gcdr = 0
        exitm
        endif
gcdm    set      gcdn
gcdn    set      gcdr
        endm
        endm
;
diskdef macro  dn,fsc,lsc,skf,bls,dks,dir,cks,ofs,k16
;;          generate the set statements for later tables
        if      nul lsc
;;          current disk dn same as previous fsc
dpb&dn equ      dpb&fsc          ;;equivalent parameters
als&dn equ      als&fsc          ;;same allocation vector size
css&dn equ      css&fsc          ;;same checksum vector size
xlt&dn equ      xlt&fsc          ;;same translate table
        else
secmax set      lsc-(fsc)        ;;sectors 0...secmax
sectors set      secmax+1        ;;number of sectors
als&dn set      (dks)/8          ;;size of allocation vector
        if      ((dks) mod 8) ne 0
als&dn set      als&dn+1
        endif
css&dn set      (cks)/4          ;;number of checksum elements
;;          generate the block shift value
blkval set      bls/128          ;;number of sectors/block
blkshf set      0                ;;counts right 0's in blkval
blkmsk set      0                ;;fills with 1's from right
        rept    16                ;;once for each bit position
        if      blkval=1
        exitm
        endif
;;          otherwise, high order 1 not found yet
blkshf set      blkshf+1
blkmsk set      (blkmsk shl 1) or 1
blkval set      blkval/2
        endm
;;          generate the extent mask byte
blkval set      bls/1024         ;;number of kilobytes/block
extmsk set      0                ;;fill from right with 1's
        rept    16
        if      blkval=1
        exitm
        endif
;;          otherwise more to shift
extmsk set      (extmsk shl 1) or 1
blkval set      blkval/2
        endm
;;          may be double byte allocation
        if      (dks) > 256
extmsk set      (extmsk shr 1)
        endif
;;          may be optional [0] in last position
        if      not nul k16
extmsk set      k16

```

```

        endif
;;      now generate directory reservation bit vector
dirrem set    dir                ;;# remaining to process
dirbks set    bls/32             ;;number of entries per block
dirblk set    0                  ;;fill with 1's on each loop
        rept    16
        if      dirrem=0
        exitm
        endif
;;      not complete, iterate once again
;;      shift right and add 1 high order bit
dirblk set    (dirblk shr 1) or 8000h
        if      dirrem > dirbks
dirrem set    dirrem-dirbks
        else
dirrem set    0
        endif
        endm
        dpbhdr dn                ;;generate equ $
        ddw      %sectors,&lt;;sec per track&gt;;
        ddb      %blkshf,&lt;;block shift&gt;;
        ddb      %blkmsk,&lt;;block mask&gt;;
        ddb      %extmsk,&lt;;extnt mask&gt;;
        ddw      %(dks)-1,&lt;;disk size-1&gt;;
        ddw      %(dir)-1,&lt;;directory max&gt;;
        ddb      %dirblk shr 8,&lt;;alloc0&gt;;
        ddb      %dirblk and 0ffh,&lt;;alloc1&gt;;
        ddw      %(cks)/4,&lt;;check size&gt;;
        ddw      %ofs,&lt;;offset&gt;;
;;      generate the translate table, if requested
        if      nul skf
xlt&dn equ    0                  ;no xlate table
        else
        if      skf = 0
xlt&dn equ    0                  ;no xlate table
        else
;;      generate the translate table
nxtsec set    0                  ;;next sector to fill
nxtbas set    0                  ;;moves by one on overflow
        gcd      %sectors,skf
;;      gcdn = gcd(sectors,skew)
neltst set    sectors/gcdn
;;      neltst is number of elements to generate
;;      before we overlap previous elements
nelts set     neltst             ;;counter
xlt&dn equ    $                  ;translate table
        rept    sectors          ;;once for each sector
        if      sectors < 256
        ddb      %nxtsec+(fsc)
        else
        ddw      %nxtsec+(fsc)
        endif
nxtsec set    nxtsec+(skf)
        if      nxtsec >= sectors
nxtsec set    nxtsec-sectors
        endif
nelts set     nelts-1

```

```

        if      nelts = 0
nxtbas set      nxtbas+1
nxtsec set      nxtbas
nelts  set      neltst
        endif
        endm
        endif ;;end of nul fac test
        endif ;;end of nul bls test
        endm
;
defds macro lab,space
lab: ds space
        endm
;
lds macro lb,dn,val
        defds lb&dn,%val&dn
        endm
;
endif macro
;; generate the necessary ram data areas
begdat equ $
dirbuf: ds 128 ;directory access buffer
dsknxt set 0
        rept ndisks ;;once for each disk
        lds alv,%dsknxt,als
        lds csv,%dsknxt,css
dsknxt set dsknxt+1
        endm
enddat equ $
datsiz equ $-begdat
;; db 0 at this point forces hex record
        endm
;

```

G. Blocking and Deblocking Algorithms

Note

This appendix consists of a cross-reference listing generated by the XREF utility from the results of assembly with MAC.

```
1      ;*****
2      ;*
3      ;*      SECTOR DEBLOCKING ALGORITHMS FOR CP/M 2.0      *
4      ;*
5      ;*****
6      ;
7      ; UTILITY MACRO TO COMPUTE SECTOR MASK
8      SMASK MACRO HBLK
9      ;; COMPUTE LOG2(HBLK), RETURN @X AS RESULT
10     ;; (2 ** @X = HBLK ON RETURN)
11     @Y SET HBLK
12     @X SET 0
13     ;; COUNT RIGHT SHIFTS OF @Y UNTIL = 1
14     REPT 8
15     IF @Y = 1
16     EXITM
17     ENDIF
18     ;; @Y IS NOT 1, SHIFT RIGHT ONE POSITION
19     @Y SET @Y SHR 1
20     @X SET @X + 1
21     ENDM
22     ENDM
23     ;
24     ;*****
25     ;*
26     ;*      CP/M TO HOST DISK CONSTANTS      *
27     ;*
28     ;*****
29     0800 = BLKSIZ EQU 2048      ;CP/M ALLOCATION SIZE</a>
30     0200 = HSTSIZ EQU 512      ;HOST DISK SECTOR SIZE
31     0014 = HSTSPT EQU 20      ;HOST DISK SECTORS/TRK
32     0004 = HSTBLK EQU HSTSIZ/128 ;CP/M SECTS/HOST BUFF
33     0050 = CPMSPT EQU HSTBLK * HSTSPT ;CP/M SECTORS/TRACK
34     0003 = SECMSK EQU HSTBLK-1 ;SECTOR MASK
35           SMASK HSTBLK      ;COMPUTE SECTOR MASK
36     0002 = SECSHF EQU @X      ;LOG2(HSTBLK)
37     ;
38     ;*****
39     ;*
40     ;*      BDOS CONSTANTS ON ENTRY TO WRITE      *
41     ;*
42     ;*****
43     0000 = WRALL EQU 0      ;WRITE TO ALLOCATED
44     0001 = WRDIR EQU 1      ;WRITE TO DIRECTORY
45     0002 = WRUAL EQU 2      ;WRITE TO UNALLOCATED
```

```

46      ;
47      ;*****
48      ;*
49      ;* THE BDOS ENTRY POINTS GIVEN BELOW SHOW THE *
50      ;* CODE WHICH IS RELEVANT TO DEBLOCKING ONLY. *
51      ;*
52      ;*****
53      ;
54      ; DISKDEF MACRO, OR HAND CODED TABLES GO HERE
55 0000 = DPBASE EQU $ ;DISK PARAM BLOCK BASE
56      ;
57      BOOT:;</a>
58      WBOOT:
59          ;ENTER HERE ON SYSTEM BOOT TO INITIALIZE
60 0000 AF      XRA A ;0 TO ACCUMULATOR
61 0001 326A01  STA HSTACT ;HOST BUFFER INACTIVE
62 0004 326C01  STA UNACNT ;CLEAR UNALLOC COUNT
63 0007 C9      RET
64      ;
65      HOME:;</a>
66          ;HOME THE SELECTED DISK
67      HOME:
68 0008 3A6B01  LDA HSTWRT ;CHECK FOR PENDING WRITE
69 000B B7      ORA A
70 000C C21200  JNZ HOMED
71 000F 326A01  STA HSTACT ;CLEAR HOST ACTIVE FLAG
72      HOMED:
73 0012 C9      RET
74      ;
75      SELDSK:
76          ;SELECT DISK
77 0013 79      MOV A,C ;SELECTED DISK NUMBER
78 0014 326101  STA SEKDSK ;SEEK DISK NUMBER
79 0017 6F      MOV L,A ;DISK NUMBER TO HL
80 0018 2600    MVI H,0
81          REPT 4 ;MULTIPLY BY 16
82          DAD H
83          ENDM
84 001A+29      DAD H
85 001B+29      DAD H
86 001C+29      DAD H
87 001D+29      DAD H
88 001E 110000  LXI D,DPBASE ;BASE OF PARM BLOCK
89 0021 19      DAD D ;HL=.DPB(CURDSK)
90 0022 C9      RET
91      ;
92      SETTRK:
93          ;SET TRACK GIVEN BY REGISTERS BC
94 0023 60      MOV H,B
95 0024 69      MOV L,C
96 0025 226201  SHLD SEKTRK ;TRACK TO SEEK
97 0028 C9      RET
98      ;
99      SETSEC:
100          ;SET SECTOR GIVEN BY REGISTER C
101 0029 79      MOV A,C
102 002A 326401  STA SEKSEC ;SECTOR TO SEEK

```

```

103 002D C9      RET
104              ;
105              SETDMA:
106              ;SET DMA ADDRESS GIVEN BY BC
107 002E 60      MOV H,B
108 002F 69      MOV L,C
109 0030 227501  SHLD DMAADR
110 0033 C9      RET</a>
111              ;
112              SECTTRAN:
113              ;TRANSLATE SECTOR NUMBER BC
114 0034 60      MOV H,B
115 0035 69      MOV L,C
116 0036 C9      RET
117              ;
118              ;*****
119              ;*
120              ;* THE READ ENTRY POINT TAKES THE PLACE OF *
121              ;* THE PREVIOUS BIOS DEFINITION FOR READ. *
122              ;*
123              ;*****
124              READ:
125              ;READ THE SELECTED CP/M SECTOR</a>
126 0037 AF      XRA A
127 0038 326C01  STA UNACNT
128 003B 3E01    MVI A,1
129 003D 327301  STA READOP ;READ OPERATION
130 0040 327201  STA RSFLAG ;MUST READ DATA
131 0043 3E02    MVI A,WRUAL
132 0045 327401  STA WRTYPE ;TREAT AS UNALLOC
133 0048 C3B600  JMP RWOPER ;TO PERFORM THE READ
134              ;
135              ;*****
136              ;*
137              ;* THE WRITE ENTRY POINT TAKES THE PLACE OF *
138              ;* THE PREVIOUS BIOS DEFINITION FOR WRITE. *
139              ;*
140              ;*****
141              WRITE:
142              ;WRITE THE SELECTED CP/M SECTOR
143 004B AF      XRA A ;0 TO ACCUMULATOR
144 004C 327301  STA READOP ;NOT A READ OPERATION
145 004F 79      MOV A,C ;WRITE TYPE IN C
146 0050 327401  STA WRTYPE
147 0053 FE02    CPI WRUAL ;WRITE UNALLOCATED?
148 0055 C26F00  JNZ CHKUNA ;CHECK FOR UNALLOC
149              ;
150              ; WRITE TO UNALLOCATED, SET PARAMETERS
151 0058 3E10    MVI A,BLKSIZ/128 ;NEXT UNALLOC RECS
152 005A 326C01  STA UNACNT
153 005D 3A6101  LDA SEKDSK ;DISK TO SEEK
154 0060 326D01  STA UNADSK ;UNADSK = SEKDSK
155 0063 2A6201  LHLD SEKTRK
156 0066 226E01  SHLD UNATRK ;UNATRK = SECTRK
157 0069 3A6401  LDA SEKSEC
158 006C 327001  STA UNASEC ;UNASEC = SEKSEC
159              ;

```

```

160          CHKUNA:
161          ;CHECK FOR WRITE TO UNALLOCATED SECTOR
162 006F 3A6C01 LDA UNACNT      ;ANY UNALLOC REMAIN?
163 0072 B7     ORA A
164 0073 CAAE00 JZ  ALLOC      ;SKIP IF NOT
165          ;
166          ; MORE UNALLOCATED RECORDS REMAIN
167 0076 3D     DCR A      ;UNACNT = UNACNT-1
168 0077 326C01 STA UNACNT
169 007A 3A6101 LDA SEKDSK      ;SAME DISK?
170 007D 216D01 LXI H,UNADSK
171 0080 BE     CMP M      ;SEKDSK = UNADSK?
172 0081 C2AE00 JNZ ALLOC      ;SKIP IF NOT
173          ;
174          ; DISKS ARE THE SAME
175 0084 216E01 LXI H,UNATRK
176 0087 CD5301 CALL  SEKTRKCMPL ;SEKTRK = UNATRK?
177 008A C2AE00 JNZ ALLOC      ;SKIP IF NOT
178          ;
179          ; TRACKS ARE THE SAME
180 008D 3A6401 LDA SEKSEC      ;SAME SECTOR?
181 0090 217001 LXI H,UNASEC
182 0093 BE     CMP M      ;SEKSEC = UNASEC?
183 0094 C2AE00 JNZ ALLOC      ;SKIP IF NOT
184          ;
185          ; MATCH, MOVE TO NEXT SECTOR FOR FUTURE REF
186 0097 34     INR M      ;UNASEC = UNASEC+1
187 0098 7E     MOV A,M      ;END OF TRACK?
188 0099 FE50   CPI CPMSPT    ;COUNT CP/M SECTORS
189 009B DAA700 JC  NOOVF     ;SKIP IF NO OVERFLOW
190          ;
191          ; OVERFLOW TO NEXT TRACK
192 009E 3600   MVI M,0      ;UNASEC = 0
193 00A0 2A6E01 LHLD UNATRK
194 00A3 23     INX H
195 00A4 226E01 SHLD UNATRK    ;UNATRK = UNATRK+1
196          ;
197          NOOVF:
198          ;MATCH FOUND, MARK AS UNNECESSARY READ
199 00A7 AF     XRA A      ;0 TO ACCUMULATOR
200 00A8 327201 STA RSFLAG     ;RSFLAG = 0
201 00AB C3B600 JMP RWOPER      ;TO PERFORM THE WRITE
202          ;
203          ALLOC:
204          ;NOT AN UNALLOCATED RECORD, REQUIRES PRE-READ
205 00AE AF     XRA A      ;0 TO ACCUM
206 00AF 326C01 STA UNACNT     ;UNACNT = 0
207 00B2 3C     INR A      ;1 TO ACCUM
208 00B3 327201 STA RSFLAG     ;RSFLAG = 1
209          ;
210          ;*****
211          ;*
212          ;*  COMMON CODE FOR READ AND WRITE FOLLOWS
213          ;*
214          ;*****
215          RWOPER:
216          ;ENTER HERE TO PERFORM THE READ/WRITE

```

```

217 00B6 AF      XRA A      ;ZERO TO ACCUM
218 00B7 327101 STA ERFLAG  ;NO ERRORS (YET)
219 00BA 3A6401 LDA SEKSEC  ;COMPUTE HOST SECTOR
220              REPT SECSHF
221              ORA A      ;CARRY = 0
222              RAR        ;SHIFT RIGHT
223              ENDM
224 00BD+B7      ORA A      ;CARRY = 0
225 00BE+1F      RAR        ;SHIFT RIGHT
226 00BF+B7      ORA A      ;CARRY = 0
227 00C0+1F      RAR        ;SHIFT RIGHT
228 00C1 326901 STA SEKHST  ;HOST SECTOR TO SEEK
229              ;
230              ; ACTIVE HOST SECTOR?
231 00C4 216A01 LXI H,HSTACT ;HOST ACTIVE FLAG
232 00C7 7E      MOV A,M
233 00C8 3601    MVI M,1    ;ALWAYS BECOMES 1
234 00CA B7      ORA A      ;WAS IT ALREADY?
235 00CB CAF200 JZ  FILHST  ;FILL HOST IF NOT
236              ;
237              ; HOST BUFFER ACTIVE, SAME AS SEEK BUFFER?
238 00CE 3A6101 LDA SEKDSK
239 00D1 216501 LXI H,HSTDSK ;SAME DISK?
240 00D4 BE      CMP M      ;SEKDSK = HSTDSK?
241 00D5 C2EB00 JNZ NOMATCH
242              ;
243              ; SAME DISK, SAME TRACK?
244 00D8 216601 LXI H,HSTTRK
245 00DB CD5301 CALL SEKTRCMP ;SEKTRK = HSTTRK?
246 00DE C2EB00 JNZ NOMATCH
247              ;
248              ; SAME DISK, SAME TRACK, SAME BUFFER?
249 00E1 3A6901 LDA SEKHST
250 00E4 216801 LXI H,HSTSEC ;SEKHST = HSTSEC?
251 00E7 BE      CMP M
252 00E8 CA0F01 JZ  MATCH  ;SKIP IF MATCH
253              ;
254              NOMATCH:
255              ;PROPER DISK, BUT NOT CORRECT SECTOR
256 00EB 3A6B01 LDA HSTWRT  ;HOST WRITTEN?
257 00EE B7      ORA A
258 00EF C45F01 CNZ WRITEHST ;CLEAR HOST BUFF
259              ;
260              FILHST:
261              ;MAY HAVE TO FILL THE HOST BUFFER
262 00F2 3A6101 LDA SEKDSK
263 00F5 326501 STA HSTDSK
264 00F8 2A6201 LHLD SEKTRK
265 00FB 226601 SHLD HSTTRK
266 00FE 3A6901 LDA SEKHST
267 0101 326801 STA HSTSEC
268 0104 3A7201 LDA RSFLAG  ;NEED TO READ?
269 0107 B7      ORA A
270 0108 C46001 CNZ READHST  ;YES, IF 1
271 010B AF      XRA A      ;0 TO ACCUM
272 010C 326B01 STA HSTWRT  ;NO PENDING WRITE
273              ;

```

```

274 MATCH:
275 ;COPY DATA TO OR FROM BUFFER
276 010F 3A6401 LDA SEKSEC ;MASK BUFFER NUMBER
277 0112 E603 ANI SECMSK ;LEAST SIGNIF BITS
278 0114 6F MOV L,A ;READY TO SHIFT
279 0115 2600 MVI H,0 ;DOUBLE COUNT
280 REPT 7 ;SHIFT LEFT 7
281 DAD H
282 ENDM
283 0117+29 DAD H
284 0118+29 DAD H
285 0119+29 DAD H
286 011A+29 DAD H
287 011B+29 DAD H
288 011C+29 DAD H
289 011D+29 DAD H
290 ; HL HAS RELATIVE HOST BUFFER ADDRESS
291 011E 117701 LXI D,HSTBUF
292 0121 19 DAD D ;HL = HOST ADDRESS
293 0122 EB XCHG ;NOW IN DE
294 0123 2A7501 LHLD DMAADR ;GET/PUT CP/M DATA
295 0126 0E80 MVI C,128 ;LENGTH OF MOVE
296 0128 3A7301 LDA READOP ;WHICH WAY?
297 012B B7 ORA A
298 012C C23501 JNZ RWMOVE ;SKIP IF READ
299 ;
300 ; WRITE OPERATION, MARK AND SWITCH DIRECTION
301 012F 3E01 MVI A,1
302 0131 326B01 STA HSTWRT ;HSTWRT = 1
303 0134 EB XCHG ;SOURCE/DEST SWAP
304 ;
305 RWMOVE:
306 ;C INITIALLY 128, DE IS SOURCE, HL IS DEST
307 0135 1A LDAX D ;SOURCE CHARACTER
308 0136 13 INX D
309 0137 77 MOV M,A ;TO DEST
310 0138 23 INX H
311 0139 0D DCR C ;LOOP 128 TIMES
312 013A C23501 JNZ RWMOVE
313 ;
314 ; DATA HAS BEEN MOVED TO/FROM HOST BUFFER
315 013D 3A7401 LDA WRTYPE ;WRITE TYPE
316 0140 FE01 CPI WRDIR ;TO DIRECTORY?
317 0142 3A7101 LDA ERFLAG ;IN CASE OF ERRORS
318 0145 C0 RNZ ;NO FURTHER PROCESSING
319 ;
320 ; CLEAR HOST BUFFER FOR DIRECTORY WRITE
321 0146 B7 ORA A ;ERRORS?
322 0147 C0 RNZ ;SKIP IF SO
323 0148 AF XRA A ;0 TO ACCUM
324 0149 326B01 STA HSTWRT ;BUFFER WRITTEN
325 014C CD5F01 CALL WRITEHST
326 014F 3A7101 LDA ERFLAG
327 0152 C9 RET
328 ;
329 ;*****
330 ;*

```

*

```

331      ;*   UTILITY SUBROUTINE FOR 16-BIT COMPARE      *
332      ;*                                              *
333      ;*****
334      SEKTRKCOMP:
335          ;HL = .UNATRK OR .HSTTRK, COMPARE WITH SEKTRK
336      0153 EB      XCHG
337      0154 216201  LXI H,SEKTRK
338      0157 1A      LDAX D      ;LOW BYTE COMPARE
339      0158 BE      CMP M      ;SAME?
340      0159 C0      RNZ      ;RETURN IF NOT
341      ; LOW BYTES EQUAL, TEST HIGH 1S
342      015A 13      INX D
343      015B 23      INX H
344      015C 1A      LDAX D
345      015D BE      CMP M ;SETS FLAGS
346      015E C9      RET
347      ;
348      ;*****
349      ;*                                              *
350      ;*   WRITEHST PERFORMS THE PHYSICAL WRITE TO      *
351      ;*   THE HOST DISK, READHST READS THE PHYSICAL  *
352      ;*   DISK.                                     *
353      ;*                                              *
354      ;*****
355      WRITEHST:
356          ;HSTDSK = HOST DISK #, HSTTRK = HOST TRACK #,
357          ;HSTSEC = HOST SECT #. WRITE "HSTSIZ" BYTES
358          ;FROM HSTBUF AND RETURN ERROR FLAG IN ERFLAG.
359          ;RETURN ERFLAG NON-ZERO IF ERROR
360      015F C9      RET
361      ;
362      READHST:
363          ;HSTDSK = HOST DISK #, HSTTRK = HOST TRACK #,
364          ;HSTSEC = HOST SECT #. READ "HSTSIZ" BYTES
365          ;INTO HSTBUF AND RETURN ERROR FLAG IN ERFLAG.
366      0160 C9      RET
367      ;
368      ;*****
369      ;*                                              *
370      ;*   UNITIALIZED RAM DATA AREAS                  *
371      ;*                                              *
372      ;*****
373      ;
374      0161      SEKDSK: DS 1      ;SEEK DISK NUMBER
375      0162      SEKTRK: DS 2     ;SEEK TRACK NUMBER
376      0164      SEKSEC: DS 1     ;SEEK SECTOR NUMBER
377      ;
378      0165      HSTDSK: DS 1     ;HOST DISK NUMBER
379      0166      HSTTRK: DS 2     ;HOST TRACK NUMBER
380      0168      HSTSEC: DS 1     ;HOST SECTOR NUMBER
381      ;
382      0169      SEKHST: DS 1     ;SEEK SHR SEC SHF
383      016A      HSTACT: DS 1     ;HOST ACTIVE FLAG
384      016B      HSTWRT: DS 1     ;HOST WRITTEN FLAG
385      ;
386      016C      UNACNT: DS 1     ;UNALLOC REC CNT
387      016D      UNADSK: DS 1     ;LAST UNALLOC DISK

```

```

388 016E      UNATRK: DS 2 ;LAST UNALLOC TRACK
389 0170      UNASEC: DS 1 ;LAST UNALLOC SECTOR
390           ;
391 0171      ERFLAG: DS 1 ;ERROR REPORTING
392 0172      RSFLAG: DS 1 ;READ SECTOR FLAG
393 0173      READOP: DS 1 ;1 IF READ OPERATION
394 0174      WRTYPE: DS 1 ;WRITE OPERATION TYPE
395 0175      DMAADR: DS 2 ;LAST DMA ADDRESS
396 0177      HSTBUF: DS HSTSIZ ;HOST BUFFER
397           ;
398           ;*****
399           ;*
400           ;* THE ENDEF MACRO INVOCATION GOES HERE *
401           ;*
402           ;*****
403 0377      END

ALLOC        00AE 164 172 177 183 203#
BLKSIZ       0800 29# 151
BOOT         0000 57#
CHKUNA       006F 148 160#
CPMSPT       0050 33# 188
DMAADR       0175 109 294 395#
DPBASE       0000 55# 88
ERFLAG       0171 218 317 326 391#
FILHST       00F2 235 260#
HOME         0008 65# 67#
HOMED        0012 70 72#
HSTACT       016A 61 71 231 383#
HSTBLK       0004 32# 33 34 35
HSTBUF       0177 291 396#
HSTDISK      0165 239 263 378#
HSTSEC       0168 250 267 380#
HSTSIZ       0200 30# 32 396
HSTSPT       0014 31# 33
HSTTRK       0166 244 265 379#
HSTWRT       016B 68 256 272 302 324 384#
MATCH        010F 252 274#
NOMATCH      00EB 241 246 254#
NOOVF        00A7 189 197#
READ         0037 124#
READHST      0160 270 362#
READOP       0173 129 144 296 393#
RSFLAG       0172 130 200 208 268 392#
RWMOVE       0135 298 305# 312
RWOPER       00B6 133 201 215#
SECMSK       0003 34# 277
SECSHF       0002 36# 220
SECTRAN      0034 112#
SEKDSK       0161 78 153 169 238 262 374#
SEKHST       0169 228 249 266 382#
SEKSEC       0164 102 157 180 219 276 376#
SEKTRK       0162 96 155 264 337 375#
SEKTRKCOMP   0153 176 245 334#
SELDSK       0013 75#
SETDMA       002E 105#
SETSEC       0029 99#
SETTRK       0023 92#

```

UNACNT	016C	62	127	152	162	168	206	386#
UNADSK	016D	154	170	387#				
UNASEC	0170	158	181	389#				
UNATRK	016E	156	175	193	195	388#		
WBOOT	0000	58#						
WRALL	0000	43#						
WRDIR	0001	44#	316					
WRITE	004B	141#						
WRITEHST	015F	258	325	355#				
WRTYPE	0174	132	146	315	394#			
WRUAL	0002	45#	131	147				

H. Glossary

address Number representing the location of a byte in memory. Within CP/M there are two kinds of addresses: **logical** and **physical**. A **physical** address refers to an absolute and unique location within the computer's memory space. A **logical** address refers to the offset or displacement of a byte in relation to a base location. A standard CP/M program is loaded at address 0100H, the base value; the first instruction of a program has a **physical** address of 0100H and a **logical** address or offset of 0000H.

allocation vector (ALV) An allocation vector is maintained in the BIOS for each logged-in disk drive. A vector consists of a string of bits, one for each block on the drive. The bit corresponding to a particular block is set to one when the block has been allocated and to zero otherwise. The first two bytes of this vector are initialized with the bytes AL0 and AL1 on, thus allocating the directory blocks. CP/M Function 27 returns the allocation vector address.

AL0, AL1 Two bytes in the disk parameter block that reserve data blocks for the directory. These two bytes are copied into the first two bytes of the allocation vector when a drive is logged in. See **allocation vector**.

ALV See **allocation vector**

ambiguous filename Filename that contains either of the CP/M wildcard characters, ? or *, in the primary filename, filetype, or both. When you replace characters in a filename with these wildcard characters, you create an ambiguous filename and can easily reference more than one CP/M file in a single command line.

American Standard Code for Information Interchange See **ASCII**

applications program Program designed to solve a specific problem. Typical applications programs are business accounting packages, word processing (editing) programs and mailing list programs.

archive attribute File attribute controlled by the high-order bit of the 13 byte (FCB + 11) in a directory element. This attribute is set if the file has been archived.

argument Symbol, usually a letter, indicating a place into which you can substitute a number, letter, or name to give an appropriate meaning to the formula in question.

ASCII American Standard Code for Information Interchange. ASCII is a standard set of seven-bit numeric character codes used to represent characters in memory. Each character requires one byte of memory with the high-order bit usually set to zero. Characters can be numbers, letters, and symbols. An ASCII file can be intelligibly displayed on the video screen or printed on paper.

assembler Program that translates assembly language into the binary machine code. Assembly language is simply a set of mnemonics used to designate the instruction set of the CPU. See **ASM** in Section 3 of this manual.

back-up Copy of a disk or file made for safekeeping, or the creation of the duplicate disk or file.

Basic Disk Operating System See **BDOS**.

BDOS Basic Disk Operating System. The BDOS module of the CP/M operating system provides an interface for a user program to the operating. This interface is in the form of a set of function calls which may be made to the BDOS through calls to location 0005H in page zero. The user program specifies the number of the desired function in register C. User programs running under CP/M should use BDOS functions for all I/O operations to remain compatible with other CP/M systems and future releases. The BDOS normally resides in high memory directly below the BIOS.

bias Address value which when added to the origin address of your BIOS module produces 1F80H, the address of the BIOS module in the MOVCPM image. There is also a bias value that when added to the BOOT module origin produces 0900H, the address of the BOOT module in the MOVCPM image. You must use these bias values with the R command under DDT or SID when you patch a CP/M system. If you do not, the patched system may fall to function.

binary Base 2 numbering system. A binary digit can have one of two values: 0 or 1. Binary numbers are used in computers because the hardware can most easily exhibit two states: off and on. Generally, a bit in memory represents one binary digit.

Basic Input/Output System See **BIOS**.

BIOS Basic Input/Output System. The BIOS is the only hardware-dependent module of the CP/M system. It provides the BDOS with a set of primitive I/O operations. The BIOS is an assembly language module usually written by the user, hardware manufacturer, or independent software vendor, and is the key to CP/M's portability. The BIOS interfaces the CP/M system to its hardware environment through a standardized jump table at the front of the BIOS routine and through a set of disk parameter tables which define the disk environment. Thus, the BIOS provides CP/M with a completely table-driven I/O system.

BIOS base Lowest address of the BIOS module in memory, that by definition must be the first entry point in the BIOS jump table.

bit Switch in memory that can be set to on (1) or off (0). Bits are grouped into **bytes**, eight bits to a byte, which is the smallest directly addressable unit in an Intel 8080 or Zilog Z80.

By common convention, the bits in a byte are numbered from right, 0 for the low-order bit, to left, 7 for the high-order bit. Bit values are often represented in hexadecimal notation by grouping the bits from the low-order bit in groups of four. Each group of four bits can have a value from 0 to 15 and thus can easily be represented by one hexadecimal digit.

BLM See **Block Mask**.

block Basic unit of disk space allocation. Each disk drive has a fixed block size (BLS) defined in its **disk parameter block** in the BIOS. A block can consist of 1K, 2K, 4K, 8K, or 16K consecutive bytes. Blocks are numbered relative to zero so that each block is unique and has a byte displacement in a file equal to the block number times the block size.

block mask (BLM) Byte value in the disk parameter block at DPB + 3. The block mask is always one less than the number of 128 byte sectors that are in one block. Note that

$$BLM = (2^{BSH}) - 1$$

block shift (BSH) Byte parameter in the disk parameter block at DPB + 2. Block shift and block mask (BLM) values are determined by the block size (BLS). Note that

$$BLM = (2^{BSH}) - 1$$

blocking and deblocking algorithm In some disk subsystems the disk sector size is larger than 128 bytes, usually 256, 512, 1024, or 2048 bytes. When the host sector size is larger than 128 bytes, host sectors must be buffered in memory and the 128-byte CP/M sectors must be blocked and deblocked by adding an additional module, the blocking and deblocking algorithm, between the BIOS disk I/O routines and the actual disk I/O. The host sector size must be an even multiple of 128 bytes for the

algorithm to work correctly. The blocking and deblocking algorithm allows the BDOS and BIOS to function exactly as if the entire disk consisted only of 128-byte sectors, as in the standard CP/M installation.

BLS Block size in bytes. See **block**.

boot Process of loading an operating system into memory. A boot program is a small piece of code that is automatically executed when you power-up or reset your computer. The boot program loads the rest of the operating system into memory in a manner similar to a person pulling himself up by his own bootstraps. This process is sometimes called a cold boot or cold start. Bootstrap procedures vary from system to system. The boot program must be customized for the memory size and hardware environment that the operating system manages. Typically, the boot resides on the first sector of the system tracks on your system disk. When executed, the boot loads the remaining sectors of the system tracks into high memory at the location for which the CP/M system has been configured. Finally, the boot transfers execution to the boot entry point in the BIOS jump table so that the system can initialize itself. In this case, the boot program should be placed at 0900H in the SYSGEN image. Alternatively, the boot program may be located in ROM.

bootstrap See **boot**.

BSH See **block shift**

BTREE General purpose file access method that has become the standard organization for indexes in large data base systems. BTREE provides near optimum performance over the full range of file operations, such as insertion, deletion, search, and search next.

buffer Area of memory that temporarily stores data during the transfer of information.

built-in commands Commands that permanently reside in memory. They respond quickly because they are not accessed from a disk.

byte Unit of memory or disk storage containing eight bits. A byte can represent a binary number between 0 and 255, and is the smallest unit of memory that can be addressed directly in 8-bit CPUs such as the Intel 8080 or Zilog Z80.

CCP Console Command Processor. The CCP is a module of the CP/M operating system. It is loaded directly below the BDOS module and interprets and executes commands typed by the console user. Usually these commands are programs that the CCP loads and calls. Upon completion, a command program may return control to the CCP if it has not overwritten it. If it has, the program can reload the CCP into memory by a warm boot operation initiated by either a jump to zero, BDOS system reset (Function 0), or a cold boot. Except for its location in high memory, the CCP works like any other standard CP/M program; that is, it makes only BDOS function calls for its I/O operations.

CCP base Lowest address of the CCP module in memory. This term sometimes refers to the base of the CP/M system in memory, as the CCP is normally the lowest CP/M module in high memory.

checksum vector (csv) Contiguous data area in the BIOS, with one byte for each directory sector to be checked, that is, **CKS** bytes. See **CKS**. A checksum vector is initialized and maintained for each logged-in drive. Each directory access by the system results in a checksum calculation that is compared with the one in the checksum vector. If there is a discrepancy, the drive is set to Read-Only status. This feature prevents the user from inadvertently switching disks without logging in the new disk. If the new disk is not logged-in, it is treated the same as the old one, and data on it might be destroyed if writing is done.

CKS Number of directory records to be checked summed on directory accesses. This is a parameter in the disk parameter block located in the BIOS. If the value of **CKS** is zero, then no directory records are checked. **CKS** is also a parameter in the `diskdef` macro library, where it is the actual number of directory elements to be checked rather than the number of directory records.

cold boot See **boot**. Cold boot also refers to a jump to the boot entry point in the BIOS jump table.

COM Filetype for a CP/M command file. See **command file**.

command CP/M command line. In general, a CP/M command line has three parts: the **command keyword**, **command tail**, and a **carriage return**. To execute a command, enter a CP/M command line directly after the CP/M prompt at the console and press the carriage return or enter key.

command file Executable program file of filetype **COM**. A command file is a machine language object module ready to be loaded and executed at the absolute address of `0100H`. To execute a command file, enter its primary filename as the command keyword in a CP/M command line.

command keyword Name that identifies a CP/M command, usually the primary filename of a file of type **COM**, or a built-in command. The command keyword precedes the command tail and the carriage return in the command line.

command syntax Statement that defines the correct way to enter a command. The correct structure generally includes the **command keyword**, the **command tail**, and a **carriage return**. A syntax line usually contains symbols that you should replace with actual values when you enter the command.

command tail Part of a command that follows the command keyword in the command line. The command tail can include a drive specification, a filename and filetype, and options or parameters. Some commands do not require a command tail.

CON: Mnemonic that represents the CP/M console device. For example, the CP/M command `PIP CON:=TEST.SUB` displays the file `TEST.SUB` on the console device. The explanation of the **STAT** command tells how to assign the logical device **CON:** to various physical devices. See **console**.

concatenate Name of the **PIP** operation that copies two or more separate files into one new file in the specified sequence.

concurrency Execution of two processes or operations simultaneously.

CONIN: BIOS entry point to a routine that reads a character from the console device.

CONOUT: BIOS entry point to a routine that sends a character to the console device.

console Primary input/output device. The console consists of a listing device, such as a screen or teletype, and a keyboard through which the user communicates with the operating system or applications program.

Console Command Processor See **CCP**

CONST: BIOS entry point to a routine that returns the status of the console device.

control character Non-printing character combination. CP/M interprets some control characters as simple commands such as line editing functions. To enter a control character, hold down the **CONTROL** key and strike the specified character key.

Control Program for Microcomputers See **CP/M**

CP/M Control Program for Microcomputers. An operating system that manages computer resources and provides a standard systems interface to software written for a large variety of microprocessor-based computer systems.

CP/M 1.4 compatibility For a CP/M 2 system to be able to read correctly single-density disks produced under a CP/M 1.4 system, the extent mask must be zero and the block size 1K. This is because under CP/M 2 an FCB may contain more than one extent. The number of extents that may be contained by an FCB is $EXM + 1$. The issue of CP/M 1.4 compatibility also concerns random file I/O. To perform random file I/O under CP/M 1.4, you must maintain an FCB for each extent of the file. This scheme is upward compatible with CP/M 2 for files not exceeding 512K bytes, the largest file size supported under CP/M 1.4. If you wish to implement random I/O for files larger than 512K bytes under CP/M 2, you must use the random read and random write functions, BDOS functions 33, 34, and 36. In this case, only one FCB is used, and if CP/M 1.4 compatibility is required, the program must use the return version number function, BDOS Function 12, to determine which method to employ.

CP/M prompt Characters that indicate that CP/M is ready to execute your next command. The CP/M prompt consists of an upper-case letter, A-P, followed by a > character; for example, A>. The letter designates which drive is currently logged in as the default drive. CP/M will search this drive for the command file specified, unless the command is a built-in command or prefaced by a select drive command: for example, B:STAT.

CP/NET Digital Research network operating system enabling microcomputers to obtain access to common resources via a network. CP/NET consists of MP/M masters and CP/M slaves with a network interface between them.

CSV See checksum vector.

cursor One-character symbol that can appear anywhere on the console screen. The cursor indicates the position where the next keystroke at the console will have an effect.

data file File containing information that will be processed by a program.

deblocking See blocking & deblocking algorithm.

default Currently selected disk drive and user number. Any command that does not specify a disk drive or a user number references the default disk drive and user number. When CP/M is first invoked, the default disk drive is drive A, and the default user number is 0.

default buffer Default 128-byte buffer maintained at 0080H in page zero. When the CCP loads a COM file, this buffer is initialized to the command tail; that is, any characters typed after the COM file name are loaded into the buffer. The first byte at 0080H contains the length of the command tail, while the command tail itself begins at 0081H. The command tail is terminated by a byte containing a binary zero value. The I command under DDT and SID initializes this buffer in the same way as the CCP.

default FCB Two default FCBs are maintained by the CCP at 005CH and 006CH in page zero. The first default FCB is initialized from the first delimited field in the command tail. The second default FCB is initialized from the next field in the command tail.

delimiter Special characters that separate different items in a command line; for example, a colon separates the drive specification from the filename. The CCP recognizes the following characters as delimiters: . : = ; < > _ , blank, and carriage return. Several CP/M commands also treat the following as delimiter characters: , [] () \$. It is advisable to avoid the use of delimiter characters and lower-case characters in CP/M filenames.

DIR: Parameter in the `diskdef` macro library that specifies the number of directory elements on the drive.

DIR attribute File attribute. A file with the `DIR` attribute can be displayed by a `DIR` command. The file can be accessed from the default user number and drive only.

DIRBUF: 128-byte scratchpad area for directory operations, usually located at the end of the BIOS. `DIRBUF` is used by the BDOS during its directory operations. `DIRBUF` also refers to the two-byte address of this scratchpad buffer in the disk parameter header at `DPbase + 8` bytes.

directory Portion of a disk that contains entries for each file on the disk. In response to the `DIR` command, CP/M displays the filenames stored in the directory. The directory also contains the locations of the blocks allocated to the files. Each file directory element is in the form of a 32-byte FCB, although one file can have several elements, depending on its size. The maximum number of directory elements supported is specified by the drive's disk parameter block value for **DRM**.

directory element Data structure. Each file on a disk has one or more 32-byte directory elements associated with it. There are four directory elements per directory sector. Directory elements can also be referred to as directory FCBs.

directory entry File entry displayed by the `DIR` command. Sometimes this term refers to a physical directory element.

disk, diskette Magnetic media used for mass storage in a computer system. Programs and data are recorded on the disk in the same way music can be recorded on cassette tape. The CP/M operating system must be initially loaded from disk when the computer is turned on. Diskette refers to smaller capacity removable floppy diskettes, while disk may refer to either a diskette, removable cartridge disk, or fixed hard disk. Hard disk capacities range from five to several hundred megabytes of storage.

diskdef macro library Library of code that when used with `MAC`, the Digital Research macro assembler, creates disk definition tables such as the **DPB** and **DPH** automatically.

disk drive Peripheral device that reads and writes information on disk. CP/M assigns a letter to each drive under its control. For example, CP/M may refer to the drives in a four-drive system as A, B, C, and D.

disk parameter block (DPB) Data structure referenced by one or more disk parameter headers. The disk parameter block defines disk characteristics in the fields listed below:

SPT is the total number of sectors per track.

BSH is the data allocation block shift factor.

BLM is the data allocation block mask.

EXM is the extent mask determined by BLS and DSM.

DSM is the maximum data block number.

DRM is the maximum number of directory entries-1.

AL0 reserves directory blocks.

AL1 reserves directory blocks.

CKS is the number of directory sectors check summed.

OFF is the number of reserved system tracks.

The address of the disk parameter block is located in the disk parameter header at `DPbase + 0AH`. CP/M Function 31 returns the **DPB** address. Drives with the same characteristics can use the same disk parameter header, and thus the same **DPB**. However, drives with different characteristics must each

have their own disk parameter header and disk parameter blocks. When the BDOS calls the `SELDSK` entry point in the BIOS, `SELDSK` must return the address of the drive's disk parameter header in register HL.

disk parameter header (DPH) Data structure that contains information about the disk drive and provides a scratchpad area for certain BDOS operations. The disk parameter header contains six bytes of scratchpad area for the BDOS, and the following five 2-byte parameters:

XLT is the sector translation table address.

DIRBUF is the directory buffer address.

DPB is the disk parameter block address.

CSV is the checksum vector address.

ALV is the allocation vector address.

Given n disk drives, the disk parameter headers are arranged in a table whose first row of 16 bytes corresponds to drive 0, with the last row corresponding to drive $n - 1$.

DKS Parameter in the `diskdef` macro library specifying the number of data blocks on the drive.

DMA Direct Memory Access. DMA is a method of transferring data from the disk into memory directly. In a CP/M system, the BDOS calls the BIOS entry point `READ` to read a sector from the disk into the currently selected DMA address. The DMA address must be the address of a 128-byte buffer in memory, either the default buffer at `0080H` in page zero, or a user-assigned buffer in the TPA. Similarly, the BDOS calls the BIOS entry point `WRITE` to write the record at the current DMA address to the disk.

DN Parameter in the `diskdef` macro library specifying the logical drive number.

DPB See disk parameter block.

DPH See disk parameter header.

DRM 2-byte parameter in the disk parameter block at `DPB + 7`. **DRM** is one less than the total number of directory entries allowed for the drive. This value is related to **DPB** bytes **AL0** and **AL1**, which allocates up to 16 blocks for directory entries.

DSM 2-byte parameter of the disk parameter block at `DPB + 5`. **DSM** is the maximum data block number supported by the drive. The product **BLS** times (**DSM**+1) is the total number of bytes held by the drive. This must not exceed the capacity of the physical disk less the reserved system tracks.

editor Utility program that creates and modifies text files. An editor can be used for creation of documents or creation of code for computer programs. The CP/M editor is invoked by typing the command `ED` next to the system prompt on the console.

EX Extent number field in an FCB. See extent.

executable Ready to be run by the computer. Executable code is a series of instructions that can be carried out by the computer. For example, the computer cannot execute names and addresses, but it can execute a program that prints all those names and addresses on mailing labels.

execute a program Start the processing of executable code.

EXM See extent mask.

extent 16K consecutive bytes in a file. Extents are numbered from 0 to 31. One extent can contain 1, 2, 4, 8, or 16 blocks. **EX** is the extent number field of an FCB and is a one-byte field at `FCB + 12`, where `FCB` labels the first byte in the FCB. Depending on the block size (**BLS**) and the maximum data block

number (**DSM**), an FCB can contain 1, 2, 4, 8, or 16 extents. The **EX** field is normally set to 0 by the user but contains the current extent number during file I/O. The term FCB folding describes FCBs containing more than one extent. In CP/M version 1.4, each FCB contained only one extent. Users attempting to perform random record I/O and maintain CP/M 1.4 compatibility should be aware of the implications of this difference. See CP/M 1.4 compatibility

extent mask (EXM) A byte parameter in the disk parameter block located at DPB + 3. The value of **EXM** is determined by the block size (**BLS**) and whether the maximum data block number (**DSM**) exceeds 255. There are $EXM + 1$ extents per directory FCB.

FCB See File Control Block.

file Collection of characters, instructions, or data that can be referenced by a unique identifier. Files are usually stored on various types of media, such as disk, or magnetic tape. A CP/M file is identified by a file specification and resides on disk as a collection of from zero to 65,536 records. Each record is 128 bytes and can contain either binary or ASCII data. Binary files contain bytes of data that can vary in value from 0H to 0FFH. ASCII files contain sequences of character codes delineated by a carriage return and line-feed combination; normally byte values range from 0H to 7FH. The directory maps the file as a series of physical blocks. Although files are defined as a sequence of consecutive logical records, these records can not reside in consecutive sectors on the disk. See also block, directory, extent, record, and sector.

File Control Block (FCB) Structure used for accessing files on disk. Contains the drive, filename, filetype, and other information describing a file to be accessed or created on the disk. A file control block consists of 36 consecutive bytes specified by the user for file I/O functions. FCB can also refer to a directory element in the directory portion of the allocated disk space. These contain the same first 32 bytes of the FCB, but lack the current record and random record number bytes.

filename Name assigned to a file. A filename can include a primary filename of one to eight characters; a filetype of zero to three characters. A period separates the primary filename from the filetype.

file specification Unique file identifier. A complete CP/M file specification includes a disk drive specification followed by a colon, *d:*, a primary filename of one to eight characters, a period, and a filetype of zero to three characters. For example, *b:example.tex* is a complete CP/M file specification.

filetype Extension to a filename. A filetype can be from zero to three characters and must be separated from the primary filename by a period. A filetype can tell something about the file. Some programs require that files to be processed have specific filetypes.

floppy disk Flexible magnetic disk used to store information. Floppy disks come in 5 1/4- and 8-inch diameters.

FSC Parameter in the *diskdef* macro library specifying the first physical sector number. This parameter is used to determine **SPT** and build **XLT**.

hard disk Rigid, platter-like, magnetic disk sealed in a container. A hard disk stores more information than a floppy disk.

hardware Physical components of a computer.

hexadecimal notation Notation for base 16 values using the decimal digits and letters A, B, C, D, E, and F to represent the 16 digits. Hexadecimal notation is often used to refer to binary numbers. A binary number can be easily expressed as a hexadecimal value by taking the bits in groups of 4, starting

with the least significant bit, and expressing each group as a hexadecimal digit, 0-F. Thus the bit value 1011 becomes 0BH and 10110101 becomes 0B5H.

hex file ASCII-printable representation of a command, machine language file.

hex file format Absolute output of ASM and MAC for the Intel 8080 is a hex format file, containing a sequence of absolute records that give a load address and byte values to be stored, starting at the load address.

HOME: BIOS entry point which sets the disk head of the currently selected drive to the track zero position.

host Physical characteristics of a hard disk drive in a system using the blocking and deblocking algorithm. The term, host, helps distinguish physical hardware characteristics from CP/M's logical characteristics. For example, CP/M sectors are always 128 bytes, although the host sector size can be a multiple of 128 bytes.

input Data going into the computer, usually from an operator typing at the terminal or by a program reading from the disk.

input/output See I/O

interface Object that allows two independent systems to communicate with each other, as an interface between hardware and software in a microcomputer.

I/O Abbreviation for input/output. Usually refers to input/output operations or routines handling the input and output of data in the computer system.

IOBYTE A one-byte field in page zero, currently at location 0003H, that can support a logical-to-physical device mapping for I/O. However, its implementation in your BIOS is purely optional and might or might not be supported in a given CP/M system. The IOBYTE is easily set using the command:

STAT logical device = physical device

The CP/M logical devices are CON:, RDR:, PUN:, and LST:; each of these can be assigned to one of four physical devices. The IOBYTE can be initialized by the BOOT entry point of the BIOS and interpreted by the BIOS I/O entry points CONST, CONIN, CONOUT, LIST, PUNCH, and READER. Depending on the setting of the IOBYTE, different I/O drivers can be selected by the BIOS. For example, setting LST:=TTY: might cause LIST output to be directed to a serial port, while setting LST:=LPT: causes LIST output to be directed to a parallel port.

K Abbreviation for kilobyte. See kilobyte.

keyword See command keyword.

kilobyte (K) 1024 bytes or 0400H bytes of memory. This is a standard unit of memory. For example, the Intel 8080 supports up to 64K of memory address space or 65,536 bytes. 1024 kilobytes equal one megabyte, or over one million bytes.

linker Utility program used to combine relocatable object modules into an absolute file ready for execution. For example, LINK-80

creates either a COM or PRL file from relocatable REL files, such as those produced by PL/I-80

LIST: A BIOS entry point to a routine that sends a character to the list device, usually a printer.

list device Device such as a printer onto which data can be listed or printed.

LISTST: BIOS entry point to a routine that returns the ready status of the list device.

loader Utility program that brings an absolute program image into memory ready for execution under the operating system, or a utility used to make such an image. For example, `LOAD` prepares an absolute `COM` file from the assembler hex file output that is ready to be executed under `CP/M`.

logged in Made known to the operating system, in reference to drives. A drive is logged in when it is selected by the user or an executing process. It remains selected or logged in until you change disks in a floppy disk drive or enter `CTRL-C` at the command level, or until a `BDOS` Function 0 is executed.

logical Representation of something that might or might not be the same in its actual physical form. For example, a hard disk can occupy one physical drive, yet you can divide the available storage on it to appear to the user as if it were in several different drives. These apparent drives are the logical drives.

logical sector See sector.

logical-to-physical sector translation table See `XLT`.

LSC: `diskdef` macro library parameter specifying the last physical sector number.

LST: Logical `CP/M` list device, usually a printer. The `CP/M` list device is an output-only device referenced through the `LIST` and `LISTST` entry points of the BIOS. The `STAT` command allows assignment of `LST:` to one of the physical devices: `TTY:`, `CRT:`, `LPT:`, or `UL1:`, provided these devices and the `IOBYTE` are implemented in the `LIST` and `LISTST` entry points of your `CP/M` BIOS module. The `CP/NET` command `NETWORK` allows assignment of `LST:` to a list device on a network master. For example,
`PIP LST:=TEST.SUB`
prints the file `TEST.SUB` on the list device.

macro assembler Assembler code translator providing macro processing facilities. Macro definitions allow groups of instructions to be stored and substituted in the source program as the macro names are encountered. Definitions and invocations can be nested and macro parameters can be formed to pass arbitrary strings of text to a specific macro for substitution during expansion.

megabyte Over one million bytes; 1024 kilobytes. See byte and kilobyte.

microprocessor Silicon chip that is the central processing unit (CPU) of the microcomputer. The Intel 8080 and the Zilog Z80 are microprocessors commonly used in `CP/M` systems.

MOVCPM image Memory image of the `CP/M` system created by `MOVCPM`. This image can be saved as a disk file using the `SAVE` command or placed on the system tracks using the `SYSGEN` command without specifying a source drive. This image varies, depending on the presence of a one-sector or two-sector boot. If the boot is less than 128 bytes (one sector), the boot begins at 0900H, the `CP/M` system at 0980H, and the BIOS at 1F80H. Otherwise, the boot is at 0900H, the `CP/M` system at 1000H, and the BIOS at 2000H. In a `CP/M` 1.4 system with a one-sector boot, the addresses are the same as for the `CP/M` 2 system – except that the BIOS begins at 1E80H instead of 1F80H.

MP/M Multi-Programming Monitor control program. A microcomputer operating system supporting multi-terminal access with multi-programming at each terminal.

multi-programming The capability of initiating and executing more than one program at a time. These programs, usually called processes, are time-shared, each receiving a slice of CPU time on a round-robin basis. See concurrency.

nibble One half of a byte, usually the high-order or low-order 4 bits in a byte.

OFF: Two-byte parameter in the disk parameter block at DPB + 13 bytes. This value specifies the number of reserved system tracks. The disk directory begins in the first sector of track OFF.

OFS: diskdef macro library parameter specifying the number of reserved system tracks. See OFF:.

operating system Collection of programs that supervises the execution of other programs and the management of computer resources. An operating system provides an orderly input/output environment between the computer and its peripheral devices. It enables user-written programs to execute safely. An operating system standardizes the use of computer resources for the programs running under it.

option One of many parameters that can be part of a command tail. Use options to specify additional conditions for a command's execution.

output Data that is sent to the console, disk, or printer.

page 256 consecutive bytes in memory beginning on a page boundary, whose base address is a multiple of 256 (100H) bytes. In hex notation, pages always begin at an address with a least significant byte of zero.

page relocatable program See PRL.

page zero Memory region between 0000H and 0100H used to hold critical system parameters. Page zero functions primarily as an interface region between user programs and the CP/M BDOS module. Note that in non-standard systems this region is the base page of the system and represents the first 256 bytes of memory used by the CP/M system and user programs running under it.

parameter Value in the command tail that provides additional information for the command. Technically, a parameter is a required element of a command.

peripheral devices Devices external to the CPU. For example, terminals, printers, and disk drives are common peripheral devices that are not part of the processor but are used in conjunction with it.

physical Characteristic of computer components, generally hardware, that actually exist. In programs, physical components can be represented by logical components.

primary filename First 8 characters of a filename. The primary filename is a unique name that helps the user identify the file contents. A primary filename contains one to eight characters and can include any letter or number and some special characters. The primary filename follows the optional drive specification and precedes the optional filetype.

PRL Page relocatable program. A page relocatable program is stored on disk with a PRL filetype. Page relocatable programs are easily relocated to any page boundary and thus are suitable for execution in a non-banked MP/M system.

program Series of coded instructions that performs specific tasks when executed by a computer. A program can be written in a processor-specific language or a high-level language that can be implemented on a number of different processors.

prompt Any characters displayed on the video screen to help the user decide what the next appropriate action is. A system prompt is a special prompt displayed by the operating system. The alphabetic character indicates the default drive. Some applications programs have their own special prompts. See CP/M prompt.

PUN Logical CP/M punch device. The punch device is an output-only device accessed through the **PUNCH** entry point of the BIOS. In certain implementations, **PUN:** can be a serial device such as a modem.

PUNCH: BIOS entry point to a routine that sends a character to the punch device.

RDR: Logical CP/M reader device. The reader device is an input-only device accessed through the **READER** entry point in the BIOS. See **PUN:**.

READ: Entry point in the BIOS to a routine that reads 128 bytes from the currently selected drive, track, and sector into the current DMA address.

READER: Entry point to a routine in the BIOS that reads the next character from the currently assigned reader device.

Read-Only (R/O) Attribute that can be assigned to a disk file or a disk drive. When assigned to a file, the Read-Only attribute allows you to read from that file but not write to it. When assigned to a drive, the Read-Only attribute allows you to read any file on the disk, but prevents you from adding a new file, erasing or changing a file, renaming a file, or writing on the disk. The **STAT** command can set a file or a drive to Read-Only. Every file and drive is either Read-Only or Read-Write. The default setting for drives and files is Read-Write, but an error in resetting the disk or changing media automatically sets the drive to Read-Only until the error is corrected. See also **ROM**.

Read-Write (R/W) Attribute that can be assigned to a disk file or a disk drive. The Read-Write attribute allows you to read from and write to a specific Read-Write file or to any file on a disk that is in a drive set to Read-Write. A file or drive can be set to either Read-Only or Read-Write.

record Group of bytes in a file. A physical record consists of 128 bytes and is the basic unit of data transfer between the operating system and the application program. A logical record might vary in length and is used to represent a unit of information. Two 64-byte employee records can be stored in one 128-byte physical record. Records are grouped together to form a file.

recursive-font procedure Code that can call itself during execution.

reentrant procedure Code that can be called by one process while another is already executing it. Thus, reentrant code can be shared between different users. Reentrant procedures must not be self-modifying; that is, they must be pure code and not contain data. The data for reentrant procedures can be kept in a separate data area or placed on the stack.

restart (RST) One-byte call instruction usually used during interrupt sequences and for debugger break pointing. There are eight restart locations, **RST 0** through **RST 7**, whose addresses are given by the product of 8 times the restart number.

R/O See Read-Only

ROM Read-Only Memory. This memory can be read but not written and so is suitable for code and pre-initialized data areas only.

RST See restart.

R/W See Read-Write.

sector In a CP/M system, a sector is always 128 consecutive bytes. A sector is the basic unit of data read and written on the disk by the BIOS. A sector can be one 128-byte record in a file or a sector of the directory. The BDOS always requests a logical sector number between 0 and (SPT - 1). This is typically translated into a physical sector by the BIOS entry point **SECTTRAN**. In some disk subsystems, the disk sector size is larger than 128 bytes, usually a power of two, such as 256, 512, 1024, or 2048 bytes.

These disk sectors are always referred to as host sectors in CP/M documentation and should not be confused with other references to sectors, in which cases the CP/M 128-byte sectors should be assumed. When the host sector size is larger than 128 bytes, host sectors must be buffered in memory and the 128-byte CP/M sectors must be blocked and deblocked from them. This can be done by adding an additional module, the blocking and deblocking algorithm, between the BIOS disk I/O routines and the actual disk I/O.

SECTTRAN: Entry point to a routine in the BIOS that performs logical-to-physical sector translation for the BDOS.

SELDSK: Entry point to a routine in the BIOS that sets the currently selected drive.

SETDMA: Entry point to a routine in the BIOS that sets the currently selected DMA address. The DMA address is the address of a 128-byte buffer region in memory that is used to transfer data to and from the disk in subsequent reads and writes.

SETSEC: Entry point to a routine in the BIOS that sets the currently selected sector.

SETTRK: Entry point to a routine in the BIOS that sets the currently selected track.

skew factor Factor that defines the logical-to-physical sector number translation in **XLT**. Logical sector numbers are used by the BDOS and range between 0 and (SPT – 1). Data is written in consecutive logical 128-byte sectors grouped in data blocks. The number of sectors per block is given by $\frac{BLS}{128}$. Physical sectors on the disk media are also numbered consecutively. If the physical sector size is also 128 bytes, a one-to-one relationship exists between logical and physical sectors. The logical-to-physical translation table (**XLT**) maps this relationship, and a skew factor is typically used in generating the table entries. For instance, if the skew factor is 6, **XLT** will be:

Logical:	0	1	2	3	4	5	6	...	25
Physical:	1	7	13	19	25	5	11	...	22

The *skew factor* allows time for program processing without missing the next sector. Otherwise, the system must wait for an entire disk revolution before reading the next logical sector. The skew factor can be varied, depending on hardware speed and application processing overhead. Note that no sector translation is done when the physical sectors are larger than 128 bytes, as sector deblocking is done in this case. See also sector, **SKF**, and **XLT**.

SKF: A diskdef macro library parameter specifying the skew factor to be used in building **XLT**. If **SKF** is zero, no translation table is generated and the **XLT** bytes in the **DPH** will be 0000H.

software Programs that contain machine-readable instructions, as opposed to hardware, which is the actual physical components of a computer.

source file ASCII text file usually created with an editor that is an input file to a system program, such as a language translator or text formatter.

SP Stack pointer. See stack.

spooling Process of accumulating printer output in a file while the printer is busy. The file is printed when the printer becomes free; a program does not have to wait for the slow printing process.

SPT: See sectors per track.

stack Reserved area of memory where the processor saves the return address when a call instruction is received. When a return instruction is encountered, the processor restores the current address on the

stack to the program counter. Data such as the contents of the registers can also be saved on the stack. The push instruction places data on the stack and the pop instruction removes it. An item is pushed onto the stack by decrementing the stack pointer (SP) by 2 and writing the item at the SP address. In other words, the stack grows downward in memory.

syntax Format for entering a given command.

SYS: See system attribute.

SYSGEN image Memory image of the CP/M system created by SYSGEN when a destination drive is not specified. This is the same as the MOVCPM image that can be read by SYSGEN if a source drive is not specified. See MOVCPM image.

system attribute (SYS) File attribute. You can give a file the system attribute by using the SYS option in the STAT command or by using the set file attributes function, BDOS Function 12. A file with the SYS attribute is not displayed in response to a DIR command. If you give a file with user number 0 the SYS attribute, you can read and execute that file from any user number on the same drive. Use this feature to make your commonly used programs available under any user number.

system prompt Symbol displayed by the operating system indicating that the system is ready to receive input. See prompt and CP/M prompt.

system tracks Tracks reserved on the disk for the CP/M system. The number of system tracks is specified by the parameter OFF in the disk parameter block (DPB). The system tracks for a drive always precede its data tracks. The command SYSGEN copies the CP/M system from the system tracks to memory, and vice versa. The standard SYSGEN utility copies 26 sectors from track 0 and 26 sectors from track 1. When the system tracks contain additional sectors or tracks to be copied, a customized SYSGEN must be used.

terminal See console.

TPA Transient Program Area. Area in memory where user programs run and store data. This area is a region of memory beginning at 0100H and extending to the base of the CP/M system in high memory. The first module of the CP/M system is the CCP, which can be overwritten by a user program. If so, the TPA is extended to the base of the CP/M BDOS module. If the CCP is overwritten, the user program must terminate with either a system reset (Function 0) call or a jump to location zero in page zero. The address of the base of the CP/M BDOS is stored in location 0006H in page zero least significant byte first.

track Data on the disk media is accessed by combination of track and sector numbers. Tracks form concentric rings on the disk; the standard IBM single-density disks have 77 tracks. Each track consists of a fixed number of numbered sectors. Tracks are numbered from zero to one less than the number of tracks on the disk.

Transient Program Area See TPA.

upward compatible Term meaning that a program created for the previously released operating system, or compiler, runs under the newly released version of the same operating system.

USER Term used in CP/M and MP/M systems to distinguish distinct regions of the directory.

user number Number assigned to files in the disk directory so that different users need only deal with their own files and have their own directories, even though they are all working from the same disk. In CP/M, files can be divided into 16 user groups.

utility Tool. Program that enables the user to perform certain operations, such as copying files, erasing files, and editing files. The utilities are created for the convenience of programmers and users.

vector Location in memory. An entry point into the operating system used for making system calls or interrupt handling.

warm start Program termination by a jump to the warm start vector at location 0000H, a system reset (BDOS Function 0), or a CTRL-C typed at the keyboard. A warm start re-initializes the disk subsystem and returns control to the CP/M operating system at the CCP level. The warm start vector is simply a jump to the WBOOT entry point in the BIOS.

WBOOT: Entry point to a routine in the BIOS used when a warm start occurs. A warm start is performed when a user program branches to location 0000H, when the CPU is reset from the front panel, or when the user types CTRL-C. The CCP and BDOS are reloaded from the system tracks of drive A.

wildcard characters Special characters that match certain specified items. In CP/M there are two wildcard characters: ? and *. The ? can be substituted for any single character in a filename, and the * can be substituted for the primary filename, the filetype, or both. By placing wildcard characters in filenames, the user creates an ambiguous filename and can quickly reference one or more files.

word 16-bit or two-byte value, such as an address value. Although the Intel 8080 is an 8-bit CPU, addresses occupy two bytes and are called word values.

WRITE: Entry point to a routine in the BIOS that writes the record at the currently selected DMA address to the currently selected drive, track, and sector.

XLT Logical-to-physical sector translation table located in the BIOS. SECTTRAN uses **XLT** to perform logical-to-physical sector number translation. **XLT** also refers to the two-byte address in the disk parameter header at DPBASE + 0. If this parameter is zero, no sector translation takes place. Otherwise this parameter is the address of the translation table.

ZERO PAGE See Page zero.

I. CP/M Error Messages

Messages come from several different sources. CP/M displays error messages when there are errors in calls to the Basic Disk Operating System (BDOS). CP/M also displays messages when there are errors in command lines. Each utility supplied with CP/M has its own set of messages. The following lists CP/M messages and utility messages. One might see messages other than those listed here if one is running an application program. Check the application program's documentation for explanations of those messages.

Message	Meaning
?	DDT. This message has four possible meanings: • DDT does not understand the assembly language instruction. • The file cannot be opened. • A checksum error occurred in a HEX file. • The assembler/disassembler was overlaid.
ABORTED	
	PIP. You stopped a PIP operation by pressing a key.
ASM Error Messages	
D	Data error: data statement element cannot be placed in specified data area.
E	Expression error: expression cannot be evaluated during assembly.
L	Label error: label cannot appear in this context (might be duplicate label).
N	Not implemented: unimplemented features, such as macros, are trapped.
O	Overflow: expression is too complex to evaluate.
P	Phase error: label value changes on two passes through assembly.
R	Register error: the value specified as a register is incompatible with the code.
S	Syntax error: improperly formed expression.
U	Undefined label: label used does not exist.
V	Value error: improperly formed operand encountered in an expression.
BAD DELIMITER	
	STAT. Check command line for typing errors.

Message	Meaning
Bad Load	CCP error message, or SAVE error message.
BDOS Err on <i>d</i> :	Basic Disk Operating System error on the designated drive: CP/M replaces <i>d</i> : with the drive specification of the drive where the error occurred. This message is followed by one of the four phrases in the situations described below.
BDOS Err on <i>d</i> : Bad Sector	This message appears when CP/M finds no disk in the drive, when the disk is improperly formatted, when the drive latch is open, or when power to the drive is off. Check for one of these situations and try again. This could also indicate a hardware problem or a worn or improperly formatted disk. Press ^C to terminate the program and return to CP/M, or press RETURN to ignore the error.
BDOS Err on <i>d</i> : File R/O	You tried to erase, rename, or set file attributes on a Read-Only file. The file should first be set to Read-Write (R/W) with the command: STAT <i>filespec</i> \$R/W.
BDOS Err on <i>d</i> : R/O	Drive has been assigned Read-Only status with a STAT command, or the disk in the drive has been changed without being initialized with a ^C. CP/M terminates the current program as soon as you press any key.
BDOS Err on <i>d</i> : Select	CP/M received a command line specifying a nonexistent drive. CP/M terminates the current program as soon as you press any key. Press RETURN or CTRL-C to recover.
Break "x" at C	ED. "x" is one of the symbols described below and c is the command letter being executed when the error occurred.
#	Search failure. ED cannot find the string specified in an F, S, or N command.
?	Unrecognized command letter c. ED does not recognize the indicated command letter, or an E, H, Q, or O command is not alone on its command line.

Message	Meaning
O	The file specified in an R command cannot be found.
Break "x" at C (continued)	
>	Buffer full. ED cannot put any more characters in the memory buffer, or the string specified in an F, N, or S command is too long.
E	Command aborted. A keystroke at the console aborted command execution.
F	Disk or directory full. This error is followed by either the disk or directory full message. Refer to the recovery procedures listed under these messages.
CANNOT CLOSE DESTINATION FILE <i>filespec</i>	
	PIP. An output file cannot be closed. You should take appropriate action after checking to see if the correct disk is in the drive and that the disk is not write-protected.
Cannot close, R/O	
CANNOT CLOSE FILES	
	CP/M cannot write to the file. This usually occurs because the disk is write-protected.
	ASM. An output file cannot be closed. This is a fatal error that terminates ASM execution. Check to see that the disk is in the drive, and that the disk is not write-protected.
	DDT. The disk file written by a W command cannot be closed. This is a fatal error that terminates DDT execution. Check if the correct disk is in the drive and that the disk is not write-protected.
	SUBMIT. This error can occur during SUBMIT file processing. Check if the correct system disk is in the A drive and that the disk is not write-protected. The SUBMIT job can be restarted after rebooting CP/M.
CANNOT READ	
	PIP. PIP cannot read the specified source. Reader cannot be implemented.
CANNOT WRITE	
	PIP. The destination specified in the PIP command is illegal. You probably specified an input device as a destination.

Message	Meaning
Checksum Error	PIP. A HEX record checksum error was encountered. The HEX record that produced the error must be corrected, probably by recreating the HEX file.
CHECKSUM ERROR LOAD ADDRESS	LOAD. File contains incorrect data. Regenerate HEX file from the source.
Command Buffer Overflow	SUBMIT. The SUBMIT buffer allows up to 2048 characters in the input file.
Command too long	SUBMIT. A command in the SUBMIT file cannot exceed 125 characters.
CORRECT ERROR, TYPE RETURN OR CTRL-Z	PIP. A HEX record checksum was encountered during the transfer of a HEX file. The HEX file with the checksum error should be corrected, probably by recreating the HEX file.
DESTINATION IS R/O, DELETE (Y/N)?	PIP. The destination file specified in a PIP command already exists and it is Read-Only. If you type Y, the destination file is deleted before the file copy is done.
Directory Full	ED. There is not enough directory space for the file being written to the destination disk. You can use the <code>OXfilespec</code> command to erase any unnecessary files on the disk without leaving the editor. SUBMIT. There is not enough directory space to write the \$\$\$SUB file used for processing SUBMITS. Erase some files or select a new disk and retry.
Disk Full	ED. There is not enough disk space for the output file. This error can occur on the W, E, H, or X commands. If it occurs with X command, you can repeat the command prefixing the filename with a different drive.
DISK READ ERROR	

Message	Meaning
	PIP. The input disk file specified in a PIP command cannot be read properly. This is usually the result of an unexpected end-of-file. Correct the problem in your file.
DISK WRITE ERROR	
	DDT. A disk write operation cannot be successfully performed during a W command, probably due to a full disk. You should either erase some unnecessary files or get another disk with more space.
	PIP. A disk write operation cannot be successfully performed during a PIP command, probably due to a full disk. You should either erase some unnecessary files or get another disk with more space and execute PIP again.
	SUBMIT. The SUBMIT program cannot write the \$\$\$SUB file to the disk. Erase some files, or select a new disk and try again.
ERROR: BAD PARAMETER	
	PIP. You entered an illegal parameter in a PIP command. Retype the entry correctly.
ERROR: CANNOT OPEN SOURCE, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Displayed if LOAD cannot find the specified file or if no filename is specified.
ERROR: CANNOT CLOSE FILE, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Caused by an error code returned by a BDOS function call. Disk might be write-protected.
ERROR: CANNOT OPEN SOURCE, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Cannot find source file. Check disk directory.
ERROR: DISK READ, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Caused by an error code returned by a BDOS function call.
ERROR: DISK WRITE, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Destination disk is full.
ERROR: INVERTED LOAD ADDRESS, LOAD ADDRESS <i>hhhh</i>	
	LOAD. The address of a record was too far from the address of the previously-processed record. This is an internal limitation of LOAD, but it can be circumvented. Use DDT to

Message	Meaning
	read the HEX file into memory, then use a SAVE command to store the memory image file on disk.
ERROR: NO MORE DIRECTORY SPACE, LOAD ADDRESS <i>hhhh</i>	
	LOAD. Disk directory is full.
Error on line <i>nnn message</i>	
	SUBMIT. The SUBMIT program displays its messages in the format shown above, where <i>nnn</i> represents the line number of the SUBMIT file. Refer to the <i>message</i> following the line number.
FILE ERROR	
	ED. Disk or directory is full, and ED cannot write anything more on the disk. This is a fatal error, so make sure there is enough space on the disk to hold a second copy of the file before invoking ED.
FILE EXISTS	
	You have asked CP/M to create or rename a file using a file specification that is already assigned to another file. Either delete the existing file or use another file specification.
	REN. The new name specified is the name of a file that already exists. You cannot rename a file with the name of an existing file. If you want to replace an existing file with a newer version of the same file, either rename or erase the existing file, or use the PIP utility.
File exists, erase it	
	ED. The destination filename already exists when you are placing the destination file on a different disk than the source. It should be erased or another disk selected to receive the output file.
** FILE IS READ/ONLY **	
	ED. The file specified in the command to invoke ED has the Read-Only attribute. ED can read the file so that the user can examine it, but ED cannot change a Read-Only file.
File Not Found	
	CP/M cannot find the specified file. Check that you have entered the correct drive specification or that you have the correct disk in the drive.

Message	Meaning
File Not Found (continued)	ED. ED cannot find the specified file. Check that you have entered the correct drive specification or that you have the correct disk in the drive. STAT. STAT cannot find the specified file. The message might appear if you omit the drive specification. Check to see if the correct disk is in the drive. FILE NOT FOUND PIP. An input file that you have specified does not exist.
Filename required	ED. You typed the ED command without a filename. Reenter the ED command followed by the name of the file you want to edit or create.
<i>hhhh??=dd</i>	DDT. The ?? indicates DDT does not know how to represent the hexadecimal value <i>dd</i> encountered at address <i>hhhh</i> in 8080 assembly language. <i>dd</i> is not an 8080 machine instruction opcode.
Insufficient Memory	DDT. There is not enough memory to load the file specified in an R or E command.
Invalid Assignment	STAT. You specified an invalid drive or file assignment, or misspelled a device name. This error message might be followed by a list of the valid file assignments that can follow a filename. If an invalid drive assignment was attempted the message Use: d:=RO is displayed, showing the proper syntax for drive assignments.
Invalid control character	SUBMIT. The only valid control characters in the SUBMIT files of the type SUB are ^A through ^Z. Note that in a SUBMIT file the control character is represented by typing the circumflex, ^, not by pressing the control key.
INVALID DIGIT	PIP. An invalid HEX digit has been encountered while reading a HEX file. The HEX file with the invalid HEX digit should be corrected, probably by recreating the HEX file.

Message	Meaning
Invalid Disk Assignment	STAT. Might appear if you follow the drive specification with anything except =R/O.
INVALID DISK SELECT	CP/M received a command line specifying a nonexistent drive, or the disk in the drive is improperly formatted. CP/M terminates the current program as soon as you press any key.
INVALID DRIVE NAME (Use A, B, C, or D)	SYSGEN. SYSGEN recognizes only drives A, B, C, and D as valid destinations for system generation.
Invalid File Indicator	STAT. Appears if you do not specify RO, RW, DIR, or SYS.
INVALID FORMAT	PIP. The format of your PIP command is illegal. See the description of the PIP command.
INVALID HEX DIGIT	LOAD. File contains incorrect HEX digit.
INVALID MEMORY SIZE	MOVCPM. Specify a value less than 64K or your computer's actual memory size.
INVALID SEPARATOR	PIP. You have placed an invalid character for a separator between two input filenames.
INVALID USER NUMBER	PIP. You have specified a user number greater than 15. User numbers are in the range 0 to 15.
<i>n?</i>	

Message	Meaning
	USER. You specified a number greater than fifteen for a user area number. For example, if you type <code>USER 18<cr></code> , the screen displays 18?.
NO DIRECTORY SPACE	ASM. The disk directory is full. Erase some files to make room for PRN and HEX files. The directory can usually hold only 64 filenames.
NO DIRECTORY SPACE <i>filespec</i>	PIP. There is not enough directory space for the output file. You should either erase some unnecessary files or get another disk with more directory space and execute PIP again.
NO FILE	DIR, ERA, REN, PIP. CP/M cannot find the specified file, or no files exist. ASM. The indicated source or include file cannot be found on the indicated drive. DDT. The file specified in an R or E command cannot be found on the disk.
NO INPUT FILE PRESENT ON DISK	DUMP. The file you requested does not exist.
No memory	There is not enough (buffer?) memory available for loading the program specified.
NO SOURCE FILE ON DISK	SYSGEN. SYSGEN cannot find CP/M either in CPMxx.com form or on the system tracks of the source disk.
NO SOURCE FILE PRESENT	ASM. The assembler cannot find the file you specified. Either you mistyped the file specification in your command line, or the filetype is not ASM.
NO SPACE	SAVE. Too many files are already on the disk, or no room is left on the disk to save the information.

Message	Meaning
No SUB file Present	
	SUBMIT. For SUBMIT to operate properly, you must create a file with filetype of SUB. The SUB file contains usual CP/M commands. Use one command per line.
NOT A CHARACTER SOURCE	
	PIP. The source specified in your PIP command is illegal. You have probably specified an output device as a source.
** NOT DELETED **	
	PIP. PIP did not delete the file, which might have had the R/O attribute.
NOT FOUND	
	PIP. PIP cannot find the specified file.
OUTPUT FILE WRITE ERROR	
	ASM. You specified a write-protected disk as the destination for the PRN and HEX files, or the disk has no space left. Correct the problem before assembling your program.
Parameter error	
	SUBMIT. Within the SUBMIT file of type SUB, valid parameters are \$0 through \$9.
PARAMETER ERROR, TYPE RETURN TO IGNORE	
	SYSGEN. If you press RETURN, SYSGEN proceeds without processing the invalid parameter.
QUIT NOT FOUND	
	PIP. The string argument to a Q parameter was not found in your input file.
Read error	
	TYPE. An error occurred when reading the file specified in the type command. Check the disk and try again. The STAT <i>filespec</i> command can diagnose trouble.
READER STOPPING	
	PIP. Reader operation interrupted.

Message	Meaning
Record Too Long	PIP. PIP cannot process a record longer than 128 bytes.
Requires CP/M 2.0 or later	XSUB. XSUB requires the facilities of CP/M 2.0 or newer version.
Requires CP/M 2.0 or new for operation	PIP. This version of PIP requires the facilities of CP/M 2.0 or newer version.
START NOT FOUND	PIP. The string argument to an S parameter cannot be found in the source file.
SOURCE FILE INCOMPLETE	SYSGEN. SYSGEN cannot use your CP/M source file.
SOURCE FILE NAME ERROR	ASM. When you assemble a file, you cannot use the wildcard characters * and ? in the filename. Only one file can be assembled at a time.
SOURCE FILE READ ERROR	ASM. The assembler cannot understand the information in the file containing the assembly-language program. Portions of another file might have been written over your assembly-language file, or information was not properly saved on the disk. Use the TYPE command to locate the error. Assembly-language files contain the letters, symbols, and numbers that appear on your keyboard. If your screen displays unrecognizable output or behaves strangely, you have found where computer instructions have crept into your file.
SYNCHRONIZATION ERROR	MOVCPM. The MOVCPM utility is being used with the wrong CP/M system.
"SYSTEM" FILE NOT ACCESSIBLE	You tried to access a file set to SYS with the STAT command.
** TOO MANY FILES **	

Message	Meaning
	STAT. There is not enough memory for STAT to sort the files specified, or more than 512 files were specified.
UNEXPECTED END OF HEX FILE	
	PIP. An end-of-file was encountered prior to a termination HEX record. The HEX file without a termination record should be corrected, probably by recreating the HEX file.
Unrecognized Destination	
	PIP. Check command line for valid destination.
Use: STAT d:=RO	
	STAT. An invalid STAT drive command was given. The only valid drive assignment in STAT is STAT d:=RO.
VERIFY ERROR	
	PIP. When copying with the V option, PIP found a difference when rereading the data just written and comparing it to the data in its memory buffer. Usually this indicates a failure of either the destination disk or drive.
WRONG CP/M VERSION (REQUIRES 2.0)	
XSUB ACTIVE	
	SUBMIT. XSUB has been invoked.
XSUB ALREADY PRESENT	
	SUBMIT. XSUB is already active in memory.
<i>Your input?</i>	
	If CP/M cannot find the command you specified, it returns the command name you entered followed by a question mark. Check that you have typed the command line correctly, or that the command you requested exists as a .COM file on the default or specified disk.

Table 9-46: CP/M Error Messages

Notes on the Zcim Project Edition

The *Zcim Project* goal is to improve the accessibility of legacy Z80 CP/M era content. As part of the project, Jim Burlingame (*jb@samplx.org*) created a *Typst* version of this manual.

You can find out more about the project at its site z80cim.org. The sources of this document are available on GitHub.

The source material is from the Tim Olmstead Memorial Digital Research CP/M Library

- Bondwell CP/M Operating System Manual
- CP/M 2.2 MANUAL
- CP/M 2.2 PDF MANUAL
- CP/M 2.2 TEX Manual
- DDT - The CP/M Debugger
- ED - The CP/M Editor
- The CPM Assembler in PDF format.

The contents of the manual were edited using the *Typst.app* site.

The figures for Section 2 (*The CP/M Editor*) were recreated as *svg* files using Inkscape.

License

The source documentation is under a license granted by the owner of the Digital Research intellectual property in an email recreated at <http://cpm.z80.de/license.html>.

The *Zcim Project* edition is under the Creative Commons Attribution 4.0 International license. **CC BY 4.0**.